

STOCHASTIC REALIZATION, FINITE QUOTIENTS, AND UNIFORM ORBIT GEOMETRY

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ABSTRACT. This complete research edition contains the structural and quantitative stochastic-realization theory and all retained mathematical refinements. Same-width, same-error stationarization under eventual approximate returns and physical-equivariant purification give equality-inclusive finite quotient criteria. The least orbit dimension supplies a uniform cap exponent for all unit directions. A fixed rational qubit experiment has an exponential exact profile, robust under exponentially small positive error. Under a full action spectral gap, spectral entropy removes the logarithmic loss in the noisy converse; least-orbit and target-orbit dimensions give sharp matching laws when they agree. An additional rational Bloch experiment has explicit spectral constants and exact profile 5^t . Its positive upper realization is constructed by a uniform rational compiler. The focused structural and quantitative articles are independently complete; this edition preserves the combined development, including arithmetic, profinite and existential higher-dimensional examples. Repeatable probes additionally give a finite-response-quotient classification for uniform causal process error and a constructive rational approximation under adaptive control. Orthogonal covariance removes the quantitative return assumption and gives joint subexponential-accuracy laws. A classical tail-capacity argument sharpens the process minimum to every error below one. The full spherical Ramanujan hypothesis further gives the entire exponential accuracy rate. A separate uniform one-pass theorem proves optimal bit-space for rational orthogonal actions with legal unit-ball outputs under the stated action-gap and cap hypotheses, including the six-rotation qubit experiment, by an explicit integer algorithm. No independent analytic pipeline closure is asserted. Positive fixed-denominator matrix rounding additionally gives finite-bit simulation of disturbing rational quantum instruments under adaptive control, and sharp density-matrix space laws in every fixed dimension under explicit spectral and cap assumptions. Reference-stable rational Choi rounding further gives explicit instrument diamond-error bounds and common adaptive-tester guarantees with arbitrary entangled references. Variable-description numerical streaming charges matrix dimension and rational coefficient lengths; stopped posterior guarantees remain probability weighted. Intrinsic instrument coordinates additionally give the exact high-resolution description exponent. On every fixed strictly positive Choi body, a reusable rational description accurate against arbitrary common adaptive testers for N calls has optimal length $\frac{1}{2}d^2(mn^2 - 1)\log_2 N + O(1)$ at fixed error and dimensions. The joint description law is now uniform in horizon and accuracy. For preparation instruments a rank-stratified rational factor code also covers all permitted singular boundaries and preserves zero outcomes. A further boundary family retains unitary data and appends a rank-bounded preparation. Its joint description coefficient separates linear-horizon coherent directions from square-root-horizon preparation directions. A rational unitary atlas supplies legal centres, and an exact retraction settles the allowed-centre question for preparation targets. Fixed-readout instruments with input-dependent outcomes additionally have a rank-boundary joint description law. A common-estimator multiplicity argument extends the half-dimensional description laws to every fixed submaximal unhalved error cap. These statements retain the distinct coherent, description and numerical-streaming resource conventions.

Complete research edition. The focused quantitative and structural articles are independently complete; this edition retains the whole active mathematical development.

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The principal additional conclusions include Theorem 35.2 (rational instrument rounding), Theorem 35.4 (adaptive reference stability), Theorem 36.3 (variable-description streaming), and Proposition 35.5 (stopped posterior control), as well as Theorem 30.1 (uniform numerical streaming space), Corollary 30.6 (rational orthogonal actions), Theorem 29.5 (exponential-accuracy crossover), Theorem 16.2 (return-free occupation), Theorem 16.5 (joint accuracy law), Theorem 3.4 (full-error process minimum), Theorem 28.1 (six-command numerical example), Theorem 15.5 (synchronized occupation), Theorem 27.2 (effective spectral example), Theorem 24.1 (all-direction orbit caps), Theorem 25.2 (explicit rational robust separation), Theorem 26.2 (constructed finite tables), Theorem 2.2 (repeatable-observation classification), and Theorem 46.1 (adaptive path-law construction).

1. INTRODUCTION

A finite stochastic register has a continuum of probability distributions. Its number of labels is therefore not determined by the cardinality of the numerical orbit it realizes. Allowing an external clock creates a further difficulty: the command rows may change at every position, and the entire realization may be redesigned at each horizon. The central question is whether these freedoms change the least width needed to preserve a fixed numerical experiment with a prescribed uniform error.

Our first theorem identifies the limiting clocked width with the stationary minimum, without a loss of labels or accuracy. The synchronization hypothesis is eventual approximation of the physical identity at every sufficiently large length. Under it, arbitrarily many cuts of width at most k force a stationary realization on k labels, even when the intervening registers are unrestricted. The proof selects approximately homogeneous composite transition tables by finite Ramsey theory. It applies to irreversible topological monoids as well as groups. It does not assume that a physical return acts trivially on the hidden register.

For compact groups a second reduction makes the stationary realization physically equivariant. Transition randomness can be removed, and hidden permutations above the physical identity can be averaged out, without increasing width or uniform error. The stationary minimum is therefore intrinsic to a finite continuous group action with randomized initialization. Combining the two reductions gives an equality-inclusive finite-quotient criterion for clocked boundedness, including positive critical radii on groups with infinitely many components. A moving-frame construction gives the exact limiting minimum in each return-length phase, again without additional labels.

1.1. The interface and its two resources. Unless a section explicitly specifies a different physical space, H is a compact Hausdorff group and $\mathcal{A}$ is a finite nonempty alphabet whose letters $u_a \in H$ generate a dense subgroup. The execution convention is

$$u_{a_1 \dots a_n} = u_{a_n} \cdots u_{a_1}.$$

There are finite nonempty seed and query sets $\mathcal{S}, \mathcal{J}$ and continuous maps $F_{s,j} : H \rightarrow Y_j$. In the structural results Y_j is a nonempty norm-compact convex subset of a real Banach space. The localization and algebraic sections specify finite-dimensional output spaces. For a categorical law, $Y_j = \Delta_{M_j}$ and the norm on differences is $\frac{1}{2} \|\cdot\|_1$, namely total variation. A prescribed joint law is one query; incompatible measurements are not assigned an artificial joint law.

Definition 1.1 (Stationary realization). A realization on k labels consists of initialization rows $\alpha_s \in \Delta_k$, row-stochastic matrices T_a of order k , and decoder values $D_j(i) \in Y_j$, all independent of time and word length. Its error is

$$\sup_{s,j,w \in \mathcal{A}^*} \|\alpha_s T_{a_1} \cdots T_{a_n} D_j - F_{s,j}(u_w)\|_j.$$

The empty word is included. Let M_ε be the least k at error at most ε , or ∞ when none exists. A permutation realization has permutation command matrices; its initialization need not be deterministic.

The clocked resource $W_{N,\varepsilon}$ instead minimizes the maximum size of the registers $S_0, \dots, S_N$ at a fixed horizon, with arbitrary row-stochastic transitions depending on time and horizon, and a terminal query decoder. Definition 7.1 states the model formally. Every advertised persistent label counts, including unused padding. Private information retained for future commands is part of the current label. The original width model does not charge the external clock, real table descriptions, their construction, exact arithmetic, or exact atomic sampling. Section 21 gives separate finite-description and fair-random-bit bounds; it does not reinterpret those resources as free computational memory. A numerical decoder vector is not an additional sampled-answer register.

For an open normal subgroup $U \leq H$ put

$$(1.1) \quad r(U) = \max_{s,j,h \in H} \text{rad}_{Y_j}(F_{s,j}(hU)), \quad \text{rad}_Y(A) = \min_{y \in Y} \sup_{x \in A} \|x - y\|.$$

The coset images are compact by continuity. The radius depends continuously on the translate, and compactness gives the displayed extrema. The restriction of the center to the legal output set is essential. The component radius R_0 uses $U = H^\circ$, whether or not this subgroup is open; it is the infimum of the finite-quotient radii. At the critical error, the relevant issue is attainment of that infimum, not a limit from larger errors.

1.2. The principal conclusions. Theorem 4.2 proves $W_{N,\varepsilon} \rightarrow M_\varepsilon$ under eventual approximate returns. Finite limits are eventually equal, and every $k < M_\varepsilon$ has a uniform narrow-cut occupation bound. Theorems 6.1 and 6.3 then give the intrinsic group-action minimum and its finite-quotient existence criterion. Theorem 6.4 treats each length phase by a time-dependent change of physical frame. Table 1 separates these quantifiers and hypotheses.

The quantitative consequences have additional hypotheses. For a specified Diophantine planar alphabet the full-subcritical matching polynomial law is retained. A fixed 2-adic interface has linear exact width and bounded width at every positive error. For a spherical interface in dimension d with a spectral-gap alphabet, Theorem 13.1 gives a block estimate; Theorem 15.5 strengthens it to the matching order $N^{(d-1)/2}$. A rational alphabet generating $\text{SO}(3)$ supplies a genuinely noncommuting example. Its spectral gap is imported from Bourgain–Gamburd, not inferred from a finite numerical test. Theorem 21.1 gives an $\exists\mathbb{R}$ -complete finite input problem, while Theorem 21.3 prices dyadic descriptions and fair random bits.

The quantitative theory also extends beyond spheres. Theorem 14.3 separates a small-ball exponent uniform over all unit feature directions from the dimension of the target orbit. For conjugation on traceless Hermitian matrices a spectral split gives the uniform exponent $2(d-1)$, including directions with repeated eigenvalues. The pure-state orbit has the same dimension. Theorem 17.3 therefore gives nearly matching width exponents $d-1$ for full legal density-matrix outputs in every dimension. Its exact upper construction stays inside the positive cone, rather than replacing the legal output set by an ambient Euclidean ball.

At an extreme target, the exact answer can be very different. Theorem 18.1 identifies the entire optimal exact width profile with the reachable orbit cardinalities. Theorem A.2 applies it to one fixed five-letter algebraic dense free alphabet in $SU(d)$: exact pure-state width is $2 \cdot 3^N - 1$, while each fixed positive subcritical error has the polynomial bounds just described. The specified seed is transcendental, not a rational-input instance. These assertions concern a full matrix-valued terminal interface, not adaptive observations or the zero-error endpoint of a single scalar readout.

The compact nonnegative-matrix group structure used in purification is classical, and is separated from the external numerical error constraints throughout Section 23. Neutral-letter

collapse for advice automata is also a relevant predecessor to stationarization. The additional assertions proved here concern the same stochastic label bound and the same closed numerical error balls, arbitrary intervening widths, and approximate rather than exact returns. Neither classical theorem is relabeled as a new result.

TABLE 1. Resources, hypotheses, and conclusions. All widths count persistent labels; all errors are worst-word numerical errors.

Object	Hypotheses	Conclusion
M_ε	Compact H ; finite dense alphabet; compact convex outputs; no returns needed	Least finite continuous H -action with randomized initialization (Theorem 6.1)
$r(U)$	U open normal; arbitrary number of components	$M_\varepsilon < \infty$ iff some $r(U) \leq \varepsilon$, equality included (Theorem 5.5)
$W_{N,\varepsilon}$	Eventual approximate returns; even an irreversible topological monoid	$W_{N,\varepsilon} \rightarrow M_\varepsilon$; finite limits eventually equal; all-profile occupation (Theorem 4.2)
R_0	Compact group and eventual approximate returns	$R_0 = \inf_U r(U)$; critical boundedness iff the infimum is attained (Theorem 6.3)
$M_{r,\varepsilon}$	Block length p , phase subgroup J ; block approximate returns	$W_{N,\varepsilon} \rightarrow M_{r,\varepsilon}$ for $N \equiv r$; no extra phase labels (Theorem 6.4)
ε_r	Finitely many components; nonempty exact-return language	Component radius in the reachable phase; earlier localized occupation proof (Theorem 9.3)
Exact boundary	Cofinite exact returns; finite-dimensional outputs	Finite translate rank yields a finite observable quotient (Theorem 10.5)
Quantitative width	Separate arithmetic, profinite, or uniform-orbit spectral-gap hypotheses	Planar and dyadic laws; sharp spherical and matrix-orbit power laws (Theorems 15.5, 17.3)
Extreme outputs	Affine group action; target orbit consists of legal extreme points	Exact profile equals orbit cardinalities; free example $2 \cdot 3^N - 1$ (Theorems 18.1, A.2)
Finite encodings	Explicit finite group and rational categorical tables	$\exists \mathbb{R}$ -complete stationary decision; dyadic precision and random-bit budgets (Theorems 21.1, 21.3)

2. REPEATABLE OBSERVATIONS AND CAUSAL REALIZATION

Terminal numerical accuracy and accuracy of a generated observation process are different requirements. Repeating a nondisturbing observation can separate arbitrarily close, but distinct, numerical responses. We now formulate this distinction within the same finite-label resource model.

Let H be a compact Hausdorff group, let $\mathcal{A}$ be a finite nonempty command alphabet generating a dense subgroup, and let $\mathcal{S}$ and $\mathcal{J}$ be finite nonempty sets. For each s, j fix a continuous law

$$p_{s,j} : H \longrightarrow \Delta_{M_j}.$$

The physical state starts at $g = e$. A command a replaces g by $u_a g$ and emits a fixed null symbol. A probe j leaves g unchanged and emits a fresh draw from $p_{s,j}(g)$, conditionally on the public past and current physical state. At each step the next command or probe may be chosen from the entire observed transcript by a randomized policy. The seed is specified at initialization. Probes are repeatable statistical observations; when $p_{s,j}$ is obtained from a density matrix, this is an oracle model, not repeated nondisturbing measurement of one unknown quantum system.

Definition 2.1 (Causal finite-label machine). At horizon N a machine has registers $S_0, \dots, S_N$, initialization rows E_s , and, for each control c and possible output y , a nonnegative matrix $K_{t,c,y} : S_{t-1} \rightarrow S_t$ such that $\sum_y K_{t,c,y}$ is row-stochastic. Its entries jointly specify the emission and next label. The matrices may depend on t, N but on the past only through the retained label and current control. Its error is

$$\sup_{s,\pi} \text{TV}(\mathbb{P}_{s,N}^\pi, \mathbb{Q}_{s,N}^\pi),$$

where the laws include the complete control–output transcript under the same policy π . Write $\text{PW}_{N,\varepsilon}$ for the minimum peak register size at error at most ε .

Every advertised label and all retained private randomness are counted. The horizon, external clock and arbitrary real tables remain free. The policy is an external experimenter, not an uncharged private memory for the machine. Outputs may influence future controls, but are not supplied again as a free history input. A sampled output register, when charged, adds the separate fixed output-alphabet cost. The error above is one budget for the entire transcript, not a per-step error budget.

For (s, g) define its future response function

$$(2.1) \quad \Theta_{s,g}(x, j) = p_{s,j}(xg), \quad x \in H, \quad \mathcal{X} = \{\Theta_{s,g} : s \in \mathcal{S}, g \in H\}.$$

Equip the finite product of continuous-function spaces with the uniform norm. Put $n = |\mathcal{X}|$ when this set is finite, and $n = \infty$ otherwise. This is a behavioral quotient, not the dimension of a linear predictive representation. In particular, infinitely many distinct predictive rows do not by themselves imply infinite nonnegative realization size for a general hidden Markov process [26, Theorem 2 and Section 4]. Repeatable observations are the additional input in the theorem below.

Theorem 2.2 (Uniform process error and finite physical actions). *For every fixed $0 \leq \varepsilon < 1$ the following are equivalent:*

- (1) $\sup_N \text{PW}_{N,\varepsilon} < \infty$;
- (2) $n < \infty$;
- (3) *some open normal subgroup $U \triangleleft H$ makes every $p_{s,j}$ constant on each coset of U ;*
- (4) *a stationary exact causal realization exists with permutation command updates and state-preserving probe emissions.*

If $n < \infty$ and $0 \leq \varepsilon < 1/2$, then

$$(2.2) \quad \text{PW}_{N,\varepsilon} = n \quad \text{for all sufficiently large } N.$$

If $n = \infty$, then, for every $\varepsilon < 1$ and every $k \geq 1$, there is $L_{k,\varepsilon} < \infty$ such that every admissible realization satisfies

$$(2.3) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_{k,\varepsilon}.$$

The same occupation conclusion holds for finite n , $\varepsilon < 1/2$ and $k < n$. Intervening registers are unrestricted. No return condition is imposed on the command alphabet: the probes themselves leave the physical state unchanged and supply the padding used in the proof.

The threshold 1 in the boundedness criterion is sharp: total variation never exceeds one, so at tolerance one a single label always suffices. The elementary one-continuation proof of (2.2) uses $\varepsilon < 1/2$. Theorem 3.4 below proves the full minimum for every $\varepsilon < 1$ by a different tail-event argument; the present proof and its stronger one-cut test remain useful in their stated range.

Lemma 2.3 (A common separating itinerary). *Choose $m \geq 2$ different functions in (2.1), represented by executable seed-prefix pairs. For every $\delta > 0$ there is one finite deterministic control itinerary with disjoint output events $A_1, \dots, A_m$ such that its target law from the i th state assigns probability at least $1 - \delta$ to A_i .*

Proof. Equality of two future response functions is preserved and reflected by left multiplication of their physical states by any $u \in H$: replace x in (2.1) by xu , a bijection of H . Positive executable products are dense in H . Indeed positive powers approximate the inverse of every generator in a compact group, so the closed positive semigroup contains the dense generated group.

List the $\binom{m}{2}$ pairs. At each stage the current two states for that pair still have different future response functions. By continuity and positive-word density, a finite executable word followed by one probe separates some coordinate of their output laws. Append that word and record the coordinate, then proceed to the next pair. This constructs one itinerary, not a family of incompatible pair-dependent suffixes. No reset or inverse word is required. Let $J = \binom{m}{2}$, and let $\mu_{i,r}$ be the selected coordinate probability at the r th probe location when the initial state is the i th representative. There is $\Delta > 0$ such that every pair of rows of this finite matrix differs by at least Δ in some coordinate.

At each location repeat the chosen probe L times. These repetitions leave the physical state fixed. The empirical coordinate frequencies $\hat{\mu}_r$ obey

$$\mathbb{P}_i\{\max_r |\hat{\mu}_r - \mu_{i,r}| \geq \Delta/3\} \leq 2J \exp(-2L\Delta^2/9).$$

For completeness, the scalar bound follows from $\mathbb{E}e^{z(B-\mathbb{E}B)} \leq e^{z^2/8}$ for a Bernoulli variable, exponential Markov inequality, and optimization in z ; a union bound gives the display. The moment bound follows by differentiating the logarithmic moment generating function twice: its second derivative is the variance of a tilted Bernoulli variable and is at most $1/4$. Choose $L \geq 1$ with the displayed upper bound at most δ . The events $A_i = \{\max_r |\hat{\mu}_r - \mu_{i,r}| < \Delta/3\}$ are disjoint, because $2\Delta/3 < \Delta$, and have the required probabilities. $\square$

Lemma 2.4 (Testing across a finite register). *Suppose m equal-length seed-prefix pairs at cut t admit an itinerary of length ℓ as in Lemma 2.3. For every $N \geq t + \ell$, an error- ε causal realization satisfies*

$$(2.4) \quad |S_t| \geq m(1 - \varepsilon - \delta).$$

If $\varepsilon + \delta < 1/2$, the stronger bound $|S_t| \geq m$ holds.

Proof. Use the same deterministic suffix for all representatives and pad its end with probes if necessary. Marginalize all prefix outputs. Write $\alpha_i(z)$ for the resulting cut-label distribution and

Q_z for the suffix-output law conditional on label z . Causality makes Q_z independent of the chosen prefix. Total variation contracts under marginalization, so correctness gives

$$1 - \delta - \varepsilon \leq \sum_z \alpha_i(z) Q_z(A_i).$$

This uses no conditional correctness assertion after a rare observed prefix. Summing in i , using $\alpha_i(z) \leq 1$, and then disjointness, yields

$$m(1 - \delta - \varepsilon) \leq \sum_z \sum_i Q_z(A_i) \leq |S_t|.$$

If $1 - \delta - \varepsilon > 1/2$, each i has a positive-mass label z_i with $Q_{z_i}(A_i) > 1/2$. Disjointness forces the labels z_i to be different. This proves the stronger assertion. $\square$

Proof of Theorem 2.2. The map $g \mapsto \Theta_{s,g}$ is continuous in the uniform norm. This follows from continuity on the compact product $H \times H$ and a finite-cover argument, uniformly in the first coordinate. Thus each seed's image is compact. Executable images are dense in it. If $\mathcal{X}$ is infinite, arbitrarily many distinct types therefore have executable representatives; if it is finite, every type has one.

Choose finitely many such representatives and pad their prefixes to a common length t_0 by any fixed probe. Their physical states are unchanged. Apply Lemma 2.3. For every $t_0 \leq t \leq N - \ell$, pad further with the same probe, marginalize its outputs, and apply Lemma 2.4. The itinerary and ℓ do not depend on t, N or on the machine.

If $n = \infty$, choose m with $m(1 - \varepsilon) > k$ and then choose $\delta > 0$ with $m(1 - \varepsilon - \delta) > k$. Every interior cut has width greater than k . At most $t_0 + \ell$ positive cuts lie outside this interval, including when it is empty. This proves (2.3), divergence, and the implication (i) $\Rightarrow$ (ii). For finite n and $\varepsilon < 1/2$, use $m = n$ and $0 < \delta < 1/2 - \varepsilon$ when $n \geq 2$. All sufficiently interior cuts require n labels. The case $n = 1$ follows from nonempty registers.

For finite $\mathcal{X}$, the maps

$$\Theta(x, j) \mapsto \Theta(xu, j), \quad u \in H,$$

define a continuous permutation action of H on $\mathcal{X}$. Continuity follows from uniform continuity of each of the finitely many functions. Its kernel U is open and normal. Evaluating at $x = e$ shows that every $p_{s,j}$ is constant on U -cosets, proving (ii) $\Rightarrow$ (iii). Conversely (iii) gives at most $|\mathcal{S}||[H : U]$ response types. More economically, use the n distinct types as labels, initialize at $\Theta_{s,e}$, update by the permutation above, and emit $\Theta(e, j)$ on probe j without changing the label. This produces the exact conditional response at each step. Induction on transcript length proves equality for every adaptive policy, hence (iv) and (i). It also supplies the upper bound n in (2.2). The occupation assertion for $k < n$ follows from the stronger testing bound just established. $\square$

Corollary 2.5 (Connected groups). *If H is connected, uniformly bounded causal width at any fixed total transcript error below one is possible exactly when every $p_{s,j}$ is constant on H . If at least one is nonconstant, every fixed width has the occupation bound (2.3).*

Proof. A continuous action of a connected group on a finite discrete set is trivial. The assertion follows from the theorem. $\square$

Remark 2.6 (Why terminal purification is insufficient). A two-label variable chosen once with equal masses and then repeatedly reported has the correct Bernoulli(1/2) marginal at every time, but its length- L law is supported on the two constant strings. Its total variation distance from L independent fair bits is $1 - 2^{1-L}$. Thus even exact numerical marginals do not supply a path-law realization. The causal theorem concerns an actual nonnegative emission instrument, not a list of separately correct decoders. It neither classifies arbitrary irreversible hidden-state systems nor asserts a measurement procedure for noncommuting quantum observables.

3. A STRONG CONVERSE FOR REPEATABLE OBSERVATION PROCESSES

The one-continuation testing bound distinguishes the full number of response types only when the error is below one half. Repeating a common diagnostic cycle gives a stronger obstruction: a finite hidden register cannot support more positive-probability observation-tail types than it has labels. The underlying tail-event principle is classical [30, 31, 32]. We give the short argument needed here, including the hidden-emission and nonhomogeneous conventions, before applying compactness to separately redesigned finite-horizon machines.

Lemma 3.1 (Tail capacity at narrow cuts). *Let $(S_t, Y_t)_{t \geq 0}$ be a hidden emission process with finite registers and finite output alphabets, whose joint next-state and emission law depends on the entire past only through S_t and the external time. Its transition kernels may be nonhomogeneous and irreversible. Suppose $|S_t| \leq k$ at infinitely many deterministic times. Then there cannot be more than k pairwise disjoint observation tail events having positive probability. Here an observation tail event belongs to $\bigcap_{t \geq 0} \sigma(Y_{t+1}, Y_{t+2}, \dots)$.*

Proof. Let $B_1, \dots, B_r$ be such events. Include the initialization variable, all past hidden labels, and all past emissions in $\mathcal{F}_t$. The Markov property for future emissions gives a version

$$\mathbb{P}(B_i \mid \mathcal{F}_t) = b_{i,t}(S_t), \quad b_{i,t}(s) = \mathbb{P}(B_i \mid S_t = s).$$

Only labels of positive probability are relevant. Since each B_i is measurable with respect to $\mathcal{F}_\infty = \sigma(\bigcup_t \mathcal{F}_t)$, upward martingale convergence gives

$$b_{i,t}(S_t) \longrightarrow \mathbf{1}_{B_i} \quad \text{almost surely and in } L^1.$$

Consequently, for every sufficiently large t and each i , some positive-probability label s_i satisfies $b_{i,t}(s_i) > 1/2$. There is one common sufficiently large time for the finite collection of events. Disjointness implies $\sum_i b_{i,t}(s) \leq 1$, so these labels are distinct. At an arbitrarily late narrow cut, $r \leq k$. This is the conditional-probability zero-one mechanism of [32, Theorem 1 and Section 5.1], applied to the emitted tail; no stationarity, mixing rate, or lower bound on state masses is used. $\square$

Fix pairwise distinct laws $P_1, \dots, P_n$ on one finite alphabet Ω . An R -step k -label instrument has one initialization row α_i for each i and common, possibly time-dependent, nonnegative emission matrices $K_{t,y}$ of size $k \times k$ with $\sum_y K_{t,y}$ row-stochastic. Let $Q_{i,R}$ be its output law, and set

$$(3.1) \quad e_{k,R} = \min_{\alpha_i, K_{t,y}} \max_{1 \leq i \leq n} \text{TV}(Q_{i,R}, P_i^{\otimes R}).$$

The minimum is attained in a finite product of compact simplexes. Initialization may be randomized and may depend on i ; subsequent kernels may not depend on an unrecorded value of i .

Proposition 3.2 (Uniform finite witnesses for product laws). *If $k < n$, then $e_{k,R}$ is nondecreasing in R and*

$$(3.2) \quad \lim_{R \rightarrow \infty} e_{k,R} = 1.$$

In particular, for every $\varepsilon < 1$ there is an integer $R(k, \varepsilon, P_1, \dots, P_n)$ for which no such instrument has error at most ε . When the laws and the tolerance are rational, one can find such an integer by a terminating sequence of finite real-algebraic feasibility decisions. No effective bound on the number of decisions, or polynomial-time claim, is made.

Proof. Truncation and contraction of total variation under marginalization show that $e_{k,R} \leq e_{k,R+1}$. Suppose its limit were some $e < 1$. Consider the compact countable product consisting of the initialization rows and all kernels $(K_{t,y})_{t \geq 1}$. For every R , impose all closed constraints $\text{TV}(Q_{i,r}, P_i^{\otimes r}) \leq e$ for $i \leq n, r \leq R$. Each finite system is feasible by attainment in (3.1). Nested compactness gives one infinite hidden process satisfying every finite constraint, with common kernels and k labels at every time.

Let Q_i denote its infinite output law and P_i^∞ the product law. The finite constraints imply $\text{TV}(Q_i, P_i^\infty) \leq e$. Indeed, for any event, approximate its indicator in L^1 under the finite measure $Q_i + P_i^\infty$ by indicators of cylinder events; these generate the product sigma-field. The bound on event-probability differences passes to the limit. Equivalently, one can apply martingale convergence to conditional probabilities under this finite measure.

For each i , let B_i be the event that empirical frequencies of all symbols converge to P_i . These are disjoint observation tail events, and the strong law gives $P_i^\infty(B_i) = 1$. Hence $Q_i(B_i) \geq 1 - e > 0$. Mix the initialization index uniformly, retaining the same common transition kernels. The resulting hidden process has output law $Q = n^{-1} \sum_i Q_i$, so $Q(B_i) > 0$ for every i . Lemma 3.1 contradicts $n > k$. This proves the limit.

For the last assertion, introduce the finitely many stochastic rows as real variables. Every word probability is a polynomial in their entries; the targets are rational. Absolute-value slack variables express each finite total-variation inequality by polynomial equalities and inequalities. Real quantifier elimination decides feasibility. Test $R = 1, 2, \dots$ until it is infeasible. The limit just proved guarantees termination. This is a finite-witness search, not an implementation of a general minimum-width algorithm. $\square$

Lemma 3.3 (A diagnostic cycle). *Suppose the response-type set $\mathcal{X}$ in (2.1) is finite, of cardinality $n \geq 2$. There is a finite deterministic control word B of positive length ℓ such that its action on $\mathcal{X}$ is the identity and its output laws $P_1, \dots, P_n$ from the n types are pairwise distinct. Successive executions of B give independent, identically distributed P_i -blocks from initial type i . Any intervening fixed probe controls may be inserted if their outputs are discarded.*

Proof. Every type has an executable seed-prefix representative, as proved in Theorem 2.2. Apply Lemma 2.3 to all n types with, for example, $\delta = 1/4$. The resulting finite deterministic itinerary T has pairwise distinct output laws: its disjoint diagnostic events have probability at least $3/4$ under the corresponding laws. Its command product induces a permutation π of the finite set $\mathcal{X}$. Let $d \geq 1$ be the order of this permutation and take $B = T^d$. Its action on types is the identity, and its output laws remain distinct because their first- T marginals are distinct.

The output law of any fixed continuation depends only on the response type, not on its chosen representative. Probes emit fresh conditionally independent observations and have no physical effect. Since B returns the type deterministically, its conditional output law after any number of previous blocks is again P_i . Induction gives the product law. Intervening probes also preserve the type; their ignored outcomes do not change the deterministic controls or the conditional block laws. The cycle is a return on the *finite response quotient*, not an assumed identity-product word in the physical group. $\square$

Theorem 3.4 (The full-error causal minimum). *In the repeatable-probe model of Definition 2.1, let $n = |\mathcal{X}| \in \mathbb{N} \cup \{\infty\}$. For every fixed $0 \leq \varepsilon < 1$,*

$$(3.3) \quad \lim_{N \rightarrow \infty} \text{PW}_{N, \varepsilon} = n.$$

When $n < \infty$, equality with n holds at every sufficiently large horizon. For every integer $k < n$ there is $L_{k, \varepsilon} < \infty$ such that every error- ε realization has

$$(3.4) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq L_{k, \varepsilon},$$

with no restriction on intervening widths. Thus the finite-response cardinality is the eventual minimum throughout the whole interval below the maximal total-variation error, not only below one half. The full-transcript and fresh nondisturbing probe assumptions are unchanged. No command return condition is imposed.

Proof. For $n = \infty$, Theorem 2.2 already proves (3.4) for every k and $\varepsilon < 1$. For $n = 1$, nonempty registers and the exact one-type machine prove the assertion. Suppose $2 \leq n < \infty$ and $k < n$.

Choose executable representatives of all types, and pad their prefixes with a fixed probe to one common length $b \geq 0$. Let B , ℓ and P_i be as in Lemma 3.3. Fix an integer R for which $e_{k,R} > \varepsilon$, supplied by Proposition 3.2.

Consider an advertised profile. Discard narrow cuts with $t < b$, and greedily select from the others $t_1 < \dots < t_J$ with gaps at least ℓ . If there are $R + 1$ selected cuts, start from each of the equal-length representatives, pad to t_1 with the same probe, and put one copy of B immediately after each of $t_1, \dots, t_R$. Fill the remaining positions before the next selected cut with that probe. Complete the horizon with any fixed controls. For each representative the retained R block outputs have target law $P_i^{\otimes R}$.

Marginalize all prefix, filler, and post-block outputs. In the proposed machine, the actual transition from one selected cut to the next, jointly with the retained block output, is a nonnegative instrument between at most k labels. Summing over the discarded emissions gives its stochastic normalization. All its entries are common to the representatives: the schedule is deterministic and the retained label is the only private past information available to a row. Intervening registers may be arbitrarily wide. The cut- t_1 distributions are allowed to be the arbitrary initialization rows α_i in (3.1). After padding unused labels, these composite kernels therefore define an R -step k -label instrument.

Whole-transcript correctness under these deterministic policies and contraction of TV under marginalization give its error at most ε . This contradicts $e_{k,R} > \varepsilon$. Thus $J \leq R$. The discarded cuts number at most b , and each selected cut accounts for at most ℓ remaining narrow cuts. Consequently one may take

$$L_{k,\varepsilon} = b + \ell R.$$

This count includes empty eligible intervals and excludes cut zero. The prefix, cycle, and witness integer depend only on the interface, k and ε , not on the horizon or machine. No correctness conditional on a rare observed prefix was used.

An exact stationary n -type machine works at every horizon by Theorem 2.2. Applying the occupation bound to $k = n - 1$ shows that a smaller peak is impossible once $N > L_{n-1,\varepsilon}$. This proves eventual equality and the theorem. $\square$

Corollary 3.5 (Strong converse below the response cardinality). *If $k < n$, the least worst-policy whole-transcript error achievable by peak width at most k tends to one as the horizon tends to infinity. At error one, one label always suffices. When the finite group action and all probe tables are explicitly rational, the proof supplies a terminating search for a valid occupation constant via the finite product-law feasibility problems.*

Proof. For each $\varepsilon < 1$, the preceding occupation bound excludes a k -peak realization for every $N > L_{k,\varepsilon}$. TV is at most one, which proves the limit and the endpoint. For explicit finite data, response types and their command permutations can be enumerated. The common finite itinerary and its permutation order can be found by finite search, its cycle laws are rational, and Proposition 3.2 finds R . The claimed bound $b + \ell R$ then follows. No efficiency conclusion follows from this termination proof. $\square$

Proposition 3.6 (Why one half is not the process boundary). *For $P_1 = \delta_0$, $P_2 = \delta_1$ and one hidden label,*

$$(3.5) \quad e_{1,R} = 1 - 2^{-R}.$$

Nevertheless, a single width-one cut, followed by two-label storage, can have worst error one half for these two product laws at arbitrarily large remaining horizons.

Proof. A one-label instrument has independent time-dependent emissions with probabilities p_t of zero. Its two target-string probabilities are $u = \prod_t p_t$ and $v = \prod_t (1 - p_t)$. The two TV errors are

$1 - u$ and $1 - v$. Since $uv \leq 4^{-R}$, their maximum is at least $1 - 2^{-R}$, attained by $p_t = 1/2$ for every t . For the last assertion, after the one-label cut choose a fair bit, store it in two labels, and emit it repeatedly. Its law is the equal mixture of the two deterministic strings and is at distance one half from each. Thus for errors above one half the theorem needs control of *repeated* narrow cuts; it does not assert the false pointwise lower bound at every interior cut. $\square$

The tail-capacity lemma is a necessary condition for arbitrary hidden emission processes, including irreversible ones. The diagnostic-cycle application, however, still uses the stated repeatable-probe model. It is neither a general HMM/POMDP realization classification nor a claim about nondisturbing repeated quantum measurements. The new conclusion is the all-error finite-horizon strong converse and occupation bound obtained from the classical tail principle, finite-product compactness, and one common physical diagnostic cycle.

4. STATIONARIZATION FROM RECURRENT NARROW CUTS

The first reduction does not require reversible physical dynamics. In this section only, replace H by a topological monoid G with identity 1, retain a finite command alphabet, and let $F_{s,j} : G \rightarrow Y_j$ be continuous. The output sets are norm-compact convex subsets of real Banach spaces. Define M_ε and $W_{N,\varepsilon}$ by the same stochastic models. Compactness of the physical monoid is not required.

Definition 4.1 (Eventual approximate returns). The alphabet has eventual approximate returns if, for every neighborhood V of 1, there is an integer $g(V) \geq 0$ such that every integer $n \geq g(V)$ is the length of a word with product in V . Cofinite exact returns mean that a single g works with product exactly 1.

The statement concerns physical products only. The hidden rows on a return word need not be close to an identity matrix. Approximation here is eventual at every length, not merely along a return subsequence.

Theorem 4.2 (Width-preserving stationarization). *Assume eventual approximate returns. For every $\varepsilon \geq 0$ and integer $k \geq 1$, the following are equivalent:*

- (1) *a stationary realization with at most k labels has error at most ε ;*
- (2) *among error- ε clocked realizations, the number of positive cuts having width at most k is unbounded.*

Consequently, if $M_\varepsilon > k$, there is $L_{k,\varepsilon} < \infty$, independent of the horizon, such that every error- ε realization satisfies

$$(4.1) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_{k,\varepsilon}.$$

All other registers may be arbitrarily large. Moreover,

$$(4.2) \quad \lim_{N \rightarrow \infty} W_{N,\varepsilon} = M_\varepsilon \quad \text{in } \mathbb{N} \cup \{\infty\}.$$

If the right side is finite, equality holds for every sufficiently large N .

The theorem preserves the proposed width, including randomized initialization. It does not assert that the microscopic rows of the clocked machine become stationary. They can be entirely unrelated at different times.

4.1. Finite witnesses and homogeneous transition tables. For a matrix A write $\|A\|_r = \max_i \sum_j |A_{ij}|$. Put $a = |\mathcal{A}|$ and $D_* = \max(1, \max_j \text{diam } Y_j)$. Let $\mathcal{X}_k$ be the compact parameter space of stationary k -label machines. For $m \geq 0$ define

$$e_{k,m} = \min_{X \in \mathcal{X}_k} \max_{s,j, |w| \leq m} \|X_{s,j}(w) - F_{s,j}(u_w)\|_j.$$

The finite maximum is continuous, so the minimum is attained. If no stationary error- ε machine exists, compactness of $\mathcal{X}_k$ and the finite intersection property give an m with

$$(4.3) \quad e_{k,m} = \varepsilon + \gamma, \quad \gamma > 0.$$

Indeed the nested closed sets defined by the constraints $|w| \leq m$ otherwise have a common point. This argument also applies when the full infinite-word error is not a continuous function of the parameters.

We shall use the finite edge-coloring theorem of Ramsey [25]. Denote by $R_q(h)$ an integer such that every coloring of the edges of a complete graph on $R_q(h)$ vertices with at most q colors has a monochromatic h -vertex subgraph. Only finiteness is essential.

Remark 4.3 (An explicit but coarse Ramsey bound). For definiteness one may take

$$(4.4) \quad R_q(h) \leq q^{qh+1} \quad (q \geq 2, h \geq 2).$$

To see this elementary bound, successively select a vertex and retain a largest color class among its remaining neighbors. With q^{qh+1} initial vertices this produces at least $q(h-1) + 1$ selected vertices, each of whose edges to later selected vertices has one color. At least h of their associated colors agree; these vertices form the required subgraph. The deliberately excessive exponent avoids any endpoint rounding issue.

Partition $[0, 1]$ into at most $\lceil k/\delta \rceil + 1$ intervals of length at most δ/k . Two tuples in the same entrywise cell of

$$\text{Stoch}(k)^{\{0\} \cup \mathcal{A}}$$

have corresponding matrices within δ in $\|\cdot\|_r$. The number of occupied cells is at most

$$(4.5) \quad q = \max(2, (\lceil k/\delta \rceil + 1)^{(a+1)k^2}).$$

Each representative used below is chosen from an occupied cell and is therefore stochastic; an entrywise rounded matrix need not be stochastic and is not used as a representative.

Proof of Theorem 4.2. The forward implication follows by using one stationary machine at every horizon. For the converse, suppose (4.3). Choose $0 < \eta < \gamma/4$. Continuity of multiplication and of the finitely many $F_{s,j}$ gives a neighborhood V of 1 with the following property. In any word of length at most m , inserting at most $m+3$ factors from V , including before and after the word, changes every output by less than η . There are only finitely many relevant command words and insertion patterns; continuity at the tuple of identity factors, followed by a finite intersection of neighborhoods, proves the assertion. This uses neither an invariant metric nor commutativity.

Let $g = g(V)$, put $b = g + 1$, and choose

$$0 < \delta \leq \min(1, \gamma/(4D_*(m+1))).$$

Take a clocked realization with so many narrow cuts that, after discarding cuts $t < g$ or $t > N - g$ and greedily retaining cuts separated by at least b , at least $R_q(m+2)$ remain. Such a choice is possible by hypothesis; at most $2g$ cuts are discarded and each retained cut accounts for at most b of the others.

Identify each retained register with a subset of $\{1, \dots, k\}$. Extend rows from dummy labels arbitrarily to stochastic rows without altering any row on the actual register. For retained cuts $t_i < t_j$ write $\ell = t_j - t_i \geq g + 1$. Choose a length- ℓ return word with product in V , and let A_{ij} be its actual composite kernel between these cuts. For each letter $c \in \mathcal{A}$, choose the word consisting of c followed by a length- $(\ell - 1)$ return word with product in V , and let $B_{ij,c}$ be its composite kernel. Thus the physical products of these words are respectively in V and of the form vu_c with $v \in V$. All the matrices are of order k , however large the intermediate registers may be.

Color the edge $\{i, j\}$, oriented from the earlier cut to the later cut, by the cell of $(A_{ij}, (B_{ij,c})_c)$. Ramsey's theorem gives $m + 2$ ordered cuts

$$t_0 < t_1 < \cdots < t_{m+1}$$

whose edges have one color. Fix a stochastic representative $(E, (B_c)_c)$ from that color. Choose fixed return words for the prefix of length t_0 and the suffix of length $N - t_{m+1}$, both with products in V . Let α_s be the actual state distribution after the prefix. Let $D_j(i)$ be the conditional expected original decoder after the suffix, given label i at its start, with arbitrary legal values on dummy labels. Convexity gives $D_j(i) \in Y_j$.

Consider the stationary machine

$$(4.6) \quad \alpha_s, \quad T_c = B_c, \quad d_j = ED_j.$$

Its decoder is legal. To test $w = c_1 \cdots c_\ell$, $0 \leq \ell \leq m$, use the first ℓ consecutive clique edges with kernels B_{i-1,i,c_i} and then the single edge from t_ℓ to t_{m+1} with kernel $A_{\ell,m+1}$. For $\ell = 0$ there is only this last edge. Add the fixed prefix and suffix. This is a genuine word of length N ; its physical product is obtained from u_w by at most $\ell + 3$ insertions from V . Its target output is consequently within η of $F_{s,j}(u_w)$.

Each of its $\ell + 1$ edge matrices differs from the corresponding matrix in (4.6) by at most δ . Stochastic multiplication contracts the ℓ^1 distance between probability rows. A telescoping expansion, and subtraction of any fixed point of Y_j from the decoder, therefore bound the difference between the actual output and $\alpha_s B_{c_1} \cdots B_{c_\ell} ED_j$ by

$$D_*(\ell + 1)\delta.$$

Wordwise correctness of the original machine implies $e_{k,m} \leq \varepsilon + \eta + D_*(m + 1)\delta < \varepsilon + \gamma$, a contradiction. The representative E is used only as part of the terminal decoder. No idempotence of E , consistency between different filler words, or hidden-state identity on a physical return was assumed.

The same argument shows that one may take

$$(4.7) \quad L_{k,\varepsilon} = 2g + (g + 1)R_q(m + 2).$$

For exact cofinite returns, use exact fillers and omit η and the neighborhood choice; g is the given exact-return bound. In general g depends on the finite witness and the continuity neighborhood. The remaining parameters depend only on the interface, the alphabet, k , and ε , never on a machine or horizon.

If $M_\varepsilon = \infty$, (4.1) for each k gives $W_{N,\varepsilon} \rightarrow \infty$. If $M_\varepsilon = K < \infty$, the stationary machine gives $W_{N,\varepsilon} \leq K$ for every N , and the bound for $k = K - 1$ gives eventual equality when $K > 1$. When $K = 1$ equality holds for every N because registers are nonempty. $\square$

Proposition 4.4 (Approximate returns without exact returns). *Let the two circle commands have angles θ, ϕ , where $1, \theta/(2\pi), \phi/(2\pi)$ are rationally independent. They have no nonempty exact-return word, but satisfy Definition 4.1.*

Proof. A length- n product has angle $n\phi + j(\theta - \phi)$, $0 \leq j \leq n$. The positive orbit of the irrational rotation $\theta - \phi$ is dense. Compactness supplies, for every accuracy, a finite initial orbit segment that is a net of the circle. All longer initial segments remain nets; rotation by $n\phi$ preserves their accuracy. Thus a product is arbitrarily close to the identity at every sufficiently large length. An exact return would be a nontrivial integer relation among $1, \theta/(2\pi), \phi/(2\pi)$. $\square$

A single irrational letter fails the eventual condition: there is only one word at each horizon. Its target can be placed in a one-label horizon-dependent decoder even when no finite stationary approximation at the same tolerance exists. Thus mere recurrence along a subsequence is not a substitute for Definition 4.1.

Remark 4.5 (Quantitative interpretation). The Ramsey estimate gives a finite occupation constant, not a useful asymptotic width rate. The homogeneous neutral representative remains inside the terminal decoder in the extraction proof. It is neither an identity transition nor an assumed idempotent.

5. STATIONARY STOCHASTIC PURIFICATION

Write $\text{Stoch}(k)$ for the compact semigroup of row-stochastic matrices of order k . The identity of a group inside this semigroup can be a singular idempotent; it must not be confused with the identity matrix.

Lemma 5.1 (An idempotent over the physical identity). *Let $\mathcal{T}$ be a compact subsemigroup of $H \times \text{Stoch}(k)$ whose projection onto H is surjective. There is an element $e = (1, E) \in \mathcal{T}$ with $E^2 = E$ and rank equal to the minimum matrix rank in $\mathcal{T}$. The monoid $e\mathcal{T}e$ is a compact group with identity e and still projects onto H .*

Proof. Only the integer ranks $1, \dots, k$ occur, so their nonempty set has a minimum which is attained. Choose (h, T) of that minimum rank r . The powers of T are bounded, since they are stochastic. Thus its complex eigenvalues have modulus at most one and all Jordan blocks on the unit circle have size one: a larger peripheral Jordan block would produce polynomially growing powers. The closure of positive powers of an element of a compact group is a group: arbitrarily large positive powers approach its identity, and the preceding powers approach its inverse. Apply this observation to the compact monothetic group generated by $(h, \lambda_1, \dots, \lambda_b)$ in $H \times (S^1)^b$, where the λ_i are the peripheral eigenvalues of T . There is a net of integers tending to infinity for which $h^n \rightarrow 1$ and all peripheral eigenvalues raised to the n th power tend to one. Along it T^n tends to the spectral projection E onto the peripheral subspace. This is a real stochastic idempotent. Its rank is at most r and at least r by minimality, so it is r . Nets avoid any metrizability assumption on H .

Every matrix B in $e\mathcal{T}e$ satisfies $B = EBE$ and has rank r . Its action on the row space of E is consequently invertible. Restriction embeds $e\mathcal{T}e$ as a compact submonoid of $H \times \text{GL}(r, \mathbb{R})$, with identity $(1, I_r)$; injectivity follows because the restriction determines $(xE)B = xB$ for every row x , by $B = EBE$, and hence determines every row of B . A compact submonoid of a topological group is a subgroup, by the same positive-power return argument. The inverse images of its inverses are still stochastic. Finally, sandwiching an element with physical coordinate h between e 's leaves that coordinate unchanged. Surjectivity follows. $\square$

Lemma 5.2 (The recurrent simplex). *If E is a stochastic idempotent of rank r , then*

$$\Delta_k E = \{p \in \Delta_k : pE = p\}$$

is a simplex with exactly r vertices. Every group of stochastic matrices with identity E permutes these vertices.

Proof. Regard E as a transition matrix. Its closed communicating classes each have a unique stationary probability row supported on that class. Every stationary row is a convex combination of these rows: mass on a transient class is zero, and stationarity within a closed irreducible class determines a one-dimensional space. For completeness, transient mass vanishes because each transient state has a positive probability of reaching a closed class in a bounded number of steps; a common bound over the finitely many states makes transient mass decrease by a fixed factor unless it is zero. The closed-class rows have disjoint supports, hence are linearly independent.

Each row of E is stationary because $E^2 = E$. Conversely, every stationary row equals its product with E . The span of the closed-class stationary rows is therefore precisely the row space of E , so there are r classes and r simplex vertices. If B has a stochastic inverse relative to E ,

multiplication by B and by this inverse are inverse affine maps of this simplex. Affine bijections permute its vertices. $\square$

Theorem 5.3 (Purification without a width or error loss). *For every compact-group interface in Definition 1.1, a stationary k -label realization of error at most ε admits a permutation realization of error at most ε on at most k labels. There is no return-word hypothesis and no restriction on the number of connected components of H .*

Proof. For the given machine form

$$\mathcal{T} = \overline{\{(u_w^{-1}, T_w) : w \in \mathcal{A}^*\}} \subset H \times \text{Stoch}(k).$$

The inverse on the physical coordinate is important: concatenation satisfies $u_{vw}^{-1} = u_v^{-1}u_w^{-1}$ and $T_{vw} = T_v T_w$. Thus $\mathcal{T}$ is a compact semigroup. Positive-word density makes its projection onto H surjective. By continuity, its every element (h, B) satisfies

$$(5.1) \quad \|\alpha_s B D_j - F_{s,j}(h^{-1})\|_j \leq \varepsilon \quad (s \in \mathcal{S}, j \in \mathcal{J}).$$

Apply Lemma 5.1 and denote the vertices of $\Delta_k E$ by $\nu_1, \dots, \nu_r$. For each letter the element

$$e(u_a^{-1}, T_a)e = (u_a^{-1}, E T_a E)$$

belongs to the group $e\mathcal{T}e$. By Lemma 5.2, $B_a = E T_a E$ permutes the ν_i . Use that permutation as the new command. If V has rows ν_i and the induced row permutation is Π_a , the precise intertwining identity is $V B_a = \Pi_a V$. Write uniquely

$$\alpha_s E = \sum_{i=1}^r \beta_s(i) \nu_i, \quad \tilde{D}_j(i) = \nu_i D_j \in Y_j.$$

The last inclusion uses convexity of the legal output set. For a nonempty word the output of the resulting r -label machine is $\alpha_s B_{a_1} \cdots B_{a_n} D_j$. For the empty word it is $\alpha_s E D_j$. In both cases the corresponding joint element lies in $\mathcal{T}$ and has physical coordinate u_w^{-1} . Equation (5.1) proves the error bound. Since $r \leq k$, the width has not increased. $\square$

The new machine need not reproduce the old machine's erroneous output word by word; it remains in the same closed error balls about the target. Nor do we assert $T_w = I$ on a physical return, invertibility of the original T_a , or deterministic initialization. These three incorrect shortcuts would invalidate the reduction.

5.1. Finite quotients and the unrestricted stationary minimum. For a tuple of permutation matrices $\sigma = (\Pi_a)_{a \in \mathcal{A}}$ of order k , set

$$\mathcal{L}_\sigma = \overline{\{(u_w^{-1}, \Pi_w) : w \in \mathcal{A}^*\}} \subset H \times \mathfrak{S}_k.$$

The topology is the product of the compact topology on H and the discrete topology on the finite group $\mathfrak{S}_k$. This is a compact subgroup. Its definition includes the possible discrepancy between physical relations and hidden-state relations.

Theorem 5.4 (An exact stationary minimum). *The value M_ε is the least positive integer k for which some tuple σ of permutations of order k , some $\beta_s \in \Delta_k$, and some $d_j(i) \in Y_j$ satisfy*

$$(5.2) \quad \|\beta_s \Pi d_j - F_{s,j}(h^{-1})\|_j \leq \varepsilon \quad \text{for all } (h, \Pi) \in \mathcal{L}_\sigma \text{ and all } s, j.$$

For fixed k there are at most $(k!)^{|\mathcal{A}|}$ discrete choices (a crude bound before simultaneous conjugacy reduction), and the remaining feasible set is compact. The optimal stationary error at any fixed width is attained.

Proof. Necessity follows from Theorem 5.3, padding by unused labels if its output has fewer than k states, and then taking the closure of word products. Sufficiency is the restriction of (5.2) to words. The probabilities and legal decoders lie in a finite product of compact sets. Each displayed inequality defines a closed subset of that product. The worst error is continuous in these parameters, uniformly in the compact group variable, so its minimum exists. There are finitely many permutation tuples. $\square$

Theorem 5.5 (Finite observable quotient criterion). *For arbitrary compact H and the stated continuous interface,*

$$M_\varepsilon < \infty \iff r(U) \leq \varepsilon \text{ for some open normal subgroup } U \leq H.$$

If such U exists, $|\mathcal{S}[[H : U]]$ deterministic-initialized permutation labels suffice. Conversely, a stationary k -label realization supplies such a U with $[H : U] \leq k!$.

Proof. Purify a stationary realization and use its joint group $\mathcal{L}_\sigma$. Put

$$U = \{h \in H : (h, I) \in \mathcal{L}_\sigma\}.$$

This is a closed normal subgroup of H : conjugate (h, I) by an element above any $g \in H$, using surjectivity of the first projection. For a fixed permutation Π , a nonempty fiber in $\mathcal{L}_\sigma$ is a coset hU . These at most $k!$ fibers cover H , so U has finite index. A closed subgroup of finite index in a compact group is open, since its complement is a finite union of closed cosets.

For each s, j and each fiber, the single legal vector $\beta_s \Pi d_j$ is within ε of every $F_{s,j}(h^{-1})$ in that fiber. Explicitly, $(hU)^{-1} = Uh^{-1} = h^{-1}U$ by normality, so inversion preserves the ordinary U -coset form. Hence $r(U) \leq \varepsilon$. Conversely, store (s, gU) , update the coset by left multiplication by u_a , and decode a minimizing center of $F_{s,j}(gU)$. Compactness supplies the centers. This gives the asserted finite realization. $\square$

For finitely many connected components, every open subgroup contains $K = H^\circ$, and K itself is open normal. Therefore $M_\varepsilon < \infty$ holds exactly when

$$\varepsilon \geq \max_{s,j,c \in H/K} \text{rad}_{Y_j}(F_{s,j}(c)),$$

without a return assumption. Executable returns enter the stronger clocked converse, not this stationary statement. The index bound $k!$ is an existence bound on a deterministic quotient, not a claim that the stationary minimum equals its index.

Proposition 5.6 (Random initialization strictly saves states). *Let $H = \mathbb{Z}/6\mathbb{Z}$, with an identity letter and a generator, one seed, and binary success probability*

$$f(n) = \frac{1}{2} + \frac{1}{4}(-1)^n + \frac{1}{4}\cos(2\pi n/3).$$

At zero error the least deterministic component-quotient width is six, whereas the unrestricted stationary stochastic minimum is five.

Proof. The values at $n = 0, \dots, 5$ are

$$(1, \frac{1}{8}, \frac{5}{8}, \frac{1}{2}, \frac{5}{8}, \frac{1}{8}).$$

This sequence has least period six. A deterministic quotient of the cyclic component action preserving these outputs must therefore have six states. For a five-state realization take the disjoint union of a two-cycle and a three-cycle, both updating the index by $i \mapsto i + 1$ modulo their respective lengths, initialize with mass $1/2$ at the zeroth vertex of each, and use decoder rows $(1, 0)$ on the two-cycle and $(1, 1/4, 1/4)$ on the three-cycle. The identity command fixes all five labels. Direct averaging gives exactly $f(n)$, with legal binary probabilities.

If a stationary realization on at most four labels existed, purification would give one whose generator is a permutation of at most four letters. Its order is one of $1, 2, 3, 4$, as follows by listing

the possible cycle lengths. Every resulting numerical sequence has a period dividing that order, contradicting the least period six. Thus five is the unrestricted minimum. $\square$

6. PHYSICAL EQUIVARIANCE AND THE COMPLETE BOUNDARY

Return now to a compact Hausdorff group H with a finite dense generating alphabet. The output sets may still be norm-compact convex subsets of real Banach spaces. Purification removes transition randomness; a further averaging removes hidden relations that have no physical counterpart.

Theorem 6.1 (Physical-equivariant purification). *Every stationary k -label realization of error at most ε admits one on $r \leq k$ labels of the form*

$$(6.1) \quad \alpha_s \rho(h^{-1}) d_j, \quad h \in H, \quad \rho : H \longrightarrow S_r \text{ a continuous homomorphism,}$$

with the same uniform error. Here permutations are represented by row-action matrices. In particular the command row is $\rho(u_a^{-1})$, all physical group relations hold on the new register, and initialization may remain random.

Proof. Apply Theorem 5.3, whose proof uses only finite convex combinations of decoder values. Denote the resulting initialization by β_s , its decoder by D_j , and its compact joint group by $\mathcal{L} \subset H \times S_k$, with the product topology and discrete permutation factor. Unused labels may be adjoined when necessary. Let P be the second projection and

$$N = \{\pi \in P : (1, \pi) \in \mathcal{L}\}.$$

Then N is normal in P . The averaging matrix $Q = |N|^{-1} \sum_{n \in N} n$ is stochastic, satisfies $Q^2 = Q$, and commutes with P . For $(h, \pi) \in \mathcal{L}$, every $(h, n\pi)$ belongs to $\mathcal{L}$. Averaging their error inequalities and using convexity of the norm ball gives

$$(6.2) \quad \|\beta_s Q \pi D_j - F_{s,j}(h^{-1})\|_j \leq \varepsilon.$$

Let $O_1, \dots, O_r$ be the N -orbits in the label set, and let v_i be the uniform probability row on O_i . The rows of Q are precisely these uniform orbit rows. Write V for the matrix with rows v_i . Every $\pi \in P$ maps N -orbits bijectively onto N -orbits and therefore induces a permutation $\bar{\pi}$ with

$$(6.3) \quad V\pi = \bar{\pi}V.$$

The induced action of each $n \in N$ is trivial. Two elements of $\mathcal{L}$ with the same first coordinate differ by an element of $\{1\} \times N$. Hence $(h, \pi) \mapsto \bar{\pi}$ descends to a homomorphism $\rho : H \rightarrow S_r$. It is continuous: the surjection from the compact group $\mathcal{L}$ onto the Hausdorff group H is a quotient map, and the original orbit action is continuous into a finite discrete group.

Set $\alpha_s(i) = \sum_{x \in O_i} \beta_s(x)$ and $d_j(i) = v_i D_j$. Then $\alpha_s V = \beta_s Q$ and $d_j = V D_j$. Equation (6.3) gives

$$\alpha_s \rho(h) d_j = \beta_s Q \pi D_j \quad ((h, \pi) \in \mathcal{L}).$$

Replace h by h^{-1} in (6.2). This proves (6.1); all decoder values are legal by convexity, and $r \leq k$. $\square$

Corollary 6.2 (Intrinsic stationary minimum). *The number M_ε is the least cardinality of a finite continuous H -set for which some probability initialization for each seed and legal decoder for each query approximate the interface within ε . It is independent of the chosen finite dense alphabet. It is unchanged by replacing every target with*

$$h \longmapsto F_{s,j}(\varphi(h)q),$$

where φ is a continuous group automorphism and $q \in H$.

Proof. Theorem 6.1 proves necessity, and restriction to command words proves sufficiency. For the last assertion, replace ρ by $\rho \circ \varphi$ and α_s by $\alpha_s \rho(q^{-1})$ in (6.1). Applying the inverse change of variables proves equality of the minima. $\square$

The finite quotient in Theorem 5.5 has a more precise description than its factorial bound. With $\mathcal{L}, P, N$ as above and $U = \{h : (h, I) \in \mathcal{L}\}$, the correspondence $(h, \pi) \mapsto (hU, \pi N)$ induces $H/U \simeq P/N$. Thus $[H : U] = |P/N| \leq k!$. The orbit action in Theorem 6.1 can have an additional kernel; no equality between this index and the minimal label count is asserted.

Theorem 6.3 (Complete clocked finite-quotient classification). *Assume eventual approximate returns on the compact group H . At every $\varepsilon \geq 0$, including a positive critical radius, the following are equivalent:*

- (1) $\sup_N W_{N,\varepsilon} < \infty$;
- (2) $M_\varepsilon < \infty$;
- (3) $r(U) \leq \varepsilon$ for an open normal subgroup U of H .

For each $k < M_\varepsilon$ the all-profile occupation bound (4.1) holds, and $W_{N,\varepsilon}$ converges to M_ε . Put

$$R_0 = \max_{s,j,h \in H} \text{rad}_{Y_j}(F_{s,j}(hH^\circ)).$$

Then $R_0 = \inf_{U \triangleleft H \text{ open}} r(U)$. Above R_0 bounded realizations exist; below it every fixed width has bounded occupation. At $\varepsilon = R_0$ bounded clocked realization exists exactly when the infimum is attained by an open normal subgroup.

Proof. The first two conditions are equivalent by Theorem 4.2. The second and third are equivalent by Theorem 5.5. That proof, including constrained center attainment, uses compactness and convexity of Y_j , not finite dimensionality.

For completeness, every open subgroup contains H° : the image of the connected identity component in a finite discrete quotient is a singleton. Thus $r(U) \geq R_0$. The quotient H/H° is profinite, so its open normal subgroups form a neighborhood basis at the identity. The inverse images U have intersection H° . Given a neighborhood O of H° in H , compactness supplies one such $U \subset O$: otherwise their intersections with the closed set $H \setminus O$ have the finite intersection property. Uniform continuity of the finite family of targets gives, for every $\eta > 0$, a neighborhood V of 1 such that $\|F_{s,j}(hkv) - F_{s,j}(hk)\|_j < \eta$ for all $h \in H, k \in H^\circ$, and $v \in V$. Choose $U \subset H^\circ V$. Every $F_{s,j}(hU)$ is then within η of $F_{s,j}(hH^\circ)$, so $r(U) \leq R_0 + \eta$. This proves the infimum formula and all remaining assertions. $\square$

This result includes the formerly separate zero and positive boundary cases. The finite-rank proof of Theorem 10.5 is retained below because it identifies an additional exact translate-space obstruction. No passage from an error strictly larger than R_0 to equality is used in Theorem 6.3.

6.1. Length phases without a width overhead. Let $p \geq 1$ be an integer and fix a command with product a . Let J be the closure of all products of lengths divisible by p . The group argument of Lemma 9.2, which uses only the block length, shows that J is open normal, H/J is cyclic of order dividing p , and all letters belong to Ja . Regard $\mathcal{A}^p$ as a finite alphabet on J . Assume that this block alphabet has eventual approximate returns. In particular this holds when the original nonempty exact-return language has greatest common divisor p .

For $0 \leq r < p$ let $M_{r,\varepsilon}$ be the intrinsic stationary minimum on J for

$$f_{r,s,j}(h) = F_{s,j}(ha^r).$$

Theorem 6.4 (Exact limiting width in each phase). *Under the preceding assumptions, for every r and $\varepsilon \geq 0$,*

$$(6.4) \quad W_{N,\varepsilon} \leq M_{r,\varepsilon} \quad (N \equiv r \pmod{p}), \quad \lim_{\substack{N \rightarrow \infty \\ N \equiv r \pmod{p}}} W_{N,\varepsilon} = M_{r,\varepsilon}.$$

Finite limits are eventually attained. If $k < M_{r,\varepsilon}$, the number of positive cuts of width at most k is bounded uniformly over that horizon phase, with no restriction on other widths. Boundedness in the phase is equivalent to the existence of an open normal subgroup $U \leq J$ such that

$$(6.5) \quad \max_{s,j,h \in J} \text{rad}_{Y_j}(F_{s,j}(hUa^r)) \leq \varepsilon.$$

Proof. Suppose first that $M_{r,\varepsilon} = K < \infty$ and choose a physical-equivariant stationary realization of f_r on K labels, with representation $\rho : J \rightarrow S_K$. For a fixed $N \equiv r \pmod{p}$, let g_t be the physical product after t commands and set

$$(6.6) \quad z_t = a^{N-t} g_t a^{-r} \in J, \quad c_{t,b} = a^{N-t} u_b a^{-(N-t+1)} \in J.$$

Normality of J and $u_b a^{-1} \in J$ justify the membership assertions. We have $z_0 = a^{N-r}$, $z_t = c_{t,b} z_{t-1}$ when the t th letter is b , and $z_N = g_N a^{-r}$. Initialize with $\alpha_s \rho(z_0^{-1})$, use the clock-dependent permutation $\rho(c_{t,b}^{-1})$, and keep decoder d_j . The row product is $\alpha_s \rho(z_N^{-1})$, so its error from $F_{s,j}(g_N) = f_{r,s,j}(z_N)$ is at most ε . This uses the free external clock but no phase register or partial-block storage. It proves the upper bound with exactly K labels.

Fix $k < M_{r,\varepsilon}$ and a cut residue $0 \leq \ell < p$. Put $b = (r - \ell) \bmod p$, chosen in $\{0, \dots, p-1\}$. Restrict the input to the fixed prefix a^ℓ , arbitrary p -letter blocks, and fixed suffix a^b . Integrate the prefix into initialization and the suffix into the decoder. The block target on J is

$$G_{\ell,s,j}(h) = F_{s,j}(a^b h a^\ell) = f_{r,s,j}(\text{Ad}_{a^b}(h) a^{\ell+b-r}).$$

Here $a^{\ell+b-r} \in J$ and conjugation by a^b is an automorphism of J . Corollary 6.2 therefore identifies its stationary minimum with $M_{r,\varepsilon}$. Theorem 4.2 on the block alphabet bounds its positive narrow cuts by a constant L_ℓ . These cuts are precisely the original cuts in residue ℓ between the prefix and suffix, apart from the initial block cut. At most $\ell + b + 1 \leq 2p$ further cuts need be charged for this residue, including small horizons shorter than the chosen prefix and suffix. Summing gives the uniform bound $\sum_{\ell=0}^{p-1} L_\ell + 2p^2$. As before, a peak at most k would force N below this bound. This proves the limit and occupation assertions. Finally apply Theorem 5.5 to f_r on J to obtain (6.5). $\square$

If the order of H/J is a proper divisor q of p , the minima and quotient-radius criteria repeat with period q : multiplication by $a^q \in J$ is a right translation of the target. The earlier phase radii have the same repetition. Neither the physical quotient order nor the return-length greatest common divisor is substituted silently for the other.

6.2. Compact outputs and finite hidden state. Theorems 4.2, 6.1, 6.3, and 6.4 hold for norm-compact convex Banach-valued outputs. A single-machine purification even needs only convexity: its new decoder values lie in the convex hull of finitely many old values. Compact output sets are used instead in finite-witness extraction, optimal-error attainment, and constrained-center existence. The representative-function and algebraic sections below retain their explicitly finite-dimensional hypotheses.

Finiteness on the hidden side has a different role. On countably many labels the full simplex is not compact in total variation; mass can escape to infinity, matrix ranks need not admit a useful finite minimum, and a recurrent simplex may have infinitely many vertices. On a general compact convex hidden state space affine automorphisms need not permute a finite vertex set: rotation of the disk already acts continuously on infinitely many extreme points. Thus these proofs give neither a finite-label purification theorem for arbitrary operator kernels nor an online observation theorem. Their terminal-output assertion is strictly the one stated in the models.

Remark 6.5 (Action and counting conventions). Probability vectors act on the right of transition matrices. Chronological row multiplication consequently corresponds to the inverse physical coordinate h^{-1} used above. The coarse deterministic quotient construction stores both the seed

and the coset; its $|\mathcal{S}[[H : U]]$ bound is not the unrestricted stochastic minimum. In the phase theorem the chosen block period p , the physical quotient order $q \mid p$, and the greatest common divisor of actual return lengths retain their separate meanings.

7. THE CLOCKED COMPONENT-RADIUS THRESHOLD

A finite stochastic register can carry infinitely many probability distributions. Thus an infinite numerical orbit is not, by itself, an obstruction to finite positive realization. A further difficulty arises when the realization may be redesigned at every horizon and its transitions may depend on a free external clock. The obstruction established here survives both freedoms. It is measured by the radius of an observable image inside its allowed output set.

For the clocked theorem in this section, let H be a compact Hausdorff group with finitely many connected components. Put $K = H^\circ$ and $C = H/K$. Then K is an open normal subgroup and C is a finite group. Fix a finite alphabet $\mathcal{A}$ with elements $u_a \in H$ whose generated group is dense in H . Our convention is

$$u_{a_1 \cdots a_n} = u_{a_n} \cdots u_{a_1}.$$

The identity e need not be an alphabet element. Instead assume the following executable return condition:

(7.1)

there is an integer $g \geq 0$ such that, for every $n \geq g$, some word of length n has product e .

An identity letter gives $g = 0$. The condition concerns physical products; the stochastic rows on a return word may perform arbitrary processing.

For nonempty finite seed and query sets $\mathcal{S}, \mathcal{J}$, let

$$F_{s,j} : H \longrightarrow Y_j$$

be continuous, where Y_j is a nonempty compact convex subset of a finite-dimensional real normed space $(V_j, \|\cdot\|_j)$. A query is selected only after the command word. A prescribed joint law of finitely many outputs is treated as one query, not as a collection of marginal constraints. In particular, a categorical interface has

$$Y_j = \Delta_{M_j} = \{p \in \mathbb{R}^{M_j} : p_i \geq 0, \sum_i p_i = 1\}, \quad \|x\|_j = \frac{1}{2} \sum_i |x_i|.$$

Then $\|p - q\|_j$ is total variation. Binary means correspond to $Y = [-1, 1]$ with $\|x\| = |x|/2$.

Definition 7.1 (Clocked realization). At horizon N choose finite nonempty registers S_t , $0 \leq t \leq N$, initialization rows E_s , row-stochastic command matrices $T_{t,a} : S_{t-1} \rightarrow S_t$, and terminal decoder maps $D_j : S_N \rightarrow Y_j$. The realized output vector on $w = a_1 \cdots a_N$ is

$$E_s T_{1,a_1} \cdots T_{N,a_N} D_j.$$

Its error is the maximum, over all seeds, length- N words, and queries, of its distance from $F_{s,j}(u_w)$. The command width is $\max_t |S_t|$, counting all declared labels, including unused padding labels. Write $W_{N,\varepsilon}$ for its least possible value at error at most ε .

All retained private information, including randomness kept for future use, is charged to the current label. The rows may depend on t and N , but not on an unrecorded input history or on the future query. The external clock and horizon, arbitrary real table descriptions, their construction and lookup, real arithmetic, and exact sampling are uncharged. This is a numerical realization invariant, not uniform workspace or program size. For a finite categorical interface, charging the terminal answer register changes the peak to $\max(W_{N,\varepsilon}, \max_j M_j)$. Thus all boundedness and growth results are unchanged; the convention that also charges a sampled binary-answer register has a minimum of two labels. A decoder vector by itself is not an additional sampled register.

For $A \subset Y_j$ define its constrained radius by

$$\text{rad}_{Y_j}(A) = \min_{y \in Y_j} \sup_{x \in A} \|x - y\|_j.$$

The minimum is attained when A is compact. The critical error is

$$(7.2) \quad \varepsilon_c = \max_{s,j,c \in C} \text{rad}_{Y_j}(F_{s,j}(c)).$$

Here a component is identified with its coset in H ; its image is compact by continuity. The requirement that the center belong to Y_j is essential for categorical outputs.

Theorem 7.2 (Intrinsic radius classification). *Under (7.1), for every $\varepsilon \geq 0$ the following are equivalent:*

- (1) $\sup_N W_{N,\varepsilon} < \infty$;
- (2) *one stationary finite-state realization with permutation command updates has error at most ε at every horizon*;
- (3) $\varepsilon \geq \varepsilon_c$.

At the boundary, at most $|\mathcal{S}||C|$ command labels suffice. If $\varepsilon < \varepsilon_c$, then for every integer $k \geq 1$ there is an integer $L_k < \infty$ such that every error- ε realization satisfies

$$(7.3) \quad \#\{t \in \{1, \dots, N\} : |S_t| \leq k\} \leq L_k.$$

All intervening widths are unrestricted. In particular, $W_{N,\varepsilon} \rightarrow \infty$.

Neither a positive lower bound on state masses nor a reachable-rank, observability, or conditioning assumption occurs in the theorem. The upper realization is deterministic until its terminal decoder. It need not have the minimum width among all stochastic realizations. Theorem 5.4, rather than a deterministic component partition, characterizes that stationary minimum.

For binary means, (7.2) reduces to one quarter of the largest componentwise oscillation. For a genuine categorical law it is generally not half its diameter; an analytic three-outcome example below has critical total-variation error $2/3$, whereas every individual event has binary threshold $1/2$. The proof uses finitely many localized vector tests rather than a separate midpoint for each coordinate.

The finite-component assumption does not assert that H is a Lie group. For example, take $H = \prod_{n \geq 1} \mathbb{R}/\mathbb{Z}$, choose an element whose every finite coordinate tuple together with 1 is rationally independent, and include this element and e as commands. The positive orbit is dense and H is connected but has no faithful finite-dimensional representation. The theorem applies to every finite continuous interface on this group. With infinitely many components the component construction need not be finite; Section 10 gives the finite-observable-quotient criterion and the strict-side clocked theorem in that setting.

There is also a quantitative counterpart. For planar rotations with a dual badly approximable s -tuple of angles and signal amplitude $0 < \rho < 1$, Theorem 12.1 proves

$$W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}) \quad \text{for every fixed } 0 \leq \varepsilon < \rho/2.$$

The upper realization is exact even when $\varepsilon > 0$. The new lower estimate reaches the entire sub-critical interval, by combining harmonic localization with the arithmetic orbit-packing mechanism. This quantitative theorem and the radius classification have distinct hypotheses: algebraic matrix entries alone do not imply the badly approximable angle condition.

7.1. Executable tests. Let $\mathcal{P} = \{u_w : w \in \mathcal{A}^*\}$. We use standard Haar averaging and the density of matrix coefficients of finite-dimensional unitary representations, in realified form [13]. No quantitative mixing assumption for a preassigned random walk is imposed.

Lemma 7.3 (Positive words and common lengths). *The semigroup $\mathcal{P}$ is dense in H , and $\mathcal{P} \cap K$ is dense in K . Every finite subset of $\mathcal{P} \cap K$ has executable representatives of a common positive length.*

Proof. For an element u of a compact group, positive powers return arbitrarily close to e with exponent at least two. Indeed cover the compact closure of its powers by finitely many translates of a sufficiently small neighborhood. Among sufficiently many powers, three lie in one member of the cover; the first and last exponents differ by at least two, and their quotient is arbitrarily close to e . This argument also covers finite-order elements by taking a sufficiently large multiple of the order. If u^r approaches e with $r \geq 2$, then $u^{r-1} = u^r u^{-1}$ approaches u^{-1} and is still a positive power. Thus u^{-1} is in the closure of positive powers. The closed semigroup $\overline{\mathcal{P}}$ therefore contains the group generated by the letters, and hence equals H . Since H has finitely many components, K is open, and a relatively open subset of K is open in H ; density gives the second assertion. For finitely many representative lengths ℓ_i , choose $B \geq \max_i \ell_i + g$, $B \geq 1$, and append return words of lengths $B - \ell_i$. The products are unchanged. $\square$

Lemma 7.4 (Finite executable gap). *Let $R : K \rightarrow O(E)$ be a continuous representation on a nonzero finite-dimensional Euclidean space, with no nonzero fixed vector. For every $k \geq 1$ there are $B \geq 1$, a probability law ν on finitely many executable length- B words with products in K , and $d \in (0, 1)$ such that*

$$(7.4) \quad \mathbb{E}_{w \sim \nu} \max_{1 \leq i \leq k} \langle v_i, R(u_w)u \rangle \leq 1 - d$$

for all unit vectors $u, v_1, \dots, v_k \in E$.

Proof. On the compact product of unit spheres consider the minimum

$$\gamma = \min_{u, v_1, \dots, v_k} \int_K \left(1 - \max_i \langle v_i, R(h)u \rangle \right) d\mu(h).$$

Repeated centers are permitted. The integrand is nonnegative. If a minimizing tuple had integral zero, full support of Haar measure would give $R(h)u \in \{v_1, \dots, v_k\}$ for every h . The orbit of u is connected, hence a singleton, contradicting the fixed-vector hypothesis. Thus $\gamma > 0$.

Partition the compact image $R(K)$ into finitely many measurable sets of small diameter in the Euclidean operator norm, and replace their Haar masses by atoms at nearby executable products in K . Cells of zero Haar mass can be discarded. The function $A \mapsto \max_i \langle v_i, Au \rangle$ is uniformly 1-Lipschitz in that norm. An approximation radius less than $\gamma/2$ therefore preserves a positive defect uniformly in the unit vectors. Lemma 7.3 gives a common length. Decreasing the defect to a number in $(0, 1)$ proves the assertion. $\square$

Lemma 7.5 (Conditional-centroid contraction). *Let $Z \in E$ be a bounded random physical vector and S the current register. Apply an independent word $w \sim \nu$ from Lemma 7.4, and let the actual block kernel generate a terminal state S' having at most k labels. Then*

$$(7.5) \quad \mathbb{E} \|\mathbb{E}[R(u_w)Z \mid S']\| \leq (1 - d) \mathbb{E} \|\mathbb{E}[Z \mid S]\|.$$

A fixed intervening return word cannot increase the same conditional norm, regardless of its intermediate or terminal widths.

Proof. Put $z_s = \mathbb{E}[Z \mid S = s]$ and write $T_w(s, i)$ for the actual composite row of the block. Since this row depends on the past only through S ,

$$\Pr(S' = i) \mathbb{E}[R(u_w)Z \mid S' = i] = \sum_s \Pr(S = s) \mathbb{E}_w T_w(s, i) R(u_w) z_s.$$

Choose v_i in the direction of each nonzero terminal centroid, arbitrarily otherwise. Pair, sum in i , and use stochasticity to bound the sum by

$$\sum_s \Pr(S = s) \mathbb{E}_w \max_i \langle v_i, R(u_w) z_s \rangle \leq (1 - d) \sum_s \Pr(S = s) \|z_s\|.$$

Zero-probability states contribute zero. No restriction was placed on registers inside the block. On a fixed return word the physical vector is unchanged at its endpoints; the new state is a stochastic image of the old one, so conditional Jensen gives nonincrease. Independence is used only at complete block boundaries. $\square$

8. REPRESENTATIVE FUNCTIONS AND LOCALIZED RADII

A real representative function on K is a matrix coefficient of a finite-dimensional continuous orthogonal representation, or a finite sum of such coefficients. Their algebra contains the constants, is closed under multiplication by tensor products, and is uniformly dense in $C(K, \mathbb{R})$ by the Peter–Weyl approximation theorem [13]. A finite collection of representative functions can be written on one space

$$\mathcal{V} = \bigoplus_{i=1}^a \text{End}(E_i), \quad \Phi(h) = (R_i(h))_{i=1}^a,$$

where the scalar product is the sum of Frobenius products. Left multiplication defines an orthogonal representation R on $\mathcal{V}$ and satisfies $\Phi(gh) = R(g)\Phi(h)$. Adjoin a trivial summand when necessary. Let

$$P = \int_K R(h) d\mu(h), \quad E = (I - P)\mathcal{V}, \quad Z(h) = (I - P)\Phi(h), \quad Q_0 = \|Z(e)\|.$$

Haar averaging is the orthogonal projection onto fixed vectors. Consequently $Z(gh) = R(g)Z(h)$, the representation on E has no fixed vectors, and $P\Phi(h) = P\Phi(e)$ for $h \in K$.

Choose and fix a coefficient vector a_p for each scalar representative function p , and a linear coefficient map A_G for each vector-valued representative function G . Their norms need not be intrinsic; fixing any such choices suffices. For an arbitrary joint distribution of $h \in K$ and a finite state S , put $Q = \mathbb{E}\|\mathbb{E}[Z(h) \mid S]\|$ and $\bar{p} = \int p d\mu$. The fixed/moving decomposition gives

$$(8.1) \quad \mathbb{E}[p(h) \mid S] = \bar{p} + \langle a_p, \mathbb{E}[Z(h) \mid S] \rangle,$$

$$(8.2) \quad \|\mathbb{E}p - \bar{p}\| \leq \|a_p\|Q, \quad \|\mathbb{E}G - \bar{G}\| \leq \|A_G\|Q.$$

Here the norm of A_G is the operator norm from the Euclidean feature space to the indicated output space. If $D : S \rightarrow Y$ and $M_Y = \max_{y \in Y} \|y\|$, then

$$(8.3) \quad \|\mathbb{E}[p(h)D(S)] - \bar{p}\mathbb{E}D(S)\| \leq M_Y \|a_p\|Q.$$

Indeed condition on S in (8.1), multiply by $D(S)$, and apply the triangle inequality. These are moment estimates, not total-variation convergence of a hidden physical distribution.

Lemma 8.1 (Finite localization of a radius). *Let $f : K \rightarrow Y$ be continuous, with Y compact and convex in a finite-dimensional normed space. Suppose $\text{rad}_Y(f(K)) > \varepsilon$. There are a vector-valued representative function G , a number $\eta > 0$, and nonnegative representative functions $p_1, \dots, p_a$ of Haar mean one such that*

$$\|G - f\|_\infty < \eta, \quad r_* := \min_{y \in Y} \max_i \|p_i \bar{G} - y\| > \delta := \varepsilon + \eta.$$

Proof. Choose $\varepsilon < t < \text{rad}_Y(f(K))$. For each $y \in Y$ some $h \in K$ has $\|f(h) - y\| > t$. The corresponding open subsets of Y cover Y , so a finite subcover yields $h_1, \dots, h_a$ with

$$\min_{y \in Y} \max_i \|f(h_i) - y\| > t.$$

The strict inequality follows also at a minimizing y , since the maximum is everywhere greater than t and Y is compact.

Fix a sufficiently small positive η . Approximate the coordinates of f uniformly by representative functions to obtain $\|G - f\|_\infty < \eta$. For each i , choose a nonzero nonnegative continuous bump supported in a neighborhood where f differs from $f(h_i)$ by less than η . Such a bump exists by normality of the compact Hausdorff space; it has positive Haar integral by full support. Approximate its square root uniformly by a representative function, square the approximation, and normalize its Haar integral. The resulting nonnegative representative function p_i has mean one, and its weighted f -mean can be made arbitrarily close to that of the bump. Thus we may ensure

$$\|p_i \bar{G} - f(h_i)\| < 3\eta.$$

The constrained radius changes by at most the maximum perturbation of the finitely many points. Taking $4\eta < t - \varepsilon$ gives $r_* > \varepsilon + \eta$. $\square$

Proposition 8.2 (Residual moment from a vector interface). *Fix the data of Lemma 8.1. Choose one feature space representing all p_i and $p_i G$, and fix their coefficient choices before defining*

$$L = \max_i (\|A_{p_i G}\| + (M_Y + \delta)\|a_{p_i}\|), \quad \Delta = r_* - \delta > 0.$$

Suppose a terminal decoder $D(S) \in Y$ obeys

$$\|\mathbb{E}[D(S) \mid w] - G(h(w))\| \leq \delta$$

for every complete deterministic word in a test law. Then $L > 0$, $E \neq \{0\}$, $Q_0 > 0$, and

$$(8.4) \quad Q \geq \Delta/L.$$

Proof. For $p = p_i \geq 0$, wordwise correctness and the triangle inequality imply

$$\|\mathbb{E}[p(h)D(S)] - \mathbb{E}[p(h)G(h)]\| \leq \delta \mathbb{E}p(h).$$

This is a conditional mean error, not a pointwise bound on the random decoder. Since $\bar{p}_i = 1$, calibration gives explicitly

$$\mathbb{E}p_i \leq 1 + \|a_{p_i}\|Q.$$

Write $y = \mathbb{E}D(S)$, which belongs to Y by convexity. Equations (8.2)–(8.3), together with $\bar{p} = 1$, give

$$\|y - p_i \bar{G}\| \leq \delta + (\|A_{p_i G}\| + (M_Y + \delta)\|a_{p_i}\|)Q.$$

Maximizing in i yields $r_* \leq \delta + LQ$. Hence $LQ \geq \Delta > 0$. In particular the moving feature and its initial norm cannot vanish: by equivariance, $Q_0 = 0$ would imply $Z(h) = 0$ for all h and therefore $Q = 0$. $\square$

8.1. Proof of Theorem 7.2. Choose (s, j, c) with component radius greater than ε . By density there is an executable prefix v with $u_v \in c$. Put $b = |v|$ and $f(h) = F_{s,j}(hu_v)$ on K . Since $Ku_v = c$, this map has the same constrained radius. Apply Lemma 8.1 and Proposition 8.2.

The moving feature is nonzero independently of any particular test law. Otherwise all represented functions would be constant on K ; since each p_i has mean one, this would force $p_i = 1$ and G constant. Then all $p_i \bar{G}$ would equal a point at distance less than η from Y , contradicting $r_* > \delta \geq \eta$.

For a fixed width k , apply Lemma 7.4 to the resulting moving feature. Let B, d be its block parameters and set

$$\chi = -\log(1 - d) > 0, \quad A = \max(1, LQ_0/\Delta).$$

Start a test word with v ; between independent length- B test blocks use fixed return words. The accumulated physical product is hu_v . Initially $h = e$, so the conditional feature norm after the

prefix equals Q_0 , regardless of the hidden-state distribution. If J block endpoints have width at most k , Lemma 7.5 gives

$$Q_N \leq Q_0(1 - d)^J.$$

The machine is correct on every deterministic word of length N . Thus any external law supported on such words inherits the conditional mean bound, with error $\delta = \varepsilon + \eta$ for G . Proposition 8.2 implies

$$(8.5) \quad J\chi \leq \log A.$$

The law depends on the advertised width profile, not on the machine's private state. When $A = 1$, (8.5) excludes even one selected narrow block; no logarithmic loss has occurred.

For clarity we give the spacing count with the full return hypothesis. Discard narrow cuts before $b + B + g$ and after $N - g$. From the remaining cuts greedily choose the first and then the next at distance at least $B + g$. Put an independent block immediately before each selected cut. The initial gap, all intervening gaps, and the final suffix have lengths at least g , so (7.1) fills them with executable returns. At most $b + B + 2g - 1$ cuts were discarded. Each chosen endpoint accounts for at most $B + g$ eligible cuts. Therefore one may take

$$(8.6) \quad L_k = \left\lceil b + B + 2g - 1 + (B + g) \frac{\log A}{\chi} \right\rceil.$$

The constants B, d depend on the fixed moving representation and k ; $\chi = -\log(1 - d)$ depends on that defect. The localization, prefix length b , and A depend only on the alphabet, interface, chosen component, and error. Consequently L_k depends on these fixed data, the return certificate g , and k , but not on N or the machine. When the eligible interval is empty the same estimate follows directly by counting. This proves occupation without restricting any register inside a block or gap. If the entire peak is at most k , then $N \leq L_k$, which proves divergence directly. Monotonicity in the horizon is not needed when an identity letter is absent.

For the upper bound, use labels $(s, c) \in \mathcal{S} \times C$, including any labels not reached at a given horizon. Initialize at (s, K) and update by left multiplication

$$(s, c) \mapsto (s, [u_a]c).$$

Choose a minimizing center of $F_{s,j}(c)$ in Y_j as decoder at (s, c) for query j . These are legal decoder values, and the error is at most (7.2). All command updates are permutations of one finite state set, independent of clock and horizon. This proves boundary attainment and the equivalence. $\square$

Proposition 8.3 (A finite return certificate). *Condition (7.1) follows if there are finitely many executable identity-product words whose positive lengths have greatest common divisor one.*

Proof. Let the lengths be $\ell_1, \dots, \ell_a$. Their nonnegative sums modulo ℓ_1 form the subgroup generated by their residues: a subsemigroup of a finite group is a subgroup. The gcd hypothesis makes this the whole residue group. Choose one nonnegative sum n_r for each residue r modulo ℓ_1 , and put $g = \max_r n_r$. Every $n \geq g$ equals its representative n_r plus a nonnegative multiple of ℓ_1 . Concatenating the corresponding return words proves the claim. $\square$

For example, the alphabet $\{R_\theta, R_{-\theta}, R_{2\pi/3}\}$, with $\theta/2\pi$ irrational, contains no identity letter but has return lengths two and three. It satisfies (7.1) with $g = 2$. A single irrational rotation letter does not: there is only one word at each horizon, and a horizon-specific constant decoder can store its numerical answer with one command label. Thus synchronization cannot be silently omitted. Nor is a bounded irreversible semigroup enough: for the identity command $I = 1$ and contraction command $a = \lambda \in (0, 1)$, the mean $\rho\lambda^{\#a(w)}$ has an exact alive/absorbed two-state realization. The group generated with inverses is not bounded. The present theorem is not a compact-semigroup classification.

9. EXACT-RETURN LANGUAGES AND LENGTH PHASES

The existence of cofinite returns is not the only useful synchronization regime. In fact an exact-return language, when nonempty, has a rigid eventual form. Here and in this section a return has positive length.

Lemma 9.1 (The return period). *Suppose*

$$\mathcal{R} = \{|w| > 0 : u_w = 1\} \neq \emptyset, \quad p = \gcd \mathcal{R}.$$

There is an integer g_p such that every multiple dn with $n \geq g_p$ belongs to $\mathcal{R}$, and no length outside $p\mathbb{N}$ belongs to $\mathcal{R}$. In particular, every nonempty exact-return language is eventually one arithmetic progression.

Proof. Concatenation makes $\mathcal{R}$ additive. A finite subset of its elements already has $\gcd p$: start with one length and successively decrease the $\gcd$ when necessary; a strictly decreasing chain of its positive divisors is finite. Divide these finitely many lengths by p . Their $\gcd$ is one, so the residue argument of Proposition 8.3 shows that their nonnegative sums contain every sufficiently large integer. Multiplying by p proves the assertion. The exclusion of other lengths is the definition of the $\gcd$. $\square$

Fix a letter with product $a \in H$, and let J be the closure of the products of words whose lengths are multiples of p , including the empty word.

Lemma 9.2 (The phase subgroup). *The group J is an open normal subgroup of H . The quotient H/J is cyclic of order dividing p , generated by aJ , and*

$$(9.1) \quad \overline{\{u_w : |w| \equiv r \pmod{p}\}} = a^r J \quad (0 \leq r < p).$$

If H/H° is finite, then $J^\circ = H^\circ$.

Proof. The compact closure defining J is a group, by the positive-power argument, and is generated densely by the block alphabet $\mathcal{A}^p$. Also $a^p \in J$. If $x = u_w$ for a word w of length p , then the executable word consisting of $p-1$ copies of the chosen letter, followed by w , followed by one more copy, has product axa^{p-1} and length $2p$. Thus $axa^{p-1} \in J$, whence $axa^{-1} \in J$. It follows that $aJa^{-1} \subset J$. Applying this inclusion p times and using $a^p \in J$ shows equality. Every letter b has $ba^{p-1} \in J$, so $ba^{-1} \in J$. Thus all letters lie in Ja , normalize J , and their products of length n lie in Ja^n . The finite union $\bigcup_{r=0}^{p-1} Ja^r$ is a closed group containing the letters, hence is H . This proves normality, finite index, and openness. Conversely, words consisting of r copies of a followed by arbitrary p -blocks have products dense in Ja^r , proving (9.1). An open subgroup contains the identity component, and its own identity component is therefore the same one. $\square$

The quotient order may be a proper divisor of p ; the phase radii repeat whenever the physical cosets repeat. The phase index records executable length, not merely the physical component quotient.

Assume now that H has finitely many components and put $K = H^\circ$. For $r \in \{0, \dots, p-1\}$ define

$$(9.2) \quad \varepsilon_r = \max_{s,j, c \in H/K: c \subset a^r J} \text{rad}_{Y_j}(F_{s,j}(c)).$$

Theorem 9.3 (The phasewise clocked threshold). *Under the sole synchronization assumption $\mathcal{R} \neq \emptyset$, for every r and $\varepsilon \geq 0$,*

$$\sup_{N \equiv r \pmod{p}} W_{N,\varepsilon} < \infty \iff \varepsilon \geq \varepsilon_r.$$

Equality is attained by a finite component-permutation realization on the indicated horizons. If $\varepsilon < \varepsilon_r$, then for each k there is a constant $L_{k,r}$, independent of N , such that every error- ε realization at $N \equiv r \pmod{p}$ satisfies

$$\#\{1 \leq t \leq N : |S_t| \leq k\} \leq L_{k,r}.$$

The intervening widths are unrestricted. Consequently the full family is bounded exactly at and above $\max_r \varepsilon_r = \varepsilon_c$; below it, divergence is asserted phase by phase, not necessarily along every horizon.

Proof. Every product of length $N \equiv r$ lies in $a^r J$. Store its component and the seed and decode component centers as before. This proves the upper bound, including equality, with at most $|\mathcal{S}||H/K|$ labels. Centers on other components may be chosen arbitrarily when only this phase is under consideration.

For the converse fix a component $c \subset a^r J$ and s, j with radius greater than ε . We must count narrow cuts in every residue class, not only at block boundaries of one phase. For each $\ell \in \{0, \dots, p-1\}$ choose $b_\ell \in \{0, \dots, p-1\}$ congruent to $r - \ell$ modulo p . By (9.1) and the openness of components there is a fixed word v_ℓ , of length $p_\ell \equiv \ell \pmod{p}$, with product in $a^{-b_\ell} c$. Indeed this component is contained in $a^\ell J$. Consider the block experiment on J with alphabet $\mathcal{A}^p$ and target

$$G_\ell(h) = F_{s,j}(a^{b_\ell} h u_{v_\ell}), \quad h \in J.$$

It is a legal compact-convex interface. Its image on $K = J^\circ$ is $F_{s,j}(c)$, since K is normal. Lemma 9.1 gives cofinite exact returns in block length for this alphabet.

Restrict the original machine to words that begin with v_ℓ , end with b_ℓ copies of the chosen letter, and have $M = (N - p_\ell - b_\ell)/p$ arbitrary p -blocks in between. When $M \geq 0$, absorb the fixed prefix into the initialization and the fixed suffix into the terminal decoder; stochastic averaging keeps that decoder in Y_j . This produces an error- ε clocked block machine for G_ℓ , with registers at original cuts $p_\ell + dm$, $0 \leq m \leq M$. Theorem 7.2 bounds its narrow cuts $1 \leq m \leq M$ by a constant depending only on k and the fixed data. The omitted cuts of residue ℓ number at most $p_\ell + b_\ell + 1$: this also charges the block cut zero, while all counted cuts remain in $1, \dots, N$. If $M < 0$, the whole horizon is bounded by $p_\ell + b_\ell$. Sum these bounds over the p residue classes. This counts all cuts $1, \dots, N$, independently of all other register widths. $\square$

The constants in this theorem depend on the fixed alphabet, return period and certificate, interface, error, width k , and chosen finite prefixes. They do not depend on the horizon or on the machine. The original constants B, d, χ, L_k in Section 8 use d for a contraction defect. The return period here is denoted p to keep the two parameters distinct.

Proposition 9.4 (Different phases can have different thresholds). *There is an alphabet with return period two for which $W_{N,0} = 1$ on every even horizon, but $W_{N,\varepsilon} \rightarrow \infty$ along odd horizons for every $0 \leq \varepsilon < \rho/2$.*

Proof. Take $H = (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z}$ and letters $(\theta, 1), (-\theta, 1)$, with $\theta/2\pi$ irrational. The generated group is dense in H . An exact return must have zero signed count and hence even length; the two-letter return gives every positive even length. Here $J = (\mathbb{R}/2\pi\mathbb{Z}) \times \{0\}$. For one binary-mean query put $F(\phi, 0) = 0$ and $F(\phi, 1) = \rho \cos \phi$, where $0 < \rho \leq 1$. The even-phase target is identically zero and needs one label. The odd-phase radius in the norm $|\cdot|/2$ is $\rho/2$. Apply Theorem 9.3. $\square$

10. INFINITE COMPONENT GROUPS AND BOUNDARY ATTAINMENT

This section retains independent finite-dimensional proofs of the strict-radius and exact translate-rank conclusions. Its output spaces are finite-dimensional. Let H again be an arbitrary compact

Hausdorff group, and set $K = H^\circ$. The quotient H/K is profinite. We use two standard compact-group consequences of this fact and of Peter–Weyl theory [13]: open normal subgroups form a neighborhood basis in a profinite group, and, for a continuous surjection $\pi : H \rightarrow G$ onto a compact Lie group, one has $\pi(K) = G^\circ$. To see the latter assertion, $\pi(K)$ is connected and normal, while $G/\pi(K)$ is a continuous group quotient of the profinite group H/K and is therefore totally disconnected. This forces $G^\circ \subset \pi(K)$; the reverse inclusion is immediate. A continuous finite-dimensional representation of a profinite group has finite image, by the closed subgroup theorem and the no-small-subgroups property of a Lie group [14]: an open normal subgroup can be placed inside a no-small-subgroups neighborhood, so its image is trivial.

Define the component radius

$$(10.1) \quad R_0 = \max_{s,j,g \in H} \text{rad}_{Y_j}(F_{s,j}(gK)).$$

Unlike $r(U)$, this quantity need not itself be realized by any finite quotient.

Lemma 10.1 (Approximation by finite quotient radii). *With $r(U)$ as in (1.1),*

$$\inf_{U \text{ open normal in } H} r(U) = R_0.$$

Proof. Every open subgroup contains K , so $r(U) \geq R_0$. Fix $\eta > 0$. Uniform continuity, for the finite family of interfaces, gives an identity neighborhood V such that

$$\|F_{s,j}(xv) - F_{s,j}(x)\|_j < \eta \quad (x \in H, v \in V, s, j).$$

There is an open normal subgroup U containing K and contained in KV , by the profinite quotient basis. Write each $u \in U$ as kv with $k \in K, v \in V$. Then $F_{s,j}(gu)$ lies within η of $F_{s,j}(gk)$. A center for the latter component image consequently covers the former coset image with an additional error at most η . Thus $r(U) \leq R_0 + \eta$. $\square$

Lemma 10.2 (A legal finite-dimensional interface approximation). *For every $\eta > 0$ there is a continuous homomorphism $\pi : H \rightarrow G \subset O(D)$ onto a compact Lie group and continuous maps $\tilde{F}_{s,j} : G \rightarrow Y_j$ such that*

$$\max_{s,j} \|F_{s,j} - \tilde{F}_{s,j} \circ \pi\|_\infty < \eta.$$

Moreover, for every $g \in H$,

$$\left| \text{rad}_{Y_j}(F_{s,j}(gK)) - \text{rad}_{Y_j}(\tilde{F}_{s,j}(\pi(g)G^\circ)) \right| < \eta.$$

Proof. Approximate all coordinates of all $F_{s,j}$ by finitely many real representative functions. The direct sum of their orthogonal representations is π , and the approximants are continuous functions on its compact image G . Choose Euclidean inner products on the finitely many output spaces. Project each approximant to the closed convex set Y_j by nearest-point projection. This projection is continuous and nonexpansive in the chosen Euclidean norm. Choose constants $a_j, b_j > 0$ with $a_j\|v\|_j \leq |v|_2 \leq b_j\|v\|_j$. An initial error less than $a_j\eta/b_j$ in $\|\cdot\|_j$ then gives a projected error less than η , by Euclidean nonexpansiveness. These constants are fixed before choosing the approximants. The closed subgroup $G \subset O(D)$ is a compact Lie group. Since $\pi(K) = G^\circ$, the two indicated images have Hausdorff distance less than η . Their constrained radii therefore differ by less than η . $\square$

Theorem 10.3 (Arbitrary compact groups on both strict sides). *For any compact-group interface, $\varepsilon > R_0$ implies a finite stationary permutation realization. At $\varepsilon = R_0$, such a realization exists exactly when $r(U) \leq R_0$ for some open normal U .*

Suppose in addition that executable exact returns have all sufficiently large lengths. If $\varepsilon < R_0$, then for each k there is a horizon-independent upper bound on the number of cuts of width at

most k , with all intervening widths unrestricted. In particular $W_{N,\varepsilon} \rightarrow \infty$. No finite-component assumption is made in either assertion.

Proof. The stationary assertions follow from Lemma 10.1 and Theorem 5.5. For the lower bound choose a component image of radius greater than ε and then $\eta > 0$ with 2η smaller than that strict difference. Apply Lemma 10.2. A machine correct for the original interface at error ε is correct for the approximating interface at error at most $\varepsilon + \eta$. The chosen component radius of that interface is greater than $\varepsilon + \eta$. The projected commands still have cofinite exact returns and generate G densely. Theorem 7.2 therefore gives the desired occupation bound for the very same registers. This argument never assumes that K is open or that executable products lying exactly in K are dense. $\square$

Theorem 6.3 now identifies this stationary boundary criterion with clocked boundedness, including $R_0 > 0$, and under the weaker approximate-return hypothesis. The zero-error boundary also admits the following independent translate-rank argument.

Lemma 10.4 (Evaluation rank and translate dimension). *For a real function f on a group, the supremum of the ranks of finite matrices $(f(y_l x_i))_{i,l}$ equals the dimension, possibly infinite, of the span of the functions $x \mapsto f(yx)$. These are left translates with the convention $(L_g f)(x) = f(g^{-1}x)$, after replacing y by g^{-1} .*

Proof. A finite-dimensional span bounds every evaluation rank. Conversely, if $f_1, \dots, f_m$ are independent functions, their evaluation vectors $(f_1(x), \dots, f_m(x))$ span $\mathbb{R}^m$: a proper span would have a nonzero annihilator, giving a linear dependence of the functions. Select m evaluation vectors forming a basis. This gives an invertible evaluation matrix and proves the reverse inequality for every finite m . $\square$

Theorem 10.5 (The exact clocked boundary). *Assume cofinite executable exact returns. The following are equivalent:*

- (1) $\sup_N W_{N,0} < \infty$;
- (2) all $F_{s,j}$ factor through one finite continuous quotient H/U , with U open normal;
- (3) there is an exact stationary finite permutation realization.

This includes groups with infinitely many connected components.

Proof. The implications from the second assertion to the third and from the third to the first are immediate. Suppose the first holds with bound k . Theorem 10.3 implies $R_0 = 0$, so every interface is constant on each K -coset.

Fix a real coordinate f of one $F_{s,j}$. For any finite families of executable prefix products x_i and suffix products y_l , use cofinite returns to pad all selected prefix words to one common length A and all suffix words to one common length B . Choose an exact realization at horizon $A + B$ of width at most k . At its cut A the matrix

$$(f(y_l x_i))_{i,l}$$

factors through the k hidden labels: the left factors are the actual state distributions after the padded prefixes and the right factors are the actual conditional terminal expectations on the padded suffixes. Hence this matrix has rank at most k . Padding does not need to preserve hidden states; it only preserves the physical products used in the displayed entries. By density and continuity the same rank bound holds for arbitrary $x_i, y_l \in H$.

It follows that the span of the left translates of f in $C(H, \mathbb{R})$ has dimension at most k . Otherwise $k + 1$ linearly independent translates would have an invertible evaluation matrix at some $k + 1$ points, contrary to the preceding rank bound. Lemma 10.4 formalizes this evaluation assertion. Equivalently it follows inductively: a nonzero function is nonzero somewhere, and a function outside the span of the preceding ones cannot have all its evaluations forced by theirs.

Take the sum V of these translate spaces over all seed–query coordinates. It is finite-dimensional and invariant under left translation. Translation is continuous in the uniform norm and preserves that norm, so it gives a continuous representation of H on V with compact image; Haar averaging makes it orthogonal. Because K is normal and every original coordinate is constant on K -cosets, K acts trivially on every translate and thus on V . The representation factors through H/K . Its image is finite by the profinite finite-dimensional representation fact above. Its kernel U is open normal. Since every original coordinate belongs to V , all interfaces are U -invariant and descend to H/U . $\square$

Proposition 10.6 (A nonattained zero-radius boundary). *On a compact group with infinitely many components, every positive error can have a bounded stationary realization while the zero-error clocked width is unbounded.*

Proof. Let $H = \mathbb{Z}_2$ be the additive group of 2-adic integers, with commands 0 and 1. Their positive products are dense, and the zero command supplies exact returns of every length. If $x = \sum_{n \geq 0} x_n 2^n$, $x_n \in \{0, 1\}$, prescribe the binary success probability

$$f(x) = \sum_{n \geq 0} \frac{2x_n}{3^{n+1}}.$$

Uniform convergence makes f continuous. It is injective: at the first differing digit the contribution $2/3^{n+1}$ exceeds the sum $1/3^{n+1}$ of all possible later differences. Since H is totally disconnected, $K = \{0\}$ and $R_0 = 0$. Truncation after m digits factors through $\mathbb{Z}/2^m\mathbb{Z}$ and has uniform error at most $\sum_{n=m}^{\infty} 2/3^{n+1} = 3^{-m}$. Thus every positive tolerance admits a finite quotient machine. But the injective function f cannot factor through any finite quotient. Theorem 10.5 excludes bounded exact clocked width, and Theorem 5.5 excludes a finite exact stationary realization. $\square$

Theorem 10.7 (Linear exact width at a profinite boundary). *For the interface of Proposition 10.6,*

$$W_{N,0} = \Theta(N).$$

More precisely, if n is a power of two with $2n - 1 \leq N$, then $W_{N,0} \geq n$, while $W_{N,0} \leq N + 1$. Every exact realization, with unrestricted intermediate widths, has at most $4k$ cuts in $\{1, \dots, N\}$ of width at most k . Nevertheless, for each $\varepsilon > 0$, $\sup_N W_{N,\varepsilon} < \infty$.

Proof. For nonnegative integers z write $v_2(z+1)$ for the exponent of two in $z+1$. Incrementing the binary expansion, with $v_2(z+1)$ trailing ones, gives

$$(10.2) \quad b(z) := f(z+1) - f(z) = \frac{5}{3} 3^{-v_2(z+1)} - 1.$$

Let $n = 2^m$, and form $B_n = (b(i+j))_{0 \leq i,j < n}$. Define a function on $\mathbb{Z}/n\mathbb{Z}$ by

$$\psi_m(x) = 3^{-v_2(x)} \ (x \neq 0), \quad \psi_m(0) = 3^{-m}.$$

Here the valuation of a nonzero residue is independent of its representative. Since $1 \leq i+j+1 \leq 2n-1$, its valuation equals that used in $\psi_m(i+j+1 \bmod n)$, including the case $i+j+1 = n$. Thus replacing row i by the old row $(-i-1) \bmod n$ changes its entry at column j to the kernel value at $j-i \bmod n$. This is the circulant with kernel $(5/3)\psi_m - 1$.

On $\mathbb{Z}/n\mathbb{Z}$ one has the elementary identity

$$\psi_m = 1 - 2 \sum_{\ell=1}^m 3^{-\ell} \mathbf{1}_{2^\ell | x}.$$

Use the eigenvector with entries $\exp(2\pi i q j / n)$; its eigenvalue is the kernel sum against $\exp(2\pi i q x / n)$. The sign is immaterial for subgroup indicators, whose transforms below are real. The Fourier

eigenvalue at frequency zero is

$$\mu_0 = \frac{2}{3}n6^{-m} > 0.$$

At frequency $q \neq 0$ it is

$$\mu_q = -\frac{10n}{3} \sum_{\substack{1 \leq \ell \leq m \\ n/2^\ell | q}} 6^{-\ell} < 0.$$

Indeed a subgroup indicator has Fourier sum $n/2^\ell$ exactly when $n/2^\ell$ divides q , and zero otherwise. For $q \neq 0$ the summand $\ell = m$ is always present. At $m = 0$ the sole eigenvalue is $2/3$. Consequently B_n is invertible for every dyadic n .

At an arbitrary command cut $t < N$, take a dyadic $n \leq \min(t+1, N-t)$. For each $i < n$ choose a prefix of length t with i increment commands. For each $j < n$ choose two suffixes of length $N-t$ with respectively $j+1$ and j increment commands. Fill all remaining positions with zero commands. The difference of their exact outputs is $b(i+j)$. Hence B_n factors through the actual register S_t : one factor consists of the prefix state rows and the other of the differences of the two conditional suffix decoder columns. Invertibility gives $|S_t| \geq n$. The difference column need not be a legal decoder; it is used only for this rank bound on two legal experiments.

Taking $t = n-1$ proves $W_{N,0} \geq n$ whenever $2n-1 \leq N$. The largest dyadic $n \leq (N+1)/2$ is greater than $(N+1)/4$, proving the linear lower order. Storing the number of increment commands uses at most $N+1$ labels and gives the exact upper bound.

For occupation, if $t < N$ and $|S_t| \leq k$, the largest dyadic integer not exceeding $\min(t+1, N-t)$ is at most k , so $\min(t+1, N-t) < 2k$. Such cuts lie within fewer than $2k$ positions of one of the two ends. Including the terminal cut gives at most $4k$ positive cuts. No width at another cut was restricted. The positive-error assertion is the finite-quotient truncation in Proposition 10.6. $\square$

This supplies a sharp quantitative law on a genuinely infinite profinite group. Its linear exact-width obstruction and bounded positive-error realizations concern one fixed continuous interface; they are not a uniform approximation statement over all continuous outputs on that group.

11. BINARY, CATEGORICAL, AND OBSERVABLE INTERFACES

Corollary 11.1 (Component oscillation for binary means). *For a finite continuous binary-mean interface $f_{s,j} : H \rightarrow [-1, 1]$, with the return hypothesis of Theorem 7.2, the exact bounded-width threshold in total variation is*

$$(11.1) \quad \varepsilon_c = \frac{1}{4} \max_{s,j,c} \left(\max_{h \in c} f_{s,j}(h) - \min_{h \in c} f_{s,j}(h) \right).$$

It is attained by a stationary permutation realization. Below it every fixed width has the occupation bound (7.3).

Proof. In the norm $|\cdot|/2$, the radius of an interval is one quarter of its length. Its midpoint lies in $[-1, 1]$. Apply Theorem 7.2. $\square$

This recovers the scalar component theorem, including all of its nonuniform quantifiers. The vector theorem does not follow by applying this corollary separately to coordinates, because the coordinate centers need not belong to the output simplex.

Proposition 11.2 (A three-outcome threshold). *On the circle let $\omega = e^{2\pi i/3}$ and, for $j = 0, 1, 2$, set*

$$p_j(\phi) = \frac{1}{9} \left| 1 + e^{i\phi} \omega^{-j} + e^{2i\phi} \omega^{-2j} \right|^2.$$

For any dense circle command alphabet satisfying the return condition, the categorical law $p = (p_0, p_1, p_2)$ has critical total-variation error $2/3$. Each separately requested binary event $\{j\}$ has threshold $1/2$.

Proof. Orthogonality of the three characters gives $\sum_j p_j = 1$; each entry is nonnegative. At $\phi = 2\pi j/3$ the vector is the simplex vertex e_j . Hence every center $q \in \Delta_3$ has maximum distance at least

$$\max_j \text{TV}(e_j, q) = 1 - \min_j q_j \geq 2/3.$$

The uniform center has distance at most $2/3$ from every point of Δ_3 , by convexity and the values at its vertices. Thus the constrained radius is exactly $2/3$. On the other hand each p_j has range $[0, 1]$, so its binary mean $2p_j - 1$ has oscillation two. Apply Theorem 7.2 and Corollary 11.1. In particular the categorical radius exceeds half the diameter, which here equals $1/2$. $\square$

Corollary 11.3 (Joint measure-once unitary laws). *Let unitary commands have compact closure H and satisfy the return condition. For a fixed unit seed ψ_s and a finite measurement $E_{j,l} \geq 0$, $\sum_l E_{j,l} = I$, prescribe the law*

$$F_{s,j}(h)_l = \langle h\psi_s, E_{j,l}h\psi_s \rangle.$$

The critical error is its componentwise simplex radius in total variation. For closure $U(d)$ and one orthonormal-basis measurement, this radius is $1 - 1/d$.

Proof. The displayed map is a continuous simplex-valued polynomial in realified matrix entries. A closed unitary matrix group has finitely many components, so Theorem 7.2 applies. For $U(d)$, every unit vector is reachable from the seed, and its squared coordinate magnitudes range over the whole simplex. The simplex vertices force radius at least $1 - 1/d$, attained by the uniform distribution. $\square$

A query selects one complete measurement, not simultaneous answers to incompatible measurements. Several compatible finite outputs, with specified correlations, instead form one joint categorical law. Language recognition at a cutpoint is a different requirement.

For linear binary interfaces on a compact matrix group, write

$$f_{s,j}(h) = q_j^T h x_s \in [-1, 1], \quad V_r = \text{span}\{h x_s : h \in H, s \in \mathcal{S}\},$$

$$V_n = \{x \in V_r : q_j^T h x = 0 \text{ for all } h \in H, j \in \mathcal{J}\}.$$

Both spaces are invariant. The observable space $V_o = V_r \cap V_n^\perp$ is the orthogonal complement of V_n inside V_r , canonically realizing the observable quotient.

Theorem 11.4 (Intrinsic zero-error criterion). *For these linear means, bounded exact clocked width is equivalent to finiteness of the generated group's image in $O(V_o)$, and to trivial action of K on V_o . An exact stationary permutation realization then exists.*

Proof. Corollary 11.1 identifies bounded exact width with constancy of all interface means on every component. Under that condition, for $k \in K$ and $g, h \in H$, normality gives

$$q_j^T h(k - I)g x_s = q_j^T (h k h^{-1}) h g x_s - q_j^T h g x_s = 0.$$

Thus $(k - I)V_r \subset V_n$. On the invariant orthogonal complement V_o the difference also lies in V_o , hence vanishes. The observable action factors through the finite component group. Conversely, if K acts trivially on V_o , the same difference is invisible and all means are componentwise constant. A finite image of the dense generated group has the same finite image closure from H , whose connected subgroup K acts trivially. The zero-dimensional quotient is included. $\square$

Corollary 11.5 (The planar threshold). *Let an orthogonal planar alphabet contain I and an irrational rotation, with any additional orthogonal letters. For signed coordinate seeds $\pm \rho e_1, \pm \rho e_2$, coordinate queries, and $0 < \rho \leq 1$, the threshold is $\varepsilon_c = \rho/2$. At and above it one command label with a fair binary decoder suffices. Below it every fixed width has a finite occupation budget.*

Proof. The identity component is $SO(2)$, and every specified mean has range $[-\rho, \rho]$ on each component of $H = SO(2)$ or $O(2)$. All midpoint decoders equal zero, so the seed/component register collapses to one label. $\square$

For the noncommuting rational alphabet

$$I, \quad R = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

one has $FRF = R^{-1} \neq R$. The rotation has infinite order: otherwise $(3 + 4i)/5$ and its inverse would be algebraic integers, whereas their rational sum $6/5$ is not an integer. This example has the sharp threshold above. We do not infer a badly approximable angle from rational matrix entries.

Proposition 11.6 (Independent toral coordinates). *Suppose a component is identified with the full product torus $(\mathbb{R}/2\pi\mathbb{Z})^m$, surjectively in independent coordinates, and a mean on it equals*

$$a_0 + \sum_{l=1}^m (a_l \cos \theta_l + b_l \sin \theta_l).$$

Its contribution to (11.1) is $\frac{1}{2} \sum_l \sqrt{a_l^2 + b_l^2}$.

Proof. Each summand has the symmetric range with amplitude $\sqrt{a_l^2 + b_l^2}$, and all extrema are simultaneously attainable in the full product. Divide the resulting oscillation by four. $\square$

On a proper subtorus, extrema must instead be taken subject to its relations. Ambient coordinates do not certify independence. Similarly, invisible directions do not create an obstruction: for $H = \{\text{diag}(R_\theta, 1)\}$, seed and query e_3 give constant mean one, whereas seed ρe_1 and query e_1 have threshold $\rho/2$.

Proposition 11.7 (Compact similarity). *The classification applies to a finite alphabet in $GL(D, \mathbb{R})$ whose generated group, including inverses, is uniformly bounded, with the executable return hypothesis retained. Conjugating to an invariant Euclidean metric changes neither output probabilities nor stochastic widths.*

Proof. Uniform bounds on matrices and inverses make the closure a compact group. The matrix $Q = \int_H h^T h d\mu(h)$ is positive definite and satisfies $A^T Q A = Q$. Hence $Q^{1/2} A Q^{-1/2}$ is orthogonal. Transport the continuous output maps through this conjugacy; for a linear mean transport x_s to $Q^{1/2} x_s$ and q_j to $Q^{-1/2} q_j$. Exact identity products and their word lengths are preserved. $\square$

11.1. A curved categorical image with distinct component centers.

Proposition 11.8. *Let $H = (\mathbb{R}/2\pi\mathbb{Z}) \times \mathbb{Z}/2\mathbb{Z}$, with an identity letter, an irrational rotation in the first factor, and a toggle of the second factor. Put*

$$q_0 = (\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{6}), \quad q_1 = (\tfrac{1}{6}, \tfrac{1}{3}, \tfrac{1}{2}), \quad t = \tfrac{1}{12},$$

and prescribe the three-outcome law

$$F(\theta, c)_l = q_{c,l} + t \cos(\theta - 2\pi l/3), \quad l = 0, 1, 2.$$

Each component image is a proper closed curve in the interior of the simplex, with unique constrained total-variation center q_c and radius t . The union of both component images has radius

1/4. Thus the sharp clocked threshold is 1/12, and the stationary minimum at that threshold is exactly two labels.

Proof. The three cosines sum to zero and every coordinate is at least 1/12, so F is a legal law in the simplex interior. The first two independent trigonometric coordinates identify a nondegenerate ellipse in its affine plane. For a zero-sum vector $z \in \mathbb{R}^3$, one has $\frac{1}{2}\|z\|_1 = \max_l |z_l|$. Each coordinate of the curve ranges over $[q_{c,l} - t, q_{c,l} + t]$. Hence for every center $y \in \Delta_3$,

$$\sup_{\theta} \text{TV}(F(\theta, c), y) = t + \max_l |q_{c,l} - y_l|.$$

This proves uniqueness of q_c and radius t . For the union, the least possible value of $\max_{c,l} |q_{c,l} - y_l|$ is 1/6, forced by the two endpoints in the first coordinate and attained by $y = (1/3, 1/3, 1/3)$. Its radius is therefore $t + 1/6 = 1/4$. A one-label stationary machine has a constant decoder and cannot reach error t ; two component labels attain it. The clocked threshold follows from Theorem 7.2. $\square$

Remark 11.9. The total-variation diameter of a probability simplex is one, so half its diameter is 1/2; the three-vertex example above has radius 2/3. In the unitary corollary, $U(d)$ is connected. These facts explain respectively why coordinate midpoints do not solve the categorical problem and why the full-unitary component is the whole group.

12. MATCHING WIDTH LAWS THROUGHOUT THE SUBCRITICAL INTERVAL

We now distinguish the dimension of the command arithmetic from the dimension of the physical orbit. Let $s \geq 1$ and let the alphabet contain the rotations $R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_s}$ and I . For the matching upper bound the alphabet is exactly these letters, optionally with the reflection $F = \text{diag}(1, -1)$. Use signed coordinate seeds of radius ρ and coordinate queries. Extra letters do not affect the lower bounds, but are not silently included in the upper construction.

For a real number x , write $\|x\|_{\mathbb{R}/\mathbb{Z}}$ for distance to the nearest integer. The dual Diophantine condition is

$$(12.1) \quad \|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b \|n\|_1^{-\tau} \quad (0 \neq n \in \mathbb{Z}^s), \quad b > 0, \quad \tau \geq s.$$

In particular $1, \alpha_1, \dots, \alpha_s$ are rationally independent. When $\tau = s$ this is the dual badly approximable condition. All implicit constants below may depend on $\alpha, b, s, \rho, \varepsilon$, but not on N or a stochastic realization.

Theorem 12.1 (Full-subcritical matching law). *Suppose (12.1) holds with $\tau = s$, and let $0 < \rho < 1$. For every fixed $0 \leq \varepsilon < \rho/2$,*

$$(12.2) \quad W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}).$$

The upper bound is achieved by an exact horizon-dependent realization. At $\varepsilon \geq \rho/2$, one command label suffices. If $\tau \geq s$ is arbitrary, the lower bound $W_{N,\varepsilon} = \Omega(N^{s/(2\tau+1)})$ still holds throughout the same subcritical interval.

The exponent for one badly approximable rotation is 1/3, not logarithmic. The statement includes zero error when the signal has strict amplitude slack $\rho < 1$. The threshold theorem continues to include $\rho = 1$, but the exact polynomial upper bound here is not asserted at that endpoint. The earlier arithmetic argument obtained this scale in a smaller noise range. The proof below supplies the missing full-noise lower estimate and states the exact upper realization in full.

Proposition 12.2 (An explicit family in every command dimension). *Let ξ be a real algebraic integer of degree $s + 1$. Then $\alpha = (\xi, \xi^2, \dots, \xi^s)$ satisfies (12.1) with $\tau = s$ and some effectively computable $b > 0$.*

Proof. For $0 \neq n \in \mathbb{Z}^s$ put $H = \|n\|_1$, and choose $n_0 \in \mathbb{Z}$ so that $z = n_0 + \sum_{l=1}^s n_l \xi^l$ has modulus $\|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}}$. The degree assumption gives $z \neq 0$. Write $\xi_1 = \xi, \dots, \xi_{s+1}$ for the conjugates and put $M_j = \max_{1 \leq l \leq s} |\xi_j|^l$. Then $|n_0| \leq HM_1 + 1/2$ and

$$\left| n_0 + \sum_{l=1}^s n_l \xi_j^l \right| \leq H(M_1 + M_j + 1/2) \quad (j \geq 2).$$

The norm of the nonzero algebraic integer z is a nonzero integer. Thus

$$|z| \geq H^{-s} \prod_{j=2}^{s+1} (M_1 + M_j + 1/2)^{-1}.$$

Choose a positive rational b below this computable positive constant and $1/2$. This proves the claim. It concerns algebraic *angles*, not algebraic entries of their rotation matrices. $\square$

12.1. Harmonic localization. For the harmonic calculation assume that the alphabet contains I, R_θ and that one seed-query pair has mean $\rho \cos \phi$ after accumulated angle ϕ , where $0 < \rho \leq 1$. Other seeds, queries, or commands impose additional constraints and do not invalidate the lower bound. Let $\lambda = e^{i\theta}$ have infinite order. Put $\delta = 2\varepsilon < \rho$, and choose an integer

$$(12.3) \quad r \geq 1, \quad r > \frac{\delta}{\rho - \delta}, \quad m = r + 1.$$

For example $r = \lfloor \delta/(\rho - \delta) \rfloor + 1$ works. Define

$$(12.4) \quad \Delta_r = 2 \left(\frac{\rho r}{r+1} - \delta \right) > 0, \quad A_r = \frac{2(1 + \rho + \delta)(2r + 1)m}{\Delta_r} > 1.$$

Lemma 12.3 (Explicit harmonic localization). *The nonnegative trigonometric polynomials*

$$p_{\pm}(\phi) = \frac{(1 \pm \cos \phi)^r}{2^{-r} \binom{2r}{r}}$$

have Haar mean one and weighted means

$$\int p_{\pm}(\phi) \rho \cos \phi \frac{d\phi}{2\pi} = \pm \frac{\rho r}{r+1}.$$

For the moving feature vector

$$Z_m(\phi) = (\cos \phi, \sin \phi, \dots, \cos m\phi, \sin m\phi),$$

any terminal realization with wordwise mean error at most δ obeys

$$(12.5) \quad \mathbb{E} \|\mathbb{E}[Z_m(\phi) \mid S]\| \geq \frac{\Delta_r}{2(1 + \rho + \delta)(2r + 1)}.$$

Proof. The identity $1 + \cos \phi = 2 \cos^2(\phi/2)$, or the constant Laurent coefficient, gives

$$C_r := \int (1 + \cos \phi)^r \frac{d\phi}{2\pi} = 2^{-r} \binom{2r}{r}, \quad C_{r+1}/C_r = \frac{2r+1}{r+1}.$$

Hence the weighted cosine mean is $C_{r+1}/C_r - 1 = r/(r+1)$. Translation by π gives the negative case.

We use the unweighted real coefficient vector in the basis $1, \cos \phi, \sin \phi, \dots$, without orthonormal Fourier rescaling. The absolute sum of the real cosine coefficients of either $p_{\pm}$, including the constant coefficient, is

$$B_r = \frac{4^r}{\binom{2r}{r}} \leq 2r + 1.$$

For p_+ its coefficients are nonnegative and their sum is $p_+(0)$; for p_- they differ only by alternating signs. The inequality follows because the central binomial coefficient is the largest of the $2r + 1$ coefficients summing to 4^r . Multiplication by $\rho \cos \phi$ has coefficient absolute sum at most ρB_r , since $\cos u \cos v = (\cos(u + v) + \cos(u - v))/2$. Thus the Euclidean norms of the moving coefficient vectors of $p_\pm$ and $p_\pm \rho \cos \phi$ are at most $2r + 1$ and $\rho(2r + 1)$, respectively. For completeness, let $c = \mathbb{E}D$, $Q = \mathbb{E}\|\mathbb{E}[Z_m | S]\|$. For either weight, the conditional coefficient identity gives $|c - \int p_\pm \rho \cos \phi| \leq \delta + (1 + \rho + \delta)(2r + 1)Q$. Subtracting the two inequalities gives (12.5). Correctness is averaged word by word before using this identity. The starting feature norm is $\sqrt{m} \leq m$; A_r deliberately uses the latter bound to obtain a rational closed form. $\square$

The integer r in (12.3) is the localization order; it is unrelated to the number s of rotation letters. For fixed subcritical error it is finite and independent of N and k .

Theorem 12.4 (A full-noise arithmetic occupation bound). *Assume (12.1) and $0 \leq \varepsilon < \rho/2$. Fix the localization parameters r, m, A_r in (12.3)–(12.4). For an integer $k \geq 1$, put*

$$L = \lceil (2k)^{1/s} \rceil, \quad B = s(L - 1).$$

Every error- ε realization of peak at most k satisfies

$$(12.6) \quad N < B \left(1 + b^{-2}(mB)^{2\tau} \log A_r \right).$$

With no restriction on other widths, the number of cuts of width at most k is at most

$$(12.7) \quad \left\lceil B - 1 + Bb^{-2}(mB)^{2\tau} \log A_r \right\rceil.$$

Proof. Use s consecutive groups of $L - 1$ command positions. In group i , choose J_i uniformly from $\{0, \dots, L - 1\}$, apply J_i copies of the i th rotation, and fill the other positions with identities. This defines an external law on length- B words with $L^s \geq 2k$ equally likely products. On the harmonic feature use

$$R_m(t) = \text{diag}(R_t, R_{2t}, \dots, R_{mt}).$$

For two distinct count vectors J, J' and any frequency $1 \leq l \leq m$, condition (12.1) and $\|l(J - J')\|_1 \leq mB$ give

$$|e^{2\pi i l(J - J') \cdot \alpha} - 1| = 2|\sin(\pi l(J - J') \cdot \alpha)| \geq 4b(mB)^{-\tau} =: \sigma.$$

Here $2\sin(\pi x) \geq 4x$ for $0 \leq x \leq 1/2$. Also $b \leq 1/2$, $m \geq 2$, and $B \geq 1$, so $\sigma \leq 1$.

For a unit vector $u = (u_1, \dots, u_m)$ in the frequency blocks, the squared distance between the two represented orbit points is

$$\sum_{l=1}^m \|u_l\|^2 |e^{2\pi i l(J - J') \cdot \alpha} - 1|^2 \geq \sigma^2.$$

An open ball of radius $\sigma/2$ about any unit center contains at most one point. At least half the L^s points therefore have distance at least $\sigma/2$ from all of any k centers. Since the inner-product defect between unit vectors equals half their squared distance, the finite word law has defect at least

$$d_k = \sigma^2/16 = b^2(mB)^{-2\tau}.$$

Apply conditional-centroid contraction to successive blocks and Lemma 12.3 to the terminal query. The starting norm is $\sqrt{m} \leq m$. Thus

$$\left\lfloor \frac{N}{B} \right\rfloor \leq \frac{\log A_r}{-\log(1 - d_k)} \leq b^{-2}(mB)^{2\tau} \log A_r,$$

which proves (12.6). The greedy endpoint count, with identity gaps and no prefix, gives (12.7). Internal registers remain unrestricted. Finally $B < s(2k)^{1/s}$, so (12.6) implies the asserted $\Omega(N^{s/(2\tau+1)})$ lower bound. $\square$

Remark 12.5. Testing a coordinate unit vector in (12.1) gives $b \leq 1/2$. In the block construction, $L = \lceil (2k)^{1/s} \rceil \geq 2$, hence $B = s(L-1) \geq 1$. The integer r is the localization order and s is the number of rotations. For the algebraic-angle example, $1, \xi, \dots, \xi^s$ are linearly independent over $\mathbb{Q}$ by the degree $s+1$ hypothesis; this ensures that the algebraic integer used in its norm bound is nonzero.

12.2. An exact polygon realization. Write $\eta_\alpha(q) = \max_i \|q\alpha_i\|_{\mathbb{R}/\mathbb{Z}}$. The following elementary construction explains why a common denominator, rather than the number of numerical orbit points, controls the upper width.

Proposition 12.6 (Exact enclosure upper bound). *Let $0 < \rho < 1$, $a = (1 + \rho)/2$, and $\gamma = \log(a/\rho) > 0$. Choose an integer $q_0 \geq 4$ with $\cos(\pi/q_0) \geq a$. If $q \geq q_0$ and*

$$(12.8) \quad 4\pi^2 N \eta_\alpha(q)/q^2 \leq \gamma,$$

then the indicated rotation alphabet, with or without F , has an exact horizon- N realization with q command labels.

Proof. Let P_q be the regular polygon with vertices $v_j = (\cos(2\pi j/q), \sin(2\pi j/q))$, $0 \leq j < q$. For each i choose a nearest integer p_i to $q\alpha_i$ and put

$$t = \pi/q, \quad h_i = 2\pi|\alpha_i - p_i/q| \leq t, \quad \Lambda_i = \frac{\cos(t - h_i)}{\cos t}.$$

Writing $n_j = (\cos((2j+1)t), \sin((2j+1)t))$, its facet description is

$$P_q = \{x : \langle n_j, x \rangle \leq \cos t \ (0 \leq j < q)\}.$$

The facet inequalities of the regular polygon give $R_{2\pi\alpha_i}P_q \subset \Lambda_i P_q$: the nearest facet normal to a rotated vertex has angular difference $t - h_i$. Further,

$$\log \Lambda_i \leq \log(1 + \tan(t)h_i) \leq \tan(t)h_i \leq 4\pi^2 \eta_\alpha(q)/q^2,$$

since $\cos h + \tan(t) \sin h \leq 1 + \tan(t)h$ and $\tan t \leq 2t$ for $0 \leq t \leq \pi/4$. Let $\Lambda = \max(1, \Lambda_1, \dots, \Lambda_s)$. The standard reflection $F = \text{diag}(1, -1)$ sends v_j to $v_{-j \pmod q}$ and preserves the chosen vertex set. The identity and reflection therefore preserve P_q , so every command U satisfies $\Lambda^{-1}UP_q \subset P_q$.

For each command and each vertex choose a probability row with barycenter $\Lambda^{-1}Uv_j$ in P_q . These are actual common command rows, not independent prefix/suffix factorizations. At most three successors suffice by planar convexity. Initialize the seed ρu_s , $u_s \in \{\pm e_1, \pm e_2\}$, with a row whose barycenter is au_s . This is possible since $a \leq \cos(\pi/q)$, the polygon's inradius. After a length- N word the mean vertex is

$$a\Lambda^{-N}U_w u_s.$$

By (12.8), $\rho\Lambda^N \leq a$. Hence the terminal columns

$$d_j(v) = \frac{\rho\Lambda^N}{a} e_j^T v$$

belong to $[-1, 1]$ and return exactly $\rho e_j^T U_w u_s$. The rows can be stationary within a horizon; the terminal scaling and chosen denominator may depend on N . No one finite all-horizon exact machine is asserted. $\square$

Lemma 12.7 (Comparable simultaneous denominators). *Assume (12.1). For every integer $Q \geq 2$ there is q with $1 \leq q \leq Q^s$, $\eta_\alpha(q) \leq 1/Q$, and*

$$q \geq cQ^{s/(\tau+1-s)}, \quad c = \left(\frac{b}{s^{\tau+1}} \right)^{s/(\tau+1-s)}.$$

In particular, for $\tau = s$ one has $cQ^s \leq q \leq Q^s$.

Proof. Partition the torus into Q^s half-open cubes and apply the pigeonhole principle to $0, \alpha, \dots, Q^s \alpha$. Their difference gives $q \leq Q^s$ and integers p_i with $|q\alpha_i - p_i| \leq 1/Q$.

Put $M = \lfloor q^{1/s} \rfloor$. The $(M+1)^s > q$ vectors in $\{0, \dots, M\}^s$ cannot all have distinct residues $n \cdot p$ modulo q , even when some nearest integers p_i are negative. Subtract two to obtain a nonzero vector n with $q \mid n \cdot p$ and $H := \|n\|_1 \leq sq^{1/s}$. Therefore

$$bH^{-\tau} \leq \|n \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq H/(qQ), \quad q^{(\tau+1-s)/s} \geq bQ/s^{\tau+1}.$$

The displayed bound follows. This argument supplies the needed quantitative transference directly; no unspecified denominator subsequence is being used. $\square$

Proof of Theorem 12.1. The lower estimate follows from Theorem 12.4. For $\tau = s$, take the constants a, γ, q_0, c above and an integer

$$Q \geq \max \left(2, (q_0/c)^{1/s}, \left(\frac{4\pi^2 N}{c^2 \gamma} \right)^{1/(2s+1)} \right).$$

Choosing the least such integer, Lemma 12.7 gives $q \leq Q^s = O(N^{s/(2s+1)})$, $q \geq q_0$, and

$$4\pi^2 N \eta_\alpha(q)/q^2 \leq 4\pi^2 N/(c^2 Q^{2s+1}) \leq \gamma.$$

Proposition 12.6 supplies the exact upper bound. At and above $\rho/2$ use the fair decoder from Corollary 11.5. This proves all claims. $\square$

The command count s is an arithmetic parameter, not a replacement for the dimension of the physical circle. The constants deteriorate as $\varepsilon \uparrow \rho/2$ through the localization degree, and as $\rho \uparrow 1$ through the enclosure slack. Both dependences are explicit in the finite inequalities. The theorem makes no uniform claim at these limiting parameters.

Corollary 12.8 (Unit amplitude at positive error). *For the exact rotation alphabet of Theorem 12.1, with or without its specified reflection, suppose the angle vector is dual badly approximable. At $\rho = 1$, for every fixed $0 < \varepsilon < 1/2$,*

$$W_{N,\varepsilon} = \Theta(N^{s/(2s+1)}).$$

Proof. The harmonic localization and orbit-packing lower bound in Theorem 12.4 permits $\rho = 1$ and every fixed $\varepsilon < 1/2$. For the upper bound choose $0 < \eta < 2\varepsilon$ and put $\rho' = 1 - \eta \in (0, 1)$, decreasing η if necessary. Apply the exact polygon construction and comparable-denominator lemma to ρ' . On every seed, word, and query its mean differs from the unit-amplitude target by at most η , so the binary total-variation error is at most $\eta/2 < \varepsilon$. Its width has the required order. This is a positive-error upper bound, not an exact realization at unit amplitude. $\square$

The case $\rho = 1$, $\varepsilon = 0$ is not decided by this corollary. In particular, taking $\eta = 0$ would destroy the strictly positive enclosure slack required by Proposition 12.6.

13. SPHERICAL EXPERIMENTS WITH A SPECTRAL GAP

The next result concerns a full vector orbit, not an abelian factor of a nonabelian system. Write σ for normalized area on S^{d-1} , where $d \geq 2$. Let a finite orthogonal alphabet contain the identity, and let μ be a probability law on its letters. Assume that its averaging operator

$$Pf(x) = \mathbb{E}_{R \sim \mu} f(Rx)$$

satisfies

$$(13.1) \quad \|P^n f\|_{L^2(\sigma)} \leq \lambda^n \|f\|_{L^2(\sigma)} \quad \left(\int f d\sigma = 0 \right), \quad 0 < \lambda < 1.$$

The physical target is $F(g) = \rho g x_0$ for a fixed $x_0 \in S^{d-1}$, with output set the closed Euclidean unit ball and error measured in Euclidean norm.

Theorem 13.1 (A spherical width law up to a logarithm). *Assume (13.1), $0 < \rho < 1$, and $0 \leq \varepsilon < \rho$. There are constants $c, C > 0$ such that*

$$(13.2) \quad c \left(\frac{N}{\log(N+2)} \right)^{(d-1)/2} \leq W_{N,\varepsilon} \leq C(N+1)^{(d-1)/2}.$$

The upper realization is exact. Every error- ε realization has at most

$$Ck^{2/(d-1)} \log(k+2)$$

positive cuts of width at most k , with other widths unrestricted. The constants may depend on $d, \lambda, \rho, \varepsilon$, but not on the horizon or the realization. At unit amplitude the same bounds hold for every fixed $0 < \varepsilon < 1$; exact unit amplitude is not included in the upper assertion.

Lemma 13.2 (Spherical quantization and executable contraction). *Under (13.1), there are $c_d > 0$ and integers $B_k \leq C_{d,\lambda} \log(k+2)$ such that*

$$\mathbb{E}_{w \sim \mu^{B_k}} \max_{i \leq k} \langle v_i, u_w u \rangle \leq 1 - c_d k^{-2/(d-1)}$$

for all unit vectors $u, v_1, \dots, v_k$. The defect may be decreased so that it lies in $(0, 1)$. When $d = 3$ one may take defect $1/(2k)$ and

$$(13.3) \quad B_k = \left\lceil \frac{\log(64k^2)}{-\log \lambda} \right\rceil.$$

Proof. Normalized spherical cap measure obeys

$$c_d h^{d-1} \leq \sigma\{x : \|x - u\|_2 < h\} \leq C_d h^{d-1} \quad (0 < h \leq 1),$$

uniformly in u . This follows by writing area in polar angle as a constant times $\sin^{d-2} \theta d\theta$ and using $h = 2 \sin(\theta/2)$; for $d = 2$ the same formula is the arc-length formula. For $f(x) = \max_i \langle v_i, x \rangle$, choose $h = ck^{-1/(d-1)} \leq 1$ so that the union of the k radius- h caps has area at most $1/2$. Outside them, $1 - f(x) \geq h^2/2$. Thus

$$(13.4) \quad \int f d\sigma \leq 1 - a_d k^{-2/(d-1)}.$$

The function f is 1-Lipschitz for chordal distance and bounded by one in absolute value.

Let K_δ average over the chordal cap of radius $\delta \leq 1$. It commutes with all rotations, fixes constants, and satisfies $\|K_\delta f - f\|_\infty \leq \delta$. The cap density at a center has L^2 norm at most $C_d \delta^{-(d-1)/2}$. Hence (13.1) and Cauchy-Schwarz imply, uniformly in the center,

$$\left| P^B f(u) - \int f d\sigma \right| \leq \delta + C'_d \delta^{-(d-1)/2} \lambda^B.$$

Take δ to be a fixed sufficiently small multiple of $k^{-2/(d-1)}$ and then $B = O_{d,\lambda}(\log(k+2))$ so that the right side is at most half the defect in (13.4). The law of the B successive letters is a finite executable word law, proving the general assertion.

For $d = 3$, a chordal cap of radius $h \leq 2$ has normalized area $h^2/4$. The union bound gives $\Pr(1 - f < t) \leq kt/2$, so integration for $0 \leq t \leq 2/k$ gives $\int (1 - f) d\sigma \geq 1/k$. A radius- δ cap density has L^2 norm $2/\delta$, and $\|f - \int f\|_2 \leq 2$. Thus the mixing error is at most $\delta + 4\lambda^B/\delta$. Choose $\delta = 1/(4k)$ and (13.3); the error is at most $1/(2k)$. $\square$

Proof of Theorem 13.1. Let $Z = u_w x_0$ and let $D(S)$ be the terminal decoder. For every deterministic word, correctness and $\|Z\|_2 = 1$ give

$$\langle \mathbb{E}[D(S) \mid w], Z \rangle \geq \rho - \varepsilon.$$

Under any external word law this implies

$$(13.5) \quad \rho - \varepsilon \leq \mathbb{E} \langle D(S), Z \rangle \leq \mathbb{E} \|\mathbb{E}[Z \mid S]\|_2,$$

since the decoder lies in the unit ball. Lemma 7.5, applied with the executable law of Lemma 13.2, contracts the last conditional norm by $1 - a_k$ at each selected narrow endpoint, where $a_k = c_d k^{-2/(d-1)} \in (0, 1)$. Initially the norm is one. Fixed identity-letter gaps cannot increase it, even though their stochastic rows may process the register. After J selected endpoints, (13.5) gives

$$J[-\log(1 - a_k)] \leq \log \frac{1}{\rho - \varepsilon}.$$

Greedy spacing by B_k as in the occupation proof discards at most $B_k - 1$ initial cuts and charges at most B_k eligible cuts to each selected endpoint. Therefore the occupation count is at most

$$(13.6) \quad B_k \left(1 + a_k^{-1} \log \frac{1}{\rho - \varepsilon} \right).$$

If every register has width at most k , this bounds N . Inverting the bound gives the lower order in (13.2). Explicitly, when $k \leq (N+1)^{d-1}$ its logarithm is $O_d(\log(N+2))$, and otherwise the asserted lower bound is automatic.

For the upper bound, choose a maximal δ -separated set of vertices on S^{d-1} containing x_0 , with $0 < \delta \leq 1/2$. The disjoint caps of radius $\delta/2$ show that it has at most $C_d \delta^{-(d-1)}$ vertices. Maximality makes it a δ -net. Its convex hull Q satisfies

$$(13.7) \quad r\mathbb{B}^d \subset Q \subset \mathbb{B}^d, \quad r = 1 - \delta^2/2,$$

because for every unit direction u some vertex v has $\langle u, v \rangle \geq 1 - \delta^2/2$. The inclusion follows from the supporting half-space characterization of a closed convex body. For each command R and vertex v_i , the point rRv_i belongs to Q . Choose barycentric coordinates of that point as the i th row of one common stochastic command matrix. The identity command is treated in the same way, with the same factor r .

Initialize at the vertex x_0 and decode vertex v_i as $(\rho/r^N)v_i$. Induction on the row means gives output exactly $\rho u_w x_0$ on every length- N word. The decoder is legal if $r^N \geq \rho$. Choose

$$\delta^2 = \min \left(\frac{1}{4}, \frac{-\log \rho}{N+1} \right).$$

Since $\log(1 - \delta^2/2) \geq -\delta^2$, the required inequality holds. The number of vertices is $O_{d,\rho}((N+1)^{(d-1)/2})$. Both rows and decoder may depend on the horizon, as permitted by the clocked model.

For $\rho = 1$ and $\varepsilon > 0$, the same lower proof applies. An exact upper realization at amplitude $\rho' = 1 - \varepsilon/2$ differs from the unit-amplitude target by $\varepsilon/2$ in Euclidean norm, proving the stated upper bound. There is no assertion of exactness at unit amplitude. $\square$

Corollary 13.3 (A rational noncommuting alphabet on $\mathrm{SO}(3)$). *Let the alphabet consist of the identity, the matrices*

$$R_x = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/5 & -4/5 \\ 0 & 4/5 & 3/5 \end{pmatrix}, \quad R_z = \begin{pmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and their inverses. For $F(g) = \rho g e_3$, with $0 < \rho < 1$ and $0 \leq \varepsilon < \rho$, the bounds are

$$c \frac{N}{\log(N+2)} \leq W_{N,\varepsilon} \leq C(N+1).$$

At error $\varepsilon \geq \rho$ a single zero decoder suffices. Every subcritical realization has at most $Ck \log(k+2)$ positive cuts of width at most k .

Proof. The angle θ of both rotations has $2 \cos \theta = 6/5$. It is not a rational multiple of 2π : otherwise $2 \cos \theta$ would be a rational algebraic integer, hence an integer. The closures of the two cyclic groups are the full circle subgroups about their distinct axes. Their Lie algebras and bracket span $\mathfrak{so}(3)$, so the generated compact closure is $\mathrm{SO}(3)$. The matrices do not commute.

Under the double cover $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$, their unit-quaternion lifts are $(2+i)/\sqrt{5}$ and $(2+k)/\sqrt{5}$. They have algebraic matrix entries and generate a dense subgroup of $\mathrm{SU}(2)$: its closed image is all of $\mathrm{SO}(3)$, so its Lie algebra has dimension three. The spectral-gap theorem of Bourgain–Gamburd [16] applies. Adding the identity with positive mass makes the symmetric averaging operator have norm strictly less than one on mean-zero L^2 . The spherical representation is a subrepresentation of the regular representation, so the uniform law on the five displayed letters satisfies (13.1) on S^2 . Apply Theorem 13.1. The radius of the full sphere ρS^2 inside the unit ball is ρ , attained at zero. $\square$

The spectral gap in this corollary is an imported theorem, not a numerical assertion extracted from finite orbit checks. This retained block estimate has a logarithmic loss. Theorem 15.5 supplies the matching linear lower order under the same spectral-gap hypothesis. It does not compute a numerical gap for this particular alphabet; Theorem 27.2 gives a separate alphabet with explicit constants.

14. A UNIFORM ORBIT CRITERION FOR QUANTITATIVE WIDTH

The spherical estimate has an extension in which the dimension of the ambient feature space is not the relevant exponent. Two geometric quantities enter: a uniform small-ball exponent for every possible conditional-centroid direction, and the covering dimension of the particular target orbit. They need not agree. Keeping this distinction is essential for matrix-valued experiments.

Let G be a compact connected Lie group with normalized Haar measure m_G and a bi-invariant Riemannian distance. Let $R : G \rightarrow O(E)$ be a continuous orthogonal representation on a finite-dimensional real Euclidean space. Fix $z_0 \in E$ of norm one, put

$$\mathcal{O} = R(G)z_0, \quad Q = \mathrm{conv}(\mathcal{O}), \quad q = \dim \mathcal{O},$$

and assume that $\mathcal{O}$ spans E and that E has no nonzero fixed vector. Thus $q \geq 1$. A finite alphabet in G contains the identity and carries a probability law μ for which

$$(14.1) \quad \|P^n f\|_2 \leq \lambda^n \|f\|_2 \quad \left(\int_G f dm_G = 0 \right), \quad Pf(g) = \mathbb{E}_{a \sim \mu} f(u_a g), \quad 0 < \lambda < 1.$$

Letters outside the support of μ are allowed. All command products and stochastic rows retain the conventions of Definition 7.1.

The uniform geometric hypothesis is that, for some $p > 0$, $A < \infty$ and $h_0 > 0$,

$$(14.2) \quad \sup_{u,v \in S(E)} m_G \{g : \|R(g)u - v\| < h\} \leq Ah^p \quad (0 < h \leq h_0).$$

It is imposed on all unit vectors u , not only on $\mathcal{O}$. A conditional mean of points in $\mathcal{O}$, after normalization, need not belong to $\mathcal{O}$.

Lemma 14.1 (From orbit small balls to executable contraction). *Under (14.1)–(14.2), constants $c, C > 0$ and integers $B_k \leq C \log(k+2)$ can be chosen so that*

$$(14.3) \quad \mathbb{E}_{w \sim \mu^{B_k}} \max_{1 \leq i \leq k} \langle v_i, R(u_w)u \rangle \leq 1 - ck^{-2/p}$$

for every $u, v_1, \dots, v_k \in S(E)$. The constant c can be decreased so that the defect lies in $(0, 1)$ for every $k \geq 1$.

Proof. Choose $h = c_0 k^{-1/p} \leq \min(h_0, 1)$ with $Ac_0^p \leq 1/2$. Outside the union of the k open balls in (14.2),

$$1 - \max_i \langle v_i, R(g)u \rangle \geq h^2/2.$$

The union has measure at most $1/2$. Consequently the Haar mean of $f(g) = \max_i \langle v_i, R(g)u \rangle$ is at most $1 - a_k$, where $a_k = c_0^2 k^{-2/p}/4$. Repeated centers are allowed. All such functions are bounded by one in absolute value and have a common Lipschitz constant L : differentiating the finite-dimensional representation bounds the derivative uniformly on unit vectors.

Write $D = \dim G$, distinct from the cap exponent p and orbit dimension q . The ball calculation uses the fixed bi-invariant Riemannian metric on G . In exponential coordinates at the identity, Haar measure has a smooth positive density. Hence a metric ball of radius $\delta \leq \delta_0$ has measure at least $c_G \delta^D$. Average $P^B f$ over this ball. Left translations are isometries, so $P^B f$ has Lipschitz constant at most L . Cauchy–Schwarz and (14.1) give

$$|P^B f(1) - \bar{f}| \leq L\delta + 2c_G^{-1/2} \delta^{-D/2} \lambda^B.$$

Take δ to be a sufficiently small fixed multiple of a_k , also bounded by δ_0 . Then choose $B_k \geq 1$ to make the second term at most $a_k/4$. The first term is at most $a_k/4$ as well. Since $\log(a_k^{-1}) = O(\log(k+2))$, this requires $B_k \leq C \log(k+2)$. Products under μ^{B_k} are genuine words of one common length. Their reverse chronological order has the same law because the letters are independent with the same law. Thus $P^{B_k} f(1)$ is the expectation in (14.3). This proves the lemma with defect $a_k/2$. $\square$

Lemma 14.2 (Quadratic inner approximation of an orbit hull). *There are constants $C_1, C_2, \delta_0 > 0$ such that, for $0 < \delta \leq \delta_0$, points $z_1, \dots, z_K \in \mathcal{O}$, including z_0 , can be chosen with*

$$(14.4) \quad K \leq C_1 \delta^{-q}, \quad (1 - C_2 \delta^2)Q \subseteq \text{conv}\{z_1, \dots, z_K\} \subseteq Q, \quad 0 < 1 - C_2 \delta^2 < 1.$$

Proof. The orbit is a compact smooth embedded homogeneous manifold. Finitely many coordinate charts with bounded derivatives and inverse derivatives give an intrinsic δ -net of size at most $C\delta^{-q}$; adjoining z_0 changes only C . Use the induced Riemannian metric on the orbit. If the functional $x \mapsto \langle b, x \rangle$ attains its maximum at $x_b \in \mathcal{O}$, its derivative along the orbit vanishes there. The second derivative along unit-speed orbit geodesics has absolute value at most $C\|b\|$, uniformly in their starting points and directions, by compactness of the unit tangent bundle and smoothness of the embedding. A net point within intrinsic distance δ of x_b therefore satisfies

$$\langle b, z_i \rangle \geq h_Q(b) - C\|b\|\delta^2, \quad h_Q(b) = \max_{x \in Q} \langle b, x \rangle.$$

Haar averaging gives $\int R(g)z_0 dm_G = 0$. Moreover 0 lies in the interior of Q in E . Otherwise a nonzero supporting functional would be nonnegative on the orbit and have Haar mean zero; continuity and full Haar support would make it vanish on the orbit, contrary to its spanning E . Thus $r\mathbb{B}_E \subset Q$ for some $r > 0$, and $h_Q(b) \geq r\|b\|$. The support inequality becomes $\max_i \langle b, z_i \rangle \geq (1 - C\delta^2/r)h_Q(b)$. The support-function criterion for inclusion of compact convex sets proves (14.4). Decrease δ_0 to obtain the last bounds. $\square$

Theorem 14.3 (Geometric width and occupation). *Assume (14.1)–(14.2). Take one seed and one query with target $F(g) = \rho R(g)z_0$, output set Q , and Euclidean error. For $0 < \rho \leq 1$ and $0 \leq \varepsilon < \rho$, put $\zeta = \rho - \varepsilon \in (0, 1]$. Every error- ε realization satisfies*

$$(14.5) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C \log(k+2) \left(1 + k^{2/p} \log \frac{1}{\zeta}\right),$$

where C depends on the geometric and spectral-gap data but not on ρ, ε, N, k or the realization. In particular, at fixed parameters,

$$(14.6) \quad W_{N,\varepsilon} \geq c \left(\frac{N}{\log(N+2)} \right)^{p/2}.$$

For $0 < \rho < 1$ there is an exact common-row realization with

$$(14.7) \quad W_{N,0} \leq C \left(1 + \frac{N+1}{-\log \rho} \right)^{q/2}.$$

For $\rho = 1$ and fixed $0 < \varepsilon < 1$, the upper order is $O_\varepsilon((N+1)^{q/2})$. When $p = q$, the lower and upper exponents agree, with the displayed logarithmic loss in the lower bound. The constrained error threshold is exactly ρ , attained by the constant zero decoder.

Proof. Put $Z = R(u_w)z_0$ and let $D(S) \in Q \subseteq \mathbb{B}_E$ be the decoder. For each deterministic word, the error bound gives $\langle \mathbb{E}[D(S) \mid w], Z \rangle \geq \rho - \varepsilon$. Under any external word law, conditioning on the register yields

$$\zeta \leq \mathbb{E} \langle D(S), Z \rangle \leq \mathbb{E} \|\mathbb{E}[Z \mid S]\|.$$

The conditional-centroid Lemma 7.5 applies to the law of Lemma 14.1. The uniformity over every direction in (14.2) is precisely what justifies this application. It contracts the conditional norm by $1 - a$ at a selected endpoint of width at most k , where $a = ck^{-2/p} \in (0, 1)$. The initial norm is one. Fixed identity-letter fillers cannot increase it, regardless of their rows or intervening widths. After J selected endpoints,

$$J[-\log(1-a)] \leq \log(1/\zeta).$$

Greedily select narrow cuts at separation at least B_k , omitting the first $B_k - 1$ cuts. Each selected endpoint accounts for at most B_k cuts. The count is at most $B_k(1 + a^{-1} \log(1/\zeta))$, proving (14.5), also when $\zeta = 1$. In that case even one selected endpoint is impossible. More precisely one may take the integer bound

$$L_k = B_k \left(1 + \left\lceil c^{-1} k^{2/p} \log(1/\zeta) \right\rceil \right).$$

For a full width bound k , either $k \geq N^{p/2}$, or $\log(k+2) \leq C_p \log(N+2)$. In the latter case the occupation inequality gives $N \leq C_\zeta \log(N+2) k^{2/p}$. These alternatives prove (14.6).

For the upper bound take the net in Lemma 14.2, and write $r = 1 - C_2 \delta^2$. The orbit hull is invariant under every command. Hence $rR(u_a)z_i$ is in the convex hull of the net points. For each of the finitely many command-vertex pairs, choose barycentric coordinates independently; no measurable selection is involved. Use them as row i of a stochastic matrix T_a , the same at every epoch within this horizon. Initialize at z_0 . The conditional row means then evolve as $r^t R(u_w)z_0$. Decode label i by $(\rho/r^N)z_i$, which belongs to Q if $r^N \geq \rho$, since $0 \in Q$. Choosing

$$\delta^2 = \min \left(\delta_0^2, \frac{1}{2C_2}, \frac{-\log \rho}{2C_2(N+1)} \right)$$

gives this inequality by $\log(1-x) \geq -2x$ for $0 \leq x \leq 1/2$. The net size gives (14.7). At unit amplitude and positive error use the exact construction for amplitude $1 - \varepsilon/2$; its discrepancy from the desired target is $\varepsilon/2$.

Finally, Haar mean zero and $\|R(g)z_0\| = 1$ imply

$$\int_G \|\rho R(g)z_0 - y\|^2 dm_G = \rho^2 + \|y\|^2.$$

Every constrained center therefore has worst error at least ρ , while $y = 0$ attains it. Below it (14.6) rules out bounded width. None of the constructions requires transition matrices to satisfy the physical group relations, and extra finitely many commands do not change the upper construction. $\square$

The geometric ingredients in Lemma 14.2 are the usual local geometry of compact homogeneous spaces; related covering estimates are developed in [15]. The additional realization statement is the common stochastic-row construction and its combination with a contraction uniform over conditional-centroid directions. A small-ball estimate restricted to the target orbit alone does not imply Theorem 14.3.

Remark 14.4 (Dependence of constants). The cap and executable-gap constants depend only on the fixed group with its bi-invariant metric, the representation, the uniform cap data (A, p, h_0) , and the declared L^2 gap λ . The hull constants also depend on the target orbit, its second derivatives, and the inradius of Q in its span E . The occupation bound further depends on $\zeta = \rho - \varepsilon$. None depends on the horizon, hidden transition rows, or masses of individual labels. The orbit is the smooth embedded quotient G/G_{z_0} and q is its real dimension. The logarithmic factor is the loss of this mixing-block argument. Theorem 15.5 removes it by a different entropy–transport argument; all statements and proofs of the block estimate remain valid.

15. SPECTRAL ENTROPY AND SHARP OCCUPATION BOUNDS

The pointwise mixing block in Lemma 14.1 costs a logarithm at each selected cut. We now replace that block argument by an entropy estimate along the selected cuts themselves. The auxiliary entropy is attached to a smoothed law of *centroid directions*; it is not the Shannon entropy of the hidden register. The normalization by centroid mass will be essential.

Let E have real dimension $m \geq 2$, let $\Sigma = \{u \in E : \|u\| = 1\}$, and let σ be normalized round measure on Σ . Write d_Σ for its geodesic distance and W_2 for the corresponding quadratic Wasserstein distance. All logarithms in this section are natural. Let $G \subseteq O(E)$ be compact, with normalized Haar measure m_G , and let λ be a probability on the command letters. On $L^2(\Sigma, \sigma)$ define

$$Pf(x) = \sum_a \lambda_a f(R(u_a)^{-1}x), \quad \Pi f(x) = \int_G f(R(g)^{-1}x) dm_G(g).$$

Here R denotes the given action; writing $G \subseteq O(E)$ identifies its image when convenient. The projection Π need not have one-dimensional range. Our spectral hypothesis is

$$(15.1) \quad \|Pf - \Pi f\|_2 \leq \kappa \|f - \Pi f\|_2, \quad 0 \leq \kappa < 1.$$

This is a bound on the whole function space, not merely on the finite-dimensional representation E . The group bound (14.1) implies (15.1), with the same κ : decompose the unitary representation on $L^2(\Sigma)$ into irreducibles; each nontrivial irreducible occurs in the regular representation, where the assumed bound holds (use the adjoint operator if needed for the inverse convention). The trivial summands are exactly the range of Π [13].

Remark 15.1 (The full spectral hypothesis). A contraction on E alone does not imply (15.1). For finitely many circle rotations, simultaneous rational approximation provides arbitrarily high characters on which the averaging eigenvalue is arbitrarily close to one, although its first harmonic may contract. Such an alphabet does not satisfy the full spectral hypothesis. This is why the present theorem does not replace the different Diophantine width laws by a spurious spherical exponent.

15.1. A compactly supported entropy potential. For $0 < h \leq 1$ set

$$(15.2) \quad K_h(u, x) = Z_h^{-1} \left(1 - \frac{\|x - u\|^2}{h^2} \right)_+^2, \quad Z_h = \int_{\Sigma} \left(1 - \frac{\|x - u\|^2}{h^2} \right)_+^2 d\sigma(x).$$

The normalizing constant is independent of u . If ν is a probability on Σ , write $K_h\nu(x) = \int K_h(u, x) d\nu(u)$. For a density f put

$$\mathcal{H}(f) = \int f \log f d\sigma, \quad \mathcal{I}(f) = \int_{\{f>0\}} \frac{\|\nabla f\|^2}{f} d\sigma, \quad 0 \log 0 = 0.$$

Lemma 15.2 (Entropy continuity for spherical mixtures). *There are constants $D_m \geq 1$ and $B_m < \infty$, depending only on m , such that every finitely supported probability ν satisfies*

$$(15.3) \quad 0 \leq \mathcal{H}(K_h\nu) \leq L_h := \log(D_m h^{-(m-1)}), \quad \mathcal{I}(K_h\nu) \leq B_m h^{-2}.$$

For any two finitely supported probabilities ν, η ,

$$(15.4) \quad |\mathcal{H}(K_h\nu) - \mathcal{H}(K_h\eta)| \leq \frac{\sqrt{B_m}}{h} W_2(\nu, \eta).$$

For $m = 3$ one can take $D_3 = 12$ and $B_3 = 24$.

Proof. Put $s = \|x - u\|^2$. Polar coordinates give a density proportional to $s^{(m-3)/2}(1 - s/4)^{(m-3)/2}$ on $0 < s < 4$. On $0 < s \leq 1$ its second factor is bounded above and below by positive dimension-dependent constants. Substitution $s = h^2 v$ therefore gives $Z_h \asymp_m h^{m-1}$. Also $\|\nabla s\|^2 = 4s - s^2$, so

$$\mathcal{I}(K_h(u, \cdot)) = \frac{4}{Z_h h^4} \int_{s < h^2} (4s - s^2) d\sigma \leq B_m h^{-2}.$$

Increasing D_m gives $K_h\nu \leq D_m h^{-(m-1)}$. Jensen's inequality gives the lower entropy bound, and the pointwise upper bound gives the upper one. Cauchy-Schwarz in a finite mixture shows that its Fisher information is at most the mixture of the individual Fisher informations.

Choose an optimal finite coupling (π_{ij}) of ν and η . Join its paired centers by shortest constant-speed spherical geodesics. Each path is $u_{ij}(t) = \exp(tA_{ij})u_i$, where A_{ij} is skew-symmetric, acts on a two-dimensional plane, and has operator norm $d_{\Sigma}(u_i, v_j)$. An antipodal pair is treated by choosing any such plane. Define

$$f_t = \sum_{i,j} \pi_{ij} K_h(u_{ij}(t), \cdot), \quad J_t(x) = \sum_{i,j} \pi_{ij} K_h(u_{ij}(t), x) A_{ij} x.$$

Rotation invariance gives $\partial_t f_t + \operatorname{div} J_t = 0$. Pointwise Cauchy-Schwarz and integration yield

$$\int_{\{f_t>0\}} \frac{\|J_t\|^2}{f_t} d\sigma \leq \sum_{i,j} \pi_{ij} d_{\Sigma}(u_i, v_j)^2 = W_2(\nu, \eta)^2.$$

The preceding Fisher bound holds at every t . To justify entropy differentiation even where f_t vanishes, replace f_t by $(f_t + \delta)/(1 + \delta)$ and J_t by $J_t/(1 + \delta)$. The kernel is C^1 , the sphere is compact, and integration by parts followed by Cauchy-Schwarz gives

$$\left| \frac{d}{dt} \mathcal{H} \left(\frac{f_t + \delta}{1 + \delta} \right) \right| \leq \frac{1}{1 + \delta} \sqrt{\mathcal{I}(f_t)} \left(\int \frac{\|J_t\|^2}{f_t} d\sigma \right)^{1/2} \leq \frac{\sqrt{B_m}}{h} W_2(\nu, \eta).$$

Here quotients at common zeros are zero; the displayed regularized identity can equivalently be obtained by smoothing the C^1 kernel. Integrate in t and let $\delta \downarrow 0$. Uniform boundedness and continuity of $x \log x$ at zero justify the limit and prove (15.4).

On $\Sigma = S^2$, the variable s has constant density $1/4$. Direct integration gives

$$Z_h = h^2/12, \quad \mathcal{I}(K_h(u, \cdot)) = 24/h^2 - 4.$$

These imply the asserted explicit constants. $\square$

The relation between entropy continuity and quadratic transport is classical; compare Polyanskiy–Wu [27, Proposition 1] for regular Euclidean densities. The proof above is included because its densities have compact support and zeros, and because the ambient sphere and the finite-mixture path, rather than Euclidean regularity, are used here.

Lemma 15.3 (Spectral entropy loss on a thin support). *Suppose (15.1) holds. Let f be a bounded probability density supported in a measurable set $A \subseteq \Sigma$ for which*

$$\Pi \mathbf{1}_A \leq \theta < 1.$$

Then

$$(15.5) \quad \mathcal{H}(f) - \mathcal{H}(Pf) \geq (1 - \kappa^2)(1 - \theta).$$

Proof. Orbital Cauchy–Schwarz gives $(\Pi\sqrt{f})^2 \leq (\Pi\mathbf{1}_A)(\Pi f)$ and hence $\|\Pi\sqrt{f}\|_2^2 \leq \theta$. The invariant and orthogonal components are preserved by P . Since $\|\sqrt{f}\|_2 = 1$,

$$\|P\sqrt{f}\|_2^2 \leq \|\Pi\sqrt{f}\|_2^2 + \kappa^2\|(I - \Pi)\sqrt{f}\|_2^2 \leq 1 - (1 - \kappa^2)(1 - \theta).$$

For nonnegative numbers z_a with $z = \sum_a \lambda_a z_a$, the relative-entropy/Hellinger inequality gives

$$\sum_a \lambda_a z_a \log z_a - z \log z \geq \sum_a \lambda_a (\sqrt{z_a} - \sqrt{z})^2.$$

For completeness this follows by summing $t \log(t/s) - t + s \geq (\sqrt{t} - \sqrt{s})^2$; continuity handles zeros. Apply it pointwise to $z_a = f(R(u_a)^{-1}x)$ and integrate. Invariance of σ and Cauchy–Schwarz give

$$\mathcal{H}(f) - \mathcal{H}(Pf) \geq 2 \left(1 - \int \sqrt{Pf} P\sqrt{f} d\sigma \right) \geq 2(1 - \|P\sqrt{f}\|_2).$$

The inequality $2(1 - \sqrt{1-d}) \geq d$ for $0 \leq d \leq 1$ completes the proof. No logarithmic Sobolev inequality or pointwise mixing estimate has been assumed. $\square$

15.2. Transport paid for by loss of centroid mass. For a unit physical random vector X and a finite register S , put

$$(15.6) \quad c_s = \mathbb{E}[X \mid S = s], \quad \alpha = \sum_s \Pr(S = s) \|c_s\|, \quad \nu = \frac{1}{\alpha} \sum_{c_s \neq 0} \Pr(S = s) \|c_s\| \delta_{c_s / \|c_s\|}.$$

Only positive-mass labels contribute. The directional law is defined when $\alpha > 0$; its weights are not, in general, the probabilities of the hidden labels.

Lemma 15.4 (Centroid-loss transport). *Apply an independent letter $a \sim \lambda$, followed by any fixed physical identity-product word, using the actual stochastic rows of the machine. Let α', ν' be the quantities at the endpoint and suppose $\alpha' \geq \zeta > 0$. If $\lambda * \nu$ denotes the mixture of the rotated laws $R(u_a)_* \nu$, then $\delta = \alpha - \alpha' \geq 0$ and*

$$(15.7) \quad W_2(\lambda * \nu, \nu')^2 \leq \frac{3\pi^2}{\zeta} \delta.$$

There is no bound on any intervening register.

Proof. Write $T_a(s, i)$ for the actual composite row, including the fixed filler. Conditional independence of the row from the unrecorded past is precisely the machine definition. Thus

$$p'_i c'_i = \sum_{s,a} p_s \lambda_a T_a(s, i) R(u_a) c_s.$$

Set $w_{sai} = p_s \lambda_a T_a(s, i) \|c_s\|$, $q_i = \sum_{s,a} w_{sai}$, and $m_i = p'_i \|c'_i\|$. For $m_i > 0$ put $v_i = c'_i / \|c'_i\|$; otherwise choose an arbitrary unit vector. With $u_{sa} = R(u_a) c_s / \|c_s\|$ for nonzero terms,

$$(15.8) \quad \sum_{s,a} w_{sai} \|u_{sa} - v_i\|^2 = 2(q_i - m_i).$$

In particular $m_i \leq q_i$, $\sum_i q_i = \alpha$, $\sum_i m_i = \alpha'$, and $\delta \geq 0$.

Normalize the flows by α . They couple $\lambda * \nu$ to $\hat{\nu} = \sum_i (q_i / \alpha) \delta_{v_i}$. Chordal and geodesic distance satisfy $d_\Sigma(u, v) \leq (\pi/2) \|u - v\|$, so this coupling has geodesic quadratic cost at most $\pi^2 \delta / (2\alpha)$. If $\delta > 0$, the excess masses give a probability χ with

$$\hat{\nu} = (\alpha' / \alpha) \nu' + (\delta / \alpha) \chi.$$

Keeping the common part and transporting the remaining part across the sphere costs at most $\pi^2 \delta / \alpha$. The triangle inequality for W_2 and $(a + b)^2 \leq 2a^2 + 2b^2$ give $W_2(\lambda * \nu, \nu')^2 \leq 3\pi^2 \delta / \alpha$. Since $\alpha \geq \alpha' \geq \zeta$, this proves the result. The case $\delta = 0$ follows from the same couplings without an excess measure. Zero centroids and zero-probability labels require no division. $\square$

15.3. The sharp lower law. Assume, for some $A \geq 1$, $p > 0$, and $h_0 > 0$, the all-direction cap bound

$$(15.9) \quad \sup_{u,v \in \Sigma} m_G \{g : \|R(g)u - v\| < h\} \leq Ah^p \quad (0 < h \leq h_0).$$

This positive-power cap bound excludes every fixed unit vector. Hence $E^G = \{0\}$ and the Haar mean of each orbit is zero; in particular $0 \in \text{conv } R(G)z_0$. The constants are fixed with the representation. For $k \geq 1$ define

$$(15.10) \quad c_0 = \min\{1, h_0, (2A)^{-1/p}\}, \quad h_k = c_0 k^{-1/p}, \quad \gamma = (1 - \kappa^2)/2, \quad L_k^{\text{ent}} = \log(D_m h_k^{-(m-1)}).$$

Theorem 15.5 (Occupation without logarithmic loss). *Let the command alphabet contain a physical identity letter, and assume (15.1) and (15.9). Fix a unit seed z_0 , let $Q = \text{conv } R(G)z_0$, and suppose the target is $\rho R(u_w)z_0$, with $0 < \rho \leq 1$ and legal decoder values in Q . Error is Euclidean error in E . For $0 \leq \varepsilon < \rho$ write $\zeta = \rho - \varepsilon$. Every horizon- N realization, including one redesigned at that horizon, satisfies*

$$(15.11) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq \left\lceil 1 + \frac{2L_k^{\text{ent}}}{\gamma} + \frac{3\pi^2 B_m}{\gamma^2 h_k^2} \frac{1 - \zeta}{\zeta} \right\rceil.$$

In particular, for fixed subcritical parameters there are constants $C_\zeta, c_\zeta > 0$ such that

$$(15.12) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C_\zeta k^{2/p}, \quad W_{N,\varepsilon} \geq c_\zeta N^{p/2}.$$

The intervening widths are unrestricted. The constants depend only on $m, A, p, h_0, \kappa, \zeta$ and the fixed round normalization.

Proof. If there are $J \geq 2$ positive cuts of width at most k , list them as $t_1 < \dots < t_J$. Use a fixed prefix up to t_1 . In each subsequent gap put one independent letter with law λ , then identity letters until the next selected cut. Finish by a fixed suffix. This is an external distribution on actual words of length N , chosen from the advertised width profile, not from the machine's private state. It does not require the machine's rows on an identity letter to be identities.

Let X be the current unit physical vector. Write α_j, ν_j for (15.6) at t_j . Wordwise correctness and $\|D_i\| \leq 1$ imply at the terminal cut

$$\zeta \leq \mathbb{E} \langle X_N, D(S_N) \rangle = \sum_i \Pr(S_N = i) \langle \mathbb{E}[X_N \mid S_N = i], D_i \rangle \leq \alpha_N.$$

The first inequality averages the conditional *mean* error on each deterministic word. It makes no assertion about a single sampled label's error. Conditional Jensen, or the flow calculation above, shows that centroid mass cannot increase along any gap or fixed suffix. Thus $1 \geq \alpha_j \geq \zeta$, and

$$\delta_j := \alpha_j - \alpha_{j+1} \geq 0, \quad \sum_{j < J} \delta_j \leq 1 - \zeta.$$

Fix $h = h_k$ and put $f_j = K_h \nu_j$. Each f_j is supported on at most k open chordal caps of radius h . The cap assumption gives $\Pi \mathbf{1}_{\{f_j > 0\}} \leq A k h^p \leq 1/2$. Lemma 15.3 therefore gives $\mathcal{H}(Pf_j) \leq \mathcal{H}(f_j) - \gamma$. Equivariance of the kernel says $Pf_j = K_h(\lambda * \nu_j)$. Lemmas 15.2 and 15.4 imply

$$\mathcal{H}(f_{j+1}) \leq \mathcal{H}(f_j) - \gamma + \frac{\pi \sqrt{3B_m}}{h \sqrt{\zeta}} \sqrt{\delta_j}.$$

Writing $M = J - 1$, summing, using $0 \leq \mathcal{H}(f_j) \leq L_h$, and applying Cauchy–Schwarz only over these M gaps, gives

$$(15.13) \quad \gamma M \leq L_h + \frac{\pi \sqrt{3B_m}}{h} \sqrt{M \frac{1 - \zeta}{\zeta}}.$$

Young's inequality bounds the second term by $\gamma M/2 + 3\pi^2 B_m (1 - \zeta)/(2\gamma h^2 \zeta)$. This proves (15.11); $J \leq 1$ is immediate. The logarithm in $L_h = O(1 + \log k)$ is additive, not multiplied by $k^{2/p}$. Since $1 + \log k = O_p(k^{2/p})$, the first bound in (15.12) follows. If the whole profile has width at most k , its N positive cuts are all counted, proving the second. $\square$

Remark 15.6 (Endpoint information and synchronization). The unrelaxed inequality (15.13) also holds. When $\zeta = 1$, it gives $J \leq 1 + L_k^{\text{ent}}/\gamma$, consistent with exponential exact width for extreme outputs. For fixed positive error it gives the polynomial law, but does not classify every joint limit $N \rightarrow \infty$, $\varepsilon \downarrow 0$.

If the identity letter is absent but physical identity-product words exist at every length at least g , greedily retain narrow cuts at spacing at least $g + 1$. Each retained cut accounts for at most $g + 1$ original narrow cuts. One independent command followed by a return of the remaining length fits each retained gap. The prefix and suffix may be any fixed words of their required lengths. The same proof multiplies the occupation bound by at most $g + 1$. Approximate returns with accuracy-dependent thresholds are not included in this quantitative extension without additional estimates.

Corollary 15.7 (Matching power laws). *In the setting of Theorem 14.3, replace its converse by (15.12). In particular, if the all-direction exponent p equals the real dimension q of the target orbit, then*

$$W_{N,\varepsilon} = \Theta((N + 1)^{q/2})$$

for every fixed $0 < \rho < 1$, $0 \leq \varepsilon < \rho$, and also for $\rho = 1$, $0 < \varepsilon < 1$. For a general fixed-point-free compact connected Lie representation one may take $p = p_*$ from Theorem 24.1; the lower and upper exponents are then $p_*/2$ and $q/2$, respectively, with no logarithmic loss in the lower bound.

Proof. Apply Theorem 15.5. The upper common-row construction in Theorem 14.3 is exact for $\rho < 1$ and has $O((N + 1)^{q/2})$ labels. At $\rho = 1$ and positive error use its amplitude-slack version. The two bounds match when $p = q$. Theorem 24.1 supplies the remaining assertion. $\square$

The endpoint $\rho = 1, \varepsilon = 0$ is not an assertion of the matching polynomial corollary. No exact leading constant or asymptotic ratio is claimed. The spectral constant is effective only when a certified value is supplied; a fully numerical example is given below.

Remark 15.8 (Return-free extension). Theorem 16.2 below removes the identity-letter and return assumptions from the occupation bound. It uses covariance under the actual deterministic physical fillers, rather than treating approximate returns as exact. The present synchronized theorem is its special case; all smoothing and transport estimates remain unchanged.

16. ENTROPY TRANSPORT WITHOUT A RETURN CONDITION

The entropy potential is invariant under a fixed physical rotation. This allows arbitrary executable words between the spectral steps; they need not return the physical system to its initial state. We first make the covariance explicit and then retain the dependence on the error instead of absorbing it into a fixed-error constant.

Keep the sphere $\Sigma = S(E)$, round probability measure, kernel K_h , and constants D_m, B_m from Lemma 15.2. For an orthogonal map B , write $B_*\nu$ for transport of a measure and $U_B f(x) = f(B^{-1}x)$ for transport of a density. Directly from the radial kernel and invariance of round measure,

$$(16.1) \quad K_h(B_*\nu) = U_B(K_h\nu), \quad H(U_B f) = H(f), \quad W_2(B_*\nu, B_*\eta) = W_2(\nu, \eta).$$

No commutation between B and the averaging operator is asserted.

Lemma 16.1 (Transport after a deterministic physical drift). *In Lemma 15.4, replace the identity-product filler by any fixed executable word, with physical orthogonal action B . If the resulting centroid masses satisfy $\alpha' \geq \zeta > 0$, then $\alpha' \leq \alpha$ and*

$$(16.2) \quad W_2(B_*(\lambda * \nu), \nu')^2 \leq 3\pi^2 \frac{\alpha - \alpha'}{\zeta}.$$

The filler may have any length and its internal registers may have arbitrary widths. A fixed physical word, with no random spectral step, also cannot increase centroid mass.

Proof. Let $T_a(s, i)$ be the actual composite row for the random letter followed by this fixed word. Conditional independence at the incoming register and orthogonality give the exact identity

$$p'_i c'_i = \sum_{s,a} p_s \lambda_a T_a(s, i) B R(u_a) c_s.$$

Use the weights $w_{s,a,i} = p_s \lambda_a T_a(s, i) \|c_s\|$ and unit vectors $BR(u_a) c_s / \|c_s\|$ in the proof of Lemma 15.4. Their weighted sum is the displayed centroid, so its squared-distance identity and both couplings are unchanged. They give (16.2). Terms with zero mass or zero centroid are omitted, exactly as in that lemma. For a fixed word, conditional Jensen and the isometry B give $\sum_i p'_i \|c'_i\| \leq \sum_s p_s \|B c_s\| = \alpha$. $\square$

Let G be a compact connected Lie group with an orthogonal representation $R : G \rightarrow O(E)$, $m = \dim E \geq 2$. Fix a unit vector z_0 whose orbit spans E , and suppose $E^G = \{0\}$. Put $Q = \text{conv}(R(G)z_0)$. The finite nonempty command alphabet need not contain the identity or any exact return. Assume that a probability law λ on the *actual letters* satisfies the full action bound

$$(16.3) \quad \|Pf - \Pi f\|_2 \leq \kappa \|f - \Pi f\|_2, \quad Pf(x) = \sum_a \lambda_a f(R(u_a)^{-1}x), \quad \kappa < 1.$$

Here $\Pi f(x) = \int_G f(R(g)^{-1}x) dg$ projects onto all invariant functions, not merely the constants. Suppose also that

$$(16.4) \quad \sup_{u,v \in \Sigma} \int_G \mathbf{1}_{\{\|R(g)u-v\| < h\}} dg \leq Ah^p \quad (0 < h \leq h_0),$$

where $p > 0$. These are global action and all-direction hypotheses; contraction on E alone is not a substitute for (16.3).

Theorem 16.2 (Return-free sharp occupation and error budget). *Under (16.3)–(16.4), consider the target $F(g) = \rho R(g)z_0$, legal decoders in Q , and Euclidean error $0 \leq \varepsilon < \rho \leq 1$. Set*

$\zeta = \rho - \varepsilon$, $c_0 = \min\{1, h_0, (2A)^{-1/p}\}$, $h = c_0 k^{-1/p}$, $\gamma = (1 - \kappa^2)/2$, $L_h = \log(D_m h^{-(m-1)})$. For any error- ε clocked realization, let $J = \#\{1 \leq t \leq N : |S_t| \leq k\}$. If $J \geq 1$ and $M = J - 1$, then

$$(16.5) \quad \gamma M \leq L_h + \frac{\pi \sqrt{3B_m}}{h} \sqrt{M \frac{1 - \zeta}{\zeta}}.$$

Consequently, as a real bound before taking an integer ceiling,

$$(16.6) \quad J \leq 1 + \frac{2L_h}{\gamma} + \frac{3\pi^2 B_m}{\gamma^2 h^2} \frac{1 - \zeta}{\zeta}.$$

No condition is imposed on return words or on intervening register widths. Constants other than ζ depend only on m, A, p, h_0, κ and the fixed spherical metric. In particular, for fixed $\zeta > 0$, $J = O_\zeta(k^{2/p})$ and $W_{N,\varepsilon} = \Omega_\zeta(N^{p/2})$.

Proof. If $J \leq 1$, the asserted real bound is immediate. Otherwise list the narrow cuts as $t_1 < \dots < t_J$. Choose an arbitrary deterministic prefix of length t_1 . In each gap put first one independent letter of law λ , and then a fixed word of length $t_{j+1} - t_j - 1$. Such a word exists at every length because the alphabet is nonempty: repeat any one of its letters. Finish with any fixed suffix of length $N - t_J$. This external word law depends on the advertised width profile, not on the machine's private state. Write B_j for the physical action of the fixed filler in gap j . The physical state at every cut remains one unit orbit point; no equality of the B_j to the identity is needed.

Let α_j, ν_j be its conditional-centroid mass and directional law at cut t_j . Terminal correctness gives $\alpha_N \geq \zeta$ by exactly the inner-product argument in Theorem 15.5. This uses wordwise correctness of the mean decoder, not correctness of each label. Lemma 16.1 and the fixed-suffix inequality imply

$$1 \geq \alpha_1 \geq \dots \geq \alpha_J \geq \zeta, \quad \sum_{j=1}^M (\alpha_j - \alpha_{j+1}) \leq 1 - \zeta.$$

There are at most k directions at each selected cut. Thus $f_j = K_h \nu_j$ is supported on at most k open chordal caps, and (16.4) gives $\Pi \mathbf{1}_{\text{supp}_+ f_j} \leq k A h^p \leq 1/2$. The kernel vanishes at each cap boundary. Lemma 15.3 and (16.1) therefore give

$$H(U_{B_j} P f_j) = H(P f_j) \leq H(f_j) - \gamma.$$

By covariance, $U_{B_j} P f_j = K_h(B_{j*}(\lambda * \nu_j))$. Entropy continuity and Lemma 16.1, with $\delta_j = \alpha_j - \alpha_{j+1}$, yield

$$H(f_{j+1}) \leq H(f_j) - \gamma + \frac{\pi \sqrt{3B_m}}{h \sqrt{\zeta}} \sqrt{\delta_j}.$$

Sum, use $0 \leq H(f_j) \leq L_h$, and apply Cauchy–Schwarz to the M nonnegative losses. This proves (16.5). The inequality $ab \leq (a^2 + b^2)/2$, with $a = \sqrt{\gamma M}$, gives (16.6). For a full peak bound k all N positive cuts are counted. Since $\log(k+1) = O(k^{2/p})$, the last conclusions follow. The constants are independent of the filler lengths, their actions, and all actual hidden tables. $\square$

Remark 16.3 (What has, and has not, been removed). The return hypothesis has disappeared from the quantitative converse, not from the general terminal stationarization theorem. The full action gap remains indispensable to this argument. A single irrational rotation letter has a unitary Koopman operator and cannot satisfy (16.3); its one-label clocked realization is therefore no counterexample. No entropy is assigned to the hidden register here: H is relative entropy of

an auxiliary smoothed directional law against normalized round measure. The classical entropy–transport principle [27] supplies neither the Markov row identity nor the width quantifiers. The logarithmic initial range L_h is retained in (16.5); it is not paid again at every narrow cut.

Corollary 16.4 (A fixed executable spectral block). *The same conclusions hold if (16.3) is supplied by a probability law on executable words of one fixed length $b \geq 1$, rather than individual letters. In (16.6) multiply the right side by b .*

Proof. Greedily select narrow cuts at separation at least b . Each selected cut accounts for at most b original cuts, including the last incomplete group of cuts. Apply the proof above with one independent spectral word at the start of every selected gap and an arbitrary fixed filler for its remaining length. There is no restriction before the first selected cut or after the last. The alphabet of spectral blocks is finite even when their internal hidden widths are unbounded. $\square$

Theorem 16.5 (A joint amplitude–accuracy law). *Assume the preceding geometric and action-gap hypotheses. Let $q = \dim R(G)z_0$, and write $W_N(\rho, \varepsilon)$ to display the amplitude dependence. Fix $\zeta_0 > 0$. Uniformly for $0 \leq \varepsilon < \rho \leq 1$ with $\rho - \varepsilon \geq \zeta_0$ and $\tau = 1 - \rho + \varepsilon > 0$, constants C_i depending only on the fixed data and ζ_0 give*

$$(16.7) \quad N \leq C_0 + C_1 \log k + C_2 \tau k^{2/p}, \quad k = W_N(\rho, \varepsilon),$$

$$(16.8) \quad W_N(\rho, \varepsilon) \leq C_3 \left(1 + \frac{N+1}{\tau}\right)^{q/2}.$$

If $p = q$ and a parameter sequence satisfies $\rho_N - \varepsilon_N \geq \zeta_0$ and $\log(1/\tau_N) = o(N)$, then

$$(16.9) \quad W_N(\rho_N, \varepsilon_N) = \Theta\left((N/\tau_N)^{p/2}\right) \quad (N \rightarrow \infty).$$

In particular, at pure amplitude and any positive subexponentially vanishing Euclidean tolerance, the matched law is $\Theta((N/\varepsilon_N)^{p/2})$. It includes polynomial and stretched-exponential accuracy sequences.

Proof. In (16.6), $1 - \zeta = \tau$ and $\zeta^{-1} \leq \zeta_0^{-1}$. Also $L_h = \log(D_m c_0^{-(m-1)}) + (m-1) \log k/p$. Counting all positive cuts proves (16.7).

For the upper bound set $\beta = \rho - \varepsilon/2$. Then $0 < \beta < 1$, $1 - \beta \geq \tau/2$, and the target change has norm at most $\varepsilon/2$. The proof of the upper construction in Theorem 14.3 uses only the invariant orbit hull and Lemma 14.2; it does not use its identity-letter or spectral assumptions. Applied at amplitude β , it gives an exact common-row machine with at most $C(1 + (N+1)/(-\log \beta))^{q/2}$ labels. The inequality $-\log \beta \geq 1 - \beta \geq \tau/2$ proves (16.8). When $\varepsilon = 0$, this is the same exact construction at $\beta = \rho < 1$; no limiting construction at $\beta = 1$ is used.

Let k_N be the minimum width. The uniform upper bound gives $\log k_N = O(1 + \log(N+1) + \log(1/\tau_N)) = o(N)$. Consequently the first two terms of (16.7) sum to at most $N/2$ for all sufficiently large N . It follows that $k_N \geq (N/(2C_2\tau_N))^{p/2}$. The matching upper bound when $p = q$ proves (16.9). The comparison constants depend only on the fixed data and ζ_0 ; the starting horizon can depend on the parameter sequence. $\square$

The least-orbit theorem supplies $p = p_*$ for each fixed-point-free compact connected Lie representation. Thus the joint law applies when $p_* = q$. The point $\tau = 0$ is precisely pure amplitude and zero error; there (16.5) instead gives $\gamma(J-1) \leq L_h$. Neither the joint theorem nor its proof determines optimal constants or the entire exponentially small transition window.

17. POSITIVE MATRIX REALIZATIONS IN EVERY DIMENSION

Let $d \geq 2$ and let $\mathcal{H}_d^0$ be the real Euclidean space of traceless Hermitian $d \times d$ matrices, with scalar product $\langle A, B \rangle = \text{tr}(AB)$ and Frobenius norm $\|\cdot\|_F$. Let

$$\mathcal{D}_d = \{D = D^* \succeq 0 : \text{tr } D = 1\}, \quad c_d = I/d, \quad r_d = \sqrt{1 - 1/d}.$$

This section concerns numerical outputs in $\mathcal{D}_d$. The requirement that every decoder value be positive with trace one is part of the model; it is not imposed merely on the final average.

Lemma 17.1 (Small balls on the complex Grassmannian). *Let P be a fixed rank- m orthogonal projection in $\mathbb{C}^d$, with $1 \leq m < d$. For Haar $U \in \text{SU}(d)$ there is $C_{d,m} < \infty$ such that*

$$\Pr\{\|UPU^* - Q\|_F < h\} \leq C_{d,m} h^{2m(d-m)} \quad (0 < h \leq 1)$$

for every rank- m projection Q . For rank one, writing $\text{dist}_{\text{pr}}(P, Q) = \sqrt{1 - \text{tr}(PQ)}$, one has $\text{dist}_{\text{pr}}(P, Q) = \|P - Q\|_F / \sqrt{2}$ and exactly

$$(17.1) \quad \Pr\{\text{dist}_{\text{pr}}(UPU^*, Q) < h\} = h^{2(d-1)} \quad (0 \leq h \leq 1).$$

Proof. Conjugation is transitive, so fix $Q = \text{diag}(I_m, 0)$. A projection sufficiently close to Q has range the graph of a unique matrix $Z \in \mathbb{C}^{(d-m) \times m}$, with

$$P(Z) = \begin{pmatrix} I \\ Z \end{pmatrix} (I + Z^* Z)^{-1} \begin{pmatrix} I & Z^* \end{pmatrix}.$$

This notation means the product of the three displayed matrices. The map and its inverse are smooth near zero, and its derivative at zero is $Z \mapsto \begin{pmatrix} 0 & Z^* \\ Z & 0 \end{pmatrix}$, of Frobenius norm $\sqrt{2}\|Z\|_F$. The invariant probability measure is normalized Riemannian volume on this compact homogeneous manifold; in this chart its density is smooth and bounded on a sufficiently small closed neighborhood. The inverse chart puts a Frobenius h -ball inside a Euclidean Ch -ball in real dimension $2m(d-m)$. This proves the bound for small h ; increasing the constant covers the remaining interval. Transitivity makes the constant independent of Q .

For rank one choose $Q = e_1 e_1^*$. A uniform unit vector can be obtained by normalizing independent standard complex Gaussian coordinates. If X is the squared modulus of the first coordinate before normalization and Y is the sum of the other squared moduli, then X is exponential of mean one and Y is an independent sum of $d-1$ such variables. For $0 < z < 1$,

$$\Pr\left\{\frac{X}{X+Y} > 1-z\right\} = \mathbb{E} \exp\left(-\frac{1-z}{z} Y\right) = z^{d-1}.$$

Put $z = h^2$. The endpoint cases follow by continuity. This proves (17.1), and also shows that dist_{pr} is the Frobenius chordal distance divided by $\sqrt{2}$. $\square$

Lemma 17.2 (A uniform conjugacy-orbit exponent). *There is $C_d < \infty$ such that*

$$(17.2) \quad \sup_{\substack{A, B \in \mathcal{H}_d^0 \\ \|A\|_F = \|B\|_F = 1}} m_G\{U : \|UAU^* - B\|_F < h\} \leq C_d h^{2(d-1)} \quad (0 < h \leq 1).$$

The exponent cannot be increased uniformly over A and B .

Proof. Write the eigenvalues of A as $a_1 \geq \dots \geq a_d$. Since their sum is zero and their squared sum is one, $a_1 - a_d \geq d^{-1/2}$. Some adjacent gap therefore satisfies

$$(17.3) \quad a_m - a_{m+1} \geq \gamma_d := \frac{1}{\sqrt{d(d-1)}}.$$

Let P_A be the spectral projection onto the first m eigenvectors. It is well defined despite possible multiplicities within either block.

Two conjugates A', A'' of A , with eigenbases x_i, y_j in this order, obey the exact identity

$$(17.4) \quad \|A' - A''\|_F^2 = \sum_{i,j} (a_i - a_j)^2 |\langle x_i, y_j \rangle|^2.$$

Expanding both traces proves it. The cross-block terms and (17.3) imply

$$(17.5) \quad \|P_{A'} - P_{A''}\|_F \leq \gamma_d^{-1} \|A' - A''\|_F,$$

since the squared norm on the left is exactly the sum of the two cross-block overlap sums.

If the set in (17.2) is nonempty, choose one member U_0 . Every other member has $\|UAU^* - U_0AU_0^*\|_F < 2h$. Thus its rank- m spectral projection lies within $2h/\gamma_d$ of $U_0P_AU_0^*$. The Haar law of UP_AU^* is the invariant Grassmannian law. Lemma 17.1 bounds the measure by $C_{d,m}(2h/\gamma_d)^{2m(d-m)}$ for $h \leq \gamma_d/2$. Since $m(d-m) \geq d-1$, and there are finitely many m , this gives (17.2) with a constant independent of all eigenvalue multiplicities. Larger h are covered by increasing C_d .

For sharpness take $A = B = (P - c_d)/r_d$ with P rank one. By (17.1), for sufficiently small h the probability is

$$(r_d^2 h^2 / 2)^{d-1}.$$

It cannot be bounded by a constant times $h^{p'}$ as $h \downarrow 0$ for any $p' > 2(d-1)$. $\square$

The spectral split, rather than an assumption that all centroids remain rank one, supplies the uniformity needed by the stochastic contraction. The conjugation representation has ambient dimension $d^2 - 1$, but its worst small-ball exponent is $2(d-1)$.

Theorem 17.3 (Unitary matrix-orbit width). *Let a finite alphabet in $SU(d)$ contain the identity and support a probability law satisfying (14.1). Fix a rank-one projection P_0 . For $0 < a \leq 1$, set*

$$(17.6) \quad F_a(U) = c_d + a(UP_0U^* - c_d), \quad Y = \mathcal{D}_d,$$

with Frobenius error. Its exact bounded-width threshold is ar_d , attained by the one-label decoder c_d . For $0 \leq \varepsilon < ar_d$, put $\zeta = a - \varepsilon/r_d$. Every realization satisfies

$$(17.7) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq C \left(1 + \log(k+1) + \frac{1-\zeta}{\zeta} k^{1/(d-1)} \right).$$

For fixed $0 < a < 1$ and $0 \leq \varepsilon < ar_d$,

$$(17.8) \quad cN^{d-1} \leq W_{N,\varepsilon} \leq C(N+1)^{d-1},$$

and the upper realization can be exact. The same orders hold for $a = 1$ and every fixed $0 < \varepsilon < r_d$.

More explicitly, for $a < 1$ and $\varepsilon = 0$ set $a' = a$; for $\varepsilon > 0$ set $a' = a - \varepsilon/(2r_d)$. Whenever $a' < 1$, an error at most $\varepsilon/2$ realization has

$$(17.9) \quad K \leq C_d \left(1 + \frac{N+1}{-\log a'} \right)^{d-1}$$

labels. In the zero-error case this realization is exact. In (17.7) the constant depends only on d and the spectral-gap data, not on the amplitude or tolerance. Extra finitely many unitary commands are allowed in both bounds.

Proof. Center at c_d and divide all outputs by r_d . The resulting legal set is

$$Q = \text{conv}\{(P - c_d)/r_d : P \text{ rank one}\} \subseteq \mathbb{B}_{\mathcal{H}_d^0}.$$

The last inclusion follows from $\text{tr } D^2 \leq 1$ for $D \in \mathcal{D}_d$. The orbit spans $\mathcal{H}_d^0$: if a traceless Hermitian A has $\text{tr}(AP) = 0$ for every rank-one P , then $v^*Av = 0$ for every v , hence $A = 0$. Conjugation-fixed matrices are scalar, so there are no fixed vectors in this traceless space. The orbit is complex projective space, of real dimension $2(d-1)$. Lemma 17.2 and Theorem 15.5, with $p = q = 2(d-1)$,

prove the lower and occupation bounds, as well as the radius statement after multiplying the norm by r_d .

We give a direct inner approximation to make positivity of the upper rows transparent. Choose a maximal δ -separated family of rank-one projections P_i for dist_{pr} , containing P_0 . Its $\delta/2$ balls are disjoint, so (17.1) gives

$$K \leq (2/\delta)^{2(d-1)}.$$

For any traceless Hermitian A , put $M = \lambda_{\max}(A)$. When $A \neq 0$, $M > 0$ and $\lambda_{\min}(A) \geq -(d-1)M$. Approximate a top eigenline by P_i at distance at most δ . Spectral decomposition then gives

$$(17.10) \quad \text{tr}(AP_i) \geq M - (M - \lambda_{\min}(A))\delta^2 \geq (1 - d\delta^2)M.$$

The trivial $A = 0$ case is automatic. Comparing support functions in $\mathcal{H}_d^0$, with $K_0 = \mathcal{D}_d - c_d$, yields

$$(1 - d\delta^2)K_0 \subseteq \text{conv}\{P_i - c_d\} \subseteq K_0.$$

Put $r = 1 - d\delta^2 > 0$. For each actual command U_b choose one row of barycentric coordinates satisfying

$$(17.11) \quad \sum_j T_b(i, j)(P_j - c_d) = rU_b(P_i - c_d)U_b^*.$$

Rows are nonnegative and sum to one by construction. Initialize at P_0 and decode label i as

$$D_i = c_d + (a'/r^N)(P_i - c_d).$$

This is in $\mathcal{D}_d$ when $r^N \geq a'$: it is a convex combination of c_d and P_i . Choosing

$$\delta^2 = \min\left(\frac{1}{4d}, \frac{-\log a'}{2d(N+1)}\right)$$

ensures this inequality. Induction in (17.11) gives mean output exactly $F_{a'}(u_w)$. The target discrepancy is $|a - a'|r_d = \varepsilon/2$ when $\varepsilon > 0$. The cardinal bound proves (17.9). The case $a = 1, \varepsilon = 0$ is deliberately excluded from this dilation construction and is treated exactly in the next section. $\square$

The family contains the qubit sphere as $d = 2$, but for $d \geq 3$ the pure-state orbit is a proper curved subset of the unit sphere in $\mathcal{H}_d^0$. Thus (17.8) does not follow by substituting the ambient dimension into the spherical theorem. The entropy argument removes the logarithmic loss. No optimal leading constant, exact asymptotic ratio, or dimension-uniform constant is claimed.

Corollary 17.4 (Algebraic generating alphabets). *Theorem 17.3 applies to the identity together with any finite symmetric algebraic-entry set generating a dense subgroup of $\text{SU}(d)$, equipped with a lazy symmetric law. Such a finite alphabet exists for every $d \geq 2$.*

Proof. The spectral gap is the imported theorem of Bourgain–Gamburd [16]. Laziness makes its one-sided gap an absolute L^2 contraction on mean-zero functions. Existence can be seen by placing two algebraic irrational-angle $\text{SU}(2)$ rotations in every adjacent pair of coordinate directions. For example, use the quaternion lifts in the explicit adjacent-block construction below within each pair. Each pair generates its full $\text{SU}(2)$ subgroup. Their Lie algebras contain the adjacent off-diagonal matrix units and their brackets, which generate $\mathfrak{su}(d)$. The connected compact closure is consequently $\text{SU}(d)$. All entries remain algebraic. The spectral-gap constant is not computed by the finite regression checks. $\square$

17.1. A finite whole-law encoding. Let $q_0 = d^2 - 1$ and choose a Frobenius-orthonormal basis $H_1, \dots, H_{q_0}$ of $\mathcal{H}_d^0$. Since $\|H_l\|_{\text{op}} \leq 1$, the $2q_0$ matrices

$$E_{l,\pm} = \frac{I \pm H_l/2}{2q_0}$$

are positive and sum to I . Put $\Phi(D)_{l,\pm} = \text{tr}(E_{l,\pm}D)$ and $Y_\Phi = \Phi(\mathcal{D}_d)$. Every entry is at least $1/(4q_0)$. The map is injective, because

$$(17.12) \quad D - c_d = \sum_l 2q_0(\Phi(D)_{l,+} - \Phi(D)_{l,-})H_l.$$

On the vector space parallel to the affine span of Y_Φ , use the norm of the reconstructed matrix on the right. Then Φ is an affine isometry, and replacing each legal matrix decoder by its image preserves every clocked or stationary width exactly. This is one whole categorical law with a compact convex legality constraint, not separate binary event tolerances. For ordinary total variation the explicit norm comparison is

$$\frac{\|D - D'\|_F}{4q_0} \leq \text{TV}(\Phi(D), \Phi(D')) \leq \frac{\|D - D'\|_F}{4\sqrt{q_0}}.$$

It follows directly by comparing the ℓ^1 and ℓ^2 norms of the basis coefficients. Enlarging the legal output set from Y_Φ to the entire simplex is a different realization problem. In particular the extreme-output argument below cannot be transferred to that enlarged set: its target vectors are interior points of the simplex.

Remark 17.5 (An explicit algebraic alphabet in every dimension). In each adjacent coordinate plane $(j, j+1)$ embed the two matrices

$$\frac{1}{5} \begin{pmatrix} -3+4i & 0 \\ 0 & -3-4i \end{pmatrix}, \quad \frac{1}{5} \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \quad 1 \leq j < d,$$

and act identically on all other coordinates. Include their inverses and the identity. Each pair has infinite-order eigenvalues, since its trace $-6/5$ is not an integer, and its two torus closures generate the adjacent $SU(2)$. The adjacent skew-Hermitian matrix units and their brackets span $\mathfrak{su}(d)$. The closed generated subgroup is therefore $SU(d)$. Bourgain–Gamburd is applied to this actual finite algebraic set, not merely to the existence of an unspecified set. The resulting spectral-gap constant is not numerically evaluated here.

The Frobenius chordal distance between rank-one projections is $\sqrt{2}$ times the projective distance used in the net construction. In the conjugacy-cap comparison, choosing one member of a nonempty target ball avoids imposing any spectral condition on its arbitrary center. All adjacent-gap and cap constants here depend on d . The amplitude $a' = a - \varepsilon/(2r_d)$ in the positive-error upper bound deliberately spends only half the available error.

18. EXTREME OUTPUTS AND THE EXACT ENDPOINT

The strict amplitude slack in the polytope construction is not a technical convenience that can simply be removed. At an extreme output, convex averaging has no room to hide a different physical state. This observation gives an exact finite-horizon formula, independently of spectral gap or synchronization.

Theorem 18.1 (The extreme-orbit formula). *Let a group act by affine bijections on a compact convex set Y . Let $x_0 \in Y$ have an orbit consisting of extreme points of Y , and let a finite nonempty command alphabet act through this group. There is one seed and one terminal query, with target $u_w x_0$. Put*

$$\mathcal{O}_t = \{u_w x_0 : |w| = t\}, \quad n_t = |\mathcal{O}_t|.$$

Every exact clocked stochastic realization at horizon N satisfies

$$(18.1) \quad |S_t| \geq n_t \quad (0 \leq t \leq N).$$

One deterministic realization attains all these inequalities simultaneously. Consequently

$$(18.2) \quad W_{N,0} = n_N.$$

Neither an identity letter nor an exact return is required.

Proof. Fix t and one suffix v of length $N - t$. Such a suffix exists because the alphabet is nonempty; at $t = N$ use the empty word. Let $D_v(i) \in Y$ be the conditional expected terminal decoder after this fixed suffix, when the cut- t label is i . Convexity guarantees that it lies in Y . For every prefix w of length t , exact correctness says

$$u_v u_w x_0 = \sum_i p_w(i) D_v(i),$$

where p_w is the actual distribution at that cut. Extremality forces $D_v(i) = u_v u_w x_0$ whenever $p_w(i) > 0$. Since the action of u_v is injective, prefixes with different physical outputs have disjoint nonempty supports in S_t . This proves (18.1), including the terminal cut.

Conversely, at cut t store the current point of $\mathcal{O}_t$. A command sends this point to its image in $\mathcal{O}_{t+1}$; the terminal decoder is the stored point. The initialization is deterministic. Finally, any one fixed command gives an injection $\mathcal{O}_t \rightarrow \mathcal{O}_{t+1}$, so n_t is nondecreasing. The peak of this realization is n_N , proving (18.2). $\square$

The same conclusion holds for an orbit on a Euclidean unit sphere with legal output set its convex hull: every orbit point is extreme by strict convexity of the ambient Euclidean ball. For density matrices, a rank-one projection is extreme. Indeed, if $P = \sum_i p_i D_i$ with $D_i \geq 0$ and $\text{tr } D_i = 1$, then $\text{tr}((I - P)D_i) = 0$ for every positive weight. Positivity forces D_i to be supported on the range of P , and hence $D_i = P$.

The cut argument fixes one suffix, whose affine action is injective. The cardinalities are nondecreasing because the nonempty alphabet contains a letter whose injective action maps each reachable set into the next one. No sampled-answer floor is included in this width.

19. EFFECTIVE ALGEBRAIC-CIRCLE BOUNDS

For algebraic unit-circle eigenvalues of infinite order, an explicit separation estimate replaces the Diophantine hypothesis of the preceding section. The parameters r, m, A_r below are those of (12.3)–(12.4); in this section r denotes the localization order, not the number of command rotations. The alphabet contains I, R_θ , and one interface mean is $\rho \cos \phi$.

Lemma 19.1 (Separated powers give a block defect). *Suppose $|\lambda^n - 1| \geq \sigma$ for every integer $1 \leq n \leq 2mk$, where $0 < \sigma \leq 1$. Write $R_m(t) = \text{diag}(R_t, R_{2t}, \dots, R_{mt})$ for the representation acting on Z_m . On this representation, the uniform law on the $2k$ words with products R_θ^j , $0 \leq j < 2k$, each padded to length $2k$, has defect at least $\sigma^2/16$ in (7.4).*

Proof. Write a unit vector as $u = (u_1, \dots, u_m)$ in its two-dimensional frequency blocks. For distinct indices $i, j \in \{0, \dots, 2k - 1\}$,

$$\|R_m(i\theta)u - R_m(j\theta)u\|^2 = \sum_{l=1}^m \|u_l\|^2 |\lambda^{l(i-j)} - 1|^2 \geq \sigma^2.$$

A ball of radius strictly less than $\sigma/2$ about any of k centers contains at most one of these $2k$ points. Thus at least k points have distance at least $\sigma/2$ from all centers (the same conclusion follows by taking the open balls of radius $\sigma/2$). For unit centers the defect is one half of the squared distance to the closest center. Averaging gives at least $(1/2)(\sigma^2/8) = \sigma^2/16$. $\square$

Theorem 19.2 (Full-noise algebraic-circle bound). *Assume λ is algebraic and not a root of unity. Let its primitive irreducible integer polynomial be*

$$P(z) = az^d + a_{d-1}z^{d-1} + \cdots + a_0, \quad a > 0.$$

Set $H_P = \max(a, |a_{d-1}|, \dots, |a_0|)$ and

$$B_P = a(1 + H_P/a)^{d-1} > 1.$$

With r, m, A_r as in (12.3)–(12.4), every error- ε realization of peak at most k satisfies

$$(19.1) \quad N < 2k \left(1 + 16 \cdot 2^{2(d-1)} B_P^{4mk} \log A_r \right).$$

Its number of command cuts $t \in \{1, \dots, N\}$ with width at most k , even if other widths are unrestricted, is at most

$$(19.2) \quad 2k - 1 + 32k \cdot 2^{2(d-1)} B_P^{4mk} \log A_r.$$

Consequently, for fixed algebraic data, ρ , and $\varepsilon < \rho/2$,

$$(19.3) \quad W_{N,\varepsilon} \geq \frac{\log N - O(\log \log N)}{4m \log B_P}.$$

All constants in the finite inequalities are explicit. If $\lambda = (a_1 + ib_1)/c$ with integers $a_1, b_1, c > 0$, $a_1^2 + b_1^2 = c^2$, and λ not a root of unity, then (19.1)–(19.3) hold with the factor $2^{2(d-1)}$ deleted and B_P replaced by c .

Proof. List the conjugates with $\lambda_1 = \lambda$. They are distinct by irreducibility in characteristic zero. None is a root of unity: otherwise irreducibility would make P a cyclotomic polynomial up to its primitive sign. The resultant

$$\text{Res}(P, z^n - 1) = a^n \prod_{j=1}^d (\lambda_j^n - 1)$$

is therefore a nonzero integer. Each conjugate has modulus at most $R_P = 1 + H_P/a$. For completeness, if $|z| > R_P$, then

$$\sum_{j=0}^{d-1} |a_j| |z|^j < \frac{H_P |z|^d}{|z| - 1} < a |z|^d,$$

so $P(z) \neq 0$. Using $|z^n - 1| \leq 2 \max(1, |z|)^n$ for each other conjugate, the resultant bound gives

$$(19.4) \quad |\lambda^n - 1| \geq 2^{-(d-1)} \left(a \prod_{j=2}^d \max(1, |\lambda_j|) \right)^{-n} \geq 2^{-(d-1)} B_P^{-n}.$$

For $1 \leq n \leq 2mk$, take $\sigma = 2^{-(d-1)} B_P^{-2mk}$ in Lemma 19.1. The block defect is at least

$$d_k = \frac{2^{-2(d-1)} B_P^{-4mk}}{16}.$$

Repeated conditional contraction and Lemma 12.3 imply

$$\left\lfloor \frac{N}{2k} \right\rfloor \leq \frac{\log A_r}{-\log(1 - d_k)} \leq 16 \cdot 2^{2(d-1)} B_P^{4mk} \log A_r.$$

This proves (19.1). The greedy occupation argument in the proof of Theorem 7.2, with $b = 0$ and block length $2k$, proves (19.2). Taking logarithms gives (19.3): fix $C > 1/(4m \log B_P)$. If $k \geq C \log N$ the claim is immediate; otherwise $\log k \leq \log C + \log \log N$, which can be absorbed in the logarithm of (19.1). The implicit constant may depend on P, ρ, ε , not on N .

In the rational-circle case, $(a_1 + ib_1)^n - c^n$ is a nonzero Gaussian integer, of modulus at least one. Thus $|\lambda^n - 1| \geq c^{-n}$ directly. Repeat the proof with $\sigma = c^{-2mk}$. Here $c > 1$ automatically. A nonminimal supplied denominator is still valid but weakens the bound. $\square$

The threshold is sharp; these width rates are not asserted sharp. The factor m records the increased localization degree as the error approaches the boundary. At smaller errors, the retained coordinate-calibration estimates can have better constants. One should use the larger of applicable lower bounds, rather than claiming numerical improvement at every parameter value.

Proposition 19.3 (An explicit nonrational algebraic example). *Let $t = (1 - \sqrt{13})/2$ and*

$$\lambda = \frac{t + i\sqrt{4 - t^2}}{2}.$$

Then $|\lambda| = 1$, λ is not a root of unity, and one can take $d = 4$ and $B_P = 8$ in Theorem 19.2.

Proof. The number t lies strictly between -2 and 2 and solves $t^2 - t - 3 = 0$. Substitution of $t = z + z^{-1}$ gives

$$P(z) = z^4 - z^3 - z^2 - z + 1.$$

Its reduction modulo two is $z^4 + z^3 + z^2 + z + 1$. It has no linear factor over $\mathbb{F}_2$ and is not divisible by the only monic irreducible quadratic $z^2 + z + 1$; hence it is irreducible. Gauss's lemma proves irreducibility over $\mathbb{Q}$. The conjugate value $(1 + \sqrt{13})/2 > 2$ gives a positive real conjugate of λ greater than one. A root of unity cannot have such a conjugate. Finally $a = H_P = 1$ and $d = 4$, so $B_P = 2^3 = 8$. $\square$

For a reproducible rational calibration, take $\rho = 1/10$ and $\varepsilon = 1/25$. Then $\delta = 2/25$, $r = 5$, $m = 6$, $\Delta_r = 1/150$, and $A_r = 23364$. Thus the rational-circle alphabet above obeys

$$N < 2k(1 + 16 \cdot 5^{24k} \log 23364).$$

This is a concrete bound inside the interval between the former small-error obstruction and the sharp threshold $1/20$. The exact coefficient and resultant checks accompanying the article verify the displayed arithmetic, not the universal theorem by finite experimentation.

20. FINITE ENCODINGS AND BOUNDARY QUOTIENTS

The general continuous theorem is not an algorithm from arbitrary real data. We now give an explicit finite input class for which its conclusions are effective. A rational orthogonal input consists of a finite alphabet in $O(D, \mathbb{Q})$ containing I , finite seed and query sets, and rational polynomials $F_{s,j,l}$ in the matrix entries specifying categorical probabilities. The promise $F_{s,j}(H) \subset \Delta_{M_j}$ can itself be checked once the compact closure is computed. Real algebraic output numbers are encoded by polynomial equations and isolating data.

We use two established algorithmic inputs: computation of a finite defining description of the Zariski closure of a finitely generated rational matrix group [21, 23], and real quantifier elimination with effective semialgebraic connected components [19]. The former is a terminating closure algorithm, not merely an enumeration of valid polynomial identities with an unknown stopping point. Compact orthogonal groups are real algebraic, so the real points of the computed closure in $O(D)$ give the Euclidean closure H [20]. The complex algebraic components must not be substituted for the real connected components used here.

Theorem 20.1 (Effective rational classification). *For a rational orthogonal input, the following are computable:*

- (1) *a semialgebraic description of every connected component of H , the finite group C , and the component of each command;*

- (2) the critical total-variation error ε_c as an exact real algebraic number, and an attaining stationary component-permutation machine with algebraic decoders;
- (3) for each algebraic $0 \leq \varepsilon < \varepsilon_c$ and integer $k \geq 1$, an integer occupation constant L_k valid for every error- ε clocked realization;
- (4) for every fixed N and algebraic $\varepsilon \geq 0$, the least width $W_{N,\varepsilon}$ and an algebraic realization attaining it.

These are terminating computability statements. No polynomial running-time bound, practical implementation of general closure/quantifier elimination, or finite-random-bit cost bound is asserted.

Proof. Compute H using the cited closure algorithm, then its real connected components and algebraic sample points. The component containing I is K . Products of sample points and semialgebraic membership tests determine the component multiplication table and the command images. In particular K is now an effectively given compact semialgebraic group.

For a component c and a query, the condition that q be a radius- t center is the real first-order formula

$$(20.1) \quad q \in \Delta_{M_j}, \quad \forall h \in c : \sum_{l=1}^{M_j} |F_{s,j,l}(h) - q_l| \leq 2t.$$

Absolute values have a finite semialgebraic description. Quantifier elimination gives the set of admissible (q, t) . Its least t exists by compactness and is real algebraic; algebraic sampling supplies an attaining q . Maximize over the finite interface and components. This proves (1)–(2), including equality at the critical error.

We detail effectivity of the occupation proof, since computing the threshold alone would not supply it. Choose a component with radius greater than the given ε , and enumerate positive words until one lies in that component. Its rational product gives the prefix used in the proof of Theorem 7.2. The resulting map $f(h) = F_{s,j}(hu_v)$ is polynomial, so it can be used directly in place of G , with no approximation error.

For any finite tensor degree, all required polynomial functions are coefficients of the explicitly given tensor representation. Its fixed subspace is

$$\{z : \forall h \in K, (R(h) - I)z = 0\}.$$

This linear subspace is effectively semialgebraic. Real algebraic sampling, successively outside the span of the vectors already chosen, gives an algebraic basis in finitely many steps. Its Gram matrix gives the orthogonal projection P exactly. Thus every polynomial Haar moment is computable algebraically from $P\Phi(e)$; numerical integration of Haar measure is unnecessary.

Enumerate finite lists of squares of rational polynomials on K , discarding zero Haar integrals and normalizing the others. For a list p_i , compute $b_i = \int p_i f d\mu$ and its constrained simplex radius by quantifier elimination. Rational polynomials are uniformly dense among real polynomial restrictions on a compact matrix group; the localization proof therefore guarantees that some enumerated list has radius strictly greater than ε . Strict algebraic comparison detects that list. Fix a positive rational Δ below the radius difference and a positive rational upper bound L for all coefficient constants in Proposition 8.2. These give the certified residual $Q \geq \Delta/L$.

Next enumerate finite rational probability laws on executable words of a common length whose products lie in K , together with rational $d \in (0, 1)$. The assertion (7.4), quantified over the finitely many unit vectors in the algebraic moving space, is semialgebraic: each maximum of finitely many inner products can be expressed by a finite disjunction. It is therefore decidable. Lemma 7.4 and sufficiently small rational perturbations of its atom weights ensure that this search terminates. Padding by the identity letter makes every finite choice executable at one common length.

The resulting B, d, L, Δ, Q_0 determine (8.6). To avoid exact comparisons involving logarithms, take a rational $A' \geq \max(1, LQ_0/\Delta)$ and use

$$\frac{\log A}{-\log(1-d)} \leq \frac{A'}{d}.$$

An integer above $b + B - 1 + BA'/d$ is a computable occupation bound. This proves (3).

For (4), fix a candidate width k and use k labels at every cut, padding smaller registers with unused labels. Stochastic rows, initialization, and categorical decoder probabilities are finitely many variables in compact simplexes. For each of the finitely many length- N words, the output vector is polynomial in these variables, and the target vector is rational. The total-variation inequalities are semialgebraic. Quantifier elimination therefore decides feasibility and supplies an algebraic point when feasible. A deterministic register storing the seed and entire prefix is a finite exact realization, so increasing k must terminate. This proves the least-width claim for each finite horizon, without claiming a finite-horizon cutoff for all stationary or nonuniform realization questions. $\square$

The distinction between an effective theorem and an implemented solver matters. The accompanying finite checks verify displayed exact examples and proof constants; they do not execute the general group-closure and quantifier-elimination construction. The finite-bit compiler for rational finite groups remains in the complete predecessor supplement with its separate hypotheses and costs. It is not used to reprice arbitrary algebraic decoders as cost-free finite-bit procedures.

20.1. The least component-quotient realization. The upper bound $|\mathcal{S}||C|$ can often be reduced. Here is an exact finite characterization of that reduction, with its scope stated explicitly. Put $X = \mathcal{S} \times C$. A *component-quotient realization* has a deterministic state of the form $\pi(s, c)$, where $\pi : X \rightarrow Q$ is a surjection; its command update is induced by $(s, c) \mapsto (s, [u_a]c)$. It is stationary and has no additional state recording a representative word within a component.

Call a partition $\mathcal{B}$ of X equivariant if

$$(s, c) \sim (s', c') \implies (s, dc) \sim (s', dc') \quad (d \in C).$$

For a block $B \in \mathcal{B}$ and query j , set

$$A_{B,j} = \bigcup_{(s,c) \in B} F_{s,j}(c).$$

Theorem 20.2 (Finite quotient minimum). *The least number of command labels in an error- ε component-quotient realization is*

$$(20.2) \quad \min \left\{ |\mathcal{B}| : \mathcal{B} \text{ equivariant and } \text{rad}_{Y_j}(A_{B,j}) \leq \varepsilon \text{ for every } B, j \right\},$$

with value $+\infty$ if there is no admissible partition. For rational orthogonal categorical inputs this minimum and an attaining quotient are computable. If $\varepsilon_c = 0$ and $\varepsilon = 0$, the coarsest admissible partition is the future-output equivalence

$$(s, c) \sim (s', c') \iff F_{s,j}(dc) = F_{s',j}(dc') \quad \text{for every } d \in C, j,$$

where componentwise constant maps are identified with their values.

Proof. An equivariant partition induces well-defined command permutations on its blocks. Choosing a minimizing center for each $A_{B,j}$ gives exactly the stated error. Conversely, the fibers of π in any component-quotient realization form an equivariant partition. Its decoder must approximate every executable product in each corresponding component. Such products are dense in that component, so continuity extends the error bound to $A_{B,j}$. Its decoder is therefore a permitted center. This proves (20.2). The command generators suffice to check equivariance, since their images generate the finite group C .

There are finitely many partitions of X . For polynomial categorical data their radius tests are of the form (20.1), with a finite disjunction over the relevant seed/component pairs, so the minimum is computable. In the exact componentwise constant case, equality of all future outputs is an equivariant equivalence relation. Every admissible partition refines it, and its own blocks have singleton output images for each query. It is therefore the unique coarsest admissible partition. $\square$

The minimization in (20.2) is a deterministic quotient minimum, distinct from the unrestricted stationary minimum in Theorem 5.4 and from the unrestricted nonuniform minimum at the boundary. Randomized encodings can identify numerical behaviors without identifying component pairs. What is classified here is exactly the finite deterministic quotient construction that attains the radius threshold. Theorem 20.1(4) separately computes the unrestricted minimum at each fixed horizon.

20.2. Stationary minimization by permutation closures.

Theorem 20.3 (Effective unrestricted stationary minimum). *For rational orthogonal commands generating H , rational polynomial categorical outputs, and algebraic $\varepsilon \geq 0$, one can compute M_ε and an attaining algebraic machine when it is finite. An identity letter is not required. At each proposed width k , at most $(k!)^{|A|}$ augmented orthogonal group closures and finite real quantifier-elimination problems suffice.*

Proof. First compute H , its real connected components, and their constrained radii as in Theorem 20.1(1)–(2); those steps do not use an identity letter. The group $H \subset O(D)$ has finitely many components. Theorem 5.5 shows that $M_\varepsilon = \infty$ below their maximum radius. Otherwise $M_\varepsilon \leq |\mathcal{S}||H/H^\circ|$.

For k between one and this upper bound enumerate all command-permutation tuples σ . The matrices

$$\text{diag}(u_a^{-1}, \Pi_a) \in O(D + k, \mathbb{Q})$$

generate a compact closure which is exactly the joint group $\mathcal{L}_\sigma$ of Theorem 5.4. Compute it by the same algebraic closure procedure. In (5.2), the unknown seed rows and decoder rows are constrained to simplexes. The mean output is bilinear in these rows, the group variable is restricted by computed defining equations whose real zero set inside the augmented orthogonal group is exactly $\mathcal{L}_\sigma$, not merely by some equations vanishing on it, and $h^{-1} = h^T$ in the physical block. Consequently all target coordinates are polynomial in the universally quantified matrix entries. Total variation is semialgebraic. Real quantifier elimination decides the resulting existential–universal formula and supplies algebraic rows whenever it is feasible. The first feasible k is the minimum by purification. The finite upper bound proves termination of the entire search. $\square$

This is a structural reduction of stationary minimization, not merely an enumeration of arbitrary stochastic transition parameters. It does not equate stationary and finite-horizon minima. Theorem 4.2 identifies their limit under its additional return hypothesis. The two-cycle/three-cycle example in Proposition 5.6 already prevents replacing the permutation-feasibility test by deterministic component partitions.

20.3. Algorithmic cost and arithmetic representation. There are three distinct layers of effectivity. First, the group-closure procedure produces explicit defining equations; real-component decomposition and algebraic sampling then identify the component table. Second, radius computation and the stationary permutation tests are finite quantifier-elimination tasks on those equations. At fixed k , the stationary seed and output variables number $k(|\mathcal{S}| + \sum_j M_j)$ before normalization equalities are eliminated; the group variables can be taken among the $(D + k)^2$ augmented matrix entries. Thus the displayed finite search is independent of a horizon parameter. Its factorial number of discrete cases and the cost of the real-algebraic subproblems should not be

mistaken for an efficient algorithm. No bound for the full closure-to-machine process in terms of only the original bit length is extracted here.

Third, occupation certificates in Theorem 20.1(3) add searches over localizer degrees and executable word laws. Their termination follows from strict analytic margins; the present proof supplies no a priori useful bound on the successful degree or word length. Fixed-horizon minimization, by contrast, has finitely many word constraints at each N but their number is exponential in N before repetitions are identified. These stages should be priced separately rather than hidden behind the word “effective.”

All algebraic Haar moments in the localizer search are represented in a real algebraic number field with isolating data, using the fixed-subspace projector and its algebraic Gram inverse. Signs and strict comparisons are exact real-algebraic comparisons; a quadrature tolerance is not a certificate. Every rational command word has a rational matrix product and hence a rational target vector for rational polynomial readouts. The orthogonal closure fact used above is precisely that the Euclidean closure of a subgroup of $O(D)$ is the real zero set, inside $O(D)$, of its vanishing polynomial ideal, as used in [20]; replacing real connected components by complex algebraic ones would give the wrong quotient.

21. FINITE DESCRIPTIONS AND DECISION COMPLEXITY

The qualitative reduction is insensitive to the encoding of real numbers. Complexity statements are not. We first specify a finite input model and then give an explicit finite-precision counterpart to label width.

Theorem 21.1 (Finite-group stationary minimization is $\exists\mathbb{R}$ -complete). *The input consists of a finite group G given by its full multiplication table, finite seed and query sets, rational categorical vectors $F_{s,j}(g)$ for every $g \in G$, a nonnegative rational tolerance ε , and an integer $k \geq 1$ in binary. Deciding whether $M_\varepsilon \leq k$ is complete, under polynomial-time many-one reductions, for the existential theory of the reals. Hardness already holds for the trivial group, one query, and zero error. In particular the rational orthogonal polynomial-interface stationary problem is $\exists\mathbb{R}$ -hard even with constant readouts and trivial dynamics.*

Proof. Let $n = |G|$. If $k \geq n|\mathcal{S}|$, a deterministic seed-group register realizes every target exactly, so the answer is yes. Otherwise k is bounded by a polynomial in the explicit input length, even though its encoding is binary.

For membership, Theorem 6.1 permits permutation matrices P_g forming a representation of G . Introduce real variables for their entries, constrain each entry x by $x(x-1) = 0$, impose every row and column sum equal to one, and impose

$$P_1 = I, \quad P_g P_h = P_{gh} \quad (g, h \in G).$$

Introduce simplex variables α_s and categorical decoder rows $d_j(i)$. For every s, j, g and output coordinate c , use a nonnegative auxiliary variable $z_{s,j,g,c}$ with

$$z_{s,j,g,c} \geq \pm((\alpha_s P_{g^{-1}} d_j)_c - F_{s,j}(g)_c), \quad \sum_c z_{s,j,g,c} \leq 2\varepsilon.$$

This is an existential real formula with rational coefficients, degree at most three, and polynomial size. Padding a representation by fixed unused labels handles an optimum strictly smaller than k . Theorem 6.1 proves equivalence with the original unrestricted stationary problem. There is no group-closure computation in this explicit finite-group input model.

For hardness, let A be a rational nonnegative matrix and let k be a proposed nonnegative rank. Delete zero rows; the all-zero case is decided directly. Normalize each remaining row by its positive row sum to obtain a rational row-stochastic matrix F . This transformation is polynomial-time and preserves nonnegative rank. Interpret the rows as seeds and the columns as the categories of

one query on the trivial group. A zero-error realization gives a stochastic factorization $F = \alpha d$ of inner dimension at most k , so the nonnegative rank is at most k .

Conversely, from any nonnegative factorization $F = UV$ delete a zero row of V together with its column of U , and let $s_i = \sum_c V_{ic} > 0$. Put $d_{ic} = V_{ic}/s_i$ and $\alpha_{bi} = U_{bi}s_i$. Then d is row-stochastic and $\sum_i \alpha_{bi} = \sum_c F_{bc} = 1$, so α is row-stochastic. Identity command rows yield an exact stationary realization. Thus M_0 equals the real nonnegative rank of F . Shitov's universality theorem, specifically its complexity consequence [24, Theorem 2], makes this decision problem $\exists\mathbb{R}$ -complete. The last assertion follows by using the one-dimensional identity matrix as the only physical command and constant rational polynomial readouts. $\square$

The upper bound is for an explicitly tabulated finite group. The hardness transfers to rational matrix input, but a polynomial-size existential description of an arbitrary compact matrix closure has not been supplied. Theorem 20.3 remains the separate termination result for that broader input class. The complexity obstruction above already occurs in randomized initialization and decoding; it is not caused solely by closure algorithms.

21.1. A finite-table resource counterpart. For a categorical interface, a dyadic table of precision b has each probability in $2^{-b}\mathbb{Z}$. Besides peak label width k , charge its binary table length, all retained working registers, and the number of fair random bits used. A horizon- N clocked implementation can store initialization, N command tables, and terminal decoders. Its probability-table length is

$$O\left(b\left(|S|k + N|\mathcal{A}|k^2 + k\sum_j M_j\right)\right).$$

A uniform implementation is one finite algorithm that produces or accesses these tables from the encoded input, horizon, and accuracy. This requirement is additional to the original nonuniform model. An arbitrary real table, without an effective encoding, is not such an algorithm.

Lemma 21.2 (Dyadic rounding of a probability row). *A probability row p with m entries has a dyadic row $\hat{p}$ of precision b such that*

$$\text{TV}(p, \hat{p}) \leq (m-1)2^{-b}.$$

One exact draw from $\hat{p}$ uses at most b independent fair bits.

Proof. Round the first $m-1$ entries down to multiples of 2^{-b} and put the remaining mass in the last entry. This is a probability row. Its total variation error equals the mass transferred to the last entry, which is at most $(m-1)2^{-b}$. For sampling, draw a uniform integer in $\{0, \dots, 2^b - 1\}$ using b bits and compare it with the integer cumulative row sums. Degenerate rows can use fewer bits. $\square$

Theorem 21.3 (Precision and random-bit budgets). *A categorical horizon- N realization of peak width k and error at most ε admits a dyadic realization on the same registers whose error is at most*

$$(21.1) \quad \varepsilon + ((N+1)(k-1) + M_* - 1)2^{-b}, \quad M_* = \max_j M_j.$$

Sampling initialization, all transitions, and one terminal answer requires at most $(N+2)b$ fair bits.

For a stationary compact-group realization of width k , the corresponding error bound can instead be made independent of N :

$$(21.2) \quad \varepsilon + (k + M_* - 2)2^{-b}.$$

The stationary dyadic realization has at most k labels, permutation command tables, table length

$$O\left(|\mathcal{A}|k \log(k+1) + b\left(|\mathcal{S}|k + k \sum_j M_j\right)\right),$$

and uses at most $2b$ fair bits in total for one experiment of any length. Given exact algebraic encodings of the selected initialization and decoder rows, the rounding and sampling are effective finite procedures.

Proof. Apply Lemma 21.2 to every row. Couple the initial draws maximally and then, as long as the two labels agree, maximally couple their next transition draws. The probability of any disagreement is at most the sum of the initialization and N row errors, hence at most $(N+1)(k-1)2^{-b}$. Conditioned on matching terminal labels, the terminal categorical rows have total variation error at most $(M_* - 1)2^{-b}$. The coupling inequality gives (21.1), uniformly over words and queries. The sample count is the number of draws times b .

For the stationary claim, first use Theorem 5.3. Its transition matrices are permutations and are already dyadic at every precision. Round only initialization and terminal categorical rows. The initialization discrepancy cannot increase under a permutation, so the same coupling gives (21.2) with no factor N . There are no random transition draws, leaving only initialization and one terminal answer. A permutation is stored as k integer images, giving the displayed description bound. Exact comparison of algebraic numbers with dyadic rationals computes the necessary floors and handles equality, after which integer cumulative sampling is immediate. $\square$

For a prescribed extra error $\eta > 0$, choosing b with the added term in (21.1) or (21.2) at most η gives an explicit precision budget. The persistent label uses $\lceil \log_2 k \rceil$ bits; an implementation of the sampler also retains a b -bit random integer and bounded indexing registers while sampling. These temporary registers are not disguised as free persistent randomness. The description and random-bit bounds do not assert a polynomial time bound for constructing an algebraic optimum. They also do not turn an exact irrational realization into an exact finite-bit one: the extra tolerance is essential in that case.

22. POSITIVE FINITE-PRECISION MATRIX DECODERS

The label-width bounds leave real table descriptions uncharged. There is nevertheless a finite-description counterpart for each supplied encoded realization. Entrywise rounding of a density matrix is unsafe: it need not preserve positivity. A Gram-factor construction avoids this problem.

Lemma 22.1 (Rational positive rounding). *Let $D \in \mathcal{D}_d$ and let $L = D^{1/2}$. Approximate the real and imaginary parts of every entry of L by dyadic multiples of 2^{-b} , each within 2^{-b} , to obtain Q . Set $\delta = \sqrt{2}d2^{-b}$. If $\delta \leq 1/4$, then*

$$(22.1) \quad \hat{D} = \frac{QQ^*}{\text{tr}(QQ^*)} \in \mathcal{D}_d \cap M_d(\mathbb{Q} + i\mathbb{Q}), \quad \|\hat{D} - D\|_F \leq 8\sqrt{2}d2^{-b}.$$

The construction is effective when D is given by exact algebraic entries. Its rational output has $O(d^2(b + \log(d+1)))$ bits for fixed encoding conventions.

Proof. Write $Q = L + E$. Then $\|E\|_F \leq \delta$ and $\|L\|_F^2 = \text{tr } D = 1$. Hence $Q \neq 0$, $\text{tr}(QQ^*) \geq (1-\delta)^2$, and

$$\|QQ^* - D\|_F \leq (2+\delta)\delta, \quad |\text{tr}(QQ^*) - 1| \leq (2+\delta)\delta.$$

The first estimate follows from the two cross terms and EE^* , and the second from $\|Q\|_F^2 - \|L\|_F^2$. Since $\|D\|_F \leq 1$, division by $\text{tr}(QQ^*)$ gives an error at most $2(2+\delta)\delta/(1-\delta)^2 \leq 8\delta$. Positivity and trace one are exact, not numerical tolerances. After multiplying Q by 2^b , the normalization is a ratio of integer Gram entries to a positive integer sum of squares. Their bit sizes are

$O(b + \log(d + 1))$. The positive square root of an algebraic positive matrix has algebraic entries and can be specified and approximated by real-algebraic arithmetic. This proves the effectivity assertion without an efficiency claim for obtaining D . $\square$

Theorem 22.2 (A supplied realization with finite tables and fair bits). *Suppose an error- ε horizon- N realization for a matrix interface uses at most k labels, and its initialization, command rows, and decoders are supplied with exact algebraic encodings. Let $\sqrt{2}d2^{-b} \leq 1/4$. There is a rational finite-table implementation, with the same persistent label widths, whose Frobenius error is at most*

$$(22.2) \quad \varepsilon + \sqrt{2}((N + 1)(k - 1) + 8d)2^{-b}.$$

Each hidden draw uses b independent fair bits, so at most $(N + 1)b$ such bits suffice. For a supplied stationary permutation realization the additional error is at most

$$(22.3) \quad \sqrt{2}(k - 1 + 8d)2^{-b},$$

and only b fair bits are needed for initialization, independently of the horizon. The numerical matrix decoder itself is not a quantum preparation or a sampled terminal answer register.

Proof. Apply Lemma 21.2 to each initialization and command row after padding to k labels. Each row changes by at most $(k - 1)2^{-b}$ in total variation. Induction, or a maximal coupling at every actual draw, bounds the terminal distribution discrepancy by $(N + 1)(k - 1)2^{-b}$; this upper bound may harmlessly exceed one. Replace each decoder by the legal rational matrix in Lemma 22.1. The Frobenius diameter of $\mathcal{D}_d$ is $\sqrt{2}$, so changing the distribution incurs at most $\sqrt{2}$ times its total-variation discrepancy. The independent decoder error is at most $8\sqrt{2}d2^{-b}$. This proves (22.2). A dyadic probability row of denominator 2^b is sampled by one uniform b -bit integer and its cumulative integer table.

For a permutation realization the command rows already have exact integer entries and need not be rounded or sampled. Only the initialization distribution and decoder are changed. The same estimate therefore gives (22.3) and the horizon-independent fair-bit bound. $\square$

The row tables and their cumulative sums require $O((N|\mathcal{A}|k^2 + |\mathcal{S}|k)b)$ bits in the general case, in addition to $O(kd^2(b + \log(d + 1)))$ bits for one matrix decoder. Temporary random integers and table indices require their own $O(b + \log(k + 1))$ workspace; this workspace is not hidden in the persistent-label invariant. Multiple queries multiply the decoder storage by their number. To implement a sampled measurement one must additionally specify and charge its sampler. The theorem compiles a supplied encoded realization; it neither computes a width minimizer nor assigns finite encodings to arbitrary unspecified real tables. Applying it after the exact inner-polytope construction uses an explicit extra error budget, not an exact finite-precision claim at an extreme zero-error endpoint.

Here an exact algebraic entry is specified by defining polynomials and real isolating intervals (for both parts of a complex entry), with compatible field operations. The positive square root is the unique positive semidefinite solution of $L^2 = D$, and its isolating data can be obtained by real-algebraic algorithms. These representation and output bit-length statements are not running-time bounds for finding the original realization or its square root. The constructed rational qubit compiler is a separate, more explicit algorithm.

23. RELATION TO NUMERICAL AUTOMATA AND POSITIVE REALIZATION

The input/output object and the order of quantifiers are decisive in comparing realization theorems. Classical probabilistic automata assign numerical acceptance probabilities to words, while stochastic-language results also study threshold recognition [5, 6]. A numerical equivalence

of two fixed machines is not the assertion that for every horizon there exists a new bounded-width approximate machine. Conversely, language equivalence need not preserve the numerical probabilities.

For weighted automata over a field, the finite-Hankel-rank characterization gives finite linear representations and identifies their minimum dimension [7]. It does not enforce nonnegative rows, stochastic normalization, or simplex-valued decoders. Our common-row contraction concerns these constraints simultaneously. The invariant observable quotient in Theorem 11.4 uses the same reachable/invisible linear reduction; its additional conclusion is a finite-group criterion for exact positive realization with the full horizon-dependent quantifier.

The classical positive-realization problem asks for a positive system representing specified input/output data, and distinguishes positive dimension from unrestricted linear dimension [3]. Invariant cones and polyhedral sections are part of that theory, not new notions introduced here. Proposition 12.6 is an explicit common invariant-enclosure construction. The new matched statement pairs it with a full-subcritical lower estimate against all clocked stochastic machines. Neither one cutwise nonnegative factorization nor a finite-rank Hankel matrix supplies that lower estimate.

Brodsky and Pippenger identify bounded-error measure-once quantum languages as group languages [9, Theorem 3.3]. Their probabilistic simulation result preserves a cutpoint language with a potentially changed cutpoint [9, Theorem 3.6]; it is not uniform additive preservation of every acceptance probability. Our categorical unitary corollary fixes an entire terminal measurement law in total variation and allows horizon-dependent stochastic approximants. The distinction is already visible in the d -outcome threshold $1 - 1/d$.

The closure theorem and strict-threshold decision method of Blondel, Jeandel, Koiran, and Portier [20], and the effective Zariski-closure algorithms [21, 23], are algorithmic premises of Section 20. Computing the closure is not a new claim of this article. The additional outputs are a constrained minimax noise threshold, an attaining categorical decoder, and a terminating construction of an all-machine occupation constant. Recent Lie-algebra-based closure computations [22] give an alternative representation of the algebraic closure, but their complex connected components still require real-component analysis for the present application.

Peter–Weyl approximation, Haar averaging, and the representative-function algebra are classical compact-group and almost-periodic-function machinery [13]. Uniform approximation by a finite sum of matrix coefficients gives a finite *linear* representation of an output function. It does not give positive stochastic rows or the repeated information loss quantified by (7.3). Similarly, Steinberg’s minimal compact topological automaton and enveloping-monoid results [12, Theorems 2.2 and 2.6] concern recognition of a language. His compact topological-monoid criterion [12, Proposition 2.8] uses clopen recognition, not a continuous simplex-valued response with a prescribed error norm. The finite-quotient radius formula is specific to groups. The stationarization theorem itself now holds for irreversible topological monoids with eventual approximate returns (Theorem 4.2).

Distribution-based bisimulation metrics provide robust comparisons for specified probabilistic automata [11]. They do not by themselves impose a minimum over all horizon-specific stochastic realizations of an external group experiment. The finite future-output equivalence in Theorem 20.2 is deliberately a component-quotient statement, rather than a claim that approximate equivalence always has a unique minimal positive stochastic realization.

Comparison object	Error or equivalence	Machine quantifier
Compact stochastic groups [1]	Idempotent and permutation structure; no external error specification	One matrix group or semigroup kernel
Advice regular languages [8]	Neutral-letter recognition	One-scan advice automata; Boolean output
Weighted series [7]	Exact scalar coefficients; linear rank	One linear representation
Group-language simulation [9]	Cutpoint language; not additive probability error	One automaton
Positive realization [3]	Exact specified numerical response	One positive system
Topological recognition [12]	Clopen language acceptance	One recognizing action
Bisimulation metrics [11]	Distributional comparison of given systems	Two specified automata
Theorem 4.2	Same closed numerical error and label bound	Arbitrarily many narrow cuts; one extracted stationary machine
Theorem 7.2	Worst-word norm error; TV for a whole categorical law	A new clocked stochastic machine at every horizon
Theorem 12.1	Binary TV, every fixed $\varepsilon < \rho/2$	Same nonuniform lower model; exact upper construction

The contribution combines stationary purification and its exact finite-quotient criterion with a sharp constrained clocked radius, phasewise arbitrary-width occupation, and, for the specified Diophantine family, a matching full-subcritical growth law. The elementary radius center, the arithmetic pigeonhole argument, the closure algorithm, and the harmonic-analysis ingredients are not asserted individually new. The accompanying theorem-level audit records the closest comparisons and their changed hypotheses; it is not a claim of exhaustive priority clearance.

23.1. The classical matrix-semigroup boundary. Flor’s description of nonnegative matrix groups [1, Theorems 2 and 3] identifies their idempotent geometry and the finiteness of bounded groups; his account explicitly credits Brown and Schwarz for earlier compact and stochastic cases. We do not claim this matrix structure as new. Theorem 5.3 uses a minimal-rank idempotent in a joint semigroup, not just in the matrix projection: its physical coordinate must be the group identity. Sandwiching each command then preserves all uniform error inequalities by closure, even when the original stochastic rows violate every nontrivial physical group relation. The resulting initialization may be random, a point essential to exact state minimization.

The stationary theorem has no return hypothesis. The clocked theorem has a different quantifier and needs synchronization; the empty-return example and the phase-dependent example show why the distinction cannot be removed by terminology. The exact-return-length reduction is an additive-semigroup argument, whereas the profinite zero-error converse uses finite Hankel rank and continuous finite-dimensional representations. The new stationarization theorem removes the reversibility requirement for the equality of limiting width and stationary width, but does not supply a finite-quotient radius formula for an arbitrary irreversible monoid.

The finite-quotient existence theorem does not say that $[H : U]$ is the optimal number of stochastic labels. The actual minimum is the permutation-feasibility invariant in Theorem 5.4; the finite cyclic example separates them. For infinitely many components, the infimum of finite-quotient radii can fail to be attained, even at zero. Finally, the exactly matched quantitative law retains its specified Diophantine alphabet. Theorem 13.1 gives the retained block estimate, strengthened to a matching power law by Theorem 15.5, for spherical representations with a

spectral gap; Corollary 13.3 gives a concrete rational noncommuting instance. The exact unit-amplitude endpoint is not covered by these upper constructions. These distinctions delimit the proved statements without weakening any of the retained conclusions.

23.2. What the reductions add. The algebraic ingredients of Lemmas 5.1 and 5.2 are the classical minimum-rank kernel mechanism, the group corner of a compact matrix semigroup, and the recurrent simplex of a stochastic idempotent. Flor’s Theorem 2 describes nonnegative idempotents; his Theorem 3 and preceding Lemma 3 give the permutation structure of bounded nonnegative groups [1]. The paper explicitly attributes earlier cases to Brown and Schwarz. The present proof adds the physical coordinate to the closure, chooses the idempotent over its identity, and tracks the closed error balls through sandwiching. Theorem 6.1 further averages the finite kernel above the physical identity and descends to an actual continuous H -action. These are specified realization reductions using classical structure, not a new classification of matrix-semigroup kernels.

For deterministic Boolean recognition, neutral-letter removal of advice is an established phenomenon. Fijalkow–Paperman [8, Theorem 10] prove the Crane Beach property for advice regular languages by padding distinguishing words and suffixes, and their account relates one-scan programs to earlier Ramsey arguments. Their Myhill–Nerode argument in fact preserves the deterministic distinguishability bound K ; preservation of a state bound alone is not claimed as new here. A stochastic k -label register has a continuum of state distributions, so this discrete class count does not apply. Theorem 4.2 instead colors approximate tuples of stochastic composite kernels, extracts one common tuple, and absorbs a final filler matrix into the decoder. It preserves the same numerical tolerance at equality and the same k , and bounds all narrow-cut occurrences with unrestricted intermediate widths. Eventual approximate returns further replace a literal neutral letter. The finite Ramsey theorem itself is classical [25].

The spherical lower estimate uses a classical spectral gap, cap measure, and smoothing; its upper estimate uses a polytope enclosing a contracted sphere. The contribution is their realization-theoretic combination with the all-profile conditional-centroid bound. The rational $SO(3)$ example invokes [16] as a deep external input. Similarly, the lower-complexity input in Theorem 21.1 is Shitov’s nonnegative-factorization universality theorem [24]; the stochastic normalization and finite-group existential upper reduction are given explicitly. Neither external theorem is claimed as part of our proof technology’s novelty.

23.3. Companion results. The companion supplement gives additional results on compatible reachable sections, one-surplus geometry, stable sections, finite-horizon certificates, word distortion, two-state duality, finite-bit compilation, and global residual hierarchies. The scalar two-extreme argument and earlier quantitative constants are also retained there. The present main proofs are self-contained apart from the classical results explicitly cited; none of these companion derivations is used to conceal a missing step in a principal theorem.

23.4. Orbit geometry, positive matrix outputs, and recognition. The homogeneous-space covering estimates and the smooth orbit geometry used in Section 14 are classical; see, for example, Szarek [15]. The spectral-gap input is Bourgain–Gamburd [16]. Dense algebraic free generators in Section 18 come from Breuillard–Gelder [17, Theorem 1.1]; the later repair and supplementary proof details are recorded in [18]. None of these results is claimed as a new group-theoretic theorem here. The realization argument requires, in addition, a contraction uniform over every conditional-centroid direction, one common positive row for each command and label, and a terminal decoder in the original legal matrix set.

Quantum finite automata already exhibit strong state-complexity separations in language recognition; Ambainis–Freivalds [10] is a relevant early comparison. Theorem A.2 instead asks a classical stochastic machine to reproduce one entire legal density-matrix vector at every input

word, permits horizon-specific redesign and a free clock, and compares exact and fixed-error width for the same interface. Its exact lower bound uses extreme points of the decoder set; it does not apply unchanged to a single acceptance probability or to an enlarged unconstrained output simplex. A finite tomography map preserves width only when its legal image and reconstructed norm are retained. This comparison identifies different quantifiers and resources, not an exhaustive priority claim about all quantum or probabilistic automata formalisms.

The spectral-entropy argument removes the logarithm from the nonabelian lower bound. Equality of lower and upper powers requires equality of least-orbit and target-orbit dimensions; no universally matching law for every representation is asserted. General online outputs, adaptive feedback, and unrestricted compact semigroup radius formulas require additional arguments. The same-width stationarization result for topological monoids remains available under its actual approximate-return hypothesis; it does not provide that missing radius formula.

24. THE LEAST ORBIT DIMENSION AND UNIFORM SMALL BALLS

The geometric hypothesis needed for a stochastic converse concerns every conditional-centroid direction, not only the prescribed target orbit. For a fixed finite-dimensional orthogonal representation of a compact connected Lie group, this hypothesis follows from a rank invariant. Throughout this section Haar measure is normalized, and the Lie algebra has a fixed invariant inner product. No constants uniform in the dimension or in the representation are asserted.

Theorem 24.1 (Uniform caps from the least orbit dimension). *Let G be a compact connected Lie group and $R : G \rightarrow O(E)$ a continuous finite-dimensional representation with $E^G = \{0\}$ and $E \neq \{0\}$. Write $D = \dim G$ and choose a basis $X_1, \dots, X_D$ of its Lie algebra, using the same symbols for their infinitesimal actions on E . Set*

$$(24.1) \quad p_* = \min_{\|u\|=1} \text{rank}[X_1 u \cdots X_D u] = \min_{\|u\|=1} \dim R(G)u.$$

Then $p_* \geq 1$, and there is $C < \infty$ such that

$$(24.2) \quad \sup_{\|u\|=1, v \in E} \mu\{g : \|R(g)u - v\| < h\} \leq Ch^{p_*} \quad (0 < h \leq 1).$$

The exponent is optimal as a uniform exponent: it cannot be replaced by any larger exponent with a constant independent of u, v, h .

Proof. The tangent space to the orbit at u is the span of the $X_i u$. An orbit of dimension zero is a singleton, since G is connected; its point would be fixed. This proves $p_* \geq 1$. Put $r = p_*$ and fix orthonormal coordinates on E .

For each unit u , some r -row, r -column minor of $[X_1 u \cdots X_D u]$ is nonzero. The maximum of the absolute values of these finitely many minors is a continuous positive function on the unit sphere. Its minimum is therefore positive. The entries are uniformly bounded. Consequently there is $\sigma > 0$ such that for every unit u one can choose r input indices and r output coordinates whose derivative matrix A_u has least singular value at least σ .

For each of the finitely many permutations of the chosen Lie algebra basis consider the product chart

$$\phi(x, y) = \exp(x_1 X_{i_1}) \cdots \exp(x_r X_{i_r}) \exp(y_1 X_{i_{r+1}}) \cdots \exp(y_{D-r} X_{i_D}).$$

All these charts are diffeomorphisms on one sufficiently small cube about zero. Their images contain a common identity neighborhood W , and their Haar densities are bounded above by a common constant. By shrinking the cube once more, uniform continuity of the derivatives on the unit sphere and the finite list of charts ensures the following. For every unit u , choose the minor above and let $\pi_u : E \rightarrow \mathbb{R}^r$ be its coordinate projection. For all (x, y) in the cube,

$$(24.3) \quad \|\partial_x(\pi_u R(\phi(x, y))u) - A_u\| \leq \sigma/2.$$

For fixed y , the map $f_{u,y}(x) = \pi_u R(\phi(x, y))u$ is injective. Indeed integration of the derivative on the segment between x and x' gives

$$\|f_{u,y}(x) - f_{u,y}(x')\| \geq \|A_u(x - x')\| - \frac{1}{2}\sigma\|x - x'\| \geq \frac{1}{2}\sigma\|x - x'\|.$$

Every singular value of its derivative is at least $\sigma/2$. The change-of-variables formula therefore bounds the r -dimensional volume of the inverse image of a ball of radius h by $\omega_r(2/\sigma)^r h^r$. A ball in E projects into a ball of the same radius in these coordinates. Integrating in y and using the bounded Haar density proves the required estimate on this chart, uniformly in u and in the center v .

Choose finitely many right translates Wg_j covering G . On such a translate apply the preceding argument to the unit vector $R(g_j)u$. For its selected minor, use the corresponding permuted chart; that chart contains W . Haar invariance and a finite sum give (24.2) for sufficiently small h . Enlarging C covers the remaining $h \leq 1$. This proof does not require a spectral decomposition of the arbitrary center v .

Finally choose u_* whose orbit has dimension r . The orbit is the smooth embedded compact homogeneous space G/G_{u_*} . The pushforward of Haar measure is its invariant probability measure, with positive smooth density in local coordinates. At u_* , sufficiently small ambient chordal balls have measure bounded below by ch^r for some $c > 0$. Thus no larger uniform exponent is possible. $\square$

Proposition 24.2 (An algebraic rank certificate). *Suppose the infinitesimal matrices X_i are supplied with exact real algebraic entries, and are known to specify the representation in Theorem 24.1. Then p_* is computable by real quantifier elimination. A positive algebraic lower bound for the largest $p_* \times p_*$ minor on the unit sphere is computable as well. If a unit seed z_0 is algebraic, its orbit dimension q is the rank of $[X_1 z_0 \cdots X_D z_0]$.*

Proof. For each integer r ask whether a unit vector exists on which all $(r+1) \times (r+1)$ minors vanish. These are finitely many polynomial conditions over real algebraic numbers; the first affirmative answer is p_* . With $r = p_*$, the sum of squares of all r -minors has a positive minimum on the unit sphere, by the first paragraph of the preceding proof. Quantifier elimination computes that minimum as a real algebraic number with isolating data. If there are M minors, their largest absolute value is at least $\sqrt{\gamma/M}$, where γ is that minimum. Algebraic rank at the supplied seed gives the last assertion. $\square$

The proposition computes the rank invariant from *supplied* infinitesimal matrices. It neither certifies that arbitrary input matrices integrate to a compact group nor computes a random-walk spectral gap.

Corollary 24.3 (Uniform width exponents without a separate cap hypothesis). *In the orbit realization model of Theorems 14.3 and 15.5, assume the declared spectral gap, but replace its small-ball hypothesis by $E^G = \{0\}$. Let p_* be (24.1), and let q be the real dimension of the unit target orbit spanning E . For fixed $0 < \rho \leq 1$ and $0 \leq \varepsilon < \rho$,*

$$W_{N,\varepsilon} \geq cN^{p_*/2}.$$

For $0 < \rho < 1$ the exact upper bound is $C(N+1)^{q/2}$; for $\rho = 1$ it holds at each fixed positive error. Every fixed width has the occupation bound of Theorems 14.3 and 15.5. Error is measured in the Euclidean norm defining the legal orbit hull.

Proof. Apply Theorem 24.1 to obtain its uniform cap estimate with exponent p_* . The lower and upper arguments of Theorems 14.3 and 15.5 then apply without alteration. $\square$

The group dimension D , least orbit dimension p_* , and target orbit dimension q need not coincide. For the standard sphere representation $p_* = q = d-1$. For conjugation on traceless Hermitian $d \times d$

matrices, an orbit with eigenvalue multiplicities $m_1, \dots, m_b$ has dimension $d^2 - \sum_i m_i^2$. A nonzero traceless matrix has $b \geq 2$, and the largest possible sum is $(d-1)^2 + 1$. Thus $p_* = 2(d-1)$, attained by a centered rank-one projection. The explicit Grassmannian estimates below also provide this exponent; they are retained as a direct geometric proof with stated dimension-dependent constants.

The converse now uses Theorem 15.5, rather than paying the pointwise-mixing block length at each narrow cut. Thus no logarithm separates the bounds when $p_* = q$. A supplied algebraic infinitesimal representation makes p_* computable; this statement does not compute a spectral gap or assert dimension-uniform constants.

25. A FIXED RATIONAL EXPERIMENT WITH AN EXPONENTIAL EXACT PROFILE

We now give fixed, explicit matrices and a rational pure seed. The freeness and trivial-stabilizer assertions have elementary finite-field proofs; no generic choice of generators or transcendental seed is used. The spectral-gap input is needed only for the noisy lower bound.

Lemma 25.1 (A quaternion free triple). *In the unit quaternions put*

$$a = (1 + 2\mathbf{i})/\sqrt{5}, \quad b = (1 + 2\mathbf{j})/\sqrt{5}, \quad c = (1 + 2\mathbf{k})/\sqrt{5}.$$

Every nonempty freely reduced word in $a^{\pm 1}, b^{\pm 1}, c^{\pm 1}$ is nonscalar. In particular these elements freely generate a group of rank three, also after passage to the projective unitary group.

Proof. Use the Hamilton relations $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1$, $\mathbf{ij} = \mathbf{k} = -\mathbf{ji}$. Modulo five these relations are respected by

$$\mathbf{i} \mapsto \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \quad \mathbf{j} \mapsto \begin{pmatrix} 0 & 1 \\ 4 & 0 \end{pmatrix}, \quad \mathbf{k} \mapsto \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

The six numerators $1 \pm 2\mathbf{i}, 1 \pm 2\mathbf{j}, 1 \pm 2\mathbf{k}$ therefore have respective images

$$(25.1) \quad \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \\ \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

They have rank one and their image lines are the six distinct points of $\mathbb{P}^1(\mathbb{F}_5)$. The kernel line of each matrix is the image line of the matrix with the opposite sign. Hence two consecutive matrices have zero product exactly when their letters are inverses. Writing each rank-one matrix as a column times a row shows that a product of them is nonzero whenever no adjacent pair is inverse.

A reduced word of length $n \geq 1$ is $5^{-n/2}Q$, where Q is an integral Hamilton quaternion of norm 5^n . Its image modulo five is nonzero by the preceding observation. Were Q scalar, its integer scalar entry t would satisfy $t^2 = 5^n$. For odd n this is impossible; for even $n \geq 2$ it forces $t = \pm 5^{n/2}$, making the image zero modulo five. This contradiction proves the assertion. Since the scalar unit quaternions are ± 1 , it proves both freeness statements. $\square$

Use the faithful complex representation

$$\rho(t + x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = \begin{pmatrix} t + xi & y + zi \\ -y + zi & t - xi \end{pmatrix}.$$

Define the following matrices, all of whose real and imaginary parts are rational:

$$(25.2) \quad U = \frac{1}{5} \begin{pmatrix} -3 + 4i & 0 \\ 0 & -3 - 4i \end{pmatrix}, \quad V = \frac{1}{5} \begin{pmatrix} -3 & 4 \\ -4 & -3 \end{pmatrix}, \quad P_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Here $U = \rho(a^2)$, $V = \rho(b^2)$, and P_0 is a rank-one density matrix. The alphabet is exactly

$$(25.3) \quad \mathcal{A} = \{I, U, U^*, V, V^*\}.$$

Theorem 25.2 (Explicit rational exact-noisy separation). *For the fixed input (25.2)–(25.3), let the terminal target be $F(g) = gP_0g^*$ and require each decoder value to be a legal density matrix in $\mathcal{D}_2$. Error is Frobenius error. At every horizon $N \geq 0$ the exact minimum command-width profile is*

$$(25.4) \quad |S_t|_{\min} = 2 \cdot 3^t - 1 \quad (0 \leq t \leq N), \quad W_{N,0} = 2 \cdot 3^N - 1.$$

In fact the same profile is necessary, and is attained by the exact machine, throughout the closed range

$$(25.5) \quad 0 \leq \varepsilon \leq 625^{-N}/16.$$

For each fixed $0 < \varepsilon < 1/\sqrt{2}$, constants $c_\varepsilon, C_\varepsilon > 0$ satisfy

$$(25.6) \quad c_\varepsilon N \leq W_{N,\varepsilon} \leq C_\varepsilon(N+1).$$

At $\varepsilon \geq 1/\sqrt{2}$ one command label suffices. The upper bound in (25.6) has a uniform rational-table and fair-bit construction, specified in Theorem 26.2.

Proof. First, $\langle U, V \rangle$ is free of rank two: a nontrivial reduced word in $a^{\pm 2}, b^{\pm 2}$ expands to a nontrivial reduced word in $a^{\pm 1}, b^{\pm 1}$, and Lemma 25.1 applies. If a member g of this subgroup fixes P_0 , then it commutes with P_0 . It consequently commutes with

$$\rho(c) = (I + 2i(2P_0 - I))/\sqrt{5}.$$

A nonempty reduced word w in a, b cannot commute with c : wcw^{-1} is reduced, has length $2|w|+1$, and cannot equal c in the free group. Thus the projective stabilizer of P_0 in $\langle U, V \rangle$ is trivial.

The subgroup is dense in $SU(2)$. Both U and V have trace $-6/5$. If either had root-of-unity eigenvalues, its trace would be a rational algebraic integer and hence an integer, a contradiction. Its powers are therefore dense in its one-dimensional torus. The Lie algebra of the closed generated subgroup contains the $\mathbf{i}$ and $\mathbf{j}$ directions and their bracket, the $\mathbf{k}$ direction. It is all of $\mathfrak{su}(2)$; connectedness of $SU(2)$ implies density.

By freeness and the trivial stabilizer, the reachable pure states are in bijection with the reduced word ball of radius t . Identity padding makes this precisely the set reachable at length t . Its size is $1 + 4 \sum_{j=0}^{t-1} 3^j = 2 \cdot 3^t - 1$. The extreme-orbit profile theorem (Theorem 18.1) gives (25.4). Its proof uses one fixed suffix at each cut, not a query-dependent suffix.

We prove the stronger assertion (25.5) directly. Write $D = (I + x \cdot \sigma)/2$, with $x \in B_3$, for the Bloch coordinates of a density matrix. Then

$$(25.7) \quad \|D - D'\|_F = \|x - x'\|/\sqrt{2}.$$

The rotations induced by U, V, U^*, V^* have entries in $25^{-1}\mathbb{Z}$. At cut t , different reachable unit Bloch vectors have coordinates in $25^{-t}\mathbb{Z}$, and hence mutual Euclidean distance at least 25^{-t} . Fix any suffix of length $N - t$. Its action is an isometry, so the same separation holds for the corresponding terminal targets. For each cut label let $z_i \in B_3$ be its conditional terminal decoder mean after this fixed suffix. For a prefix with pure target n , error at most ε gives

$$\left\| \sum_i p_i z_i - n \right\| \leq \sqrt{2}\varepsilon, \quad \sum_i p_i \|z_i - n\|^2 \leq 2 \left(1 - \left\langle \sum_i p_i z_i, n \right\rangle \right) \leq 2\sqrt{2}\varepsilon.$$

Some positive-mass label is therefore within $\sqrt{2\sqrt{2}\varepsilon}$ of n . Under (25.5), twice this radius is at most $\sqrt{\sqrt{2}/2} 25^{-N} < 25^{-t}$. These closed balls are disjoint for different prefixes. They require distinct labels and prove the pointwise lower bound at every cut. Deterministic exact storage gives the simultaneous upper bounds.

For the positive-error converse, apply the spectral-gap theorem of Bourgain–Gamburd [16, Theorem 1] to the *displayed* algebraic generators U, V . The density just proved verifies its

hypothesis. The symmetric walk becomes an absolute L^2 contraction off the constants after making it lazy, for example with mass $1/2$ at I and $1/8$ at each other letter. The legal density-matrix width theorem in dimension two gives the lower bound in (25.6). The rational construction proved next gives its upper bound without using a spectral gap. The center $I/2$ is at distance $1/\sqrt{2}$ from every pure state, which gives the final threshold assertion; the lower bound excludes bounded width at any smaller fixed error. $\square$

This is a finite-input separation, not merely an existence statement in an arbitrary-real advice model. Every length- N target has real and imaginary entries with denominator dividing $2 \cdot 25^N$, hence bit length $O(N + 1)$. Neither the exponential exact lower bound nor its robust extension uses a spectral-gap theorem. The spectral-gap constant in the noisy converse remains non-effective in the present proof; no numerical value is inferred from finite word tests. The logarithmic interval in (25.6) is stated explicitly rather than suppressed.

26. A UNIFORM RATIONAL QUBIT COMPILER

The preceding example also admits a constructive upper realization. The compiler below constructs its rows rather than assuming that an optimal realization has already been supplied. Its persistent labels and finite table resources are accounted for separately.

Lemma 26.1 (A rational spherical inner hull). *For $m \geq 2$ an integer, let $\mathcal{V}_m$ consist of the distinct points*

$$(26.1) \quad v_{j,\ell,\eta} = \eta \frac{(2jm, 2\ell m, m^2 - j^2 - \ell^2)}{m^2 + j^2 + \ell^2}, \quad -m \leq j, \ell \leq m, \quad \eta \in \{-1, 1\}.$$

Put $r = 1 - m^{-2}$ and add the vertex 0 as a distinguished label. Then

$$rB_3 \subseteq \text{conv}(\mathcal{V}_m) \subseteq B_3, \quad K := 1 + |\mathcal{V}_m| \leq 1 + 2(2m + 1)^2 \leq 13m^2.$$

The points have $O(\log(m+1))$ -bit rational coordinates. For any supplied rational rotation $A \in SO(3)$ and any label v , the point rAv has a rational barycentric expression using 0 and at most three members of $\mathcal{V}_m$. Such an expression can be found by finitely enumerating triples and solving rational 3×3 systems.

Proof. The stereographic parametrization $\psi(s, t) = (2s, 2t, 1 - s^2 - t^2)/(1 + s^2 + t^2)$ has differential norm at most two. Every point in the closed northern hemisphere has a preimage in the unit disk, hence in $[-1, 1]^2$. Rounding its coordinates to the nearest multiples of $1/m$ gives a point of (26.1) within distance $\sqrt{2}/m$. Negation treats the southern hemisphere. For every unit n there is thus $v \in \mathcal{V}_m$ with $\langle n, v \rangle = 1 - \|n - v\|^2/2 \geq 1 - m^{-2}$. The support-function criterion proves the inner inclusion. The stated cardinality bound follows since $m \geq 2$; duplicates are discarded.

The polytope contains a neighborhood of zero. The ray through a nonzero rAv reaches its boundary in a face. Triangulate a facet containing that face using its vertices. A triangle containing the ray endpoint, together with zero, gives the desired expression, allowing zero coefficients on lower-dimensional faces. Its three nonzero vertices are linearly independent: their affine plane is a supporting plane not through zero. Enumerate all such triples, solve for their three coefficients, and retain any solution with nonnegative coefficients and sum at most one. The remaining mass is assigned to zero. Cramer's rule gives rational coefficients; for $v = 0$ use the point mass at zero. All arithmetic is finite, and no selection theorem is needed. $\square$

Theorem 26.2 (Constructed finite-input upper realizations). *Let the alphabet consist of a fixed finite list of rational Bloch rotations, and let the seed be the first coordinate unit vector. These*

include the explicit experiment (25.2). For a rational $0 < \beta < 1$ and horizon $N \geq 0$, choose

$$(26.2) \quad m = \max \left(2, \left\lceil \sqrt{\frac{N}{1-\beta}} \right\rceil \right), \quad r = 1 - m^{-2}.$$

There is an exactly correct rational stochastic realization of the matrix target $F_\beta(g) = I/2 + \beta(gP_0g^* - I/2)$ on at most $13m^2$ command labels. Each command row has at most four nonzero entries and is independent of the command epoch within the chosen horizon.

For any rational $0 < \varepsilon < 1/\sqrt{2}$, the pure target F_1 admits an error- ε realization on $O(1 + N/\varepsilon)$ labels with rational legal matrix decoders and dyadic rows. It can be sampled with at most Nb fair bits, where it suffices to take

$$(26.3) \quad b = \left\lceil \log_2 \frac{24(N+1)}{\varepsilon} \right\rceil.$$

Both constructions are uniform algorithms polynomial in N , $(1-\beta)^{-1}$ or ε^{-1} , and the input bit lengths. This is not a polynomial bound in the binary length of N or in the logarithm of the inverse accuracy.

Proof. The net includes the first coordinate unit vector, so initialization is deterministic. For each letter A and each net label v_i , select the rational row in Lemma 26.1 whose barycentric mean is rAv_i . The zero label stays at zero. Induction shows that the mean Bloch vector of the label at time t is $r^t A_{a_t} \cdots A_{a_1} e_1$. At label v_i use the matrix decoder

$$(26.4) \quad D_i = \frac{1}{2} \left(I + \frac{\beta}{r^N} v_i \cdot \sigma \right).$$

Bernoulli's inequality and (26.2) give $r^N \geq 1 - N/m^2 \geq \beta$. Thus every decoder is positive semidefinite of trace one. Its expectation is exactly F_β , for every length- N word; the same rows work for all those words. Formula (26.4) also gives $I/2$ at the zero label.

For the pure target choose $\eta = \min(\varepsilon, 1/2)$ and $\beta = 1 - \eta$. The amplitude loss contributes Frobenius error $\eta/\sqrt{2} \leq \varepsilon/\sqrt{2}$. Keep the rational decoders and round each sparse probability row to denominator 2^b without enlarging its support: round down the first at most three masses and give the last mass the remainder. Its total-variation error is at most $3 \cdot 2^{-b}$. Stochastic contraction bounds the terminal distribution discrepancy by $3N2^{-b}$. The Frobenius diameter of $\mathcal{D}_2$ is $\sqrt{2}$; therefore the additional error is at most $3\sqrt{2}N2^{-b} \leq \varepsilon/4$ under (26.3). Since $1/\sqrt{2} + 1/4 < 1$, the total is strictly less than ε . One uniform b -bit integer samples each row. There is no initialization draw and no sampled quantum output in this count.

For completeness we give finite construction costs. Enumerating the net uses $O(m^2)$ rational points. For each of $|\mathcal{A}|K$ rows, enumerating at most $\binom{K-1}{3}$ triples and solving fixed-size systems uses polynomially many rational operations. Cramer's rule shows that the exact row coefficients have bit length $O(\log(m+1) + B_{\mathcal{A}})$, where $B_{\mathcal{A}}$ bounds the supplied rotation-entry bit lengths. The denominator r^N has bit length $O(N \log(m+1))$. Expanded decoders therefore have $O(B_\beta + (N+1) \log(m+1))$ bits each. A naive full enumeration needs $O(|\mathcal{A}|K^4)$ fixed-size rational solves, each of polynomial bit cost. This proves the algorithmic assertion as well as the bit bounds. $\square$

Sparse row tables require $O(|\mathcal{A}|K(B_{\mathcal{A}} + \log(K+1) + \log(m+1)))$ bits before rounding, and $O(|\mathcal{A}|K(b + \log(K+1)))$ bits after rounding. Expanded matrix decoders require $O(K(B_\beta + (N+1) \log(m+1)))$ bits. The persistent label uses $\lceil \log_2 K \rceil$ bits; the horizon counter, temporary sampler, and arithmetic working storage are separately charged. Unlike a statement only about output bit length, the preceding proof gives a terminating polynomial table-construction algorithm for this specific family. It does not solve general minimum-width realization or certify a spectral gap. The full legal density-matrix constraint is maintained at every rounding step.

Proposition 26.3 (Supplied rational unit seeds). *Theorem 26.2 remains valid for any supplied rational unit Bloch seed, with the label bound replaced by $13m^2 + 1$.*

Proof. Add the seed to the rational unit net if it is absent. The inner-ball inclusion is unchanged, every contracted rotated vertex still has an exact rational sparse row, and initialization is now deterministic at the added seed. The same induction and terminal rescaling apply. $\square$

The implementation accepts the supplied finite rotation list and unit seed as exact rational data. Orthogonality, determinant one and unit seed norm are checked by rational identities before table allocation.

27. A RATIONAL EXPERIMENT WITH EXPLICIT SPECTRAL CONSTANTS

The original rational pair in Theorem 25.2 now has a sharp linear fixed-error law, but its spectral constant is still supplied existentially by Bourgain–Gamburd. We give a second finite input for which every constant in a useful linear lower bound is numerical. The deep input here is the classical Lubotzky–Phillips–Sarnak theorem, not a new spectral-gap calculation.

Let $\sigma_1, \sigma_2, \sigma_3$ be the Pauli matrices in their usual order, and put $D(x) = (I + \sum_j x_j \sigma_j)/2$ for $x \in \mathbb{B}_3$. Thus $D(\mathbb{B}_3) = \mathcal{D}_2$ and $\|D(x) - D(y)\|_F = \|x - y\|/\sqrt{2}$. Consider the following *rational Bloch rotations*:

$$(27.1) \quad A_1 = \frac{1}{5} \begin{pmatrix} 5 & 0 & 0 \\ 0 & -3 & -4 \\ 0 & 4 & -3 \end{pmatrix}, \quad A_2 = \frac{1}{5} \begin{pmatrix} -3 & 0 & 4 \\ 0 & 5 & 0 \\ -4 & 0 & -3 \end{pmatrix}, \quad A_3 = \frac{1}{5} \begin{pmatrix} -3 & -4 & 0 \\ 4 & -3 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Take the alphabet

$$\mathcal{A}_{\text{LPS}} = \{I, A_1, A_1^\top, A_2, A_2^\top, A_3, A_3^\top\}$$

and the rational pure seed e_1 , so $D(e_1) = P_0$ as in (25.2). The induced unitary lifts can be taken as $(I - 2i\sigma_j)/\sqrt{5}$. They are algebraic, not rational matrices; the rational input in (27.1) describes the physical Bloch action directly. This is a different alphabet from the rational unitary pair of Theorem 25.2.

Lemma 27.1 (The classical six-rotation spectral bound). *Let P_{LPS} average uniformly over the six nonidentity letters in $\mathcal{A}_{\text{LPS}}$. On normalized S^2 ,*

$$(27.2) \quad \|P_{\text{LPS}}\|_{L_0^2(S^2) \rightarrow L_0^2(S^2)} = \frac{\sqrt{5}}{3}.$$

Their generated subgroup is dense in $SO(3)$.

Proof. The matrices in (27.1) are the adjoint actions of the unit Hamilton quaternions $(1 + 2\mathbf{i})/\sqrt{5}, (1 + 2\mathbf{j})/\sqrt{5}, (1 + 2\mathbf{k})/\sqrt{5}$. Together with their inverses they are exactly the norm-five set in the Lubotzky–Phillips–Sarnak theorem [28, Theorems 1.3 and 1.5]; see the precise restatement [29, Theorem 1.1], which gives $2\sqrt{p}/(p+1)$ for this set at $p = 5$. This proves (27.2). The cited theorem uses the theory of automorphic forms and Deligne’s theorem; no finite matrix enumeration replaces it.

Density can also be checked directly. Each rotation has cosine $-3/5$. If its angle were a rational multiple of 2π , twice this cosine would be a rational algebraic integer, contrary to $-6/5 \notin \mathbb{Z}$. Its powers are dense in its circle subgroup. The three coordinate-axis circle subgroups generate $SO(3)$, as follows either from their infinitesimal generators and brackets or from Euler rotations. Hence the closed generated subgroup is $SO(3)$. $\square$

Theorem 27.2 (An effective linear noisy law and an exponential endpoint). *For the fixed input (27.1), the terminal target is $D(A_w e_1)$, every decoder belongs to $\mathcal{D}_2$, and error is Frobenius error. For $0 < \varepsilon < 1/\sqrt{2}$ put $\zeta = 1 - \sqrt{2}\varepsilon$. Every admissible realization satisfies*

$$(27.3) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq 1 + 6 \log(12k) + 648\pi^2 \frac{1-\zeta}{\zeta} k \leq \frac{6500}{\zeta} k.$$

In particular,

$$(27.4) \quad \frac{\zeta N}{6500} \leq W_{N,\varepsilon} \leq 52 + \frac{104N}{\varepsilon}.$$

For rational ε in this interval the upper machine has uniformly constructed rational legal decoders and dyadic stochastic rows, each supported on at most four labels. It uses at most Nb fair bits for hidden transitions, where $b = \lceil \log_2(24(N+1)/\varepsilon) \rceil$.

At every horizon $N \geq 0$, the simultaneous exact minimum profile is

$$(27.5) \quad |S_t|_{\min} = 5^t \quad (0 \leq t \leq N).$$

The same profile remains necessary and is attained throughout

$$(27.6) \quad 0 \leq \varepsilon \leq 25^{-N}/16.$$

For $\varepsilon \geq 1/\sqrt{2}$, one label suffices.

Proof. For a spherical chordal cap of radius $h \leq 1$, normalized area is exactly $h^2/4$. Consequently the support of a mixture of k kernels K_h occupies at most $kh^2/4$ of each group orbit. Take $h = k^{-1/2}$. In Lemma 15.3 we may take

$$\theta = 1/4, \quad \kappa^2 = 5/9, \quad \gamma = (1 - \kappa^2)(1 - \theta) = 1/3.$$

Lemma 15.2 gives $L_h = \log(12k)$ and $B_3 = 24$. The proof of Theorem 15.5, with these constants and the normalization $\zeta = 1 - \sqrt{2}\varepsilon$, gives the first inequality in (27.3) as a real bound before taking an integer ceiling. For $k \geq 1$, $\log(12k) \leq 3k$, and $\pi^2 < 10$. Thus its right side is at most $19k + 6480(1 - \zeta)k/\zeta \leq 6500k/\zeta$. Counting all positive cuts proves the lower bound in (27.4).

For the upper bound apply the rational compiler of Theorem 26.2 to the displayed seven-letter list and the first-axis seed. At rational tolerance take $\beta = 1 - \varepsilon/2$ and $m = \max\{2, \lceil \sqrt{2N/\varepsilon} \rceil\}$. It gives an exact rational realization of $D(\beta A_w e_1)$ on at most $13m^2 \leq 52 + 52N/\varepsilon$ labels, with sparse common rows. The target change has Frobenius norm $\varepsilon/(2\sqrt{2})$. The stated dyadic precision adds at most $3\sqrt{2}N2^{-b} \leq \sqrt{2}\varepsilon/8$, so the total is less than ε . This proves the looser advertised numerical upper bound and the fair-bit statement. For a real tolerance choose a rational $\tilde{\varepsilon} \in (\varepsilon/2, \varepsilon)$ and use the same construction at $\tilde{\varepsilon}$; the bound $52 + 104N/\varepsilon$ follows. The algorithmic assertion uses rational input, not an oracle for an unencoded real number.

We next prove the exact profile independently of the spectral bound. Lemma 25.1 shows that the three norm-five quaternion generators freely generate modulo scalars, hence their adjoint rotations form a free group on a, b, c . The stabilizer of e_1 in this free group is exactly $\langle a \rangle$. Indeed, a rotation fixing e_1 commutes with the rotation a about that axis. In a free group the centralizer of this generator is its cyclic subgroup: write a reduced word outside $\langle a \rangle$ as $a^r v a^s$, where v is nonempty and begins and ends with neither a nor a^{-1} . Its conjugate of a reduces to $a^r v a v^{-1} a^{-r}$, which is not a . The converse, that every power of a fixes e_1 , is immediate.

The orbit is therefore indexed by the right cosets of $\langle a \rangle$. Each coset has a unique reduced representative with no terminal $a^{\pm 1}$, obtained by removing the maximal terminal power of a . It has least word length in that coset. At length $\ell \geq 1$ there are $4 \cdot 5^{\ell-1}$ such representatives. Identity padding gives exactly the ball of these cosets at command length t , whose cardinality is $1 + 4 \sum_{\ell=1}^t 5^{\ell-1} = 5^t$. Theorem 18.1 applies because pure states are extreme in $\mathcal{D}_2$, proving (27.5) at every cut; deterministic orbit storage attains the entire profile at once.

For robustness, distinct cut- t Bloch targets have coordinates in $5^{-t}\mathbb{Z}^3$ and hence distance at least 5^{-t} . One fixed suffix of length $N - t$ preserves these distances. If $z_i \in \mathbb{B}_3$ is the conditional terminal decoder mean from cut label i , wordwise error at a pure target n implies

$$\sum_i p_i \|z_i - n\|^2 \leq 2\sqrt{2}\varepsilon.$$

Some positive-mass label is within $\sqrt{2\sqrt{2}\varepsilon}$ of n . Under (27.6), twice this radius is strictly less than $5^{-N} \leq 5^{-t}$. The corresponding closed balls for distinct prefix orbit points are disjoint, and so require distinct labels. The exact construction attains this bound. No fixed positive tolerance is asserted to lie in (27.6) for all horizons. Finally $D(0) = I/2$ is at Frobenius distance $1/\sqrt{2}$ from every pure state, proving the one-label assertion. $\square$

Remark 27.3 (What is effective). The input, orbit counts, robust tolerance, spectral norm, constants in (27.3)–(27.4), and rational compiler are explicit. The constant 6500 is conservative; it is not an optimal leading constant. The spectral norm is imported from [28, 29], not inferred from finite harmonic tests. For the earlier five-letter rational-unitary example, Theorem 15.5 removes the logarithm but does not supply a numerical Bourgain–Gamburd constant. Neither statement claims a complete two-parameter phase diagram as the error tends to zero. The finite-table and fair-bit resources refer to a classical hidden label and a numerical matrix output, not quantum-state preparation.

28. THE SIX-COMMAND SPHERE AND VARIABLE ACCURACY

The full spectral measure in Lemma 27.1 already lives on six nonidentity commands. The following result uses exactly those six commands for the experiment itself, with no idle instruction.

Theorem 28.1 (An explicit no-idle experiment). *Let A_1, A_2, A_3 be the rational matrices in (27.1). Take exactly*

$$\mathcal{A}_6 = \{A_1, A_1^\top, A_2, A_2^\top, A_3, A_3^\top\}, \quad x_0 = e_1, \quad F(w) = D(A_w e_1).$$

Every decoder must lie in $\mathcal{D}_2$, and error is Frobenius error. For $0 < \varepsilon < 1/\sqrt{2}$, put $\zeta = 1 - \sqrt{2}\varepsilon$ and $\theta = (1 - \zeta)/\zeta$. Then every admissible width profile satisfies

$$(28.1) \quad \#\{1 \leq t \leq N : |S_t| \leq k\} \leq 1 + 6 \log(12k) + 648\pi^2 \theta k.$$

For $N \geq 2$, the minimum peak width obeys the explicit joint bounds

$$(28.2) \quad \min \left\{ \frac{e^{(N-1)/12}}{12}, \frac{N-1}{1296\pi^2\theta} \right\} \leq W_{N,\varepsilon} \leq \min\{5^N, 52 + 104N/\varepsilon\}.$$

At every horizon the exact simultaneous cut profile is 5^t , $0 \leq t \leq N$, and the same profile is necessary and attained for $0 \leq \varepsilon \leq 25^{-N}/16$. At $\varepsilon \geq 1/\sqrt{2}$, one label suffices. For any fixed $\varepsilon_0 < 1/\sqrt{2}$ and sequence $0 < \varepsilon_N \leq \varepsilon_0$ with $\log(1/\varepsilon_N) = o(N)$,

$$(28.3) \quad W_{N,\varepsilon_N} = \Theta(N/\varepsilon_N).$$

The constants in this last comparison can be taken independent of the sequence; its starting horizon can depend on the sequence.

Proof. The classical full spherical norm is $\sqrt{5}/3$ for the uniform law on exactly $\mathcal{A}_6$ [29, Theorem 1.1]. Use the drift proof of Theorem 16.2, taking $h = k^{-1/2}$ and the sharper sphere constants $\gamma = 1/3$, $L_h = \log(12k)$, $B_3 = 24$ as in Theorem 27.2. This gives (28.1), with arbitrary deterministic commands in the gaps. It imports no uncomputed spectral constant.

For a full width bound k , at least one of $6 \log(12k)$ and $648\pi^2 \theta k$ is at least $(N-1)/2$. Inverting the corresponding inequality gives the lower bound in (28.2). The sparse rational construction

in Theorem 27.2 applies unchanged to the six-letter sublist; no row for the identity is needed. It gives the second upper bound, with the same rational-decoder and dyadic-row guarantees for rational tolerances.

For the exact profile, recall the independently proved free-group and stabilizer calculation in Theorem 27.2. The orbit is the right-coset space of $\langle a \rangle$ in the free group on a, b, c , where $a = A_1$. Every coset has a unique shortest representative w with no terminal $a^{\pm 1}$. If its length is $\ell \leq t$, the group word $wa^{t-\ell}$ has length t and the same action on e_1 . In chronological execution this means applying the a 's first; they fix the seed. There is no cancellation when w is nonempty, and the empty representative is padded by a^t . Thus even without an identity letter, the length- t reachable orbit is exactly the radius- t coset ball. Its size is 5^t . The extreme-orbit theorem gives the exact profile and deterministic orbit storage gives the upper bound 5^N .

The denominator-five lattice and fixed-suffix isometry argument of Theorem 27.2 does not require identity padding once the orbit count is established. It gives the same robust interval. The center $I/2$ gives the final constant-width assertion.

Finally $\theta \leq C_{\varepsilon_0} \varepsilon_N$, while $\log(N/\varepsilon_N) = o(N)$. The exponential alternative in the minimum on the left of (28.2) eventually exceeds a constant times N/ε_N . The other alternative already has that lower order. The upper bound proves (28.3). $\square$

Remark 28.2 (Three separate accuracy regimes). The exact count and its robust interval, the explicit two-parameter bounds (28.2), and the subexponential-accuracy asymptotic (28.3) are different statements. For example, $\varepsilon_N = N^{-a}$ gives width $\Theta(N^{a+1})$, and $\varepsilon_N = e^{-N^b}$, $0 < b < 1$, gives $\Theta(Ne^{N^b})$. Neither assertion extrapolates the exact plateau to a fixed positive error or resolves its full crossover at $\log(1/\varepsilon_N) \asymp N$. The constant 6500 remains a safe fixed-error simplification of (28.1), not an optimal leading constant. The seven-letter experiment and the older rational-unitary pair remain separate retained examples.

29. NONBACKTRACKING ENTROPY AND EXPONENTIAL ACCURACY

The estimates for subexponential inverse accuracy do not determine the width when $\log(1/\varepsilon)$ is comparable to the horizon. In this section a Ramanujan bound on a symmetric spherical action determines that exponential rate. The averaging polynomials and the relative-entropy affinity inequality are classical. The point is to retain their logarithmic strength in the centroid transport budget.

29.1. The logarithmic entropy defect. We continue to use the spherical kernel, entropy and invariant projection from Section 15. In the next lemma the projection may have more than one-dimensional range.

Lemma 29.1 (Affinity form of the entropy defect). *Let T be a finite convex combination of measure-preserving rotations and suppose $T\Pi = \Pi T = \Pi$ and $\|T(I - \Pi)\|_{2 \rightarrow 2} \leq \kappa < 1$. If a bounded probability density f is supported in A and $\Pi \mathbf{1}_A \leq \theta < 1$, then*

$$(29.1) \quad \mathcal{H}(f) - \mathcal{H}(Tf) \geq -\log(\kappa^2 + (1 - \kappa^2)\theta).$$

Proof. Write $Tf = \sum_j a_j f_j = g$, discarding zero coefficients. All f_j have integral one and the same entropy as f . The standard order-1/2 Rényi bound [33, Theorem 3], with its elementary Jensen proof, is

$$D(u\|v) = -2 \int u \log \sqrt{v/u} d\sigma \geq -2 \log \int \sqrt{uv} d\sigma.$$

Approximation or the extended-value convention handles zeros. Here $g \geq a_j f_j$, so each divergence needed below is finite. Convexity of $-\log$, followed by Cauchy–Schwarz, gives

$$\begin{aligned} \mathcal{H}(f) - \mathcal{H}(g) &= \sum_j a_j D(f_j \| g) \\ &\geq -2 \log \sum_j a_j \int \sqrt{f_j g} d\sigma \geq -2 \log \|T\sqrt{f}\|_2. \end{aligned}$$

Orbital Cauchy–Schwarz, exactly as in Lemma 15.3, gives $\|\Pi\sqrt{f}\|_2^2 \leq \theta$. Orthogonality of the invariant and orthogonal parts then gives $\|T\sqrt{f}\|_2^2 \leq \kappa^2 + (1 - \kappa^2)\theta$. This proves the claim. In contrast with its linearized version, (29.1) can grow with the length of an averaging block. $\square$

29.2. A spherical Ramanujan hypothesis. Fix $m \geq 2$, $p = m - 1$, a unit seed $z_0 \in \mathbb{R}^m$, and an alphabet of $2r$ indexed commands, $r \geq 2$, consisting of orthogonal maps $A_1, A_1^{-1}, \dots, A_r, A_r^{-1}$. Put $q = 2r - 1$. The legal output set is the closed unit ball $\mathbb{B}_m$, the target is $A_w z_0$, and error is Euclidean. Assume that the uniform command average P on $L^2(S^{m-1}, \sigma)$ satisfies

$$(29.2) \quad \|P|_{L_0^2}\|_{2 \rightarrow 2} \leq \frac{2\sqrt{q}}{q+1}.$$

This is a hypothesis on the *whole* spherical action, not just on $\mathbb{R}^m$. It implies that its compact group closure is transitive on S^{m-1} : otherwise averaging a continuous function distinguishing two compact orbits would give a nonconstant invariant function. In particular, the Haar orbit law is round measure and satisfies

$$(29.3) \quad \sigma\{x : \|x - u\| < h\} \leq A_m h^p \quad (0 < h \leq 1), \quad A_m \geq 1.$$

The harmless choice $A_m \geq 1$ will simplify the formulas. No identity letter or return assumption is made. The indexed inverse relation is used in counting formal words; no freeness assumption is needed for the following upper spectral bound. Reversing a formal nonbacktracking word preserves its law, so the chronological product convention gives the same averaging operator. The polynomial formula below does not replace a positive word average by a signed command kernel.

Let P_b average uniformly over the $(q+1)q^{b-1}$ formal words of length b having no adjacent inverse pair, for $b \geq 1$. Define

$$(29.4) \quad a_q = \frac{q-1}{q+1}, \quad \eta_b = (b + a_q)^2 q^{-b}.$$

Lemma 29.2 (Classical radial-polynomial estimate). *Under (29.2), $\|P_b|_{L_0^2}\|^2 \leq \eta_b$. In particular $\eta_b \leq 1/4$ for every $b \geq 4$ and $q \geq 3$.*

Proof. Let $A = (q+1)P$, and let S_b be the unnormalized formal word sum in the action. Cancellation of the first inverse pair gives

$$S_0 = I, \quad S_1 = A, \quad S_2 = A^2 - (q+1)I, \quad S_b = AS_{b-1} - qS_{b-2} \quad (b \geq 3).$$

The exceptional coefficient $q+1$ at length two is essential. With U_j the Chebyshev polynomial of the second kind and $U_{-1} = 0$, induction yields, for $b \geq 1$,

$$S_b = q^{b/2} \left[U_b \left(\frac{A}{2\sqrt{q}} \right) - q^{-1} U_{b-2} \left(\frac{A}{2\sqrt{q}} \right) \right],$$

where the second term is zero when $b = 1$. The operator A is self-adjoint and its spectrum on L_0^2 is in $[-2\sqrt{q}, 2\sqrt{q}]$. The elementary identity $U_j(\cos t) = \sum_{\ell=0}^j e^{i(j-2\ell)t}$ implies $|U_j(x)| \leq j+1$ on $[-1, 1]$. Dividing by $(q+1)q^{b-1}$ gives the bound $(b + a_q)q^{-b/2}$, including $b = 1$. For $b \geq 4$, the square of this bound is strictly decreasing in b : its successive ratio is at most $(1 + 1/4)^2/3 < 1$. At

$b = 4, q = 3$ its square is $(4 + 1/2)^2/81 = 1/4$, while for $q \geq 5$ it is at most $25/625 < 1/4$. This proves the non-strict bound. $\square$

The use of radial polynomials is not a new spectral theorem. For the LPS rotations, Pinochet Lobos–Pittet prove the sharper exact norm $(1 + a_q b)q^{-b/2}$ [29, Theorem 1.2]. We use the slightly larger bound (29.4) because the displayed polynomial argument proves it directly from (29.2).

29.3. A finite block inequality for arbitrary width profiles. Keep D_m, B_m from Lemma 15.2; here the entropy range exponent $m - 1$ equals the cap exponent p . For an error $0 \leq e < 1$ put $\xi = e/(1 - e)$.

Theorem 29.3 (Block entropy budget). *Let $t_0 < \dots < t_M$ be command cuts, each of width at most k , with $t_{j+1} - t_j \geq b$, where $b \geq 4$. Cut zero is allowed in this intermediate statement. Every error- e realization satisfies*

$$(29.5) \quad Md_b \leq \log \frac{D_m A_m k}{\eta_b} + C_m \left(\frac{A_m k}{\eta_b} \right)^{1/p} \sqrt{M\xi}, \quad d_b = -\log(2\eta_b - \eta_b^2), \quad C_m = \pi\sqrt{3B_m}.$$

All registers between these cuts are unrestricted. For a peak-width- k realization, one may take $M = \lfloor N/b \rfloor$. If instead J positive cuts have width at most k , one may take $M = \lceil J/b \rceil - 1$ whenever this number is positive.

Proof. Choose $h = (\eta_b/(A_m k))^{1/p} \leq 1$. At each selected cut the smoothed directional law is supported in at most k caps; its round support measure is at most η_b . By Lemmas 29.1 and 29.2, applying an independent P_b word lowers its entropy by at least d_b .

Use a deterministic prefix to t_0 . In each gap apply a fresh independent formal nonbacktracking word of length b , then any fixed word filling the rest of the gap. Use a fixed suffix after t_M . These are executable words of the original horizon, irrespective of possible cancellations between different blocks. Conditional on the incoming hidden label and the whole block word, the actual composite row is a stochastic kernel. The proof of Lemma 16.1 applies verbatim to this finite word law, not just to a one-letter law: its weighted flow identity sums over the word in place of the letter. Consequently the entropy correction at gap j is at most

$$\frac{\pi\sqrt{3B_m}}{h} \sqrt{\frac{\alpha_j - \alpha_{j+1}}{1 - e}}.$$

The deterministic filler rotates the potential and costs no entropy; no commutation with P_b is required. Wordwise terminal mean correctness and legal decoder norm at most one give $\alpha_j \geq 1 - e$. At the first selected cut the physical vector is deterministic, so $\alpha_0 = 1$, even with randomized initialization. Thus the losses sum to at most e . Telescoping the entropy, using its range $[0, \log(D_m h^{-p})]$, and then Cauchy–Schwarz proves (29.5).

For a peak bound take $t_j = jb$ and fill the final suffix. For J positive narrow cuts, greedily keep the first and then the next at distance at least b . Each selected cut accounts for at most b integer cuts, so there are at least $\lceil J/b \rceil$ selected cuts. Use the first $\lceil J/b \rceil$ of them. This proves the last claim without including cut zero in J . $\square$

Corollary 29.4 (Explicit inversion). *Put $H = Md_b - \log(D_m A_m / \eta_b)$. If $H > 0$, then for any $0 < \lambda < 1$,*

$$(29.6) \quad k \geq \min \left\{ \exp((1 - \lambda)H), \quad \frac{\eta_b}{A_m} \left(\frac{\lambda H}{C_m \sqrt{M\xi}} \right)^p \right\}.$$

When $e = 0$, the second term is interpreted as infinity and the stronger bound $k \geq \exp H$ holds.

Proof. Subtract $\log(D_m A_m / \eta_b)$ in (29.5). Either $\log k \geq (1 - \lambda)H$, or its remaining transport term is greater than λH . Solving the latter inequality proves the result. When $\xi = 0$ that term vanishes. $\square$

29.4. Uniform crossover on the exponential scale.

Theorem 29.5 (Spherical exponential-accuracy law). *Under (29.2), fix $0 < e_0 < 1$. There are constants $C, C' > 0$, depending only on the alphabet, m , and e_0 , such that for all $N \geq 1$ and $0 \leq e \leq e_0$,*

$$(29.7) \quad \exp[-C\sqrt{N \log(N+2)}] \min\{q^N, ((N+1)/e)^{p/2}\} \leq W_{N,e} \leq C' \min\{q^N, ((N+1)/e)^{p/2}\}.$$

The convention is $1/0 = \infty$. Moreover, if an error- e realization has $J \geq 1$ positive cuts of width at most k , then

$$(29.8) \quad k \geq \exp[-C\sqrt{J \log(J+2)}] \min\{q^J, ((J+1)/e)^{p/2}\}.$$

In particular, if $0 \leq e_N \leq e_0$ and $N^{-1} \log(1/e_N) \rightarrow a \in [0, \infty]$, then

$$(29.9) \quad \lim_{N \rightarrow \infty} \frac{\log W_{N,e_N}}{N} = \min\{\log q, pa/2\}.$$

Proof. We prove the lower estimate using a parameter n equal to N for peak width and to J for the occupation statement. For sufficiently large n set

$$b = \left\lceil \sqrt{n \log(n+2)} \right\rceil, \quad \lambda = n^{-1/2}.$$

Then $b \geq 4$, $b = o(n)$, and the respective choices of M in Theorem 29.3 satisfy $n/b - 1 \leq M \leq n/b$. By (29.4),

$$-\log \eta_b = b \log q - 2 \log(b + a_q), \quad d_b \geq b \log q - 2 \log(b + a_q) - \log 2.$$

It follows, with constants independent of e, n, k , that

$$\begin{aligned} H &\geq n \log q - C_1(b + (n/b) \log(b+1) + 1) \\ &\geq n \log q - C_2 \sqrt{n \log(n+2)}, \quad H \geq \tfrac{1}{2} n \log q \end{aligned}$$

for all sufficiently large n . The first term in (29.6) is therefore at least $q^n \exp[-C_3 \sqrt{n \log(n+2)}]$. For $e > 0$, use $\xi \leq e/(1-e_0)$, $M \leq n/b$, $\eta_b \geq q^{-b}$ and $\lambda H \geq \frac{1}{2} \sqrt{n} \log q$. The second term in that formula is at least

$$c q^{-b} b^{p/2} / e^{p/2} \geq ((n+1)/e)^{p/2} \exp[-C_4 \sqrt{n \log(n+2)}].$$

The last inequality absorbs only constants and powers of n , besides q^{-b} . If $e = 0$, apply the last sentence of Corollary 29.4. Increasing C handles the finitely many smaller n , since the minimum in either statement is at most q^n and every register has at least one label.

For the upper bound, formal cancellation leaves a reduced word of length at most t after t commands. Thus at most $1 + (q+1) \sum_{j=0}^{t-1} q^j \leq C_q q^t$ physical products need be stored. Deterministic exact storage gives the first bound; it does not require physical freeness. For $e > 0$, the spherical inner-hull construction in Theorem 14.3 realizes the smaller amplitude $1 - e/2$ exactly with at most $C_{\text{net}}(1 + (N+1)/e)^{p/2}$ labels. Its bias is $e/2$ and its decoders remain in $\mathbb{B}_m$. The same conclusion follows directly by taking a spherical δ -net with $\delta^2 \asymp e/(N+1)$, placing each contracted rotated vertex in its inner hull, and rescaling the terminal decoder within the available amplitude slack. Transitivity established above makes the target orbit the whole sphere. Taking the better of the two machines proves the upper estimate. Finally divide logarithms in (29.7) by N ; the loss term is $o(N)$, proving (29.9), including $a = \infty$. $\square$

Remark 29.6 (What the crossover determines). The estimate is uniform in the error and determines the entire exponential rate, including the saturation rate. Its factor $\exp(O(\sqrt{N \log(N+2)}))$ is not asserted optimal. It does not give uniform multiplicative Θ bounds through the transition,

optimal leading constants, or the largest interval of exact equality with the orbit count. The subexponential-accuracy matching theorem 16.5 remains stronger in its own regime.

29.5. The fixed rational six-rotation experiment.

Corollary 29.7 (Explicit rational crossover). *For the six nonidentity rational Bloch commands of Theorem 28.1, seed e_1 , legal decoders $\mathcal{D}_2$ and Frobenius error, fix $0 < \varepsilon_0 < 1/\sqrt{2}$. Uniformly for $N \geq 1$ and $0 \leq \varepsilon \leq \varepsilon_0$,*

$$(29.10) \quad \exp[-C\sqrt{N \log(N+2)}] \min\{5^N, (N+1)/\varepsilon\} \leq W_{N,\varepsilon} \leq \min\{5^N, 52 + 104N/\varepsilon\}.$$

Consequently $\log(1/\varepsilon_N)/N \rightarrow a$ implies $\log W_{N,\varepsilon_N}/N \rightarrow \min\{a, \log 5\}$. At every cut the exact minimum is still 5^t for $\varepsilon \leq 25^{-N}/16$. No claim of exact saturation at $a = \log 5$ is made.

Proof. The affine Bloch identification multiplies Frobenius error by $\sqrt{2}$. Thus apply Theorem 29.5 with $m = 3$, $p = 2$, $q = 5$, and $e = \sqrt{2}\varepsilon$. The full-action Ramanujan hypothesis is precisely the imported LPS norm of Lemma 27.1; the nonbacktracking estimates above are for this same six-command alphabet. Constant factors of $\sqrt{2}$ are absorbed in C . The exact and rational-compiler upper bounds and the robust pointwise statement are Theorem 28.1. $\square$

For finite lower certificates on this experiment one may use its exact spherical cap area, rather than $A_3 \geq 1$. If $b \geq 4$, $M = \lfloor N/b \rfloor$, $\eta_b = (b + 2/3)^2 5^{-b}$, and $\xi = \sqrt{2}\varepsilon/(1 - \sqrt{2}\varepsilon)$, take $h = 2\sqrt{\eta_b/k} \leq 1$. The same proof yields

$$(29.11) \quad Md_b \leq \log(3k/\eta_b) + 3\pi\sqrt{2M\xi k/\eta_b}.$$

A strict violation excludes width k . This is a sufficient exclusion test, not a decision procedure for exact minimum width. The supplied `accuracy_profile.py` checks such violations with rational logarithm intervals and rational upper bounds for $\sqrt{2}$ and π ; it never treats a finite enumeration of words or harmonics as a proof of the imported spectral norm.

30. UNIFORM ONE-PASS COMPUTATION WITH CHARGED BIT-SPACE

The preceding label-width invariant permits arbitrary real transition tables. A lower bound for that invariant also constrains finite-bit algorithms, but an upper bound for it need not be implementable at a comparable cost. We now give a separate uniform result in which all integer storage and arithmetic are charged. Throughout this section the physical alphabet is exactly $\mathcal{A}_6$ in Theorem 28.1; in particular, no idle command is added. All logarithms are natural except those explicitly denoted by $\log_2$.

30.1. Computational convention and the space theorem. A numerical streaming algorithm has a fixed finite program. It reads parameters $N \geq 2$ and $L \geq 2$, followed by N commands, in one pass. A consumed command cannot be consulted again. No input-dependent information may be stored in a readable output tape, an external clock, or an uncharged address. After the last command it returns a rational matrix $\hat{F}(w) \in \mathcal{D}_2$. For randomized algorithms every returned matrix is required to be legal, and our correctness condition is the weaker, mean condition

$$(30.1) \quad \sup_{w \in \mathcal{A}_6^N} \|\mathbb{E}\hat{F}(w) - D(A_w e_1)\|_F \leq 2^{-L}.$$

Internal computation terminates almost surely between successive commands and at the output stage. Randomness is fresh; any retained random seed is part of the configuration.

Writable bit-space includes the data registers, temporary integer storage, retained parameter information, work-tape addresses and control information. Equivalently, one may charge a binary encoding of the complete private configuration. To relate these conventions, a fixed program with s work bits and a fixed number of work heads has at most $C(s+1)^{d_{2cs}}$ configurations, for

constants depending only on its machine convention. The prescribed position of the one-pass input head at the t th command boundary contains no input-dependent information. A private address that does contain such information must be counted. The parameters are not an advice table; they contain only the two integers N, L .

Theorem 30.1 (Uniform streaming space). *There is one deterministic numerical streaming algorithm for the six rational rotations such that, for every $N \geq 2$ and $L \geq 2$, every output is legal, the pointwise error is at most 2^{-L} , and the writable bit-space is*

$$(30.2) \quad O(\min\{N, L + \log_2(N + 1)\}).$$

It uses no real advice, no random sampling, and no oracle arithmetic. Writing $b_ = \min\{N, L + \lceil \log_2 N \rceil + 2\}$, its command updates can be performed in $O(b_*)$ bit operations each with $O(b_*)$ scratch space. Parameter scanning is charged separately by its input length; a binary rational output uses $O(b_*)$ bits.*

Conversely, there is an absolute $c > 0$ such that every one-pass algorithm satisfying (30.1) and having at most 2^B complete private configurations at each command boundary satisfies

$$(30.3) \quad B \geq c \min\{N, L + \log_2(N + 1)\}.$$

There is no running-time restriction in this converse. It applies even to arbitrary horizon-dependent stochastic updates. Consequently every fixed-program randomized bit algorithm has the same lower order of writable space, with machine-dependent constants. Deterministic and randomized uniform space therefore have the common order $\Theta(\min\{N, L + \log_2(N + 1)\})$ for this numerical task.

The spectral input for the converse is the full spherical Koopman norm $\sqrt{5}/3$ of the six-letter average, not the norm of its action on $\mathbb{R}^3$. The upper algorithm uses only rational orthogonality and does not need a spectral assumption. The theorem is a consequence of the entropy occupation law together with an explicit finite-arithmetic construction, not a new proof of the LPS spectral theorem.

30.2. From complete configurations to stochastic width.

Lemma 30.2 (Boundary reduction). *If a one-pass algorithm has at most k private configurations at every command boundary and satisfies mean error ε , its boundary laws define a clocked numerical realization of peak width at most k and the same error.*

Proof. Take the configurations at boundary t as S_t . Include every piece of retained information, whether or not it is called a data register. Starting from a boundary configuration and a supplied next command, integrate over all internal computation and fresh randomness up to the next boundary. Almost-sure termination gives a stochastic row of total mass one. The row depends only on that configuration and command, with the fixed parameters and boundary index permitted as public parameters. Consumed input and old random bits are inaccessible unless recorded in the configuration. This gives precisely the Markov rows of Definition 7.1.

At the terminal boundary, replace any randomized output computation by its conditional mean matrix. This mean is in $\mathcal{D}_2$ by convexity and compactness, and is independent of the discarded input conditional on the complete configuration. Thus the numerical decoder is legal and has exactly the mean output in (30.1). No running-time or exact-sampling assumption is introduced in this reduction. $\square$

Proposition 30.3 (A finite bit-accuracy obstruction). *For the six-command experiment, suppose a correct realization has at most 2^B complete configurations at every positive cut and error $0 \leq \varepsilon < 1/\sqrt{2}$. Put*

$$a_B = 1 + 6 \log(12 \cdot 2^B), \quad c_B = 648\pi^2 2^B.$$

Then

$$(30.4) \quad N \leq a_B + c_B \frac{\sqrt{2}\varepsilon}{1 - \sqrt{2}\varepsilon}.$$

In particular, if $d = N - a_B > 0$, then

$$(30.5) \quad \varepsilon \geq \frac{d}{\sqrt{2}(c_B + d)}.$$

At zero error the exact profile gives the stronger bound $B \geq N \log_2 5$.

Proof. Apply (28.1) to all N positive cuts with $k = 2^B$ and $\zeta = 1 - \sqrt{2}\varepsilon$. This gives (30.4), including its limiting zero-error case. Solving $d \leq c_B \sqrt{2}\varepsilon / (1 - \sqrt{2}\varepsilon)$ gives (30.5). For the last assertion use the exact terminal width 5^N in Theorem 28.1 instead of the weaker entropy estimate. $\square$

Lower bound in Theorem 30.1. Let k be the maximum number of boundary configurations. Since $\varepsilon = 2^{-L} \leq 1/4$, Proposition 30.3, or (28.1) with k , implies, for absolute positive constants C_0, C_1, C_2 ,

$$(30.6) \quad N \leq C_0 + C_1 \log k + C_2 \varepsilon k.$$

For $N \geq 2C_0$, either $\log k \geq N/(4C_1)$ or $k \geq N/(4C_2\varepsilon)$. Taking logarithms and enlarging an absolute additive constant yields

$$\log_2 k \geq c_0 \min\{N, \log_2(N/\varepsilon)\} - C_3.$$

The additive constant can be absorbed uniformly. Indeed $k \geq 2$ for all $N \geq 2$ in the stated error range: the all- A_1 word ends at e_1 , whereas $N - 1$ copies of A_1 followed by A_2 end at $A_2 e_1 = (-3/5, 0, -4/5)$. Their density matrices have Frobenius distance $\sqrt{8/5} > 1/2$. A single constant mean decoder cannot be within $1/4$ of both. If $\min\{N, \log_2(N/\varepsilon)\}$ is below a fixed large constant, $\log_2 k \geq 1$ suffices; above that constant it absorbs C_3 in the preceding estimate. The finitely bounded range $N < 2C_0$ is handled in the same way. Since $k \leq 2^B$, this proves (30.3). Finally the fixed-program configuration count stated above gives $\log_2 k = O(s + 1)$, so the same conclusion holds for work-tape space. This argument neither assumes uniform transition tables nor assumes that an individual sampled decoder approximates the physical target. $\square$

30.3. A legality-preserving integer algorithm. For $b \geq 0$, define coordinatewise truncation toward zero by

$$T_b(x)_i = 2^{-b} \operatorname{sgn}(x_i) \lfloor 2^b |x_i| \rfloor.$$

Unlike rounding to the nearest grid point, this operation preserves the closed Euclidean unit ball.

Lemma 30.4 (Directed rounding). *For every $x \in \mathbb{R}^d$,*

$$\|T_b(x)\| \leq \|x\|, \quad \|T_b(x) - x\| \leq \sqrt{d} 2^{-b}.$$

If $R_1, \dots, R_N$ are orthogonal, $x_t = R_t x_{t-1}$ with $\|x_0\| \leq 1$, and $v_0 = x_0$ is on the 2^{-b} grid, then $v_t = T_b(R_t v_{t-1})$ satisfies

$$(30.7) \quad \|v_t\| \leq 1, \quad \|v_t - x_t\| \leq t\sqrt{d} 2^{-b}.$$

Proof. Each absolute coordinate decreases under truncation and changes by less than 2^{-b} . Squaring and summing gives the first two estimates. Orthogonality and the triangle inequality give $\|v_t - x_t\| \leq \sqrt{d} 2^{-b} + \|v_{t-1} - x_{t-1}\|$. Induction proves both assertions in (30.7). $\square$

Write $A_a = M_a/5$, where M_a is one of the six fixed integer matrices from (27.1) and their transposes, multiplied by five. Set

$$(30.8) \quad b = L + \lceil \log_2 N \rceil + 2.$$

When $b < N$, initialize $n = (2^b, 0, 0) \in \mathbb{Z}^3$. On command a , compute

$$(30.9) \quad z = M_a n, \quad n'_i = \text{sgn}(z_i) \lfloor |z_i|/5 \rfloor, \quad n \leftarrow n'.$$

The current numerical vector is $v = n/2^b$. The update is exactly $T_b(A_a v)$; in particular it does not use an approximate matrix. The common denominator is kept fixed.

When $b \geq N$, instead initialize $n = e_1$ and $d = 1$ and use the exact updates

$$(30.10) \quad n \leftarrow M_a n, \quad d \leftarrow 5d.$$

After t commands, $n/d = A_{a_t} \cdots A_{a_1} e_1$ and $d = 5^t$. No history of the commands is retained in either mode.

Upper bound in Theorem 30.1. In the fixed-grid mode, Lemma 30.4 gives legality at every step and

$$\|D(v_N) - D(x_N)\|_F = \frac{\|v_N - x_N\|}{\sqrt{2}} \leq \sqrt{3/2} N 2^{-b} < 2^{-L}.$$

The exact mode is legal and has zero error. In both modes the output is a rational density matrix. For example, if $v = n/d$, one may output

$$(30.11) \quad D(v) = \frac{1}{2d} \begin{pmatrix} d + n_3 & n_1 - in_2 \\ n_1 + in_2 & d - n_3 \end{pmatrix}.$$

The guarantee $\|n\| \leq d$ implies positive semidefiniteness. There is no square root, eigenvalue computation, projection oracle, or physical quantum preparation in the algorithm.

In the fixed-grid mode each coordinate needs $b + O(1)$ bits, including its sign. Every entry of M_a has absolute value at most five and each row has absolute sum at most seven. Thus the temporary integer z_i and the new coordinate also require only $b + O(1)$ bits. A constant number of buffers suffices. Addition, multiplication by these fixed integers, and division of a nonnegative integer by five are executable by binary digit scans with $O(b)$ bit operations and $O(b)$ storage. Truncation of a negative coordinate uses its sign and absolute value; rounding toward negative infinity would not give this algorithm.

In the exact mode $\|n\| = d = 5^t$, so all numerators, the denominator, and intermediate results have $O(t + 1)$ bits. Their updates use $O(t + 1)$ bit operations and storage. Taking $b < N$ for the first mode and $b \geq N$ for the second yields (30.2) and the asserted command time. Formula (30.11) can be written as signed binary integers over one common denominator in $O(b_*)$ time and space. Conversion to a decimal presentation is not needed for the bit-operation bound.

For completeness, parameter handling does not require storing a very large requested precision. Read N and store it in binary. While reading L , maintain its value capped at N by the recurrence $\ell \leftarrow \min\{N, 2\ell + \delta\}$ for each binary digit δ . If the cap is reached, the exact mode is already determined. Otherwise $L < N$ is stored exactly, and (30.8) is computed with $O(\log N)$ auxiliary bits. This also works with any fixed input radix. Reading the entire header takes its actual input length; a straightforward implementation uses $O(\log N)$ work bits and $O(\log N)$ bit operations per header digit. A counter validating that exactly N commands were read costs $O(\log N)$ additional bits, which are within the asserted bound. No command-time external clock is needed. $\square$

Corollary 30.5 (The zero-error and arbitrary-tolerance regimes). *Exact one-pass numerical realization of this fixed rational experiment has bit-space $\Theta(N)$. For every real $0 < \varepsilon \leq 1/4$, a sufficient binary accuracy parameter $L = \lceil \log_2(1/\varepsilon) \rceil$ gives optimal space order*

$$\Theta(\min\{N, \log(N + 1) + \log(1/\varepsilon)\}).$$

Thus the order is logarithmic at fixed positive error and linear once $\log(1/\varepsilon)$ is of order N or larger.

Proof. Use the exact profile 5^N for the first lower bound and (30.10) for its upper bound. For positive tolerance, the lower proof applies directly with that tolerance in (30.6). The chosen L differs from $\log_2(1/\varepsilon)$ by less than one and supplies the upper bound. $\square$

30.4. Rational orthogonal actions. The computational consequence is not restricted to the numerical LPS example. Legal unit-ball outputs allow the same argument for any fixed rational orthogonal alphabet satisfying the two geometric hypotheses used by the entropy theorem.

Corollary 30.6 (Uniform space for rational orthogonal actions). *Let $\mathcal{A} \subset O(\mathbb{Q}^m)$ be a fixed finite alphabet, let G be its compact generated group, and let $x_0 \in \mathbb{Q}^m$ be a unit vector. Assume the full action gap (16.3) and the all-direction cap bound (16.4) with a positive exponent p . The dimension, alphabet, rational denominators, seed, spectral constant and cap constants are fixed independently of N, L . Require numerical decoder values to belong to the closed unit ball of $\mathbb{R}^m$. In the one-pass model, with Euclidean mean error at most 2^{-L} , the optimal uniform bit-space order is*

$$\Theta(\min\{N, L + \log_2(N + 1)\})$$

for $L \geq 2$ and N above a constant depending only on the fixed input. There is no identity or return hypothesis. The upper bound is deterministic with pointwise error. Exact computation has order N .

Proof. The lower proof of Theorem 16.2 uses the legal output set only through $\|D_i\| \leq 1$ in the terminal centroid estimate. It therefore applies unchanged to unit-ball decoders, with $\rho = 1$ and $\zeta = 1 - \varepsilon$. Its finite budget gives

$$N \leq C_0 + C_1 \log k + C_2 \varepsilon k^{2/p} \quad (0 \leq \varepsilon \leq 1/4).$$

The same dichotomy as in (30.6) gives $\log k \geq c \min\{N, \log(N/\varepsilon)\} - C'$. For sufficiently large N this lower bound absorbs its additive constant uniformly, since $\log(N/\varepsilon) \geq \log(4N)$. At zero error the finite budget directly gives $\log k \geq cN - C'$. The complete-configuration reduction then applies.

For the upper bound write $A_a = M_a/D$ and $x_0 = n_0/d_0$ with fixed integer data and positive common denominators. At grid precision b initialize $v_0 = T_b(x_0)$ and perform the integer update $n'_i = \text{sgn}((M_a n)_i) \lfloor |(M_a n)_i|/D \rfloor$. The initial error contributes one additional rounding term, so Lemma 30.4 gives $\|v_N - x_N\| \leq (N + 1)\sqrt{m} 2^{-b}$ and legal outputs at all prefixes. Choose $b = L + \lceil \log_2(N + 1) \rceil + \lceil \log_2 \sqrt{m} \rceil + 2$. Alternatively retain the exact vector with denominator $d_0 D^t$. Since the dimension and integer coefficients are fixed, these modes use respectively $O(b)$ and $O(N + 1)$ bits, including scratch and parameter storage. Choose the smaller mode. The capped parameter scan and fixed-integer digit operations from the preceding proof apply without change. No orbit-covering algorithm or construction of a real transition table is used. $\square$

This corollary requires unit-ball decoders. It does not assert that coordinatewise truncation preserves an arbitrary orbit hull or the positive-semidefinite cone in higher matrix dimension. The Bloch ball is exactly the legal qubit output set, which is why the preceding explicit result gives legal density matrices.

30.5. Relation to finite automata and positive realization. The passage from a configuration to a stochastic label in Lemma 30.2 is elementary, as is directed finite-precision arithmetic. The quantitative content is that the entropy obstruction continues to force optimal uniform bit-space even when arbitrary randomization and unlimited internal running time are allowed. This is not a reinterpretation of free real-table lookup as an implemented algorithm. The upper program is independently specified. The logarithm of width records fewer asymptotic details than width

itself: the present matching space order does not remove the subexponential multiplicative loss in Theorem 29.5.

The invariant-cone characterization of positive realization, stated by Benvenuti–Farina [3, Theorem 2], concerns exact stationary nonnegative realizations of a rational transfer function. Our lower bound instead concerns all-word, horizon-dependent approximation by common stochastic rows. The affine Bloch recursion has constant real linear dimension, but its negative matrix entries do not make it a positive finite-label realization. A finite-bit implementation of that recursion must pay for its accuracy.

Ambainis–Watrous [35, Section 2 and Theorem 1] allow a two-way input head and a fixed-dimensional quantum register in a language recognition problem. Panduranga Rao–Vinay [36, Theorem 1] construct complex-weighted two-way automata for one-sided language recognition. Their transitions are not nonnegative stochastic rows with a legal density-matrix decoder. Neither comparison identifies algebraic amplitudes with finitely many classical configurations. The model here has a one-pass classical input, no physical quantum register, and a numerical output required for every command word. We do not claim a language-recognition separation or an extension to repeated quantum measurements.

31. POSITIVE FINITE-PRECISION MATRIX STATES

Coordinate truncation preserves a Euclidean ball, but not the cone of positive matrices. We give a replacement whose output is a density matrix in every dimension. The correction is explicit; no spectral projection or membership oracle is used in an update. Write

$$\mathcal{D}_d = \{\rho \in \mathbb{C}^{d \times d} : \rho = \rho^*, \rho \geq 0, \text{tr } \rho = 1\}, \quad \|X\|_1 = \text{tr } \sqrt{X^* X}.$$

All dimensions and all physical input data below are fixed independently of the horizon and the precision. Trace norm is not divided by two.

31.1. A fixed-denominator positive rounding map. Fix an integer $B \geq 1$. For $\rho \in \mathcal{D}_d$, form a Hermitian matrix $Q_B(\rho)$ as follows. Truncate the first $d - 1$ diagonal entries and the real and imaginary parts of every strictly upper triangular entry toward zero on the grid $B^{-1}\mathbb{Z}$. Reflect the upper triangle by conjugation, and set the last diagonal entry equal to one minus the sum of the other diagonal entries. Thus $\text{tr } Q_B(\rho) = 1$. Define

$$(31.1) \quad \mathcal{R}_B(\rho) = \frac{BQ_B(\rho) + 2dI}{B + 2d^2}, \quad \delta_B = \frac{6d^2}{B}.$$

Lemma 31.1 (Positive rounding with fixed denominator). *For every $\rho \in \mathcal{D}_d$, the matrix $\mathcal{R}_B(\rho)$ belongs to $\mathcal{D}_d$ and*

$$(31.2) \quad \|\mathcal{R}_B(\rho) - \rho\|_1 \leq \delta_B.$$

Its numerator is a Gaussian-integer Hermitian matrix of trace $B + 2d^2$. If $\rho = P/z$ with $P \in \mathbb{Z}[i]^{d \times d}$, $P \geq 0$ and $z = \text{tr } P > 0$, its numerator is computable by integer multiplication, addition and signed division, using $O_d(\log(B + 1) + \log(z + 1))$ bits of scratch space.

Proof. Set $E = Q_B(\rho) - \rho$. The first $d - 1$ diagonal errors have absolute value less than B^{-1} ; the last is at most $(d - 1)B^{-1}$. Each off-diagonal real or imaginary error has absolute value less than B^{-1} . Consequently

$$\|E\|_F^2 \leq \frac{(d - 1) + (d - 1)^2 + 2d(d - 1)}{B^2} = \frac{3d(d - 1)}{B^2}.$$

In particular $\|E\|_{\text{op}} \leq 2d/B$ and $\|E\|_1 \leq d\|E\|_{\text{op}} \leq 2d^2/B$. Since $\rho \geq 0$, the matrix $Q_B(\rho) + (2d/B)I$ is positive. Its trace is $1 + 2d^2/B$, proving positivity and trace one in (31.1). With

$$\eta = 2d/B,$$

$$\mathcal{R}_B(\rho) - \rho = \frac{E + \eta(I - d\rho)}{1 + d\eta}.$$

The estimate $\|I - d\rho\|_1 \leq 2d$ proves (31.2). The bounds deliberately leave slack.

For rational input, the independent entries of $BQ_B(\rho)$ are $\text{sgn}(a)[B|a|/z]$, where a is the corresponding integer part of P . The last diagonal entry is B minus the other diagonal numerators. Adding $2d$ to each diagonal gives the displayed common denominator. Positivity of P gives $|P_{ij}| \leq \sqrt{P_{ii}P_{jj}} \leq z$. Thus a fixed number of integer buffers of the asserted length suffices. Binary long division uses linear storage and quadratic bit operations in that length. $\square$

The positivity correction cannot simply be omitted. For $\rho = \frac{1}{13} \begin{pmatrix} 4 & 6 \\ 6 & 9 \end{pmatrix}$ and $B = 3$, trace-preserving entry truncation gives $Q_B(\rho) = \frac{1}{3} \begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$, whose determinant is negative.

The map is not a quantum operation and is not asserted to be affine. It is a numerical update on a stored matrix. Its positivity and its uniform error are the only properties needed below.

31.2. The trace-norm estimate for an instrument. An instrument with outcomes in a finite set $\mathcal{Y}$ is a family of completely positive maps $(\Phi_y)_{y \in \mathcal{Y}}$ whose sum is trace preserving. The elementary contraction below is standard quantum-information theory [37, Corollary 3.40]; a proof is included to specify the norm and the treatment of zero outcomes.

Lemma 31.2 (Contraction and positive branch repair). *For every Hermitian X ,*

$$(31.3) \quad \sum_y \|\Phi_y(X)\|_1 \leq \|X\|_1.$$

For positive X , put

$$\overline{\mathcal{R}}_B(X) = \begin{cases} (\text{tr } X) \mathcal{R}_B(X / \text{tr } X), & \text{tr } X > 0, \\ 0, & X = 0. \end{cases}$$

Then $\text{tr } \overline{\mathcal{R}}_B(X) = \text{tr } X$ and

$$(31.4) \quad \|X - \overline{\mathcal{R}}_B(X)\|_1 \leq \delta_B \text{tr } X.$$

Proof. Use the Jordan decomposition $X = X_+ - X_-$, with $X_{\pm} \geq 0$ and $\text{tr } X_+ + \text{tr } X_- = \|X\|_1$. Positivity and the triangle inequality give

$$\sum_y \|\Phi_y(X)\|_1 \leq \sum_y \text{tr } \Phi_y(X_+ + X_-) = \text{tr}(X_+ + X_-).$$

Equation (31.4) is (31.2) multiplied by $\text{tr } X$. A positive matrix of trace zero is zero, so no normalization of a zero-probability branch is required. $\square$

Neither conclusion requires a lower bound on a nonzero outcome probability. Complete positivity is needed for the physical interpretation of the maps; positivity and preservation of the total trace already prove the displayed contraction.

Remark 31.3 (Diagonal coordinates before positive repair). For a density input, the specific Q_B above also has its final diagonal entry in $[0, 1]$: the other diagonal entries are nonnegative and their sum is at most $\sum_{i < d} \rho_{ii} \leq 1$. This scalar observation does not imply positivity of Q_B ; the rank-one example above has an indefinite rounded matrix despite nonnegative diagonal entries. The buffer repairs the matrix, not a missing scalar diagonal bound. The constants in the repair error are convenient upper bounds and are not optimized.

32. ONE-PASS SIMULATION OF DISTURBING QUANTUM INSTRUMENTS

We now allow genuine state disturbance. This is a constructive finite-horizon statement, not a finite-state classification. It is separate from the classification by repeatable nondisturbing probes.

32.1. Fixed physical data and operational error. Fix $d \geq 2$, a finite command set $\mathcal{A}$, finite outcome sets, and an instrument $(\Phi_{a,y})_y$ for each $a \in \mathcal{A}$. Assume a fixed Gaussian-rational Kraus description

$$(32.1) \quad \Phi_{a,y}(X) = q^{-2} \sum_r M_{a,y,r} X M_{a,y,r}^*, \quad M_{a,y,r} \in \mathbb{Z}[i]^{d \times d}, \quad \sum_{y,r} M_{a,y,r}^* M_{a,y,r} = q^2 I$$

with one fixed positive integer q . The initial density matrix is P_0/z_0 , where P_0 is Gaussian integral and positive, and $z_0 = \text{tr } P_0 > 0$. Rationality here refers to this supplied finite description; no assertion that every rational superoperator has a Gaussian-rational Kraus decomposition is needed.

At each step an external classical controller chooses a command as a function of the public history, possibly using its own randomness. The instrument emits y and updates the physical state accordingly. The simulator receives the same kind of command and emits one outcome immediately. It cannot reread consumed commands or emitted outcomes; any information retained privately is charged. A common controller is used when comparing the two experiments. Its memory is not simulator memory, and no access to an unrecorded external history is granted to the simulator.

For a complete control–outcome history h at time t , let A_h be the true subnormalized state, of trace equal to the probability of h . Let $\hat{A}_h$ be the analogous simulator matrix. Define

$$(32.2) \quad \mathfrak{D}_t = \sum_h \|A_h - \hat{A}_h\|_1.$$

This is the trace norm of the difference of the block-diagonal classical–quantum states, not an average conditioned on a rare observed history. Histories of probability zero contribute the zero matrix. Classical transcript total variation is at most $\mathfrak{D}_t/2$. The same bound holds after any common, history-dependent terminal measurement. The compared interface consists of the classical transcript and its numerical matrix blocks. We do not claim total variation of binary matrix encodings, nor simulation of an arbitrary external entangled quantum memory.

Theorem 32.1 (Charged simulation of rational instruments). *For every fixed input (32.1) there is one randomized classical one-pass program which, for all $N \geq 2$ and $L \geq 2$, simulates every N -step adaptive control policy and satisfies*

$$(32.3) \quad \sup_{\text{policies}} \mathfrak{D}_t \leq 2^{-L} \quad (0 \leq t \leq N).$$

Its stored and returned numerical state is always in $\mathcal{D}_d$. Writing

$$s_* = \min\{N, L + \lceil \log_2(N+1) \rceil\},$$

its writable space is $O_{\text{input}}(s_)$, equivalently $O(d^2 s_*)$ with constants allowed to depend on the fixed Kraus data. Command processing uses $O_{\text{input}}(s_*^2)$ expected bit operations and $O_{\text{input}}(s_*)$ expected fair bits per step. Every step terminates almost surely, with a worst-case space bound even during rejection sampling. No exact-probability sampling oracle is used.*

Exact simulation of the entire adaptive experiment, including its conditional numerical matrix states, is possible in $O_{\text{input}}(N)$ writable bits. No positivity bound on outcome probabilities, reversibility, physical return, or spectral gap is assumed. For one-outcome instruments the program is deterministic.

The finite program and physical data are fixed before N, L are supplied. Parameter scanning is charged by its actual input length. The theorem concerns a classical numerical simulation, not preparation of a quantum state. It gives no uniform error bound for normalized states conditional on arbitrary rare histories.

32.2. An integer realization of every step. For approximate simulation choose

$$(32.4) \quad b = L + \lceil \log_2(6d^2(N+1)) \rceil, \quad B = 2^b, \quad m = B + 2d^2.$$

Store $\mathcal{R}_B(P_0/z_0)$ as P/m with an integral Hermitian numerator P . Given command a , form

$$(32.5) \quad S_y = \sum_r M_{a,y,r} P M_{a,y,r}^*, \quad s_y = \text{tr } S_y, \quad T = \sum_y s_y = q^2 m.$$

Emit y with probability s_y/T . Only positive s_y can be emitted. Replace the state by $\mathcal{R}_B(S_y/s_y)$, using the divisions in Lemma 31.1. Its denominator is again exactly m .

These rational probabilities have an explicit fair-bit implementation. Let $\ell = \lceil \log_2 T \rceil$, draw ℓ independent fair bits as an integer $J \in \{0, \dots, 2^\ell - 1\}$, and retry if $J \geq T$. If $T = 1$ take $J = 0$ without drawing a bit. Accept the unique outcome whose cumulative interval of integer lengths $(s_y)_y$ contains J . Acceptance probability is greater than $1/2$, except that for a power of two it is one. Thus the expected number of bits is at most 2ℓ . Discarded trials are not stored, and their count is not needed. Zero-length intervals can never be selected.

There is also an exact mode. Start from P_0 and, on each command, form $(S_y)_y$ by (32.5), now with $T = q^2 \text{tr } P$. Sample the same integer intervals and set $P \leftarrow S_y$. The numerical state is $P/\text{tr } P$. This update is precisely the normalized physical branch, while the numerator itself is never divided. After t commands,

$$(32.6) \quad 1 \leq \text{tr } P \leq z_0 q^{2t}.$$

It is an integer because all input matrices are Gaussian integral. The upper bound follows from $\text{tr } S_y \leq \sum_y \text{tr } S_y = q^2 \text{tr } P$. All entries have modulus at most $\text{tr } P$ by positivity.

32.3. Uniform control of the whole path.

Proof of Theorem 32.1. In approximate mode, each stored state is legal by Lemma 31.1, and every sampling row has total mass one by (32.1). For an arbitrary common controller, write $\pi(a | h)$ for its next command law. The true and approximate subnormalized updates are

$$A_{hay} = \pi(a | h) \Phi_{a,y}(A_h), \quad \hat{A}_{hay} = \pi(a | h) \overline{\mathcal{R}}_B(\Phi_{a,y}(\hat{A}_h)).$$

This expression agrees with the actual emitted probabilities and normalized successor matrices. For each history, the triangle inequality and Lemma 31.2 imply

$$\sum_{a,y} \|A_{hay} - \hat{A}_{hay}\|_1 \leq \|A_h - \hat{A}_h\|_1 + \delta_B \text{tr } \hat{A}_h.$$

Summing over histories and using $\sum_h \text{tr } \hat{A}_h = 1$ gives $\mathfrak{D}_{t+1} \leq \mathfrak{D}_t + \delta_B$. Initial rounding contributes $\mathfrak{D}_0 \leq \delta_B$. Hence

$$\mathfrak{D}_t \leq (t+1)\delta_B \leq (N+1)6d^2/B \leq 2^{-L}.$$

This proof works uniformly for every controller. A controller with private randomness may be conditioned on its seed, followed by integration; the displayed bound is independent of that seed. Taking traces gives the classical total-variation assertion. Appending a terminal instrument and applying (31.3) gives the assertion about terminal measurements. No division by a small target probability has appeared.

In grid mode, the numerator P and the denominator m have $O_{\text{input}}(b+1)$ bits per entry. The fixed integral Kraus products, the outcome traces, the cumulative intervals, and the integer J have the same order. The numerators in the signed divisions have at most $2b + O_{\text{input}}(1)$ bits. A constant number, depending on the fixed physical data, of matrix buffers suffices. Elementary long division gives quadratic bit time and linear scratch space. Rejection sampling uses at most $O_{\text{input}}(b+1)$ space in every trial, and has the stated expected bit and time bounds.

Exact mode reproduces every branch probability and conditional state exactly. Equation (32.6), together with the fixed Kraus coefficients, bounds all stored and temporary integers by $O_{\text{input}}(t+1)$ bits. Thus the exact simulation uses linear space and has the stated conservative quadratic time bound per step.

Use grid mode when $b < N$ and exact mode otherwise. The choice is made before allocating B : store N , scan L with its value capped at N , and compute the threshold with binary integer lengths. A precision string of arbitrary length is consumed without being stored. A counter verifying the number of commands uses $O(\log N)$ bits. The fixed-radix numerical matrix output has $O_{\text{input}}(s_*)$ bits and requires no readable transcript tape. These observations complete the charged-space proof. $\square$

Corollary 32.2 (Rational channels with legal matrix outputs). *A fixed finite family of Gaussian-rational Kraus channels and a fixed rational initial density matrix admit one deterministic one-pass algorithm with pointwise trace-norm error at most 2^{-L} at every prefix, legal outputs, and space $O_{\text{input}}(\min\{N, L + \log(N+1)\})$. Exact output uses $O_{\text{input}}(N)$ bits.*

Proof. There is one outcome per command, so its probability is one and no random bits are needed. The single block in (32.2) is the difference of the two numerical matrices. Apply Theorem 32.1. $\square$

The contraction and hybrid argument are classical. Their use here is to give a fully specified positive integer simulator with bounded space, exact rational emissions and no rare-outcome assumption. This constructive theorem does not replace the repeatable-probe hypothesis in the finite-response-quotient classification.

33. SHARP SPACE WITH HIGHER-DIMENSIONAL DENSITY-MATRIX OUTPUTS

Positive rounding removes the restriction of the uniform upper construction to a Bloch ball. We retain the spectral hypotheses in the converse; none is needed by the algorithm.

Theorem 33.1 (Sharp matrix-output space). *Fix a finite family of Gaussian-rational unitary matrices in $U(d)$ and a Gaussian-rational pure initial state ρ_0 . Let G be the compact closure of their conjugation action on the real Euclidean space*

$$E = \{H \in \mathbb{C}^{d \times d} : H = H^*, \text{tr } H = 0\}, \quad \|H\| = \|H\|_F.$$

Assume a probability law supported on the given commands has a full Koopman action gap on the unit sphere of E , off all invariant functions, and an all-direction cap estimate (16.4) of exponent $p > 0$. The dimension, matrices, seed, rational denominators, cap constants and spectral constant are fixed independently of N, L .

For a single fixed-program classical one-pass simulator, assume almost-sure termination between commands and at final output, and require a legal numerical output $\hat{\rho}(w) \in \mathcal{D}_d$ on every run and only the mean condition

$$(33.1) \quad \sup_{|w|=N} \|\mathbb{E}\hat{\rho}(w) - U_w \rho_0 U_w^*\|_F \leq 2^{-L}.$$

For every such fixed program, a matching space lower bound holds for $L \geq 2$ and all sufficiently large N , with constants and threshold allowed to depend on that program and the fixed physical data. A single uniform algorithm attains the upper bound. In this fixed-program asymptotic sense, the optimal writable bit-space has order

$$(33.2) \quad \Theta_{\text{input}}(\min\{N, L + \log_2(N+1)\}).$$

The lower bound permits arbitrary randomization and unlimited internal running time. A deterministic integer program attains the upper bound with pointwise trace-norm error. Exact computation has order N . No idle or return assumption is made.

Proof. Put $c = I/d$ and $r_d = \sqrt{(d-1)/d}$. For every density matrix D ,

$$\|D - c\|_F^2 = \text{tr}(D^2) - 1/d \leq r_d^2;$$

equality holds for a pure state. Thus the physical target $X_w = (U_w \rho_0 U_w^* - c)/r_d$ is unit, and every normalized legal decoder $(D - c)/r_d$ has norm at most one.

The complete-configuration reduction of Lemma 30.2 works with $\mathcal{D}_d$ in place of $\mathcal{D}_2$: almost-sure termination gives stochastic rows, and conditional means of legal outputs remain legal by compact convexity. The induced rows may depend on the public boundary index, which only enlarges the lower-bound comparison class. The proof of Theorem 16.2 uses the decoder set only through the unit-norm bound. With error ε/r_d and $0 \leq \varepsilon \leq 1/4$, its finite budget therefore gives

$$(33.3) \quad N \leq C_0 + C_1 \log k + C_2 \varepsilon k^{2/p}$$

for a simulator with at most k complete private configurations at every positive boundary. Constants depend only on the fixed data. Indeed the residual mass is $1 - \varepsilon/r_d$, bounded away from zero uniformly in this error range.

For $N \geq 2C_0$, either $\log k \geq N/(4C_1)$ or $k^{2/p} \geq N/(4C_2\varepsilon)$. Taking logarithms yields

$$\log_2 k \geq c \min\{N, L + \log_2(N+1)\} - C'.$$

Since the minimum tends to infinity with N , uniformly for $L \geq 2$, the additive constant is absorbed above a fixed physical-data-dependent horizon. For a fixed finite program, the complete-configuration count is at most $C(s+1)^{a2^{bs}}$ when s work bits are used, with fixed machine-dependent constants. After absorbing the fixed finite-control factor C , this proves the work-space lower bound above a program-dependent horizon. No bound uniform over arbitrarily large uncharged finite programs is asserted. At zero error (33.3) gives $\log k \geq cN - C'$.

Regard each unitary as a one-outcome rational Kraus instrument. Corollary 32.2 supplies the deterministic trace-norm upper bound and the exact fallback. Trace norm dominates Frobenius norm. The algorithm does not use the spectral or cap constants, nor construct a stochastic orbit-hull table. $\square$

Corollary 33.2 (An explicit family in every fixed dimension). *For every fixed $d \geq 2$, take the adjacent-coordinate rational unitaries displayed in Section 17, their inverses and the identity, and take $\rho_0 = e_1 e_1^*$. Legal numerical density matrix simulation satisfies (33.2), with dimension-dependent constants.*

Proof. The density argument in that section shows that the generated compact group is $SU(d)$. All entries are algebraic. The Bourgain–Gamburd theorem [16], alternatively the Benoist–de Saxcé extension [38, Theorem 1.2], gives a regular-representation spectral gap for this fixed dense algebraic set. Symmetrization and laziness give a contraction norm strictly less than one. The identity is already an allowed command, so this is an averaging measure on actual letters. Peter–Weyl decomposition transfers that norm to the action off its invariant functions. For conjugation on traceless Hermitian matrices, the fixed space is zero. The least orbit dimension calculation in Theorem 24.1 gives $p_* = 2(d-1)$ and its uniform cap estimate. Theorem 33.1 applies. $\square$

This corollary does not supply a numerical spectral constant uniformly in d . It uses an explicit rational alphabet, not an unspecified existence argument. The construction for a given alphabet is effective without knowing its spectral gap; that gap is a premise of the lower proof only.

Corollary 33.3 (Sharpness for an instrument family containing an expanding unitary subsystem). *Suppose the fixed rational instrument family of Theorem 32.1 contains the unitary commands and pure seed of Theorem 33.1. Among simulators that additionally return a legal numerical matrix and satisfy the classical–quantum error requirement (32.3), the optimal space has order (33.2).*

Proof. The instrument simulator gives the upper bound. For the lower bound, restrict the controller to a deterministic word of unitary commands. Their outcome sets are singletons, so the error requirement implies (33.1). Apply the preceding theorem. This is not a lower bound for transcript-only simulators without a legal numerical matrix interface. $\square$

34. STRICT CUTPOINTS AND NUMERICAL SIMULATION

State cost for recognition and numerical precision are different invariants. Recent results make it necessary to state the distinction at theorem level, rather than relying on the words “classical simulation.”

Chen–Wu [39, Definition 2.1 and Theorems 3.2, 4.4] define simulation by equality of strict-cutpoint languages, allowing the simulator’s cutpoint to change. Their August 2026 version gives the exact worst-case cost $n^2 + 1$ for n -state one-way general quantum automata, $n \geq 2$. In [40, Theorems 11, 14 and Corollary 15] they give order cq^2 for $c, q \geq 2$ in a model with c classical states and a q -dimensional quantum register, and order n^2 for the measure-once model. These are relevant positive results. The first paper’s current title is *The State Cost of Classical Simulation of One-Way General Quantum Finite Automata*; its version and exact bound, rather than the earlier title or only its quadratic order, are used here.

Their stochasticization preserves a sign, not the numerical acceptance function. In particular, [39, Proposition 3.1] contains an attenuation factor $\delta\eta^{|w|} > 0$ multiplying a zero-cutpoint linear output. It can tend to zero exponentially without changing the recognized language. The following elementary construction isolates this point.

Proposition 34.1 (Attenuation preserves a strict cutpoint). *Let a probabilistic automaton have acceptance function $f(w)$, and fix $0 < \beta < 1$ and $0 \leq \lambda < 1$. Adding two absorbing states gives an automaton with acceptance function*

$$(34.1) \quad f_\beta(w) = \lambda + \beta^{|w|}(f(w) - \lambda).$$

It recognizes the same strict-cutpoint language at λ . If some length- N word has $|f(w) - \lambda| \geq c > 0$, its numerical error on that word is at least $(1 - \beta^N)c$.

Proof. On every letter, each old state follows its original stochastic row with total probability β . The remaining mass goes to an absorbing accepting state with probability $(1 - \beta)\lambda$ and to an absorbing rejecting state with probability $(1 - \beta)(1 - \lambda)$. The old terminal functional is unchanged. After t letters the surviving old distribution is β^t times the original distribution, while the absorbed accepting mass is $\lambda(1 - \beta^t)$. This proves (34.1). The factor $\beta^{|w|}$ is positive, proving language equality. Subtracting $f(w)$ proves the last assertion. $\square$

A legal density-matrix approximation gives a numerical guarantee for every supplied effect: if $0 \leq E \leq I$ and $\rho, \sigma \in \mathcal{D}_d$, then

$$|\operatorname{tr} E(\rho - \sigma)| \leq \frac{1}{2} \|\rho - \sigma\|_1.$$

Indeed the positive and negative parts of $\rho - \sigma$ have equal trace, and an effect takes a value between zero and that trace on either part. Therefore an approximation preserves a threshold decision whenever its numerical error is strictly smaller than the distance from the cutpoint. Without a margin, numerical approximation alone does not give exact language equivalence.

The prepare–test and sign-rank methods in [40] concern finite prefix–suffix sign patterns. Such patterns are unchanged by positive attenuation. Our entropy converse instead requires a calibrated residual mean and a legal convex decoder; attenuation can destroy that calibration. Conversely, its width lower bound cannot be transferred to unrestricted cutpoint-language simulation. Constant real linear dimension, fixed PFA state count and finite writable bit-space are not identified with one another. No quantum language separation, resolution of the Chen–Wu state-cost problem,

or unconditional lower bound for arbitrary noisy channels is claimed. The new positive process construction is an upper theorem for a different, explicitly specified numerical and transcript interface.

35. RATIONAL INSTRUMENTS STABLE UNDER EXTERNAL REFERENCES

A rounding operation on density matrices need not be affine. In particular, the map of Lemma 31.1 cannot be tensored with the identity on an unknown reference system. We therefore round a *description of the instrument*, once and independently of its input, rather than round the state on which it acts. The resulting maps are completely positive and jointly trace preserving. All constants below are explicit in the input and output dimensions and the number of outcomes.

35.1. An integral repair of Choi data. Write M_d for the complex matrices of order d . An instrument from M_d to M_n with m outcomes is a family of completely positive maps Φ_y whose sum is trace preserving. We use the input-first, *unnormalized* Choi convention

$$(35.1) \quad J_y = \sum_{i,j=1}^d |i\rangle\langle j| \otimes \Phi_y(|i\rangle\langle j|), \quad J_y \geq 0, \quad \sum_{y=1}^m \text{Tr}_n J_y = I_d.$$

Here Tr_n traces the output factor; in particular $\sum_y \text{tr} J_y = d$, not one. These are the classical Choi criteria [37, Theorems 2.22 and 2.26]. The channel with the outcome retained is $\mathcal{F}(X) = \bigoplus_y \Phi_y(X)$. We write $\|\cdot\|_\diamond$ for the completely bounded trace norm.

Lemma 35.1 (Choi trace norm and reference stability). *For any linear map $\Lambda : M_d \rightarrow M_n$ with unnormalized Choi matrix $J(\Lambda)$,*

$$d^{-1} \|J(\Lambda)\|_1 \leq \|\Lambda\|_\diamond \leq \|J(\Lambda)\|_1.$$

Consequently the diamond distance of two instruments, with the outcome retained, is at most the sum of the trace norms of their Choi-block differences.

Proof. Applying $\text{id}_d \otimes \Lambda$ to the normalized maximally entangled state gives the lower bound. For the upper bound, let u, v be unit vectors in $\mathbb{C}^r \otimes \mathbb{C}^d$. With $\Omega = \sum_i e_i \otimes e_i$, write $u = (A \otimes I_d)\Omega$ and $v = (B \otimes I_d)\Omega$. The matrices $A, B : \mathbb{C}^d \rightarrow \mathbb{C}^r$ have Hilbert–Schmidt norm one and therefore operator norm at most one. Thus

$$(\text{id}_r \otimes \Lambda)(|u\rangle\langle v|) = (A \otimes I_n)J(\Lambda)(B^* \otimes I_n)$$

has trace norm at most $\|J(\Lambda)\|_1$. The trace-norm unit ball is the convex hull of rank-one partial isometries (including zero as a convex combination of opposite such matrices). A singular-value decomposition of an arbitrary input matrix therefore proves the same induced trace-norm bound, independently of r . Taking the supremum over r proves the assertion. For an instrument the Choi matrix is, up to a fixed permutation of factors, the direct sum of its blocks. $\square$

Let B be a positive integer. The supplied approximations T_y are Hermitian matrices with entries in $B^{-1}\mathbb{Z}[i]$. Assume that every independent real diagonal coordinate, and every independent real and imaginary off-diagonal coordinate, differs from that of J_y by at most B^{-1} . The T_y need not be positive or satisfy the marginal constraint. Put

$$(35.2) \quad R = I_d - \sum_y \text{Tr}_n T_y, \quad Q_y = T_y + \mathbf{1}_{\{y=1\}} R \otimes |e_1\rangle\langle e_1|,$$

$$(35.3) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad \tilde{J}_y = \frac{BQ_y + cI_{dn}}{D}.$$

The choice of the first outcome and first output coordinate is merely a fixed convention. Any fixed outcome and output basis coordinate give the same legality and uniform error bound. If specified

zero outcomes must remain exactly zero, remove them before applying the theorem and restore them afterwards; m then counts only the repaired outcomes.

Theorem 35.2 (Reference-stable rational instrument rounding). *For every $d, n, m, B \geq 1$, the matrices in (35.3) are Choi matrices of an instrument. Their numerators are positive semidefinite matrices with Gaussian-integer entries, their common denominator is D , and*

$$(35.4) \quad \|\mathcal{F} - \tilde{\mathcal{F}}\|_{\diamond} \leq \sum_y \|J_y - \tilde{J}_y\|_1 \leq \frac{3mndc}{B + mnc} \leq \frac{K(d, n, m)}{B}, \quad K(d, n, m) = 6mn^2d^2(1 + mn).$$

The construction uses integer additions, multiplication by specified integers, and the supplied grid entries. It uses no spectral factorization, positivity projection, or rational Kraus factorization. Its complete output description has

$$O(md^2n^2 \log(B + mnc + 1))$$

bits, with an absolute implicit constant for the fixed encoding.

Proof. Set $E_y = T_y - J_y$. Counting the real coordinates gives $\|E_y\|_{\text{op}} \leq \|E_y\|_F \leq 2nd/B$. For a Hermitian E , the inequalities $-\|E\|_{\text{op}}I \leq E \leq \|E\|_{\text{op}}I$ imply $\|\text{Tr}_n E\|_{\text{op}} \leq n\|E\|_{\text{op}}$. It follows that

$$\|R\|_{\text{op}} \leq 2mn^2d/B, \quad \|Q_y - J_y\|_{\text{op}} \leq c/B.$$

Since $J_y \geq 0$, the numerator $BQ_y + cI_{dn}$ is positive. All its entries are Gaussian integers. The partial trace constraint is exact, because

$$\sum_y \text{Tr}_n(BQ_y + cI_{dn}) = (B + mnc)I_d.$$

The Choi criteria therefore give a completely positive instrument, including when some of the original outcomes are zero.

Let $F_y = Q_y - J_y$. The identity

$$D(\tilde{J}_y - J_y) = BF_y + cI_{dn} - mncJ_y$$

and $\|F_y\|_1 \leq dnc/B$ give

$$D \sum_y \|\tilde{J}_y - J_y\|_1 \leq mndc + mndc + mnc \sum_y \text{tr } J_y = 3mndc.$$

Lemma 35.1 proves the norm assertion. Finally each diagonal entry of a positive numerator is at most D , by the summed partial trace constraint. Positivity bounds every off-diagonal modulus by the geometric mean of two diagonal entries, hence also by D . Encoding $m(dn)^2$ entries and the common denominator proves the bit bound. All intermediate entries have polynomially bounded bit length in the stated parameters and the supplied data. $\square$

For exact rational input, directed truncation supplies T_y by integer division. For uncertain real input, the hypothesis is a certified error bound on the *final grid coordinates*; an uncertified floating estimate is not a premise of the theorem. Although B may be a power of two, $D = B + mnc$ generally is not. The output is rational, not in general dyadic. The construction may replace a zero outcome by a small nonzero one; its mass is charged in (35.4). Known zero outcomes can instead be removed before rounding and restored as zero blocks afterwards, with m counting the remaining outcomes.

Corollary 35.3 (A finite-description instrument net). *For fixed d, n, m and $0 < \delta \leq 1$, the preceding construction yields a finite diamond- δ net of legal instruments. Each member has an integer description of length*

$$O(md^2n^2 \log(2 + K(d, n, m)/\delta + mnc)).$$

This is a deterministic description net, not a statistically learned net. A particular member is computable in polynomial bit time from rational input data and the grid precision; enumeration of the entire net is not required.

Proof. Every Choi diagonal is at most one by (35.1), and positivity bounds every entry modulus by one. There are finitely many possible directed grid truncations at $B = \lceil K/\delta \rceil$. Apply the theorem to those arising from valid instruments. Computation of one truncation and its repair uses a polynomial number of elementary integer operations on polynomial-length operands. $\square$

The CP/TP constraints and adding a scalar identity buffer are not new principles. Projected least-squares process tomography already repairs Choi estimates and uses identity mixing to finish a TP-preserving positivity correction [42, Section 4]. Here the particular integer formula supplies an explicit denominator and a deterministic diamond-error budget from coordinate errors. No nearest-projection, statistical sample-complexity, or rank-optimality claim is made.

35.2. Adaptive use and stopped observations. A tester may retain an arbitrary finite-dimensional quantum memory, start with a state entangled across that memory and the instrument input, and insert arbitrary channels and finite-outcome measurements between uses. The command port is classical: its value may be selected from the public history and measured tester memory. Coherent access to a superposition of commands is not an interface assumed here. The *same* tester is used in the two compared experiments.

Theorem 35.4 (Adaptive reference-stable precision budget). *Suppose that at use t , uniformly over permitted commands and public histories, $\|\mathcal{F}_{t,a} - \tilde{\mathcal{F}}_{t,a}\|_{\diamond} \leq \delta_t$. For every common tester with a classical command port (not coherent superpositions of commands), every initial state, and every public stopping rule $\tau \leq N$, the final states, with the stopped history and tester memory retained, satisfy*

$$(35.5) \quad \|\Omega_{\tau} - \tilde{\Omega}_{\tau}\|_1 \leq \sum_{t=1}^N \delta_t.$$

The bound is independent of the tester-memory dimension. In particular, rounding each command by Theorem 35.2 with $B \geq NK(d, n, m)2^L$ gives stopped joint-state error at most 2^{-L} and transcript total-variation error at most 2^{-L-1} .

Proof. Replace the instruments one use at a time. Just before the replaced use, both members of a neighboring hybrid pair have the same joint state. Decomposing a classical command/history register into blocks, the diamond bound applies to each positive block and is multiplied by its trace. These traces sum to one. The error introduced at this use is therefore at most δ_t , including any quantum reference. All subsequent operations in the hybrid pair are the same channels; complete trace-norm contraction bounds their final difference by the same quantity. Summing the N hybrid differences proves (35.5). For a stopping rule retain an absorbing halt flag, use the identity on halted blocks, and finish all experiments at use N . A public history-dependent stopping rule is then part of the same tester. The same argument applies, without conditioning on a possibly rare stopping history. Taking traces and dividing by two gives the total-variation assertion. $\square$

The hybrid principle is classical; see [37, Section 3.3, equation (3.306)] and the operational strategy norms of [43]. The conclusion here is a uniform finite-description approximation by *genuine instruments*. It is not a classical device that accepts an unknown physical entangled state and prepares its output. A numerical simulation of a specified joint state must charge the joint matrix dimension. Nor does the nonlinear state repair in the preceding section become a channel by this theorem.

Proposition 35.5 (Posterior control at a stopping time). *Let $A_h = p_h \rho_h$ and $\hat{A}_h = q_h \hat{\rho}_h$ be the positive blocks at a common bounded stopping time, with both total traces one. At zero simulated mass assign any density matrix to $\hat{\rho}_h$. If $\sum_h \|A_h - \hat{A}_h\|_1 \leq \delta$, then*

$$(35.6) \quad \sum_h p_h \|\rho_h - \hat{\rho}_h\|_1 \leq 2\delta, \quad \Pr_p\{\|\rho_h - \hat{\rho}_h\|_1 > \eta\} \leq \min\{1, 2\delta/\eta\}.$$

The tail estimate holds for every $\eta > 0$. For any event of true probability $\alpha > 0$, the two pooled conditional states have distance at most $\min\{2, 2\delta/\alpha\}$, with an arbitrary conditional state when the simulated event has probability zero.

Proof. Writing $e_h = \|A_h - \hat{A}_h\|_1$, trace contraction gives $|p_h - q_h| \leq e_h$. For positive p_h ,

$$p_h \|\rho_h - \hat{\rho}_h\|_1 \leq e_h + |p_h - q_h| \leq 2e_h.$$

The inequality remains valid if $q_h = 0$. Sum and apply Markov's inequality. Applying the same calculation to the sums over an event gives the final assertion; two density matrices have distance at most two. Zero true mass makes no contribution. $\square$

The proposition also applies to the numerical history metric of Theorem 32.1. For its stopped version, freeze every halted block and apply the same contraction and mass-weighted repair argument only to live blocks. At each use their total mass is at most one, so the bound remains $(N+1)\delta_B$. Thus the theorem provides a high-probability posterior guarantee at a chosen stopping time without a probability floor or a union bound over all times. It does not assert simultaneous accuracy of every rare normalized posterior. This distinction is relevant when comparing finite-bit history approximation with numerical quantum filtering [41, Section II, equations (10)–(13)].

36. UNIFORM STREAMING FROM RATIONAL CHOI DESCRIPTIONS

The preceding instrument description is not supplied with a Kraus factorization over the Gaussian rationals. Computing such a factorization is unnecessary for classical numerical trajectories. We now allow the matrix dimension and the entire rational instrument to be input to one program, rather than fixed constants of that program.

36.1. Description and exact update. Fix an encoding of finite lists of signed binary integers. A valid description consists of $d \geq 2$, a positive semidefinite initial matrix with Gaussian-integer entries P_0 with trace $z_0 > 0$, and a finite command list. For each command a it gives a positive integer D_a and positive semidefinite matrices with Gaussian-integer entries $C_{a,y}$ of order d^2 such that

$$(36.1) \quad \sum_y \text{Tr}_d C_{a,y} = D_a I_d.$$

A zero outcome is simply a zero Choi matrix. The associated instrument has input-first, unnormalized Choi blocks $C_{a,y}/D_a$ and initial state P_0/z_0 . Let S be the total encoded length, including dimensions, delimiters, command identifiers and all matrix entries. The entire description is retained and charged. In particular the longest identifier is at most S bits and bounds the parser's temporary token store; a free external read-only description would define a different space model. For a stored numerator P define

$$(36.2) \quad (S_{a,y})_{\alpha\beta} = \sum_{i,j=1}^d (C_{a,y})_{i\alpha,j\beta} P_{ij}.$$

The indices in this formula are important: no conjugation or transposition of P_{ij} is inserted.

Lemma 36.1 (Integral Choi trajectory update). *In the input-first, unnormalized convention $J(\Phi) = \sum_{i,j} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|)$, for positive semidefinite P with Gaussian-integer entries the matrices in (36.2) are positive semidefinite matrices with Gaussian-integer entries and*

$$\sum_y \text{tr } S_{a,y} = D_a \text{tr } P.$$

The conditional outcome probabilities are $\text{tr } S_{a,y} / (D_a \text{tr } P)$, and the next normalized state is $S_{a,y} / \text{tr } S_{a,y}$ on a positive-mass outcome.

Proof. The Choi convention gives $S_{a,y} = D_a \Phi_{a,y}(P)$, so positivity follows from complete positivity, and integrality follows from the entry formula. Taking traces and using (36.1) gives the identity. The scalar factors D_a and any accumulated previous factors cancel from both probabilities and normalized states. A positive matrix of trace zero is zero and is never sampled. $\square$

Put $D_* = \max_a D_a$ and, using logarithms to base two in this section,

$$(36.3) \quad \begin{aligned} H &= \lceil \log_2(D_* + 1) \rceil, & \ell_0 &= \lceil \log_2(z_0 + 1) \rceil, \\ Q &= \ell_0 + NH, & b &= L + \lceil \log_2(6d^2(N + 1)) \rceil, & u &= \min\{Q, b\}. \end{aligned}$$

The description, once validated, may be kept read-only; charging its entire length S is included in the bound below.

Lemma 36.2 (Reduced Schur entries and bit length). *Let A be a Hermitian matrix of order q with Gaussian-integer entries, whose real and imaginary parts have absolute value at most 2^h . In exact Schur elimination, let I be the ordered indices of the positive pivots already eliminated. Every remaining entry is*

$$(36.4) \quad S_{ij} = \frac{\det A[I \parallel i, I \parallel j]}{\det A[I, I]},$$

where $I \parallel i$ appends i in that order, and the empty principal determinant is one. Reduced numerators and denominators have $O(q(h + \log(q + 1)))$ bits. Hermitian positive semidefiniteness is therefore decidable by this elimination in polynomial bit time and space in the explicit input length. No optimized polynomial exponent is asserted.

Proof. When the leading pivot t is positive, block congruence reduces

$$\begin{pmatrix} t & b^* \\ b & C \end{pmatrix} \geq 0 \iff C - bb^*/t \geq 0.$$

A negative pivot rejects. At a zero diagonal, each 2×2 principal minor forces the corresponding row and column to vanish; otherwise reject, and if they vanish delete that row and column. Positive pivots make $A[I, I]$ positive definite. The bordered determinant identity

$$\det \begin{pmatrix} A[I, I] & A[I, j] \\ A[i, I] & A_{ij} \end{pmatrix} = \det A[I, I] (A_{ij} - A[i, I] A[I, I]^{-1} A[I, j])$$

proves (36.4), including after zero rows are skipped. The specified ordering prevents a spurious permutation sign.

A minor of order $k \leq q$ has determinant modulus at most $k^{k/2}(\sqrt{2}2^h)^k$, by the product of its row norms. Its real and imaginary parts are integers with $O(k(h + \log(k + 1)))$ bits. The principal denominator is a positive integer. Cancelling common factors cannot enlarge these lengths. Each arithmetic operation on reduced Gaussian-rational entries thus has polynomial operand size. There are polynomially many operations and at most q^2 stored entries; ordinary multiplication, division and the Euclidean algorithm give the stated bit bounds. For rational rather than integral input, first clear a common positive denominator. Its bit length is at most the sum of the encoded denominator lengths, so this preliminary step also has polynomial cost. $\square$

Theorem 36.3 (Variable-description numerical simulation). *There is a single classical randomized program with the following property. Given any valid description above and $N, L \geq 2$, it consumes an adaptive command stream once, emits each outcome immediately, and retains a legal numerical density matrix. Against every common classical adaptive controller, its subnormalized history error at every prefix, and at every common public stopping time bounded by N , is at most 2^{-L} .*

Description validation has polynomial bit time and polynomial working space in S . After validation, the maximum writable space, including temporary arithmetic and the stored description, is

$$(36.5) \quad O(S + d^2[H + \log(d + 1) + u] + \log(N + 1)),$$

with an absolute implicit constant for the fixed machine encoding. The total peak space is the maximum of this bound and the stated polynomial preprocessing bound. Expected bit time per command is $\text{poly}(S)(1 + u)^2$ and expected fresh fair-bit consumption is $O(H + u + \log(d + 1))$. The space bound holds on every rejection trial; running time is not bounded on every such execution. The theorem requires neither a spectral gap nor a Kraus factorization.

Proof. In grid mode use $B = 2^b$ and retain the integer numerator of $R_B(\rho)$ from Lemma 31.1; its trace is always $B + 2d^2$. At a command compute the branch integers in (36.2), sample an outcome from their exact integer traces, and repair the chosen normalized branch back to the same state denominator. The sampler uses fresh binary words of length $\lceil \log_2(D_a \text{tr } P) \rceil$, rejecting words outside the required integer interval. Its acceptance probability is greater than one half, except for the power-of-two case where it is one. Rejected words are discarded, not recorded. Branch weights can be computed sequentially; a second scan finds and recomputes the selected branch. Thus the number of retained matrices does not grow with the number of outcomes.

The proof of Theorem 32.1 uses positivity and (36.1), not rational Kraus entries. It therefore applies verbatim at the level of its stated block recursion: the initial error is at most $6d^2/B$, and each subsequent use adds at most $6d^2/B$, weighted by total simulated branch mass. It gives

$$\sum_h \|A_h - \hat{A}_h\|_1 \leq (t + 1)6d^2/B \leq 2^{-L} \quad (t \leq N).$$

The outcome probabilities used here are computed from the approximate stored state, not from the unknown exact state. The block recursion accounts for that difference. Freezing halted branches proves the same stopped bound as in Proposition 35.5.

In exact mode retain the selected unnormalized matrix without repair. This numerator represents a path block up to omitted common scalar denominators. Those factors cancel in every conditional branch probability and normalized output; they are not reconstructed or stored merely to restore a redundant normalization. Positivity and the trace identity imply

$$1 \leq \text{tr } P_t \leq z_0 D_*^t < 2^{\ell_0 + tH}.$$

Every entry modulus is bounded by the trace. This mode has zero approximation error. Choose it when $Q \leq b$, and choose grid mode otherwise. The switch is an order bound, not an exact minimization of small-instance memory.

For space accounting, each Choi numerator entry has modulus at most D_a , by positivity and (36.1). Each summand in (36.2) is a product of integers of lengths at most H and $u + O(\log(d + 1))$. An accumulator over d^2 summands needs at most $H + u + O(\log(d + 1))$ bits. A constant number of $d \times d$ matrices, traces, indices, division buffers and one rejection word suffice. Initial grid rounding reads the initial entries sequentially; its extra operand length ℓ_0 is at most the charged input length S . Counters and address arithmetic give (36.5). Classical multiplication and long division on w -bit integers take $O(w^2)$ bit operations and $O(w)$ scratch. Scanning the explicit outcome list a fixed number of times gives the claimed polynomial factor in S ; powers of H are

absorbed there. The sampling bound follows from the rejection interval length and the expected number of trials, which is below two. A single nonzero branch requires no random bits.

Validation uses the reduced-rational Schur procedure of Lemma 36.2, not the unreduced fixed-dimensional scaled recursion. Its ratios-of-minors formula proves polynomial operand lengths. Hermitian symmetry, the exact partial trace identities and positive initial trace are checked separately. No pivot permutation is needed: a zero leading diagonal of a positive semidefinite matrix forces its entire row and column to vanish. This establishes a separate polynomial preprocessing bound, not a validator matching the subsequent streaming-space bound.

Parameters are parsed before commands. The precision L may be read with saturation at Q : once that threshold is reached, exact mode is selected and its exact value is unnecessary. The time to read all parameter digits is charged as preprocessing in their actual length; no equally long precision integer is retained. A binary counter checks the command count. The numerical output consists of signed binary integer entries and their common denominator; writing it requires no additional matrix-sized precision store. $\square$

Corollary 36.4 (Known joint systems and two error budgets). *Suppose the entire joint system to be simulated, including a specified reference and all specified controller operations, has a rational Choi description on dimension d_{joint} . Theorem 36.3 applies with that dimension and the full description length charged. If its instruments were first compiled by Theorem 35.2, the reference-stable description error and the numerical history error add. In particular, allocating $\varepsilon/2$ to each gives total history error at most ε .*

Proof. Use the joint Choi maps and joint initial state as input to the uniform program. Theorem 35.4 bounds the first replacement, and Theorem 36.3 bounds the numerical simulation of the replaced experiment. The triangle inequality gives the sum. An unspecified quantum tester remains covered by the *instrument-description* comparison only; a classical numerical simulation of it is not thereby supplied at no cost. $\square$

Theorem 36.3 is an upper bound with variable data. The lower bounds of Theorem 33.1 retain their own fixed-program quantifiers, numerical matrix interface, and full-action spectral and cap hypotheses. No dimension-uniform spectral constant, no sharp lower bound for arbitrary noisy transcript-only interfaces, and no classification of disturbing instruments follows from the upper construction. Conversely, a rational Choi input is enough to execute the construction even when a Gaussian-rational Kraus list is not supplied. This is a representation-level enlargement of the constructive theorem, not an alteration of the inherited lower model.

37. INTRINSIC DESCRIPTION ENTROPY OF INSTRUMENTS

We now determine the high-resolution description exponent, rather than only an upper bound on the number of stored matrix entries. The objects being described remain genuine quantum instruments. A binary description of such an object is not a physical classical implementation of it.

Fix positive integers d, n, m , put $q = dn$, and use the input-first, unnormalized Choi convention of Section 35. Let

$$\mathcal{C} = \mathcal{C}_{d,n,m} = \{(J_y)_{y=1}^m : J_y \in \text{Herm}(q), J_y \geq 0, \sum_y \text{Tr}_n J_y = I_d\}.$$

The corresponding outcome-retaining channel is denoted by $\mathcal{F}_J$. On tuples of Hermitian matrices write

$$|X|_2 = \left(\sum_y \|X_y\|_F^2 \right)^{1/2}, \quad \|X\|_\diamond = \|\mathcal{F}_X\|_\diamond.$$

Here $\mathcal{F}_X$ is the linear map defined by the Choi blocks, whether or not they are positive. Distances in this section are unhalved norms. The instrument diamond norm means exactly the diamond norm of this flagged direct-sum channel, not the sum of the individual outcome-map diamond norms. All m labeled outcome slots are part of the body. A legal η -description code is a finite set of codewords with one member of $\mathcal{C}$ assigned to each codeword, such that every target is within diamond distance η of a decoded member. Each codeword selects one fixed memoryless instrument, not a target-dependent random mixture with uncharged mixing probabilities. A rational code requires the decoded matrices to have rational real and imaginary parts. Dimensions, the outcome names and the requested tolerance are fixed public parameters in the code-length statements. Their encodings are additional data when they vary.

Lemma 37.1 (Affine dimension and an interior ball). *The affine hull of $\mathcal{C}$ has real dimension*

$$(37.1) \quad s = mq^2 - d^2 = d^2(mn^2 - 1).$$

Let $V = \{X : \sum_y \text{Tr}_n X_y = 0\}$ and $J_y^\circ = I_q/(mn)$. For $X \in V$,

$$(37.2) \quad |X|_2 \leq \sum_y \|X_y\|_1 \leq d\|X\|_\diamond, \quad \|X\|_\diamond \leq \sqrt{mq}|X|_2.$$

The $\|\cdot\|_\diamond$ -ball in V of radius $r_0 = (dmn)^{-1}$, translated by J° , is contained in $\mathcal{C}$. Moreover $\mathcal{C} - J^\circ$ lies in the diamond ball of radius two.

Proof. The map from $\text{Herm}(q)^m$ to $\text{Herm}(d)$ given by summing output partial traces is surjective: an arbitrary $H \in \text{Herm}(d)$ is the image of the tuple with first block $H \otimes |e_1\rangle\langle e_1|$ and other blocks zero. Its kernel has codimension d^2 . All blocks of J° are positive definite, so small perturbations in that kernel remain legal. This proves (37.1) and that V is the translation space of the full affine hull, not merely of a containing affine space.

The first two inequalities follow from Lemma 35.1 and the direct-sum Choi matrix. The last follows from that lemma, $\|X_y\|_1 \leq \sqrt{q}\|X_y\|_F$, and Cauchy–Schwarz over y . If $\|X\|_\diamond \leq r_0$, every block has operator norm at most $|X|_2 \leq 1/(mn)$, which proves positivity of $J^\circ + X$. The affine constraint is unchanged. Finally a channel has diamond norm one, so the distance between any two outcome-retaining channels is at most two. This also follows from complete trace-norm contractivity of channels, as in [37, Corollary 3.40]. $\square$

The lower Choi estimate from Section 35 thus has a specific converse use: it transfers a diamond ball into the positive Choi body. Neither this step nor the dimension count uses a spectral gap.

Theorem 37.2 (Sharp high-resolution description exponent). *Suppose $s > 0$. Let $\mathcal{N}_\diamond(\eta)$ and $\mathcal{N}_\diamond^\mathbb{Q}(\eta)$ be the smallest cardinalities of legal and rational legal η -description codes for $\mathcal{C}$. For $0 < \eta \leq 1$,*

$$(37.3) \quad \max\{1, (r_0/\eta)^s\} \leq \mathcal{N}_\diamond(\eta) \leq \mathcal{N}_\diamond^\mathbb{Q}(\eta) \leq (1 + 8/\eta)^s.$$

In particular, for fixed d, n, m , both optimal fixed-length binary code lengths are

$$(37.4) \quad s \log_2(1/\eta) + O_{d,n,m}(1) \quad (\eta \downarrow 0).$$

When $s = 0$, $m = n = 1$, the unique instrument is the trace map and the code length is zero.

Proof. Equip V with Lebesgue measure and the diamond norm restricted to V . By Lemma 37.1, the target set contains a ball of radius r_0 . A union of M balls of radius η covering this ball has volume at most $M\eta^s$ times the unit-ball volume. Hence $M \geq (r_0/\eta)^s$. Centres anywhere in the affine hull give the same lower bound.

For the upper bound, take a maximal set of points of $\mathcal{C}$ whose mutual distances are greater than $\eta/2$. Balls of radius $\eta/4$ about them are disjoint up to null boundaries and lie in the ball of radius $2 + \eta/4$ about J° . The number of points is therefore at most $(1 + 8/\eta)^s$, and maximality

makes them an $\eta/2$ -net. Rational legal instruments are dense, by the explicit construction in Theorem 37.3 below. Replace each centre by a rational legal point within $\eta/2$. This yields a rational η -net of the same cardinality. This dependence is not circular: the construction below is proved directly from the Choi constraints.

A codebook with M words can be indexed by $\lceil \log_2 M \rceil$ bits, and a code of at most b fixed-length bits has at most 2^b words. Taking logarithms in (37.3) proves (37.4). The singleton case follows directly from the constraint $J_1 = I_d$. $\square$

This is an elementary volumetric theorem for the precise legal Choi body and norm. The volume comparison is not a new general covering principle. The coefficient s is sharp in the high-resolution limit with dimensions fixed; the estimate does not assert optimal constants in a simultaneous dimension–accuracy limit.

37.1. An intrinsic rational coordinate code. Fix an outcome y_* and an output basis index α_* . The independent real coordinates of a Hermitian matrix are its real diagonal entries and the real and imaginary parts of its strictly upper-triangular entries. Omit all d^2 such coordinates belonging to the submatrix

$$((J_{y_*})_{i\alpha_*,j\alpha_*})_{1 \leq i,j \leq d}.$$

There are exactly s remaining real coordinates, in a fixed order. Every coordinate of a legal target has absolute value at most one: positivity and the marginal identity bound each diagonal by one, and $|J_{uv}|^2 \leq J_{uu}J_{vv}$ bounds the off-diagonal entries.

For $B \in \mathbb{N}$, truncate the s retained coordinates toward zero on the grid $B^{-1}\mathbb{Z}$, producing digits $z_1, \dots, z_s \in [-B, B] \cap \mathbb{Z}$. Fill the omitted submatrix by the exact marginal identity:

$$(37.5) \quad (Q_{y_*})_{i\alpha_*,j\alpha_*} = \delta_{ij} - \sum_{(y,\alpha) \neq (y_*,\alpha_*)} (Q_y)_{i\alpha,j\alpha}.$$

All summands on the right were retained. Lower-triangular entries are Hermitian copies, not additional digits.

Theorem 37.3 (Exact intrinsic codec). *Let*

$$(37.6) \quad c = 2nd(1 + mn), \quad D = B + mnc, \quad K = 3mndc.$$

The matrices

$$(37.7) \quad \hat{J}_y = (BQ_y + cI_q)/D$$

are positive semidefinite with Gaussian-integer numerators, satisfy $\sum_y \text{Tr}_n \hat{J}_y = I_d$, and obey

$$(37.8) \quad \|\mathcal{F}_J - \mathcal{F}_{\hat{J}}\|_{\diamond} \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/D \leq K/B.$$

They can be encoded by one integer in $[0, (2B+1)^s)$, using at most $\lceil \log_2((2B+1)^s) \rceil$ bits. Thus $B = \lceil K/\eta \rceil$ achieves the leading coefficient s in (37.4) by a direct coordinate code, without enumerating a net. For rational input, encoding, decoding and validity testing have polynomial bit complexity in the explicit input and output lengths. This is not a workspace bound for a trajectory simulator.

Proof. Non-omitted independent-coordinate errors are at most $1/B$. An omitted coordinate is reconstructed from at most $mn - 1$ errors, so its error is at most $(mn - 1)/B$. In particular every independent real error is at most mn/B . Counting the Hermitian copies gives the safe estimate

$$\|Q_y - J_y\|_{\text{op}} \leq \|Q_y - J_y\|_F \leq 2qmn/B \leq c/B.$$

The affine constraint is exact by (37.5). Therefore $BQ_y + cI_q \geq 0$ and its summed output partial trace is $(B + mnc)I_d$. With $E_y = Q_y - J_y$,

$$D(\hat{J}_y - J_y) = BE_y + cI_q - mncJ_y.$$

The sum of the trace norms of each of the three terms is bounded by $mndc$, using $\sum_y \text{tr } J_y = d$ for the last term. Lemma 35.1 proves (37.8).

Set $M = 2B + 1$. The integer with base- M digits $z_j + B$ determines all retained coordinates, and (37.5) then determines the omitted coordinates. A fixed number of digits is used, including leading zeros. Only integers whose reconstructed, buffered blocks are positive are valid codewords. They can be recognized by the exact Schur test in Lemma 36.2. Every code produced from a legal target is valid by the preceding proof. If a total decoder is required, assign each invalid word the canonical rational instrument J° ; unused binary strings can be treated in the same way. The valid codebook already covers all targets. In the singleton case the construction decodes the trace map from the empty string.

For rational data the coordinate digits are obtained by exact signed integer division. Matrix filling, buffering and base conversion are finite integer operations with polynomial operand lengths. Exact PSD validation has polynomial cost by Lemma 36.2. No eigenvector computation, rational Kraus factorization, random draw, statistical sample or floating-point decision is involved. For an arbitrary real target, the coordinate construction proves existence and density; it is not an algorithm for recovering unknown real input from finitely many unverified measurements. $\square$

Remark 37.4 (Known zero outcomes and public headers). If a specified set of outcomes is known to be zero, remove it before coding and restore it afterwards. With $m' \geq 1$ active outcomes the intrinsic dimension is $d^2(m'n^2 - 1)$, and m' replaces m in the constants. If the active set is not a fixed public parameter, its mask must also be encoded. The generic construction does not preserve zeros without this operation. The choice of (y_*, α_*) affects neither the dimension nor the bounds. Headers for dimensions, B , outcomes and the chosen coordinate convention are not part of a fixed-parameter codeword; a self-contained file carries and charges them separately.

38. DESCRIPTION ENTROPY UNDER REPEATED ADAPTIVE USE

The number of calls changes the precision needed in a reusable instrument description. On a strictly positive Choi body the relevant scale is $N^{-1/2}$, not the N^{-1} scale supplied by a general one-slot hybrid. We prove a finite, multiparameter coding statement against arbitrary common adaptive quantum testers.

For two instruments $J, L \in \mathcal{C}_{d,n,m}$, define

$$(38.1) \quad d_N(J, L) = \sup_{\mathbf{T}} \|\Omega_J^{\mathbf{T}} - \Omega_L^{\mathbf{T}}\|_1.$$

Here $\mathbf{T}$ is the same causal tester in both experiments, uses the same memoryless instrument at most N times, and may have arbitrary finite-dimensional quantum memory, initially entangled references, intermediate quantum channels and measurements, and classical feedback. Fresh ancillary systems of arbitrary finite dimension are allowed: each tester has finite-dimensional registers at every slot, but no common dimension bound is imposed in the supremum. Outcomes and final memory are retained. A public stopping time bounded by N is allowed. A changed input dimension can be handled by a tester channel between calls. There is no coherent superposition of command ports, no tester informed of the hidden choice J versus L , and no stopping based on inaccessible simulator state. Taking suprema of trace distances gives the triangle inequality. In particular $d_N \leq 2$. Public stopping is included by an absorbing flag, as in Theorem 35.4.

For a fixed a with $0 < a < 1/(mn)$ let

$$\mathcal{C}_a = \{J \in \mathcal{C} : J_y \geq aI_{dn} \text{ for every } y\}.$$

This is a Choi eigenvalue margin, not a spectral gap for an action or a mixing hypothesis on trajectories. Its affine dimension is the same $s = d^2(mn^2 - 1)$ as before.

Lemma 38.1 (Two-point classical programme bound). *Suppose $J_y, L_y \geq \lambda I_{dn}$ for every outcome and $r = \|J - L\|_2 \leq \lambda$, where $\lambda > 0$. Then, for every $N \geq 1$,*

$$(38.2) \quad d_N(J, L) \leq \min\{2, 2\sqrt{N}r/\lambda\}.$$

The same bound covers arbitrary initially entangled references and bounded public stopping in (38.1).

Proof. There is nothing to prove for $r = 0$. Put $K = (J + L)/2$ and $U = (J - L)/r$. Since $U \in V$ and each $\|U_y\|_{\text{op}} \leq 1$,

$$K^+ = K + \lambda U, \quad K^- = K - \lambda U$$

are legal instruments. With

$$p_{\pm} = \frac{1}{2} \pm \frac{r}{4\lambda}, \quad q_{\pm} = \frac{1}{2} \mp \frac{r}{4\lambda},$$

we have $J = p_+K^+ + p_-K^-$ and $L = q_+K^+ + q_-K^-$. All four probabilities are at least $1/4$. The elementary inequality $\log x \leq x - 1$ implies

$$D(p\|q) \leq \sum_{b \in \{+, -\}} \frac{(p_b - q_b)^2}{q_b} \leq \frac{2r^2}{\lambda^2}.$$

Choose the programme signs independently in advance for all N slots. Conditional on a sign string, a fixed tester acts on exactly the same sequence of genuine quantum instruments in the two experiments. It therefore defines one channel from the classical sign string to the final quantum–classical state. This remains true with adaptive feedback: conditioning fixes the instruments, not the tester’s outcomes. If the tester stops early, the unused signs are ignored. Contractivity and the classical product identity give

$$\|\Omega_J^T - \Omega_L^T\|_1 \leq 2 \text{TV}(p^{\otimes N}, q^{\otimes N}) \leq \sqrt{2ND(p\|q)} \leq 2\sqrt{N}r/\lambda.$$

The middle bound is the usual Pinsker inequality. To recall an elementary proof, coarse-grain two distributions to the set where the first is larger. The log-sum inequality reduces relative entropy to binary relative entropy. Its second derivative in the first argument is $1/[x(1-x)] \geq 4$, so integration about its minimum proves $D \geq 2 \text{TV}^2$. Taking the supremum over testers proves the assertion. $\square$

The classical-programme idea is an established method in quantum metrology; Demkowicz-Dobrzański, Kołodyński and Guţă [47, Equations (4)–(6), Methods] use parameter-dependent mixtures of fixed channels to bound distinguishability. The lemma above gives the finite two-point trace-norm estimate needed here, including common adaptive testers and public stopping. It is not a claim that the mixture method or the square-root scale is a new metrological principle. The operations $K^{\pm}$ remain quantum operations, not classical physical substitutes for unknown input.

Lemma 38.2 (Repeated Choi tests). *For any two instruments $J, L \in \mathcal{C}$, with $t = \|J - L\|_2$,*

$$(38.3) \quad d_N(J, L) \geq 2 - 4 \exp\left(-\frac{Nt^2}{8d^2}\right).$$

A negative right-hand side is simply a vacuous lower bound.

Proof. At each call supply half of a normalized maximally entangled state and retain its other half and the outcome. The resulting density matrices are $A = d^{-1} \oplus_y J_y$ and $B = d^{-1} \oplus_y L_y$. They

have trace one. The projector on the positive part of $A - B$ gives a binary measurement whose probability gap is

$$g = \frac{1}{2} \|A - B\|_1 \geq t/(2d).$$

The same binary measurement is made on each of the N independent outputs. With a threshold halfway between the two means, each wrong-side probability is at most $\exp(-Ng^2/2)$. Indeed for a Bernoulli variable Z , the second derivative of $\log \mathbb{E}e^{u(Z-\mathbb{E}Z)}$ is the variance of a tilted Bernoulli variable and is at most $1/4$. The value and first derivative at zero vanish; integration and Chernoff's inequality give $\Pr\{N^{-1} \sum Z_i - \mathbb{E}Z \geq v\} \leq e^{-2Nv^2}$, and likewise for the other tail. Set $v = g/2$. The total variation distance between the two product experiments is at least $1 - 2e^{-Ng^2/2}$; the unhalved trace norm of their classical outputs is twice this. This particular nonadaptive tester is allowed in (38.1), and the result follows. The tester may depend on the pair J, L , as is permitted when lower-bounding a metric; it is the same tester for the two members of that pair. $\square$

Theorem 38.3 (Sharp horizon exponent for interior descriptions). *Fix d, n, m with $s > 0$, a margin $0 < a < 1/(mn)$ and a tolerance $0 < \delta < 1$. Let $\mathcal{M}_N(\delta; a)$ be the minimum number of legal decoded instruments whose d_N -balls of radius δ cover $\mathcal{C}_a$. A decoded instrument need not itself have margin a . Write $\mathcal{M}_N^{\mathbb{Q}}$ when rational decoded instruments are required. Set*

$$R = \frac{1}{mn} - a, \quad c_\delta = \sqrt{8d^2 \log \frac{4}{1-\delta}}, \quad B_N = \left\lceil \frac{K}{a} \left(2 + \frac{4\lceil \sqrt{N} \rceil}{\delta} \right) \right\rceil,$$

where K is from (37.6). Then

$$(38.4) \quad \max\{1, (R\sqrt{N}/c_\delta)^s\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s.$$

Consequently the minimum fixed-length description satisfies

$$(38.5) \quad \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) = \frac{s}{2} \log_2 N + O_{d,n,m,a,\delta}(1),$$

and the same formula holds without the rationality restriction. The upper bound is achieved by the exact intrinsic codec applied once at grid B_N ; its decoded instrument is reused at every call.

Proof. The tuple-Frobenius ball in V of radius R about J° is contained in $\mathcal{C}_a$. Choose a maximal set of points in that ball with pairwise distances greater than $c_\delta/\sqrt{N}$. Balls of that radius about the selected points cover the original ball. Euclidean volume comparison therefore gives at least $(R\sqrt{N}/c_\delta)^s$ points. By Lemma 38.2, their pairwise d_N distances are greater than

$$2 - 4e^{-c_\delta^2/(8d^2)} = 1 + \delta > 2\delta.$$

One d_N -ball of radius δ cannot cover two of these points, regardless of the location of its legal centre. This proves the lower bound.

For the upper bound use Theorem 37.3. It gives

$$r = |J - \hat{J}|_2 \leq \sum_y \|J_y - \hat{J}_y\|_1 \leq K/B_N \leq \min\{a/2, a\delta/(4\sqrt{N})\}.$$

Every block of $\hat{J}$ is consequently at least $(a/2)I$, and both J and $\hat{J}$ satisfy Lemma 38.1 with $\lambda = a/2$ and $r \leq \lambda$. Its conclusion gives $d_N(J, \hat{J}) \leq 4\sqrt{N}r/a \leq \delta$. There are at most $(2B_N + 1)^s$ possible digit strings. All codewords arising from targets in $\mathcal{C}_a$ decode legally. The remaining codewords may be rejected or assigned J° ; this does not affect the cover. Since $B_N = O_{d,n,m,a,\delta}(\sqrt{N})$, taking logarithms in (38.4) proves (38.5). $\square$

Remark 38.4 (What is charged). The lower bound concerns the number of possible reusable mathematical instrument descriptions, not the mutable memory of a simulator that receives and stores the entire instrument as input. It applies to all full-dimensional strictly positive instrument

families above, without an expanding unitary subsystem. It therefore complements, rather than replaces, the hypothesis-sensitive streaming-space lower bounds. For variable a, δ, d, n, m , the displayed finite inequalities remain valid; (38.5) holds with those parameters fixed. The estimates of this section alone assert neither an optimal uniform dimension constant nor a complete joint N - δ asymptotic. The latter is supplied, for fixed dimensions and margin, by Theorem 39.3. The rational implementation takes rational a, δ and an asserted margin, which it can verify exactly. No samples of an unknown channel are supplied or required.

Proposition 38.5 (The boundary is quantitatively different). *There is no constant C such that $d_N(J, L) \leq C\sqrt{N}|J - L|_2$ holds for every pair of qubit channels and every N . In particular the strict margin in Theorem 38.3 cannot be removed from its proof by setting $a = 0$.*

Proof. Let $U_\theta = \text{diag}(1, e^{i\theta})$ and compare its unitary channel to the identity channel. For their unnormalized Choi matrices,

$$|J(U_\theta) - J(I)|_2 = 2\sqrt{2}|\sin(\theta/2)| \leq \sqrt{2}|\theta|.$$

At $\theta = \pi/N$, sequentially applying the unknown channel N times to $|+\rangle$ gives respectively $|-\rangle$ and $|+\rangle$. The final trace distance is two. The proposed upper bound would tend to zero. The general additive diamond hybrid remains valid at this boundary; the proposition asserts no global covering exponent for boundary or rank-constrained instrument classes. $\square$

38.1. Description, learning and physical simulation. A reusable rational description is different from a protocol that replaces quantum communication by classical messages. Naik, Zartab, Gisin and Banik [44, Theorems 2 and 5] show a finite-classical-message obstruction for the perfect qubit channel in a reference-sensitive joint-measurement model, and finite-resource possibility for noisy depolarizing channels. Their receiver retains a quantum system; the task is physical simulation. Here the decoded objects and the vertices in Lemma 38.1 are genuine quantum instruments. Our numerical algorithms instead start from specified matrices. Neither construction supplies a classical physical device accepting unknown entangled input.

Unknown-channel learning charges channel queries used to infer a classical description. Mele and Bittel [45, Theorem I.1] prove dimension-optimal query complexity at fixed diamond accuracy; Oufkir and Girardi [46] give further accuracy-dependent query lower bounds. The input here is already specified, and the lower bound in (38.4) counts descriptions, not the queries needed to select one from unknown data. The elementary packing and testing ingredients are shared methods, not a priority claim that those learning theorems are consequences of this code.

39. JOINT HORIZON AND ACCURACY IN DESCRIPTION LENGTH

The exponential test in Lemma 38.2 resolves pairs at fixed error, but its lower bound is vacuous when the two experiments are very close. A bounded statistic gives the local estimate needed to retain the accuracy parameter. Throughout this section $N \geq 1$ and the adaptive distance d_N is the unhalved final-state trace distance maximized over one common tester, as in (38.1). The unknown device is the same memoryless instrument at every call; commands are classical, and stopping is public and bounded by N .

Lemma 39.1 (A local binary test). *For $0 \leq p, q \leq 1$,*

$$(39.1) \quad \text{TV}(\text{Bern}(p)^{\otimes N}, \text{Bern}(q)^{\otimes N}) \geq \frac{3}{32} \min\{\sqrt{N}|p - q|, 1\}.$$

The estimate is uniform up to the endpoints of the probability interval.

Proof. Assume $p < q$ and first suppose $q - p \leq N^{-1/2}$. Put $b = N(p + q)/2$ and let $f : \mathbb{R} \rightarrow [0, 1]$ be the nondecreasing function which is affine with slope $(8\sqrt{N})^{-1}$ on $[b - 4\sqrt{N}, b + 4\sqrt{N}]$, zero

to its left, and one to its right. For $S \sim \text{Bin}(N, t)$, differentiating the finite binomial polynomial gives the exact identity

$$\frac{d}{dt} \mathbb{E}_t f(S) = N \mathbb{E}_t \{f(T+1) - f(T)\}, \quad T \sim \text{Bin}(N-1, t).$$

It holds on $(0, 1)$ and extends continuously to both endpoints. For $t \in [p, q]$, the distance from $\mathbb{E}_t T$ to b is at most $\sqrt{N}/2 + 1$. Also $\text{Var}_t T \leq (N-1)/4$. Chebyshev's inequality gives

$$\Pr_t\{|T - \mathbb{E}_t T| \leq \sqrt{N}\} \geq 3/4.$$

On this event, both T and $T+1$ belong to the affine interval: their distance to b is at most $3\sqrt{N}/2 + 2 \leq 4\sqrt{N}$. Since f is nondecreasing everywhere,

$$\frac{d}{dt} \mathbb{E}_t f(S) \geq \frac{3}{32} \sqrt{N}.$$

Integration from p to q , and the characterization of total variation by $[0, 1]$ -valued statistics, prove the assertion in this case. The statistic depends only on the count, which is a function of the full binary string.

If $q - p > N^{-1/2}$, apply the same construction to p and $q_0 = p + N^{-1/2} < q$. Couple Bernoulli samples by common uniform random variables. Monotonicity of f then gives $\mathbb{E}_q f(S) \geq \mathbb{E}_{q_0} f(S)$, and the preceding lower bound is $3/32$. The cases $p = q$ and $q < p$ follow by equality and symmetry. $\square$

The proof uses only binomial differentiation and a bounded statistic; it requires neither a normal approximation nor a lower variance bound. In particular its constant does not deteriorate for a rare outcome.

Corollary 39.2 (Local repeated Choi separation). *For the instrument body $\mathcal{C}_{d,n,m}$, let $t = |J - L|_2$. Then*

$$(39.2) \quad d_N(J, L) \geq \frac{3}{16} \min \left\{ \frac{\sqrt{N}}{2d} t, 1 \right\}.$$

Proof. Use the same normalized Choi inputs as in Lemma 38.2. The positive spectral projection of $d^{-1} \oplus_y (J_y - L_y)$ defines a binary test with probability gap at least $t/(2d)$. Repeating this test independently and applying Lemma 39.1 gives (39.2); classical trace distance is twice total variation. This is one admissible nonadaptive tester. For each pair it is the same tester in both experiments, and it is allowed in the supremum defining d_N . $\square$

Theorem 39.3 (The joint interior coding law). *Fix d, n, m with $s = d^2(mn^2 - 1) > 0$ and $0 < a < 1/(mn)$. For all $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(39.3) \quad \max \left\{ 1, \left(\frac{R\sqrt{N}}{32d\delta} \right)^s \right\} \leq \mathcal{M}_N(\delta; a) \leq \mathcal{M}_N^{\mathbb{Q}}(\delta; a) \leq (2B_N + 1)^s,$$

where $R = 1/(mn) - a$ and B_N is the explicit grid in Theorem 38.3. Consequently the optimal legal fixed-length description, with or without rational decoded matrices, has length

$$(39.4) \quad \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O_{d,n,m,a}(1),$$

uniformly on this whole range of N and δ . The decoded instrument is reused unchanged, and no spectral-gap hypothesis occurs.

Proof. The tuple-Frobenius ball of radius R about J° in V is contained in $\mathcal{C}_a$. Take a maximal set in this ball whose pairwise Euclidean distances are greater than $h = 32d\delta/\sqrt{N}$. Maximality and s -dimensional volume give at least $(R/h)^s$ points. Corollary 39.2 separates each pair by more

than 2δ in d_N : before the minimum saturates its value is greater than 3δ , and after saturation it is $3/16 > 2\delta$. Thus no legal centre of a d_N -ball of radius δ can cover two packing points, even if that centre has no positive eigenvalue margin. This proves the lower bound.

The intrinsic rational codec and its classical-programme estimate in Theorem 38.3 give the upper bound with the same B_N . Since $N \geq 1$ and $\delta \leq 1/32$, there is a constant $C = C(d, n, m, a)$ such that $2B_N + 1 \leq C\sqrt{N}/\delta$. The lower bound and this inequality give (39.4), including rounding a logarithm up to a fixed-length binary code. When the volumetric lower expression is less than one, its logarithm is negative; the nonnegative code length still gives the same uniform additive estimate. Neither bound charges the queries of a learning procedure. $\square$

The earlier fixed-error theorem is retained. The new assertion is the uniform additive remainder in (39.4), which is independent of both the horizon and the accuracy. It does not assert sharp dimension-dependent constants, nor is it a lower bound for the mutable workspace of the variable-description trajectory algorithm.

40. BOUNDARY-UNIFORM CODING OF PREPARATION INSTRUMENTS

Strict positivity is essential to the two-point programme argument, as Proposition 38.5 shows. Nevertheless it is not necessary for a complete coding law in an irreversible family. We consider all outcome-retaining instruments that erase their input and prepare a specified quantum–classical state. This family contains zero outcomes and arbitrary rank-deficient output blocks.

For $m, n \geq 1$ put

$$\mathcal{S}_{m,n} = \{(\sigma_y)_{y=1}^m : \sigma_y \in \text{Herm}(n), \sigma_y \geq 0, \sum_y \text{tr } \sigma_y = 1\}.$$

Write $\sigma = \bigoplus_y \sigma_y$. For any input dimension $d \geq 1$, the associated instrument is

$$(40.1) \quad \mathcal{R}_{\sigma,y}(X) = \text{tr}(X)\sigma_y, \quad J_y(\mathcal{R}_\sigma) = I_d \otimes \sigma_y.$$

The output label y is retained. Its conditional quantum state is $\sigma_y/\text{tr } \sigma_y$ when that trace is positive. This is a specified mathematical quantum instrument; a description of it is not a numerical algorithm acting on an unknown physical input.

Lemma 40.1 (Reduction to a product-state experiment). *For arbitrary $\sigma, \tau \in \mathcal{S}_{m,n}$,*

$$(40.2) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_\tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

Here the tester may have arbitrary quantum memory, initial reference entanglement, feedback and a public stopping time at most N .

Proof. Supply N independent program states σ to one fixed processor. At the next device call the processor discards the incoming device system and returns the next program state, including its classical outcome. Between calls it performs the prescribed tester operations. This reproduces the experiment exactly, even when the incoming system is entangled with the tester memory: the incoming system is traced out in (40.1). Unused program states are ignored after a public stop. The processor is the same channel for σ and τ , so trace-norm contraction gives the upper bound. Conversely, a tester can make exactly N calls and retain every outcome and quantum output. Its final state is the displayed product, which proves equality. The processor is a quantum channel, not a claimed finite-classical-message physical simulator. $\square$

40.1. Triangular factors and a rational code. Every $\sigma \in \mathcal{S}_{m,n}$ has triangular factors L_y with $\sigma_y = L_y L_y^*$ and nonnegative real diagonal entries. The usual no-pivot Cholesky procedure also covers singular matrices: a zero diagonal pivot forces the entire residual row and column to vanish and contributes a zero factor column. A block of rank r therefore has exactly r positive pivots. All factors together satisfy $\sum_y \|L_y\|_F^2 = 1$.

Fix public bounds $0 \leq r_y \leq n$, not all zero, and write

$$(40.3) \quad D = \sum_{y=1}^m r_y(2n - r_y), \quad v = D - 1, \quad \mathcal{S}_r = \{\sigma \in \mathcal{S}_{m,n} : \text{rank } \sigma_y \leq r_y\}.$$

If the positive pivot positions of L_y are $T_y \subseteq \{1, \dots, n\}$, record only the real diagonal and the real and imaginary coordinates strictly below each such pivot. Their total number is

$$h = \sum_y \sum_{j \in T_y} \{1 + 2(n - j)\} \leq D.$$

The resulting real coordinate vector $x \in \mathbb{R}^h$ has norm one. The inequality follows by placing each block's pivots as early as possible. It holds also when the actual ranks are smaller than the public bounds.

Choose a coordinate j of maximal absolute value, put $a = |x_j|$, and set $w = x/a$. For an integer $B \geq 1$, take

$$(40.4) \quad z_j = B \operatorname{sgn}(x_j), \quad z_i = \operatorname{sgn}(x_i) \lfloor B|w_i| \rfloor \quad (i \neq j).$$

Insert these integers in the recorded factor positions to obtain Gaussian-integer triangular matrices Z_y ; all unrecorded columns are zero. Define

$$(40.5) \quad T = \sum_y \|Z_y\|_F^2, \quad \hat{\sigma}_y = Z_y Z_y^* / T.$$

Since $|z_j| = B$, the denominator satisfies $B^2 \leq T \leq hB^2$.

Theorem 40.2 (A boundary-uniform rational factor code). *The output (40.5) lies in $\mathcal{S}_r$, preserves every zero target outcome, and does not increase the rank of any individual block. For all $N \geq 1$,*

$$(40.6) \quad d_N(\mathcal{R}_\sigma, \mathcal{R}_{\hat{\sigma}}) \leq \min \left\{ 2, \frac{4\sqrt{N(h-1)}}{B} \right\} \leq \min \left\{ 2, \frac{4\sqrt{Nv}}{B} \right\}.$$

At fixed m, n, r, B a fixed-length code needs at most

$$(40.7) \quad mn + \lceil \log_2(2D(2B+1)^v) \rceil$$

bits. For rational target blocks, its encoder and decoder use exact rational and integer arithmetic, including integer square roots; no algebraic-number oracle, positive eigenvalue margin, or Kraus factorization is required.

Proof. The product $Z_y Z_y^*$ is positive semidefinite, and normalization makes the sum of its traces exactly one. Its number of possibly nonzero columns is no larger than that of L_y , so its rank does not exceed the target rank. When an outcome is zero it has no recorded columns and remains exactly zero.

Put $z' = z/B$. Its j th coordinate agrees with that of w ; every other coordinate differs by less than $1/B$. Both $\|w\|$ and $\|z'\|$ are at least one. The elementary normalization estimate gives

$$(40.8) \quad \left\| x - \frac{z}{\|z\|} \right\| \leq \frac{2\|w - z'\|}{\|w\|} \leq \frac{2\sqrt{h-1}}{B}.$$

Use the normalized vectors

$$|L\rangle = \sum_{y,i,j} (L_y)_{ij} |y, i\rangle |y, j\rangle, \quad |\hat{L}\rangle = T^{-1/2} \sum_{y,i,j} (Z_y)_{ij} |y, i\rangle |y, j\rangle.$$

Their reduced states on the first factor are σ and $\hat{\sigma}$. Their vector distance is the left side of (40.8). For any unit vectors u, v , diagonalization on their two-dimensional span gives

$$\| |u\rangle\langle u|^{\otimes N} - |v\rangle\langle v|^{\otimes N} \|_1 = 2\sqrt{1 - |\langle u, v \rangle|^{2N}} \leq 2\sqrt{N} \|u - v\|.$$

Indeed $1 - t^N \leq N(1 - t)$ on $[0, 1]$ and $1 - |\langle u, v \rangle|^2 \leq 2(1 - \Re \langle u, v \rangle)$. Taking partial traces and using Lemma 40.1 proves (40.6). These are standard purification and product-fidelity estimates; see [37, Section 3.2]. Their short proof specifies the normalization used here.

The pivot sets use mn bits. Given them, there are at most $2h$ anchor/sign choices and $(2B+1)^{h-1}$ remaining digit strings. Pad their index to the maximum length in (40.7). An invalid header may be rejected, or sent by a total mathematical decoder to a fixed legal state. Valid factor words always decode positively, even if they did not arise from this encoder.

For rational input, maintain the exact rational Schur residual A . At a positive pivot $u = A_{jj}$, the factor diagonal is $\sqrt{u}$ and a lower entry is $A_{ij}/\sqrt{u}$. Thus the square of each of its real coordinates is rational, respectively u , or $(\Re A_{ij})^2/u$, or $(\Im A_{ij})^2/u$; its sign is known. Updating the residual uses rational arithmetic only. Choose the largest squared coordinate $b_j > 0$ by exact comparisons. If another coordinate has squared value b_i , then

$$\lfloor B|w_i| \rfloor = \text{isqrt} \left(\left\lfloor B^2 b_i / b_j \right\rfloor \right).$$

Consequently no square root of a rational number must be represented. The ratios-of-minors bounds in Lemma 36.2 give polynomial operand lengths during this preprocessing. Integer square root, products and mixed-radix coding have polynomial bit complexity in the explicit input and output lengths. Encoding workspace and the expanded decoded matrix are not identified with the compressed payload. $\square$

40.2. Sharp description laws on the boundary. Let $\mathcal{P}_N(\delta; \mathbf{r})$ be the minimum number of legal preparation instruments whose d_N -balls of radius δ cover $\mathcal{S}_{\mathbf{r}}$; the decoded centre may have any output rank. Add a superscript $\mathbb{Q}$ when rational centres are required. For an upper construction our centres will obey the same rank bounds.

Theorem 40.3 (Rank-stratified joint coding law). *Fix $m, n, \mathbf{r}$ as in (40.3). If $v > 0$, there exist constants $c, C > 0$, depending only on these fixed data, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(40.9) \quad \max\{1, (c\sqrt{N}/\delta)^v\} \leq \mathcal{P}_N(\delta; \mathbf{r}) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C(\sqrt{N}/\delta)^v.$$

The optimal legal and rational fixed-length descriptions therefore have length

$$(40.10) \quad \frac{v}{2} \log_2 N + v \log_2(1/\delta) + O_{m,n,\mathbf{r}}(1)$$

uniformly in N and δ . The statement has no spectral-gap or positive-eigenvalue-margin hypothesis. It includes every rank-deficient boundary allowed by the public bounds. If $v = 0$, the family is a singleton and its optimal description length is zero.

Proof. For the upper bound take $B = \lceil 4\sqrt{Nv}/\delta \rceil$ in Theorem 40.2; for rational implementation it suffices to replace $\sqrt{Nv}$ by its integer ceiling. Its error is at most δ , and its codebook cardinality is at most $2^{mn} 2D(2B+1)^v$. This has the asserted upper order uniformly on the stated parameter range.

For completeness we construct a full-dimensional local patch for the lower bound. For every block with $r_y > 0$, take an n by r_y lower-trapezoidal factor: entries above the first r_y diagonal positions vanish, and these diagonal entries are positive real numbers. These factors have D independent real coordinates. Impose $\sum_y \|L_y\|_F^2 = 1$ and work near the point with all these diagonal entries equal to $(\sum_y r_y)^{-1/2}$ and all other entries zero. An open patch of this unit sphere is a smooth v -dimensional manifold. The map

$$(L_y)_y \mapsto (L_y L_y^*)_y$$

is one-to-one there and has a smooth inverse on its image. To see the inverse explicitly, the leading r_y by r_y principal block is positive definite and determines its unique positive-diagonal Cholesky

factor A_y . The bottom-left block is $B_y A_y^*$, so it determines B_y by multiplication by $(A_y^*)^{-1}$. These operations are smooth near the chosen point. When $r_y = 0$, the entire block is identically zero and contributes no coordinate.

Choose a closed coordinate ball of sufficiently small positive radius inside this patch. The chart and its inverse have bounded derivatives on a slightly larger neighborhood; hence distances between parameters in the ball are bounded above and below by constant multiples of the tuple-Frobenius distance between their states. More explicitly one may bound the derivative of the inverse on a convex neighborhood of the chosen positive principal blocks, where the same Cholesky formulas extend smoothly. A maximal Euclidean packing of this parameter ball at scale $A\delta/\sqrt{N}$ therefore has at least $(c\sqrt{N}/\delta)^v$ points whose state distances exceed $32\delta/\sqrt{N}$, after increasing the fixed A .

The binary argument of Corollary 39.2, applied directly to the trace-one block states (input dimension one), separates each pair by more than 2δ after N copies. By Lemma 40.1 the same separation holds in d_N . One radius- δ ball, regardless of the rank of its legal centre, cannot contain two packing points. This proves the lower bound and then (40.10). Finally $D = 1$ requires $n = 1$ and exactly one bound $r_y = 1$; all other bounds vanish. The sole permitted state has unit mass on that outcome, as asserted. $\square$

Corollary 40.4 (All ranks, with an explicit lower constant). *For the full family $\mathcal{S}_{m,n}$ put $v = mn^2 - 1 > 0$. Then*

$$(40.11) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{64mn\delta} \right)^v \right\} \leq \mathcal{P}_N(\delta) \leq \mathcal{P}_N^{\mathbb{Q}}(\delta) \leq 2^{mn} 2mn^2 (2B + 1)^v, \quad B = \left\lceil \frac{4\sqrt{N}v}{\delta} \right\rceil.$$

In particular a general n -dimensional prepared state has coefficient $n^2 - 1$, while the class of pure prepared states has coefficient $2n - 2$ in (40.10). A classical m -outcome law has coefficient $m - 1$, including probabilities equal to zero.

Proof. The upper bound is Theorem 40.2 with every $r_y = n$. For the lower bound, the affine space of block states has dimension $mn^2 - 1$. Its tuple-Frobenius ball of radius $1/(2mn)$ about the blocks $I_n/(mn)$ lies in $\mathcal{S}_{m,n}$. Repeat the packing proof of Theorem 39.3 with input dimension one. For pure states take $m = 1, r_1 = 1$ in Theorem 40.3; for classical laws take $n = 1$. $\square$

Remark 40.5 (Operational and priority distinctions). Input erasure is decisive in (40.2). No analogous product-state reduction has been proved here for an arbitrary disturbing instrument; coherent accumulation for unitary channels remains governed by Proposition 38.5. The present theorems classify reusable description lengths in the stated preparation families, not general simulator workspace. Constants remain dimension dependent.

Quantum population compression starts with physical copies of an unknown state and compresses their quantum information. Yang, Bai, Chiribella and Hayashi [48] obtain a half-parameter-count logarithmic memory law in that different task and show a limitation on purely classical compression of noncommuting families. Here the encoder is given a classical matrix description, and each codeword specifies a state or preparation instrument. It need not infer that word from unknown copies. The common half-dimension scale, purification inequalities and finite-dimensional covering methods are classical antecedents, not claimed inventions. The contribution established here is a uniform two-parameter legal-description law with a rational, zero-outcome-preserving, rank-nonincreasing encoder and matching lower bounds throughout the declared boundary families.

41. RETRACTION OF COVERING CENTRES FOR PREPARATION TARGETS

The centres in a covering problem need not belong to the family being covered. For preparation targets this distinction can be removed exactly, without a small-error or rank assumption. Throughout this section a centre is one memoryless instrument, reused at every call; it is not a device with an additional persistent environment.

Theorem 41.1 (Preparation-centre retraction). *Let $\mathcal{C} = (\mathcal{C}_y)_{y=1}^m$ be any instrument from M_d to M_n , and set*

$$(41.1) \quad \tau_y = \mathcal{C}_y(I_d/d) = d^{-1} \text{Tr}_{\text{in}} J_y(\mathcal{C}), \quad \mathfrak{R}\mathcal{C} = \mathcal{R}_\tau.$$

Then $\mathfrak{R}$ maps legal instruments to preparation instruments, fixes every preparation instrument, and preserves rational Choi data. For every $N \geq 1$ and every $\sigma \in \mathcal{S}_{m,n}$,

$$(41.2) \quad d_N(\mathcal{R}_\sigma, \mathfrak{R}\mathcal{C}) \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Here d_N is the unhalved trace distance optimized over the common adaptive testers of Lemma 40.1. Consequently, for any subset of preparation targets and any radius $\delta > 0$, the minimum covering cardinality is unchanged when arbitrary legal memoryless centres are restricted to preparation centres. The assertion also holds when centres are required to have rational Choi entries.

Proof. Complete positivity gives $\tau_y \geq 0$, and trace preservation gives $\sum_y \text{tr} \tau_y = 1$. The formula is rational and is the identity on the preparation family, so $\mathfrak{R}^2 = \mathfrak{R}$. Feed the state I_d/d into each of N calls to $\mathcal{C}$, use a fresh input at the next call, and retain all outputs and outcome labels. The resulting state is $\tau^{\otimes N}$, since the centre is memoryless. The same tester applied to $\mathcal{R}_\sigma$ returns $\sigma^{\otimes N}$. Therefore

$$\|\sigma^{\otimes N} - \tau^{\otimes N}\|_1 \leq d_N(\mathcal{R}_\sigma, \mathcal{C}).$$

Lemma 40.1 identifies the left side with the left side of (41.2). Retracting every centre in any cover preserves coverage and cannot increase cardinality. The reverse inequality follows because preparation centres are among the legal centres. The same argument preserves rationality. $\square$

The retraction does not preserve a target rank bound: for example the identity channel retracts to the maximally mixed preparation. Thus the centre restriction in Theorem 41.1 is to the full preparation family, not necessarily to the target rank stratum. The rank-constrained rational upper centres of Theorem 40.2 remain a separate, stronger construction. The theorem concerns the geometry of mathematical descriptions. Its maximally mixed test is a quantum experiment, not a finite-classical-message simulation of unknown inputs.

The reduction mechanism for input-erasing channels has a direct antecedent in Cooney–Mosonyi–Wilde [50]. Their discussion of two replacer channels reduces channel divergences to state divergences; their principal results concern asymptotic discrimination and strong-converse exponents for a channel versus a replacer. Lemma 40.1 uses the same input-erasure mechanism in an exact finite-use trace-norm identity, whereas Theorem 41.1 concerns the allowed centres of an entire covering problem. No exact nonadaptive optimality for an arbitrary non-replacer channel against a replacer is inferred from that identity. The processor in the product identity is quantum, and input erasure, rather than entanglement breaking alone, is its essential hypothesis.

42. COHERENT AND PREPARATION DIRECTIONS AT A SINGULAR BOUNDARY

The preparation theorem has a square-root horizon scale. Reversible quantum information can instead accumulate a coherent discrepancy at a linear scale. We now exhibit both scales in a single non-input-erasing family, and determine their combined description cost.

Fix $d \geq 2$, $n, m \geq 1$ and public rank bounds $0 \leq r_y \leq n$, not all zero. For $U \in U(d)$ and $\sigma \in \mathcal{S}_r$ define an instrument from M_d to M_{dn} by

$$(42.1) \quad \mathcal{F}_{U,\sigma,y}(X) = UXU^* \otimes \sigma_y.$$

Its outcome register is retained. Global phase is immaterial, so the unitary parameter lies in $PU(d)$. Put

$$u = d^2 - 1, \quad v = \sum_y r_y(2n - r_y) - 1.$$

The Choi matrix of each outcome is, up to a fixed tensor permutation, $|U\rangle\rangle\langle\langle U| \otimes \sigma_y$. Its rank is $\text{rank } \sigma_y$, not $d \text{rank } \sigma_y$ as in the input-erasing family. These instruments transmit the data system unitarily, including its entanglement with a reference. Their preparation supports may change and may lie on any of the permitted singular strata. Nevertheless, the outcome probabilities do not depend on the input. This is a specified boundary family, not the whole body of disturbing instruments.

Let $\mathcal{C}_N(\delta; \mathbf{r})$ denote its covering number in d_N , with centres allowed to be any legal memoryless instrument on the same input, output and outcome spaces. Let $\mathcal{C}_N^{\mathbb{Q}}$ impose the additional condition that the centres have Gaussian-rational Choi entries. A target is supplied as U and its preparation blocks; no unknown-device learning interface is used.

Theorem 42.1 (Joint coding at a coherent-preparation boundary). *There are $0 < c \leq C < \infty$, depending only on the fixed dimensions and rank bounds, such that, for every $N \geq 1$ and $0 < \delta \leq 1/32$,*

$$(42.2) \quad c \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v \leq \mathcal{C}_N(\delta; \mathbf{r}) \leq \mathcal{C}_N^{\mathbb{Q}}(\delta; \mathbf{r}) \leq C \left(\frac{N}{\delta} \right)^u \left(\frac{\sqrt{N}}{\delta} \right)^v.$$

The metric is the unhalved final-state trace norm, maximized over common adaptive quantum testers using the same memoryless instrument at most N times, with arbitrary finite references, quantum memory, feedback and bounded public stopping. In particular, the optimal fixed-length reusable description has length

$$(42.3) \quad (u + v/2) \log_2 N + (u + v) \log_2(1/\delta) + O_{d,n,m,\mathbf{r}}(1).$$

The upper centres can be chosen within the family (42.1), with rational unitary matrices and rational preparation blocks. They do not increase a target block rank and preserve its zero outcomes. A finite exact encoder exists for supplied rational unitaries and rational preparation blocks. Description length is not its workspace or running time, and decoding describes a quantum instrument, not a classical physical processor for unknown entangled inputs.

42.1. The two metric scales. For $U, V \in U(d)$ write

$$(42.4) \quad \chi(U, V) = \left(1 - \frac{|\text{tr}(U^*V)|^2}{d^2} \right)^{1/2}, \quad \Delta_N(\sigma, \tau) = \|\sigma^{\otimes N} - \tau^{\otimes N}\|_1.$$

The first expression is the chordal distance between normalized Choi rays; it is a metric after quotienting global phase.

Lemma 42.2 (Anisotropic adaptive comparison). *For the instruments (42.1),*

$$(42.5) \quad \max\{\min\{N\chi(U, V)/\sqrt{2}, 1\}, \Delta_N(\sigma, \tau)\} \leq d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}),$$

$$(42.6) \quad d_N(\mathcal{F}_{U,\sigma}, \mathcal{F}_{V,\tau}) \leq \min\{2, 2dN\chi(U, V) + \Delta_N(\sigma, \tau)\}.$$

All lower testers are permitted to depend on the pair being compared.

Proof. Discarding the data-system outputs gives the product-state lower bound. For the unitary bound, discard the preparation systems and outcome labels, and apply the same known operation U^* to the data output after each call. The two resulting data operations are I and $W = U^*V$. If $e^{i\phi_j}$ are the eigenvalues of W , then

$$\chi(U, V)^2 = \frac{2}{d^2} \sum_{i,j} \sin^2((\phi_i - \phi_j)/2).$$

There is an eigenvalue pair whose principal half-angle $\theta \in [0, \pi/2]$ satisfies $\sin \theta \geq \chi(U, V)/\sqrt{2}$. Prepare the equal superposition of its orthogonal eigenvectors. After k calls its two possible pure

states have trace distance $2|\sin(k\theta)|$. If $N\theta \leq \pi/2$, take $k = N$ and use $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. The result is at least $N \sin \theta$. If $N\theta > \pi/2$, take $k = \lceil \pi/(4\theta) \rceil \leq N$. Then $\pi/4 \leq k\theta \leq 3\pi/4$, giving distance at least $\sqrt{2}$. The case $\theta = 0$ is immediate. This proves the first minimum in (42.5); the two lower witnesses need not coincide.

For the upper bound first change U to V , keeping σ fixed. The unnormalized Choi difference of the unitary channels has trace norm $2d\chi(U, V)$ and bounds their diamond distance. Appending a fixed preparation state does not increase that norm. A slot-by-slot hybrid and trace contraction for the common tester therefore cost at most $2dN\chi(U, V)$. Next hold V fixed and change σ to τ . All calls to V and all tester operations form one common quantum processor consuming N preparation states; unused copies are discarded at a public stop. This costs at most $\Delta_N(\sigma, \tau)$, not the sum of N one-copy trace distances. Triangle inequality gives (42.6). $\square$

Finite-use amplification of unitary differences is classical; see Acín [51] and Duan–Feng–Ying [52]. The elementary estimate above is included to specify its uniform constant, its pair-dependent controls and its compatibility with bounded stopping. Neither the existence of unitary discrimination nor the real dimension $d^2 - 1$ is claimed as a new ingredient.

42.2. A finite rational atlas for unitary channels. For $\ell \geq 1$, a signed stereographic chart on S^ℓ has the form

$$(42.7) \quad S_{j,\xi}(t)_j = \xi \frac{1 - \|t\|^2}{1 + \|t\|^2}, \quad S_{j,\xi}(t)_i = \xi \frac{2t_i}{1 + \|t\|^2} \quad (i \neq j), \quad \xi \in \{-1, 1\}.$$

For a target $q \in S^\ell$, choose j of largest absolute coordinate, $\xi = \text{sgn}(q_j)$ and $t_i = \xi q_i / (1 + |q_j|)$. Then $|t_i| \leq 1$ and $S_{j,\xi}(t) = q$. Direct expansion gives

$$(42.8) \quad \|S_{j,\xi}(t) - S_{j,\xi}(t')\| = \frac{2\|t - t'\|}{\sqrt{(1 + \|t\|^2)(1 + \|t'\|^2)}}.$$

Rounding each coordinate towards zero at denominator B thus gives a rational unit vector at distance at most $2\sqrt{\ell}/B$. Every decoded chart word is on the sphere, regardless of whether it is the canonical encoding of a target.

Lemma 42.3 (Rational unitary code). *Put $L = d(d-1)/2$ and $K_d = 3L + 2(d-1)$. There is an explicitly enumerable set $\mathcal{U}_B \subset U(d, \mathbb{Q}(i))$ satisfying*

$$(42.9) \quad |\mathcal{U}_B| \leq 6^L 4^{d-1} (2B+1)^{d^2-1}, \quad \sup_U \min_{V \in \mathcal{U}_B} \chi(U, V) \leq K_d/B.$$

For rational U , membership in any prescribed rational squared-error ball $\chi(U, V)^2 \leq b$ is decidable by rational arithmetic alone.

Proof. The usual complex Givens elimination, in a fixed order of index pairs, expresses every unitary as a product of L embedded matrices

$$\begin{pmatrix} c & -\bar{s} \\ s & c \end{pmatrix}, \quad c \in \mathbb{R}, \quad c^2 + |s|^2 = 1,$$

and a diagonal unitary. For completeness, a column pair (a, b) is sent to one nonzero coordinate using $c = |a|/(|a|^2 + |b|^2)^{1/2}$ and $s = cb/a$ when $a \neq 0$; the cases $a = 0$ or $(a, b) = 0$ are obtained by $c = 0$ or by the identity. Applying the adjoint rotation eliminates the second coordinate. Successive eliminations give an upper triangular unitary, necessarily diagonal. Removing the phase of its first diagonal entry leaves $d-1$ free phases. All factors are taken in the corresponding fixed reversed order when reconstructing U .

Each Givens matrix has an S^2 parameter $(c, \Re s, \Im s)$; its operator-norm change equals the Euclidean change of that vector. Each remaining phase has an S^1 parameter. Apply (42.7) at grid B to these $L + d - 1$ factors. There are 6 chart headers for each S^2 and 4 for each S^1 , with

$2L + d - 1 = d^2 - 1$ grid digits altogether. Telescoping a product of unitaries gives, up to a global phase, operator error at most

$$(2\sqrt{2}L + 2(d - 1))/B \leq K_d/B.$$

For any fixed phase z of modulus one, $\chi(U, V)^2 \leq d^{-1}\|U - zV\|_F^2 \leq \|U - zV\|_{\text{op}}^2$. This proves (42.9). The formula (42.4) has rational square for rational U, V , proving the last assertion. Enumeration of these finite words followed by the squared-distance test is an exact encoder whenever the radius is at least K_d/B . This does not assert a fast nearest-centre algorithm. $\square$

42.3. The joint covering theorem.

Proof of Theorem 42.1. For the upper bound use

$$B_U = \lceil 4dK_dN/\delta \rceil, \quad B_P = \max\{1, \lceil 8\sqrt{Nv}/\delta \rceil\}.$$

Lemma 42.3 makes the first term in (42.6) at most $\delta/2$. The factor code in Theorem 40.2 makes the second term at most $\delta/2$. Its finite pivot headers and anchor code give at most $C_{n,m,r}(2B_P + 1)^v$ legal images. If $v = 0$ the preparation family is a singleton and no such digits are needed. Multiplication of the two cardinalities proves the upper bound and all legality, rank and zero-outcome assertions. For rational data, search the finite unitary list using the rational test $\chi(U, V)^2 \leq \delta^2/(16d^2N^2)$; the preceding estimate guarantees a result. The preparation encoder is already exact. Decoded Choi entries are rational products of the two rational outputs. The finite search and the expanded matrices are not charged to the reusable payload.

For the converse, a neighborhood of the identity in $PU(d)$ contains a quantitatively bi-Lipschitz u -dimensional Euclidean patch for the metric χ . Here is a direct verification. Use traceless Hermitian coordinates H near zero and the ray of e^{iH} . The differential of its normalized Choi projection at zero vanishes only when H is scalar, hence only when $H = 0$. On a sufficiently small convex coordinate ball its differential is uniformly closer than half its least singular value to that at zero. Integration along a line segment gives both Lipschitz inequalities. Since the projection Frobenius distance is $\sqrt{2}\chi$, the asserted metric comparison follows. Volumetric packing therefore gives at least $c_d(N/\delta)^u$ unitaries separated by more than $4\delta/N$, with a smaller fixed constant covering the coarse-scale cases as well.

The maximal-rank preparation patch used in Theorem 40.3 has real dimension v and is bi-Lipschitz in tuple Frobenius norm. Pack it at separation greater than $32\delta/\sqrt{N}$, obtaining at least $c_{n,m,r}(\sqrt{N}/\delta)^v$ points. The local binary bound in Corollary 39.2, applied to the preparation state (input dimension one), separates their N -copy states by more than 2δ . Indeed the bound at that scale is $(3/16)\min\{16\delta, 1\}$, which is 3δ before saturation and at least $3/16 > 2\delta$ afterwards. For $v = 0$ use its single point.

Take the Cartesian product of these packings. Two elements with different unitaries are separated by (42.5), since $\min\{2\sqrt{2}\delta, 1\} > 2\delta$. Two elements with the same unitary and different preparation blocks are separated by the other lower witness. Thus a ball of radius δ about any legal centre can contain at most one product point. This proves the lower bound even for centres outside the family. Taking base-two logarithms and rounding a fixed-length word length upward changes the answer by at most a fixed constant, proving (42.3). $\square$

The coefficient $u + v/2$ reflects two different reuse scales, rather than half of the total parameter dimension $u + v$. No spectral-gap assumption or positive eigenvalue margin is used. The constants are nevertheless fixed-dimensional. Nor is this a classification of every tangent direction of an arbitrary instrument: coupling an input-dependent measurement to its disturbance is not part of (42.1).

42.4. **An explicit qubit codec.** For $q = (q_0, q_1, q_2, q_3) \in S^3$ put

$$(42.10) \quad U(q) = \begin{pmatrix} q_0 + iq_3 & q_2 + iq_1 \\ -q_2 + iq_1 & q_0 - iq_3 \end{pmatrix}.$$

The points q and $-q$ describe the same channel. Choose a largest absolute coordinate, flip that common sign to make it positive, and use (42.7) with $\ell = 3$ and $\xi = 1$. Its header has four possibilities, and its body has exactly three signed digits.

Corollary 42.4 (Exact rational coherent-preparation description). *For a supplied rational unit quaternion and rational preparation blocks, the preceding three-digit chart, at $B_U = \lceil 16N/\delta \rceil$, combined with the preparation codec at tolerance $\delta/2$, gives a legal rational instrument within d_N error δ . Its fixed-length description has*

$$(3 + v/2) \log_2 N + (3 + v) \log_2(1/\delta) + O_{n,m,r}(1)$$

bits, with no increase of a block rank and exact preservation of zero outcomes. It uses only integer and rational arithmetic, including the integer square roots in the preparation code. Its output is a mathematical instrument description, not physical classical simulation.

Proof. For quaternion matrices, $\|U(q) - U(q')\|_{\text{op}} = \|q - q'\|$. The unitary diamond error is at most twice this quantity. Thus the chart's coherent contribution is at most $4\sqrt{3}N/B_U < \delta/2$. Its squared upper certificate is $48N^2/B_U^2$. The preparation contribution is at most $\delta/2$. Triangle inequality proves the claim; one may certify its square by $2(e_U^2 + e_P^2)$, not by $e_U^2 + e_P^2$. The chart returns an exactly unit quaternion, and the preparation decoder returns positive blocks of total trace one. Formula (42.1) therefore gives exact complete positivity and trace preservation, including on singular inputs. The chart has at most $4(2B_U + 1)^3$ words, which gives the stated length after adding the inherited preparation payload. $\square$

The reference encoder replays both deterministic encoders when verifying a supplied target. Bare decoding checks only syntax, rational legality and the declared construction budget. It cannot certify proximity to an unspecified target. In particular, reference stability concerns the quantum instrument defined by the decoded Choi matrices, not access to a classical processor with equivalent action on an unknown reference-entangled state.

43. COVERINGS THROUGHOUT THE SUBMAXIMAL ERROR RANGE

A pairwise separation argument at radius δ requires distances larger than 2δ . Since our adaptive distance is an unhalved trace norm, this argument cannot by itself reach $\delta \geq 1$. We instead bound the number of targets that one centre can cover. The same measurement, rather than a different binary test for each pair, will be used in this part of the converse. Dimensions are fixed throughout. For a target family $\mathcal{F}$, write $\mathcal{M}_N(\delta; \mathcal{F})$ for its least covering cardinality in d_N with arbitrary legal memoryless centres, and add a superscript $\mathbb{Q}$ when the decoded Choi blocks must be rational.

Lemma 43.1 (Multiplicity of a covering centre). *Let $J_1, \dots, J_K$ be targets and let one common tester with at most N calls have disjoint classical events $A_1, \dots, A_K$ such that $\Pr_{J_i}(A_i) \geq 1 - \eta$. If $0 \leq \eta < 1$ and $0 \leq \delta < 2(1 - \eta)$, every covering of these targets by M legal centres at adaptive radius δ satisfies*

$$(43.1) \quad M \geq (1 - \eta - \delta/2)K.$$

A centre need not belong to the target family. Indeed the conclusion still holds for a centre specified by a legal N -slot process with private memory, provided its distance to each assigned target controls this same tester.

Proof. Let Q be the tester's output distribution at a centre. For each target assigned to that centre, trace contraction under the terminal measurement gives $Q(A_i) \geq 1 - \eta - \delta/2$. The division by two is required because classical total variation is half the unhalved trace norm. Disjointness gives at most $(1 - \eta - \delta/2)^{-1}$ assigned targets. Sum this bound over centres. No product structure, unbiasedness or independence is required of Q . $\square$

Lemma 43.2 (One common finite-dimensional estimator). *Put $b = m(dn)^2$. On $\mathcal{C}_{d,n,m}$ there is, for every $N \geq 1$, one nonadaptive tester and a Hermitian tuple estimator $\hat{J}$ satisfying*

$$(43.2) \quad \sup_{J \in \mathcal{C}_{d,n,m}} \mathbb{E}_J |\hat{J} - J|_2^2 \leq \frac{C_0}{N}, \quad C_0 = 2d^2b^2.$$

The estimator need not be positive or trace preserving. The tester and estimator are independent of the unknown target and of any later choice of a packing set.

Proof. One normalized maximally entangled input produces the flagged Choi state $\Omega_J = d^{-1} \bigoplus_y J_y$. Choose a Hilbert–Schmidt orthonormal Hermitian basis $H_1, \dots, H_b$ of the block-diagonal space. Every H_j has operator norm at most one. Thus $(I + H_j)/2, (I - H_j)/2$ is a binary POVM, and its signed outcome has mean $\text{tr}(H_j \Omega_J)$ and variance at most one. For $N \geq b$, use $k = \lfloor N/b \rfloor$ independent Choi inputs for each of the b measurements. There are at most N calls. Let $\hat{a}_j$ be the corresponding empirical signed mean and set $\hat{\Omega} = \sum_j \hat{a}_j H_j$. Orthonormality gives

$$\mathbb{E}_J \|\hat{\Omega} - \Omega_J\|_F^2 = \sum_j \text{Var}(\hat{a}_j) \leq b/k \leq 2b^2/N.$$

Set $\hat{J} = d\hat{\Omega}$, interpreted as a tuple. For $N < b$ use $\hat{J} = 0$ and make no calls. Then $|J|_2^2 = d^2 \text{tr} \Omega_J^2 \leq d^2 \leq C_0/N$. This proves every asserted range, without a rank or eigenvalue-margin condition. The binary POVMs are block diagonal in the public outcome, so coherent access to different classical flags is not needed. $\square$

This elementary informationally complete estimation argument is a standard finite-dimensional tomography construction; compare the process estimation setting of [42]. Here it is used only to build a common lower-bound witness. It does not constitute a new optimal-learning claim, and the description encoder below is still supplied its target.

Theorem 43.3 (A full-range covering lower bound). *Let $\mathcal{F} \subseteq \mathcal{C}_{d,n,m}$ contain the image of a nonempty a -dimensional Euclidean ball under a bi-Lipschitz map for tuple Frobenius distance, with $a > 0$. For every fixed $0 < \delta_* < 2$ there is $c > 0$, depending on this patch, the fixed dimensions and δ_* , such that*

$$(43.3) \quad \mathcal{M}_N(\delta; \mathcal{F}) \geq cN^{a/2} \quad (N \geq 1, 0 < \delta \leq \delta_*).$$

Centres may be arbitrary legal memoryless instruments. The same lower bound holds under the relaxation to legal N -slot centres described in Lemma 43.1.

Proof. Set $\eta = (2 - \delta_*)/4 > 0$ and $r_N = (C_0/(\eta N))^{1/2}$. The previous lemma and Markov's inequality give $\Pr_J\{|\hat{J} - J|_2 \leq r_N\} \geq 1 - \eta$ for every target. A volumetric packing of the fixed patch supplies at least $c_1 r_N^{-a}$ points at pairwise tuple distance greater than $2r_N$. The constant can be decreased to cover coarse scales: a singleton packing always exists. The estimator events consisting of the closed r_N -balls at these points are disjoint. Lemma 43.1 applies because $1 - \eta - \delta/2 \geq 1 - \eta - \delta_*/2 = \eta$. It gives $\mathcal{M}_N \geq \eta c_1 r_N^{-a}$, as required. In particular one centre may cover several packing points when the error is large; the proof bounds that multiplicity rather than asserting it is one. $\square$

Corollary 43.4 (Full-range interior and preparation laws). *Fix $0 < \delta_* < 2$. In Theorem 39.3, with fixed d, n, m, a and $s = d^2(mn^2 - 1) > 0$, both real and rational legal-centre covering numbers satisfy*

$$(43.4) \quad \log_2 \mathcal{M}_N(\delta; a) = \log_2 \mathcal{M}_N^{\mathbb{Q}}(\delta; a) + O(1) = \frac{s}{2} \log_2 N + s \log_2(1/\delta) + O(1)$$

uniformly for $N \geq 1$ and $0 < \delta \leq \delta_$. The constants may depend on δ_* and the positive margin a . For the preparation family in Theorem 40.3, the same extension holds with s replaced by $v = \sum_y r_y(2n - r_y) - 1$, with fixed dimensions and public output-rank bounds. Its constants require no positive eigenvalue margin. If $v = 0$ the family is a singleton and needs no payload.*

Proof. For $0 < \delta \leq 1/32$ these are the inherited joint laws. The interior ball and maximal-rank preparation chart used in their proofs are bi-Lipschitz patches of dimensions s and v , respectively. Theorem 43.3 therefore supplies the lower term $(s/2) \log_2 N + O(1)$, or $(v/2) \log_2 N + O(1)$, on the remaining range $1/32 < \delta \leq \delta_*$. On that range $\log_2(1/\delta)$ is bounded above and below by constants depending only on δ_* . For the upper bound use the old code at accuracy $1/32$, which is also accurate to δ . This proves a uniform additive remainder for both kinds of centres. If $\delta_* \leq 1/32$ there is no remaining range. $\square$

The cap $\delta_* < 2$ is substantive: at unhalved error two every nonempty family is covered by a single legal centre. Nothing above is uniform as $\delta_* \uparrow 2$, or along a varying error sequence converging to two. The exact zero-error endpoint is also separate from the positive-error coding law. The older coherent/preparation theorem retains its own $\delta \leq 1/32$ range; the present common tomography argument gives the half-dimension scale, not its coherent N^{-1} scale.

44. INPUT-DEPENDENT BOUNDARY INSTRUMENTS

Fix an orthonormal basis $e_1, \dots, e_d$ of the input and the associated complete dephasing map $\mathcal{D}(X) = \sum_x \langle e_x, X e_x \rangle |e_x\rangle \langle e_x|$. For positive semidefinite $n \times n$ matrices σ_{xy} satisfying $\sum_y \text{tr } \sigma_{xy} = 1$ for every x , define

$$(44.1) \quad \Phi_{\sigma,y}(X) = \sum_{x=1}^d \langle e_x, X e_x \rangle \sigma_{xy}, \quad 1 \leq y \leq m.$$

These instruments measure a fixed input basis and then prepare an outcome-dependent state. Their public outcome probabilities $\sum_x \langle e_x, X e_x \rangle \text{tr } \sigma_{xy}$ may depend on the input. Measurement destroys input coherences; the family is not in general input erasing, and it is not the coherent tensor family of Theorem 42.1.

The Choi blocks and their ranks are

$$(44.2) \quad J_y = \bigoplus_{x=1}^d \sigma_{xy}, \quad \text{rank } J_y = \sum_x \text{rank } \sigma_{xy}.$$

In particular the marginal constraint is exactly one unit-trace condition per input row. Fix public integers $0 \leq r_{xy} \leq n$, with $\sum_y r_{xy} > 0$ in every row, and let $\mathcal{F}_r$ be the family with $\text{rank } \sigma_{xy} \leq r_{xy}$. Put

$$(44.3) \quad v_x = \sum_y r_{xy}(2n - r_{xy}) - 1, \quad V = \sum_x v_x.$$

The ranks here are conditional output-state ranks, not independent rank bounds on arbitrary Choi blocks. The input basis, dimensions and rank bounds are fixed, not learned from an unknown device.

Theorem 44.1 (Rank-boundary coding with input-dependent outcomes). *Fix the preceding data and $0 < \delta_* < 2$. If $V > 0$, the least covering cardinality by arbitrary legal memoryless centres and the least cardinality with rational decoded Choi blocks both obey*

$$(44.4) \quad \log_2 \mathcal{M}_N(\delta; \mathcal{F}_r) = \frac{V}{2} \log_2 N + V \log_2(1/\delta) + O(1)$$

uniformly for $N \geq 1$ and $0 < \delta \leq \delta_$. The error is the unhalved final-state trace distance over arbitrary common quantum testers with finite references, quantum memory, feedback and public stopping bounded by N . A rational upper code has decoded centres in (44.1); it never increases any conditional block rank and preserves every zero block exactly. Its encoder on rational target data uses integer and rational arithmetic only. If $V = 0$ the family is a known singleton.*

The remainder depends on the fixed dimensions, rank bounds and δ_ , but not on N , δ or a smallest positive output eigenvalue. Payload length is distinct from encoder workspace and expanded decoder storage. The decoded object is a mathematical quantum instrument, not a physical classical simulator on an unknown reference-entangled input.*

44.1. Centres and the adaptive metric. Write $\Omega_x = \bigoplus_y \sigma_{xy}$ for the flagged state of one input row. A useful general observation first identifies the allowed centres.

Proposition 44.2 (Idempotent input retraction). *Let Π be a fixed completely positive trace-preserving idempotent input map and let every target satisfy $\Phi\Pi = \Phi$. Then*

$$(44.5) \quad d_N(\Phi, K\Pi) \leq d_N(\Phi, K)$$

for every legal memoryless centre K and every N . Thus covering numbers with all legal centres equal those with centres fixed by right composition with Π , at every radius. For $\Pi = \mathcal{D}$, the retracted centre has rows $\tau_{xy} = K_y(|e_x\rangle\langle e_x|)$ and is of the form (44.1). In the prescribed coordinate basis this operation preserves rationality. It need not preserve the target's rank bounds.

Proof. For any tester of Φ against $K\Pi$, insert Π on its input immediately before each use of the unknown device. This is a legal common tester of Φ against K : in the first hypothesis $\Phi\Pi = \Phi$, and in the second it implements $K\Pi$. References, feedback and stopping are unaffected. Taking the supremum proves (44.5). Idempotence says that $K\Pi$ is fixed by right composition with Π . Map every covering centre this way; cardinality cannot increase. The reverse covering inequality follows from inclusion of allowed centre classes. For dephasing, the displayed rows are positive and have unit total trace, since K is an instrument. They are diagonal input blocks of the original Choi matrices. $\square$

This is a data-processing retraction, not a rank-preserving projection. It generalizes the preparation-centre observation of Theorem 41.1, whose fixed input map discards its input and supplies one chosen state. In the converse of Theorem 44.1, no rank promise is imposed on centres.

Lemma 44.3 (Row witnesses and a common programme). *For two instruments of (44.1),*

$$(44.6) \quad \max_x \|\Omega_x^{\otimes N} - \Lambda_x^{\otimes N}\|_1 \leq d_N(\Phi_\sigma, \Phi_\tau) \leq \left\| \bigotimes_x \Omega_x^{\otimes N} - \bigotimes_x \Lambda_x^{\otimes N} \right\|_1.$$

If unit purifications a_x, b_x of the corresponding row states are chosen and $\eta_x = \|a_x - b_x\|$, then

$$(44.7) \quad d_N(\Phi_\sigma, \Phi_\tau) \leq 2\sqrt{N \sum_x \eta_x^2}.$$

For one use there is the exact identity $d_1 = \max_x \|\Omega_x - \Lambda_x\|_1$.

Proof. For the first inequality repeatedly feed $|e_x\rangle\langle e_x|$ and retain all flagged outputs. For the second, prepare, for each of the N slots, one copy of every row state. A common processor measures the actual input in the prescribed basis and uses only the programme state with that outcome x . The remaining row states in that slot are discarded. If the input is entangled with the tester's memory, its conditional memory operator is retained by this measurement; appending the selected row state gives exactly (44.1) on that joint input. The unobserved input outcome is internal to the processor, not new feedback granted to the tester. All later operations and the discard of unused slots at a stop form one common CPTP processor. Trace contraction proves the upper bound.

Purify all these programme states by the chosen vectors. For unit vectors, $1 - |\langle a_x, b_x \rangle|^2 \leq \eta_x^2$. The elementary product inequality $1 - \prod t_j \leq \sum (1 - t_j)$ for $t_j \in [0, 1]$ gives pure-programme trace distance squared at most $4N \sum_x \eta_x^2$. Partial trace and the common processor give (44.7). Finally, for an arbitrary one-use input with reference, the difference is $\sum_x \xi_R^x \otimes (\Omega_x - \Lambda_x)$, where $\xi_R^x \geq 0$ and $\sum_x \text{tr} \xi_R^x = 1$. The triangle inequality gives the one-use upper maximum, and a basis input attains it. $\square$

The product programme consumes up to dN row states in the proof, whereas the actual tester makes at most N instrument calls. It is only a comparison device, not an equality of these resource counts. Neither inequality in (44.6) is asserted to be an equality for general N . In particular the absence of an asymptotic adaptive advantage in a binary discrimination exponent does not imply a finite-use metric equality; compare [49, Sections II and V].

44.2. Exact rational decoding and the covering proof.

Lemma 44.4 (A common-grid row code). *For rational target data, apply the singular factor encoder of Theorem 40.2 separately in every input row at a common integer grid $B \geq 1$. If h_x is that row's actual real factor-coordinate count, the decoded instrument satisfies*

$$(44.8) \quad d_N(\Phi, \tilde{\Phi})^2 \leq \frac{16N \sum_x (h_x - 1)}{B^2} \leq \frac{16NV}{B^2}.$$

There are at most $C_{d,n,m,r}(2B+1)^V$ legal decoder images. The conditional block ranks do not increase; consequently no outcome Choi rank increases either. All zero conditional blocks remain zero. The expanded common denominator is at most $B^{2d} \prod_x h_x$ and thus has $O_{d,n,m}(1 + \log B)$ bits.

Proof. The inherited anchor construction gives a unit factor purification in row x within distance $2\sqrt{h_x - 1}/B$ of the target's chosen unit factor vector. Apply (44.7). Trace normalization is performed separately in each row, so (44.2) is exactly trace preserving. The inherited zero-pivot proof, rational squared coordinates and integer-square-root ratio calculation apply row by row. Their finite masks and anchor headers have fixed cardinality; the row body has at most v_x signed grid coordinates. Multiplying these counts gives the asserted exponent V . Only valid decoded words count. The block-rank and zero statements hold row by row and pass to the direct sum in (44.2). Row x has normalization denominator $T_x \leq h_x B^2$. A common denominator is the least common multiple of the T_x , bounded by their product. This proves the final assertion. $\square$

Proof of Theorem 44.1. For the upper bound take $B = \max\{1, \lceil 4\sqrt{NV}/\delta \rceil\}$. The last lemma gives error at most δ and at most $C(2B+1)^V$ centres, hence the upper description law for $\delta \leq \delta_* < 2$. Real targets admit the same factor rounding construction; the decoded matrices are still rational. For supplied rational targets, no irrational factor is stored: the exact inherited square-comparison algorithm determines the digits. Thus the real-family rational-centre upper bound does not restrict the targets to a countable subclass.

For the lower bound choose, in every row with $v_x > 0$, the gauge-fixed maximal-rank factor patch used in Theorem 40.3. Its inverse is the local leading-block Cholesky map. Rows with $v_x = 0$ are fixed. A Cartesian product of sufficiently small closed coordinate balls contains a V -dimensional ball. The map to the tuple (σ_{xy}) , and hence to J , is bi-Lipschitz; direct sum is an isometry for tuple Frobenius norm. This supplies one fixed V -dimensional patch with no claim of a global factor chart. For $\delta \leq 1/32$ pack it at separation greater than $32d\delta/\sqrt{N}$. Corollary 39.2 gives distances larger than 2δ , so even an unrestricted legal centre covers at most one packing point. This gives $c(\sqrt{N}/\delta)^V$ centres. For $1/32 < \delta \leq \delta_*$ use Theorem 43.3 on the same patch and the boundedness of $\log(1/\delta)$ on that interval. These lower bounds apply also to rational centres, which form a smaller class. Taking logarithms, with the usual ceiling for fixed-length words, completes the proof. If $V = 0$, each nonempty row consists of a specified one-dimensional unit block; its location is fixed by the public rank bounds. Thus the whole family is a known singleton. $\square$

For example, with scalar output $n = 1$ and all rank bounds equal to one, (44.4) has $V = d(m-1)$: it is a boundary coding law for the rows of a classical stochastic matrix. With one quantum output and full row ranks, $V = d(n^2 - 1)$; with one pure output per row, $V = d(2n - 2)$. Mixed and pure conditional states need not commute. These dimensions are standard rank-stratum dimensions; the theorem identifies their joint coding scale under the declared adaptive metric. The measured input basis is fixed. No law for varying measurement bases, unknown instruments, or all irreversible quantum dynamics follows from this result.

45. COMPARISON AND BOUNDARY GEOMETRY

45.1. Adaptive discrimination and description. Salek–Hayashi–Winter [49, Sections II–V] study asymptotic binary channel-discrimination exponents. Their results include equal adaptive and nonadaptive exponents for classical–quantum channels, and separations for some entanglement-breaking channels. Our object is instead a covering number of a supplied family in the finite-use metric d_N . Its lower packing may use a different binary test for each pair; it does not supply a universal measurement which identifies every codeword. The exact reduction in Lemma 40.1 uses input erasure, not merely entanglement breaking, and implies no general absence of an adaptive advantage for entanglement-breaking channels.

Cooney–Mosonyi–Wilde [50] is a direct antecedent for the input-erasure mechanism. Its two-replacer discussion and asymptotic channel-versus-replacer exponents are distinct from an exact finite-use covering problem. The retraction of Theorem 41.1 concerns admissible centres, not an improvement to those discrimination exponents. The coherent amplification used here likewise has the established unitary discrimination antecedents [51, 52].

The two-point programme method has a metrological antecedent in [47]. Likewise the half-parameter scale has antecedents in quantum population compression [48]; there the input consists of unknown physical copies and the memory interface may include quantum storage. Here the encoder is given classical matrix data and the decoder specifies an operation or state. These are different coding tasks, even when a logarithmic coefficient has the same form.

Mele–Bittel [45] and Oufkir–Girardi [46] investigate learning unknown channels in diamond distance. The present encoder receives its target and asserts no query or sample bound. The reference-sensitive physical classical-simulation problem of Naik–Zartab–Gisin–Banik [44] is different again: a rational description with a diamond guarantee remains a description of a quantum operation, not a finite-classical-message implementation accepting an unknown entangled input. No bound here is asserted for the hardware dimension of a universal programmable processor.

45.2. Ranks, tangent directions and remaining joint regimes. For a general instrument J , let P_y project onto the support of J_y . The one-sided tangent cone of the legal instrument body is

$$T_{\mathcal{C}}(J) = \overline{\{t(L - J) : t \geq 0, L \in \mathcal{C}\}}.$$

Every element H of this cone satisfies the marginal constraint and $P_y^\perp H_y P_y^\perp \geq 0$. These necessary conditions identify where positivity becomes active; by themselves they assert no reuse modulus. A boundary classification must examine the metric of repeated use along support-changing directions and along reversible coherent directions, rather than count affine coordinates alone. Proposition 38.5 shows that a coherent unitary phase can have scale N^{-1} , whereas the preparation family in Theorem 40.3 has an $N^{-1/2}$ coding scale even at rank-deficient outputs. In that preparation family the Choi ranks are d rank σ_y , and the rank bounds in the theorem are output ranks. They are not arbitrary Choi-rank strata of all instruments.

Theorem 42.1 now gives a joint law in the non-input-erasing tensor family $UXU^* \otimes \sigma_y$. It includes a full projective unitary manifold and support-changing preparation blocks, but no input-dependent measurement probabilities. Its mixed coefficient $u + v/2$ is established by two explicit adaptive moduli and matching product packings, not by a tangent-dimension count alone. These qualifications are essential: it is not a classification of all coherent/noisy couplings.

Theorems 39.3 and 40.3 settle the joint N, δ dependence in their specified fixed-dimensional families. They do not assert dimension-uniform constants, a uniform limit as the positive margin tends to zero, or a full tangent-cone classification for general disturbing instruments. The older exponential-accuracy width result retains its multiplicative gap in the separate label-width resource. None of these description laws is a mutable-workspace converse for general variable-input trajectory simulation.

45.3. Input dependence and a common full-range witness. The fixed-readout family of Theorem 44.1 is a classical–quantum channel with a fixed quantum input measurement. Its N -slot programme selects conditional row states and therefore handles input-dependent probabilities. This is within the channel families considered by Salek–Hayashi–Winter [49, Sections II and V] for asymptotic binary discrimination. Their exponent comparisons are direct antecedents, but are not a finite-use family covering number or a rational rank-aware code. We do not infer equality between the row-product upper bound and the adaptive distance, nor a finite-use prohibition on adaptive advantage. The outcome/state rows may be noncommuting and have zero blocks. The measured basis remains fixed, so a varying measurement basis and general coherent/noisy couplings are not parameterized by V .

The common-estimator argument in Section 43 uses standard informationally complete tomography [42] and an elementary event-counting inequality. Its point is to control covering multiplicity at error greater than one, where the earlier requirement of pairwise distance greater than 2δ is unavailable. This new witness is deliberately common to all packing points; the local small-error proof still uses pair-dependent binary tests. Neither witness turns the supplied-description code into an unknown-instrument learning result. The full-range extension concerns half-dimensional coding scales and does not extend the coherent N^{-1} scale by fiat. The author-side comparisons here identify premises and conclusions; they do not constitute an independent exhaustive priority review.

46. A RATIONAL COMPILER FOR ADAPTIVE OBSERVATION PROCESSES

The common-row construction also yields a causal approximation, but its proof requires control of the full observation law. Correct terminal means alone are not enough. We give that additional argument here.

The physical state is $x \in B_3$, initialized at a supplied rational $x_0 \in S^2$. A finite rational rotation list $\{A_a\} \subset SO(3)$ acts by $x \mapsto A_a x$ and emits a null symbol. A finite probe list has rational affine

probability laws

$$(46.1) \quad q_{j,y}(x) = b_{j,y} + \langle \ell_{j,y}, x \rangle, \quad \sum_y b_{j,y} = 1, \quad \sum_y \ell_{j,y} = 0, \quad b_{j,y} - \|\ell_{j,y}\| \geq \eta > 0.$$

A probe emits a fresh draw and leaves x unchanged. Controls may depend on the entire observed transcript. A simulator has finite retained labels and a nonnegative joint emission/transition matrix for each control and output; these matrices may depend on horizon and epoch. Let $\text{PW}_{N,\varepsilon}$ denote its minimum peak label count for one total-variation budget ε on the entire transcript, uniformly over all adaptive randomized policies. This paragraph defines a causal model independently of the terminal matrix-output model.

Put $A = \max_j \sum_y \|\ell_{j,y}\|^2$ and $C_0 = \max(3, \max_j (M_j - 1))$. The lower probability bound in (46.1) is used only for the relative-entropy upper estimate, not for the finite-testing converse below.

Theorem 46.1 (Adaptive rational and fair-bit realization). *For rational $0 < \varepsilon < 1$ and integer $N \geq 0$, choose*

$$(46.2) \quad m = \max \left(2, \left\lceil \sqrt{\frac{4AN^2}{\eta\varepsilon^2}} \right\rceil \right), \quad r = 1 - m^{-2}, \quad b = \max \left(0, \left\lceil \log_2 \frac{2C_0 \max(1, N)}{\varepsilon} \right\rceil \right).$$

There is an error- ε causal realization on at most $13m^2 + 1$ labels. Before probability rounding every rotation row is rational, has at most four nonzero entries, and has barycentric mean $rA_a v$. Probe emissions leave the label unchanged. After rounding, all emission and transition probabilities are dyadic, with at most b fair bits per control and Nb bits in total. The tables are stationary within the chosen horizon.

An exact rational algorithm constructs these tables in time polynomial in the numerical parameters $N, \varepsilon^{-1}, A/\eta$, the explicit input size, and the input bit lengths. In particular, for fixed probe and rotation lists,

$$(46.3) \quad \text{PW}_{N,\varepsilon} = O(1 + N^2/\varepsilon^2).$$

The bit count now includes the actual finite probe outputs as well as hidden transitions; it does not describe quantum-state preparation.

Proof. Take the rational net V_m from Lemma 26.1, including zero, and add x_0 if needed. This adds at most one label and cannot shrink the convex hull. Initialize deterministically at x_0 . For every rotation and label v , the same exact triple search supplies a row with mean $rA_a v$. At probe j , emit y with probability $q_{j,y}(v)$ and retain v . All points lie in B_3 .

Fix an arbitrary policy. On the same finite trajectory space of controls, outputs and hidden labels, introduce an auxiliary law P : hidden labels follow the just constructed rotation kernels, but every probe emits $q_{j,y}(X)$ from the true physical state X , independently of the hidden label conditional on the public history. Its public marginal is exactly the target experiment. Let Q be the constructed simulator law. The policy kernels, initial label and hidden rotation kernels are the same under P, Q ; only probe emissions differ.

Under P , if the current physical and hidden points are X, V , both X and every nonzero hidden vertex are unit vectors. At a rotation step, for $V \neq 0$,

$$\begin{aligned} \mathbb{E}[\|V' - A_a X\|^2 \mid X, V, a] &\leq 2 - 2r\langle V, X \rangle \\ &= r\|V - X\|^2 + 2(1 - r). \end{aligned}$$

For $V = 0$ choose its row to remain at zero; the same upper bound holds because the left side is one and the right side is $2 - r \geq 1$. The added seed is a unit vector, so these two cases exhaust

the state set. This is why a general interior seed has not been silently added to the theorem. A probe changes neither point. Iteration, with the control selected adaptively, gives

$$(46.4) \quad \mathbb{E}_P \|V_t - X_t\|^2 \leq 2t/m^2.$$

The inequality is unconditional; no posterior error bound after an unlikely observation is assumed.

For probability vectors p, q with $q_y \geq \eta$, the inequality $\log z \leq z - 1$ gives

$$D(p\|q) \leq \sum_y (p_y - q_y)^2 / q_y.$$

Consequently the conditional divergence at a probe is at most $A\|X - V\|^2/\eta$. The finite-trajectory relative-entropy chain rule cancels the common policy and rotation kernels. Applying (46.4) at the probe times gives

$$(46.5) \quad D(P\|Q) \leq \frac{2AN^2}{\eta m^2}, \quad \text{TV}(P_{\text{pub}}, Q_{\text{pub}}) \leq \sqrt{D(P\|Q)/2} \leq \frac{N}{m} \sqrt{A/\eta} \leq \varepsilon/2.$$

Here marginalization can only decrease divergence. These elementary entropy facts apply to the joint classical trajectory, not to a quantum state. One may verify the total-variation inequality by partitioning into the set where $P \geq Q$ and its complement: the log-sum inequality reduces divergence to a Bernoulli divergence, whose second derivative in its first parameter is at least four. This yields $D(P\|Q) \geq 2\text{TV}(P, Q)^2$. The chain rule follows by factoring each trajectory probability into its conditional kernels. All probe probabilities in Q are positive, so absolute continuity causes no exception. If $A = 0$, every probe is constant and the divergence is zero.

Round each rotation row and each probe row by rounding all but its last positive mass down to denominator 2^b and assigning the remainder to the last. A row supported on d points changes in total variation by at most $(d-1)2^{-b}$. Thus every one-step joint kernel changes by at most $C_0 2^{-b}$. Coupling the two label-transcript processes until their first discrepancy proves, for every adaptive policy, an additional path error at most $NC_0 2^{-b} \leq \varepsilon/2$. Combining this with (46.5) proves the asserted error. Each rounded row can be sampled by one uniform b -bit integer and cumulative integer masses; zero or deterministic rows may use fewer bits.

The net and an added rational unit seed are explicit. For every command-vertex pair the exact enumeration in Lemma 26.1 examines at most $\binom{k-1}{3}$ triples, where $k \leq 13m^2 + 1$. The total is $O(|\mathcal{A}|k^4)$ constant-size rational linear-algebra operations, in addition to the probe tables. All determinants, comparisons and row identities involve rational numbers with polynomially bounded bit lengths in the supplied data and $\log m$; dyadic rounding adds b bits. The computation of m and b can use integer comparisons, not floating-point decisions. This proves the stated uniform, numerical-parameter complexity. It is a construction, not a minimum-width algorithm, and its exhaustive fallback need not be practical for large horizons. $\square$

46.1. An effective process converse for the rational free experiment. Keep the rational rotations and seed of (25.2)–(25.3), now exposing the three nondisturbing binary probes

$$(46.6) \quad q_{j,+}(x) = (1 + \lambda x_j)/2, \quad q_{j,-}(x) = (1 - \lambda x_j)/2, \quad j = 1, 2, 3,$$

where $0 < \lambda < 1$ is rational. This defines a fixed rational classical experiment. The probing law is not an implementation of nondisturbing quantum measurement. Let PW refer to the causal model above, not to terminal density-matrix width W .

Theorem 46.2 (Effective repeated-observation lower bound). *Let $0 \leq \varepsilon < 1$ and $0 < \delta < 1 - \varepsilon$. For an integer $t \geq 0$ put*

$$(46.7) \quad L_t = \max \left(1, \left\lceil 18\lambda^{-2} 625^t \log(6/\delta) \right\rceil \right).$$

At every horizon $N \geq t + 3L_t$, any error- ε causal machine has

$$(46.8) \quad |S_t| \geq (1 - \varepsilon - \delta)(2 \cdot 3^t - 1).$$

If $\varepsilon + \delta < 1/2$, it has the stronger lower bound $|S_t| \geq 2 \cdot 3^t - 1$. More generally, the same bounds hold at every cut u with $t \leq u \leq N - 3L_t$ after padding the prefixes by probes. In particular, for each fixed $0 < \varepsilon < 1$, there are explicit positive constants $c_{\varepsilon, \lambda}, C_{\varepsilon, \lambda}$ such that, for all sufficiently large N ,

$$(46.9) \quad c_{\varepsilon, \lambda} N^{\log 3 / \log 625} \leq \text{PW}_{N, \varepsilon} \leq C_{\varepsilon, \lambda} (N + 1)^2.$$

The converse uses no spectral-gap theorem. Its exponent is not asserted to be sharp.

Proof. The rational freeness and stabilizer argument in Theorem 25.2 gives $2 \cdot 3^t - 1$ different reachable pure Bloch vectors at time t . Their coordinates lie in $25^{-t}\mathbb{Z}^3$. Hence each distinct pair differs in some coordinate by at least 25^{-t} , and the associated plus-probabilities in (46.6) differ by at least $\lambda 25^{-t}/2$. Use one common continuation: repeat each of the three probes L_t times. Hoeffding's elementary Bernoulli bound and a union bound give

$$\mathbb{P}_x \left\{ \max_j |\hat{q}_j - q_{j,+}(x)| \geq \lambda 25^{-t}/6 \right\} \leq 6 \exp(-L_t \lambda^2 625^{-t}/18) \leq \delta.$$

The empirical neighborhoods for the different targets are disjoint. At a cut label z , let Q_z be the law of this same continuation. Marginalizing prefix outputs and using total-variation correctness shows that the continuation law from each prefix gives its own neighborhood probability at least $1 - \varepsilon - \delta$. Writing this law as $\sum_z \alpha_x(z) Q_z$ and summing over the disjoint neighborhoods yields (46.8). If the probability exceeds $1/2$, each neighborhood needs a different label assigning it probability greater than $1/2$, giving the stronger conclusion. The proof does not condition on the success of a previous observation. Padding by arbitrary fixed probes leaves the target states unchanged and proves the assertion at later cuts. Unused final controls may also be probes. A terminal cut without enough remaining observations is not claimed to satisfy this lower bound.

For the power lower bound take $\delta = (1 - \varepsilon)/2$ and $B = 54\lambda^{-2} \log(6/\delta) + 4$. Then $t + 3L_t \leq B625^t$. For $N \geq B$, choose $t = \lfloor \log_{625}(N/B) \rfloor$. Since $3^t \geq (N/B)^{\log 3 / \log 625}/3$ and $2 \cdot 3^t - 1 \geq 3^t$, (46.8) gives, for example, $c_{\varepsilon, \lambda} = (1 - \varepsilon)B^{-\log 3 / \log 625}/6$. The upper bound follows from Theorem 46.1 with $\eta = (1 - \lambda)/2$ and $A = \lambda^2/2$. For irrational requested ε , construct at any rational tolerance between $\varepsilon/2$ and ε . All constants can thus be bounded from the displayed data without computing a group spectral gap. $\square$

The tests in this theorem consume horizon. The full-state terminal exponential profile and the causal lower bound therefore express different requirements. Conversely, the causal upper construction controls one joint transcript law under adaptive control, not just its one-time marginals. These process statements do not by themselves sharpen the terminal spectral-gap converse. Its logarithmic loss is removed separately by Theorem 15.5.

APPENDIX A. AN EXISTENTIAL HIGHER-DIMENSIONAL ENDPOINT

A.1. A fixed nonabelian family with an exponential exact endpoint. The next example uses two established inputs, explicitly separated from the realization argument. Breuillard and Gelander prove that a dense subgroup of a connected semisimple real Lie group contains a dense free subgroup on two generators [17, Theorem 1.1]; their later correction supplies repairs and details for other parts of the original proof [18]. The spectral-gap input for dense algebraic unitary generators is [16]. No algorithm for finding free generators, or for certifying a spectral-gap constant, is asserted here.

Lemma A.1 (A line with trivial algebraic projective stabilizer). *Fix a real number τ transcendental over $\mathbb{Q}$ and put*

$$v_\tau = \frac{(1, \tau, \dots, \tau^{d-1})^\top}{(\sum_{j=0}^{d-1} \tau^{2j})^{1/2}}, \quad P_\tau = v_\tau v_\tau^*.$$

An invertible matrix with algebraic entries that preserves the line $\mathbb{C}v_\tau$ is scalar.

Proof. If $Mv_\tau = \lambda v_\tau$, the eigenvalue λ is algebraic because it is a root of the characteristic polynomial of the algebraic matrix M . Each row of $(M - \lambda I)(1, \tau, \dots, \tau^{d-1})^\top = 0$ is a polynomial identity in τ with algebraic coefficients. Transcendence over $\mathbb{Q}$ implies transcendence over the field of algebraic numbers: an algebraic relation over a finite number field would make τ algebraic over $\mathbb{Q}$ by transitivity of algebraic extensions. Every row polynomial therefore has all coefficients zero. Thus $M = \lambda I$. $\square$

One may take a fixed computable transcendental such as $\tau = \sum_{n=1}^{\infty} 10^{-n!}$. For completeness, its truncations have rational denominator $10^{m!}$ and strictly positive tail at most $2 \cdot 10^{-(m+1)!}$. For a nonzero integer polynomial A of degree D , a rational p/q not among its finitely many roots satisfies $|A(p/q)| \geq q^{-D}$. If $A(\tau) = 0$, a bounded derivative near τ would instead bound $|A(p/q)|$ by a constant times $q^{-(m+1)}$ along these truncations, a contradiction for large m . This also records precisely why the chosen seed is not part of the rational-input specialization.

Theorem A.2 (Exponential exact width and polynomial noisy width). *For every $d \geq 2$, with a transcendental seed and existentially selected generators, there is a fixed five-letter algebraic unitary alphabet*

$$\{I, U, U^{-1}, V, V^{-1}\} \subset SU(d)$$

and a fixed rank-one seed P_τ as above such that the full matrix interface $F(g) = gP_\tau g^$, with legal decoder set $\mathcal{D}_d$, has*

$$(A.1) \quad W_{N,0} = 2 \cdot 3^N - 1 \quad (N \geq 0).$$

For each fixed $0 < \varepsilon < r_d = \sqrt{1 - 1/d}$, its Frobenius-error width instead satisfies

$$(A.2) \quad c_{d,\varepsilon} \left(\frac{N}{\log(N+2)} \right)^{d-1} \leq W_{N,\varepsilon} \leq C_{d,\varepsilon} (N+1)^{d-1}.$$

At $\varepsilon \geq r_d$ one label suffices. For the same alphabet and seed, every fixed amplitude $0 < a < 1$ admits the exact polynomial upper bound in (17.9) and the corresponding lower bound in (17.8).

Proof. The subgroup of $SU(d)$ consisting of matrices with algebraic entries is dense. One direct verification is to approximate a unitary matrix by a nonsingular matrix with entries in $\mathbb{Q} + i\mathbb{Q}$, apply Gram–Schmidt, and multiply its final column by the inverse determinant of the resulting unitary matrix. All entries produced are algebraic; continuity near the original unitary matrix gives density. Apply [17, Theorem 1.1] to this subgroup. It contains algebraic U, V freely generating a dense free group Γ of rank two. The scalar matrices in $SU(d)$ form a finite group. Since a free group is torsion-free, Γ contains no nonidentity scalar matrix. Lemma A.1 now shows that the orbit map $\Gamma \rightarrow \mathcal{D}_d, g \mapsto gP_\tau g^*$, is injective.

With the identity available for padding, the products at length N are exactly the elements of reduced free-word length at most N . There are four choices for the first letter of a nonempty reduced word and three for each subsequent letter. Thus the ball has cardinality

$$1 + 4 \sum_{j=1}^N 3^{j-1} = 2 \cdot 3^N - 1.$$

Apply Theorem 18.1 to obtain (A.1). In fact the entire pointwise minimum profile is $|S_t| = 2 \cdot 3^t - 1$.

Assign a symmetric positive law to the four nonidentity commands and positive mass to the identity. The algebraic dense-generator theorem [16], with laziness, gives (14.1). Consequently Theorem 17.3 applies to this same fixed alphabet and seed. It yields (A.2), the threshold r_d , and the asserted strict-amplitude conclusion. $\square$

This is a full legal-matrix-output statement. It does not resolve the zero-error unit-amplitude problem for a single scalar planar readout: that readout need not expose every orbit point as an extreme decoder value. Nor does it identify a sampled measurement outcome with a density matrix stored in a classical label. The distinction between a prescribed convex output set and a larger simplex is essential here.

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